

WALK. BIKE. CONNECT

glendale **active** **transportation plan**

the way we move

Glendale
ARIZONA

Final Report
June 2019

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Project Leadership Team

Patrick Sage, AICP, Project Manager
Purab Adabala, Transportation Planning Administrator
Marilu Garcia Ramirez, Transportation Project Manager
Trevor Ebersole, Director of Transportation
Dory Ludwig, Traffic Education Program Manager
Debbie Albert, PE, City Traffic Engineer
Kiran Guntupalli, PE, Principal Engineer

City of Glendale Departments

Attorney's Office
Community Services
Economic Development
Parks and Recreation
Planning
Transportation
Glendale Citizen Active Transportation Advisory Committee
Glendale Transportation Oversight Committee

Study Partners

- Arizona Department of Transportation
- City of Avondale
- City of Buckeye
- City of El Mirage
- City of Goodyear
- City of Litchfield Park
- Maricopa Association of Governments
- Maricopa County Department of Public Health
- Maricopa County Department of Transportation
- City of Peoria
- City of Phoenix
- City of Surprise
- Valley Metro

Public Participants

Thank you to the residents of the City of Glendale for their participation in this planning process and their passion for improving the place they call home.

Consultant Team

JACOBS

Phyllis Davis, Project Manager
Vamshi Yellisetty
Kyle Benne
Alex Quintero
Andrew McIntyre

Y2K ENGINEERING

Yung Koprowski, PE

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1. PLAN OVERVIEW

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In 2018, the City of Glendale (Glendale) was awarded funding through Maricopa Association of Government (MAG) to complete the City's first Active Transportation Plan (ATP). The ATP outlines a vision to help the City develop a safe and connected active transportation network to encourage healthy and efficient travel throughout the City and region.

Glendale Overview

Originally settled in the late 1800s and incorporated in 1910, Glendale has evolved into a thriving, full-service community with historic charm, beautiful neighborhoods, robust recreational amenities, and some of the nation's finest sports and entertainment facilities. Located in the Phoenix metropolitan area's "West Valley", Glendale is in one of the fastest growing areas in the nation, Maricopa County, and is Arizona's sixth largest city. Glendale's Municipal Planning Area (MPA) covers nearly 92 square miles and borders six different jurisdictions. Approximately 32 percent of the MPA is located within unincorporated Maricopa County.

GLENDALE AT A GLANCE

- Founded in 1892 by William John Murphy
- Sixth largest city in Arizona
- Sports mecca home to the NFL's Arizona Cardinals, NHL's Arizona Coyotes, annual Fiesta Bowl, and the BCS Championship Game
- Home to more than 70 parks, totally over 2,000 acres of park and conservation land
- Known as the Arizona's "Antique Capital"

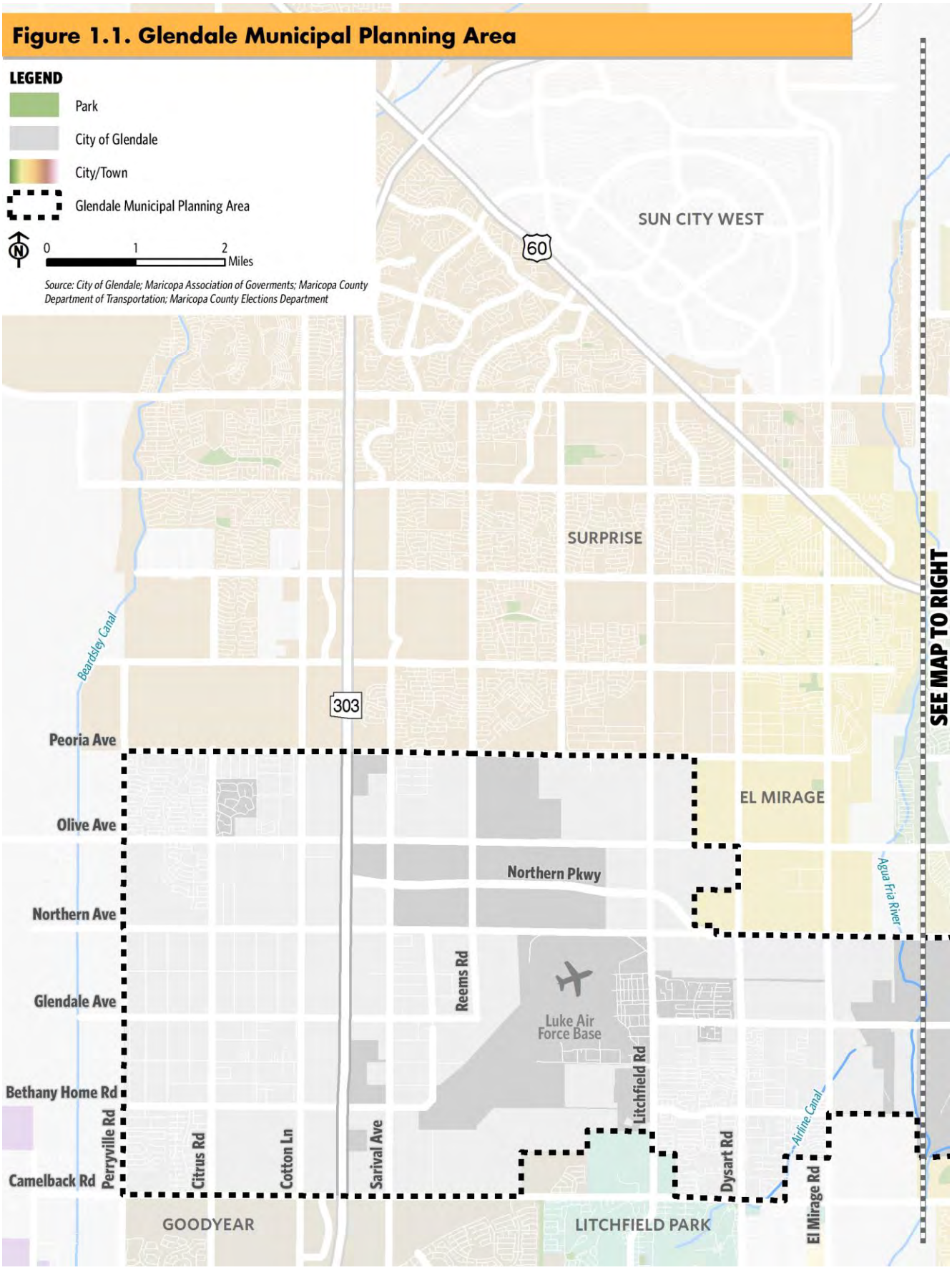
As illustrated in Figure 1.1, the Glendale ATP study area includes the entire Glendale MPA. The ATP primarily focuses on transportation facilities owned, operated, and/or maintained by Glendale and how those facilities connect to and intersect with facilities located in other agencies. Other agencies may include state, local and neighboring jurisdictions, and quasi-government agencies operating facilities adjacent to or crossing Glendale roadways.

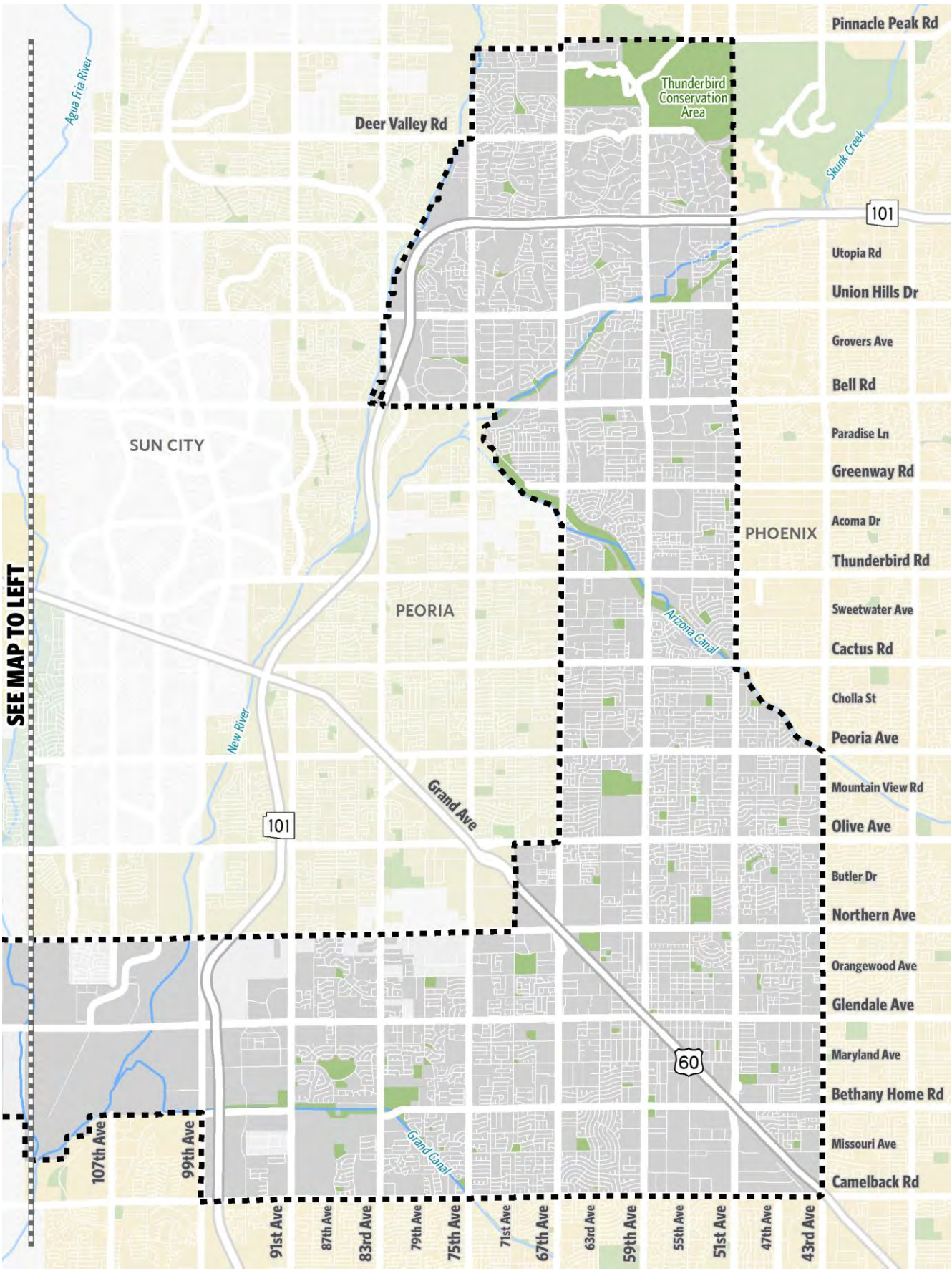
Purpose of the ATP

Glendale initiated the ATP to develop a comprehensive planning guide that outlines a blueprint for a complete and accessible active transportation network for all people regardless of their age or ability. The ATP ultimately serves as an action plan that guides decisions and investments for improving, expanding, and connecting active transportation within Glendale. The actions and strategies identified in the ATP enable Glendale to strategically overcome active transportation barriers, support growth, and increase transportation options.

The ATP supersedes the 1990 *Bicycle Plan of the City of Glendale.* While the 1990 Bicycle Plan focused only on creating bicycle infrastructure improvements, the ATP explores pedestrian, bicycle, equestrian, and transit facility needs to accommodate all users regardless of their age or ability. Since the 1990s, Glendale has worked to enhance and expand its active transportation network; however, new developments and changing travel behaviors are altering how and why people travel. Careful and strategic planning is needed to ensure that Glendale's active transportation network meets current and future needs.

Figure 1.1. Glendale Municipal Planning Area




Pinnacle Peak Rd
Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
Skunk Creek
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
SUN CITY
Paradise Ln
Greenway Rd
PHOENIX
Acoma Dr
Thunderbird Rd
SEE MAP TO LEFT
PEORIA
Arizona Canal
Sweetwater Ave
Cactus Rd
Cholla St
Peoria Ave
New River
Grand Ave
101
Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
60
Maryland Ave
Bethany Home Rd
Grand Canal
Missouri Ave
Camelback Rd
107th Ave
99th Ave
91st Ave
87th Ave
83rd Ave
79th Ave
75th Ave
71st Ave
67th Ave
63rd Ave
59th Ave
55th Ave
51st Ave
47th Ave
43rd Ave

What is Active Transportation?

Active transportation includes any self-propelled, human-powered mode of transportation that engages people in active participation, including walking, biking, jogging, skateboarding, inline skating, and the use of assistive mobility devices. Just as roads connect motorists to destinations, active transportation networks allow the public to travel without the use of a car. Often referred to as "non-motorized transportation" the term active transportation is preferred because it suggests a stronger connection between transportation choices and healthy lifestyles.

Active transportation is an important component to the overall mobility of a community and region, providing a low-cost and accessible form of transportation. Investing in walking, biking, and other forms of active transportation results in a more balanced, accessible transportation system. Active transportation benefits also include the alleviation of socioeconomic and health disparities, the support of economic diversity and prosperity, and assists in creating a more livable and sustainable community.

MOBILITY
Provides transportation choices for persons, regardless of age, sex, ability, or economic status

COMMUNITY
Improves livability, connects communities, and fosters economic development

HEALTH
Enables people to incorporate physical activity into their daily lives

COMFORT
Fosters active travel through connected and comfortable facilities

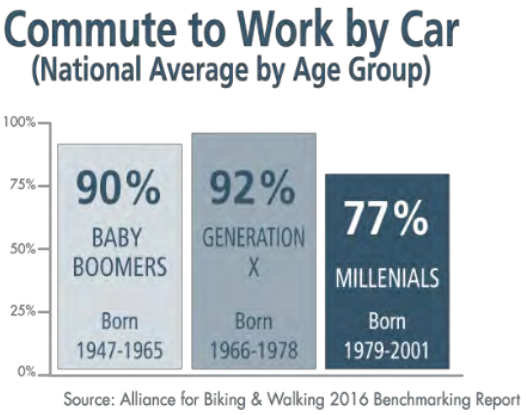
ENVIRONMENT
Lowers vehicle dependency and in turn improve air quality

Why Active Transportation is Important

There are many benefits from active transportation that contribute positively to our quality of life. Infrastructure investments in trails, sidewalks, traffic-calming, and public transit support active travel and increase routine physical activity, which improves health, the quality of the environment and livability of communities. The following are just a few of the benefits of active transportation.

Transportation Choices

Emerging technologies, social trends, and travel behaviors are altering how people travel. Several studies have shown that millennials (those born between 1979 and 2001) are driving less, owning fewer cars, and/or not getting a driver's license. The 2017 National Household Travel Survey found that 42 billion miles were walked and biked in the US – imagine roadway congestion if those miles had been driven instead!



Source: Alliance for Biking & Walking 2016 Benchmarking Report

Economic Benefits

Active transportation investments provide numerous economic benefits including lower transportation costs for individuals; savings to public agencies and jurisdictions from less wear and tear on streets; a greater ability for public agencies and jurisdictions to attract new residents and employers; and a potential boost in tourism. The Arizona Department of Transportation's "Economic Impact Study of Bicycling in



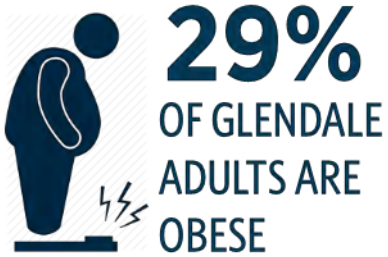
Source: AAA Cost of Driving (2017)

Arizona" reported that $88 million in economic benefits are generated each year from out-of-state bicycle tourists visiting Arizona. In Lodi, California streetscape improvements attracted 60 new stores to the area, lowering the vacancy rate to 6 percent from 18 percent and resulting in a 30% increase in sales tax revenues!



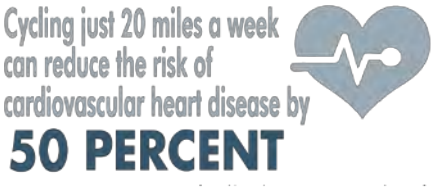
Community Livability

Baby boomers, retirees, and millennials are increasingly moving to locations where they can walk or ride a bike to access their daily needs. Research conducted by the Bureau of Transportation Statistics shows that nearly two-thirds of homebuyers consider the walkability of an area in their purchase decision. Furthermore, employers are increasingly looking for locations that offer their employees walkable and bikeable communities in which to live, work, and play.



Active Lifestyles and Healthy Community

Public health officials recognize the connection between mental and physical health and the built environment. Lack of physical activity is associated with increased risk of many health problems, particularly obesity, diabetes, and heart disease. By definition, active transportation allows people to integrate physical activity into everyday life by enabling them to walk or bike to their destinations. Implementing walking and biking facilities creates access to places where residents can be physically active and provides more opportunities for social interaction and community cohesion that have positive impacts for mental health.





Environmental

By providing facilities for people to walk or bike instead of traveling by vehicle, active transportation can help address a number of environmental challenges. Research shows that approximately 60 percent of vehicle pollution happens within the first few minutes. Replacing these short vehicle trips with walking and biking trips can not only reduce car related emissions, but also reduce noise pollution and congestion. Other environmental benefits include energy savings, less water pollution, reduced dependency on fossil fuels, and even reduced pressure to develop agricultural and open spaces.

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How We Plan

The development of the ATP began in February 2018 and was completed in the following six steps:

- **Developed the Goals and Objectives.** Using input and feedback from Glendale staff, the Glendale Citizens Active Transportation Advisory Committee (CATAC), and the public, goals and objectives were established for the ATP.
- **Understand Current Conditions.** Conducted a comprehensive inventory of existing land use, socioeconomics, safety, roadway, and active transportation conditions within the study area.
- **Identified System Issues, Needs, Gaps and Opportunities.** Identified gaps, assessed pedestrian and bicycle friendliness of the network, and predicted potential nonmotorized travel demand through a comprehensive system assessment.
- **Identified and Evaluated Potential Improvement Solutions.** Identified, evaluated, and prioritized issues, needs, and potential facility improvements.
- **Developed Phased Improvement Plan.** Compiled final recommendations into a phased implementation plan.



The development of the ATP was framed by the League of American Bicyclist's six "E's": Engineering, Education, Encouragement, Enforcement, Evaluation, and Equity. Community feedback also guided the overall direction of the ATP. The study included collaboration with a group of stakeholders and local public agency representatives. To engage the public and stakeholders, outreach including stakeholder workshops, open houses, online outreach in the form of a web map was conducted.

Who We Plan For

We plan for everyone. The ATP examines facility needs to accommodate all user types and levels of comfort, including children, commuters, those without access to a vehicle, casual bicyclists, recreational walkers, and other vulnerable users.

Pedestrians

Walking is the most common form of transportation, as every trip begins and ends by foot. At some point in the day, everyone is a pedestrian. The range of pedestrian abilities is fairly straightforward and can be categorized into the following:



UTILITARIAN WALKER

Walking to complete daily errands, such as shopping, medical appointments, to visit friends/family, etc. Utilitarian walkers need:

- Connected and accessible sidewalks and paths
- Direct connections from residential areas to shopping and services
- Access to transit centers and bus stops
- Shelter and shade at bus stops



COMMUTER WALKER

Persons that walk or utilize transit as their primary means of transportation to work. Commuter walkers need:

- Direct access routes to major employment centers
- Connected and accessible sidewalks and paths
- Access to transit centers and bus stops
- Shelter and shade at bus stops



RECREATION/FITNESS WALKER

Those running, jogging, enjoying a leisurely stroll, or walking their dog. Recreation/Fitness walkers need:

- Low stress facilities (i.e., wide sidewalk, sidewalk on local road, shared use path, etc.)
- Direct connections to residential areas
- Access to parks, open space, and trails



ON SMALL WHEELS

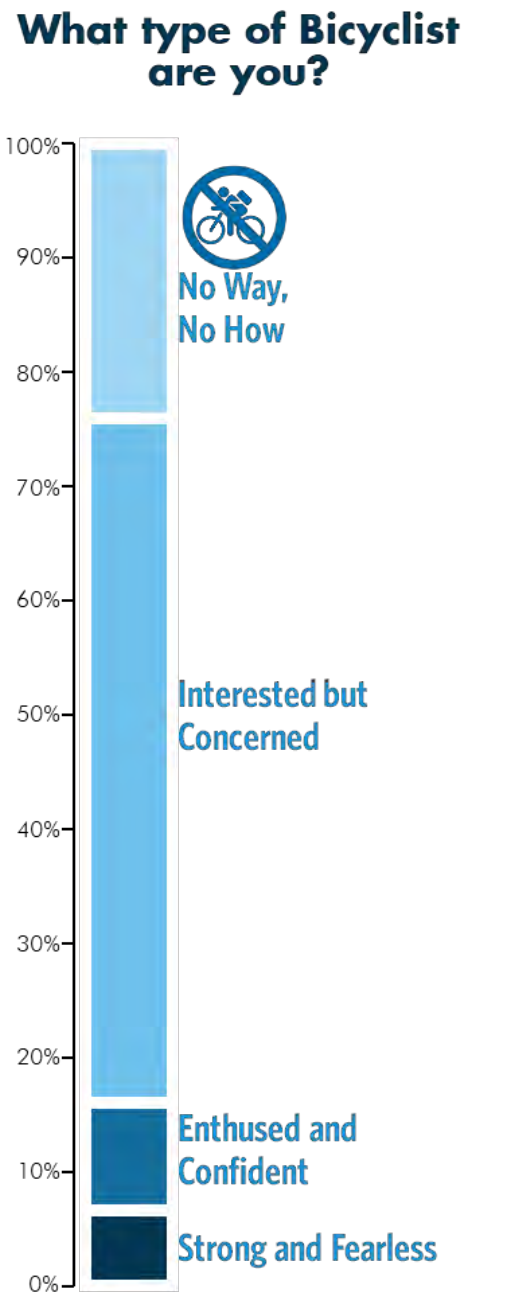
Persons utilizing scooters, skateboards, inline skates, etc. to travel faster than by foot. Walkers on small wheels need:

- Connected sidewalks and paths
- Firm and stable surfaces in good condition
- Smooth sidewalk/street transitions
- Gradual cross slopes at driveways

Bicyclists

Bicyclist skill level greatly influences expected speeds and behavior, both in separated bikeways and on shared roadways. Bicycle infrastructure should accommodate as many user types as possible, with decisions for separate or parallel facilities based on providing a comfortable experience for the greatest number of people. To understand a potential user's comfort and willingness to ride a bike on a given roadway, Roger Geller developed a bicyclist classification system for the city of Portland, Oregon. This concept was later defined for the nation by Jennifer Dill and Nathan McNeil at Portland State University. The classification system categorizes the general population by their different needs and biking comfort levels given roadway conditions. The bicycle classification system categories include the following four groups:

- **Strong and Fearless (~6% of the population).** Characterizes a person who considers bicycling a strong part of their identity and will ride anywhere regardless of roadway conditions or weather. This type of bicyclist may ride faster than other users, prefers direct routes, and will typically choose roadway connections over shared use paths and trails.
- **Enthused and Confident (~9% of the population).** These bicyclists are comfortable riding on roads with motor vehicle traffic but prefer to use low traffic stress roads or shared use paths.
- **Interested but Concerned (~60% of the population).** This group represents the majority of the population and is comprised of persons who are interested in bicycling or enjoy riding a bike but are afraid to ride on roadways. Through experience, education and encouragement, people in this category may become "enthused and confident" riders.
- **No Way, No How (~ 25% of the population).** Persons in this category are not interested in bicycling due to their inability to bicycle, perceived safety issues, lack of interest, or other constraints. Through education, some people in this category may eventually ride a bike.



Source: Dill & McNeil (2012) "Four Types of Cyclists?"

ATP Goals, Objectives, and Strategies

Glendale envisions a complete active transportation network that addresses local and regional issues and needs and provide multimodal transportation options for people of all ages and abilities. In the spirit of a united and coordinated effort, the ATP used goals, objectives, and strategies developed by Glendale that guides all long-term transportation planning by the City.

- A **Goal** is what the City of Glendale wants to achieve over the long term.
- An **Objective** is a specific, measurable statement that marks progress towards the goal.
- A **Strategy** is a continual action that guides decision-making to achieve the objectives and goals of the Plan.

Table 1.1 outlines the City's overall transportation goals. Tables 1.2-1.3 outline active transportation specific objectives and strategies developed by the City.

Table 1.1. Transportation Goals

Goals	
1	Provide for and sustain a mix of transportation modes that can meet the continuing needs of increasing personal mobility, enhancing the quality of life of citizens, and improving the movement of goods consistent with the City of Glendale's vision and values.
2	Provide transportation infrastructure that promotes economic development initiatives within the undeveloped Municipal Planning Area and supports infill development opportunities within "established" Glendale.
3	Comprehensively plan for all locally significant modes of transportation and work towards their interconnection. Coordinate with other jurisdictions and agencies that either influence or are affected by Glendale's transportation planning efforts.
4	Develop a balanced land-use and transportation system that reflects the overall need to minimize automobile travel and maximizes the opportunities for transportation alternatives such as public transportation and non-motorized travel modes.
5	Maintain, upgrade, or develop existing and future transportation systems in a way that renders them, safe, functional, flexible, environmentally acceptable and aesthetically pleasing.
6	Manage the transportation system to provide for an optimum level of mobility for the greatest number of persons while addressing mobility for those in need.
7	Improve safety in all transportation modes through timely maintenance of existing infrastructure, development of new infrastructure, enforcement of access and design standards, and continuing public education.
8	Provide a safe, reliable, coordinated, and accessible local transit system that operates effectively and continues to offer a viable choice of travel for residents and visitors to the City of Glendale.

Source: Moving Glendale Ahead

Table 1.2. Active Transportation Objectives

Objectives	
0.1	Complete an Active Transportation Plan by 2022.
0.2	Restripe 50% of appropriate arterial streets to include bike lanes by 2022; restripe remaining 50% by 2032.
0.3	Complete one north/south bicycle corridor by 2022 and one by 2032; complete one east/west bicycle corridor by 2032 and one by 2042.
0.4	Construct a bicycle/pedestrian specific facility to cross Grand Avenue by 2042.
0.5	Construct sidewalks adjacent to the 10 top highest ridership bus stops by 2032, where feasible, and 10 additional bus stops by 2042.

Source: Moving Glendale Ahead

Table 1.3. Active Transportation Strategies

Strategies	
S.1	Where possible, revise city standards to require that appropriate bicycle and pedestrian facilities are constructed as part of all new transportation infrastructure projects.
S.2	Ensure connectivity between bicycle and pedestrian facilities and public transportation network.
S.3	Develop and maintain bicycle and pedestrian master plans.
S.4	Where possible, comply with Glendale street standards and build sidewalks on both sides of all new roadways.
S.5	Ensure that existing bike lane striping on arterial streets is maintained/replaced when pavement management activities are performed.
S.6	Apply to the League of American Bicyclist's Bicycle Friendly Community Program with the goal of being recognized as a bicycle friendly community.
S.7	Provide a continuous regional network of safe and convenient bikeways connected to other transportation modes and local bikeways systems. Integrate multi-use paths with on-street bikeways where possible.
S.8	Provide ongoing maintenance of bicycle facility surfaces to maintain smooth surfaces free of potholes and debris.
S.9	Integrate multi-use paths with on-street bikeways where possible.
S.10	Coordinate with local jurisdictions, ADOT, and MAG to identify high-frequency, bicycle-related crash locations and improvements to address safety concerns in these locations.
S.11	All newly constructed or reconstructed bikeway facilities when practical will be constructed to meet or exceed the requirements of the Manual of Uniform Traffic Control Devices (MUTCD), American Association of Street and Highway Transportation Officials (AASHTO) and the Americans with Disabilities Act (ADA).
S.12	All new development (including major employment centers, recreation facilities, schools, ·transportation centers, etc.) will be encouraged to construct bicycle facilities (including routes, lanes, paths and parking), as appropriate, that are internal or adjacent to the development.
S.13	Review the constructability of bicycle facilities whenever streets are constructed, reconstructed or rehabilitated.
S.14	Promote an education and safety program for bicyclists, pedestrians, and motorists.

Source: Moving Glendale Ahead

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2. GLENDALE TODAY

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About Glendale

Situated in the heart of the Sonoran Desert in south-central Arizona, Glendale's landscape varies between low desert valleys and developed urbanized areas. Over 94 percent of the MPA is privately owned, with the rest of the MPA being comprised of military, Arizona State Trust Land, and other land holders. Numerous water bodies transverse Glendale, including the New River, Agua Fria River, Skunk Creek, Arizona Canal, Grand Canal, and the Airline Canal.

General Character Areas

Glendale's development patterns and community characteristics vary greatly. Broadly speaking, Glendale can be generally divided into four distinctive areas (see Figure 2.1) characterized by their development patterns and existing active transportation networks.

Northern Developed Area

General Location: MPA area north of Peoria Avenue
General Characteristics: Mix of residential developments and key commercial corridors. Blend of young families, middle aged persons, and retirees. Large "snowbird" population.
Key Activity Centers: Arrowhead Towne Center, Thunderbird Conservation Area, Thunderbird Medical Complex, and Midwestern University.


Arrowhead Towne Center

Central Core

General Location: MPA area south of Peoria Avenue and east of 75th Avenue
General Characteristics: Older, established downtown Glendale with a mixture of single- and multi-family residential areas and large industrial areas.
Key Activity Centers: Downtown Glendale, Glendale Community College


Downtown Glendale

Loop 101 Area

General Location: MPA area west of 75th Avenue to the Agua Fria River
General Characteristics: Developing area with a blend of master planned residential areas, major entertainment and commercial districts, and vast open agricultural lands.
Key Activity Centers: State Farm Stadium, Westgate Entertainment District

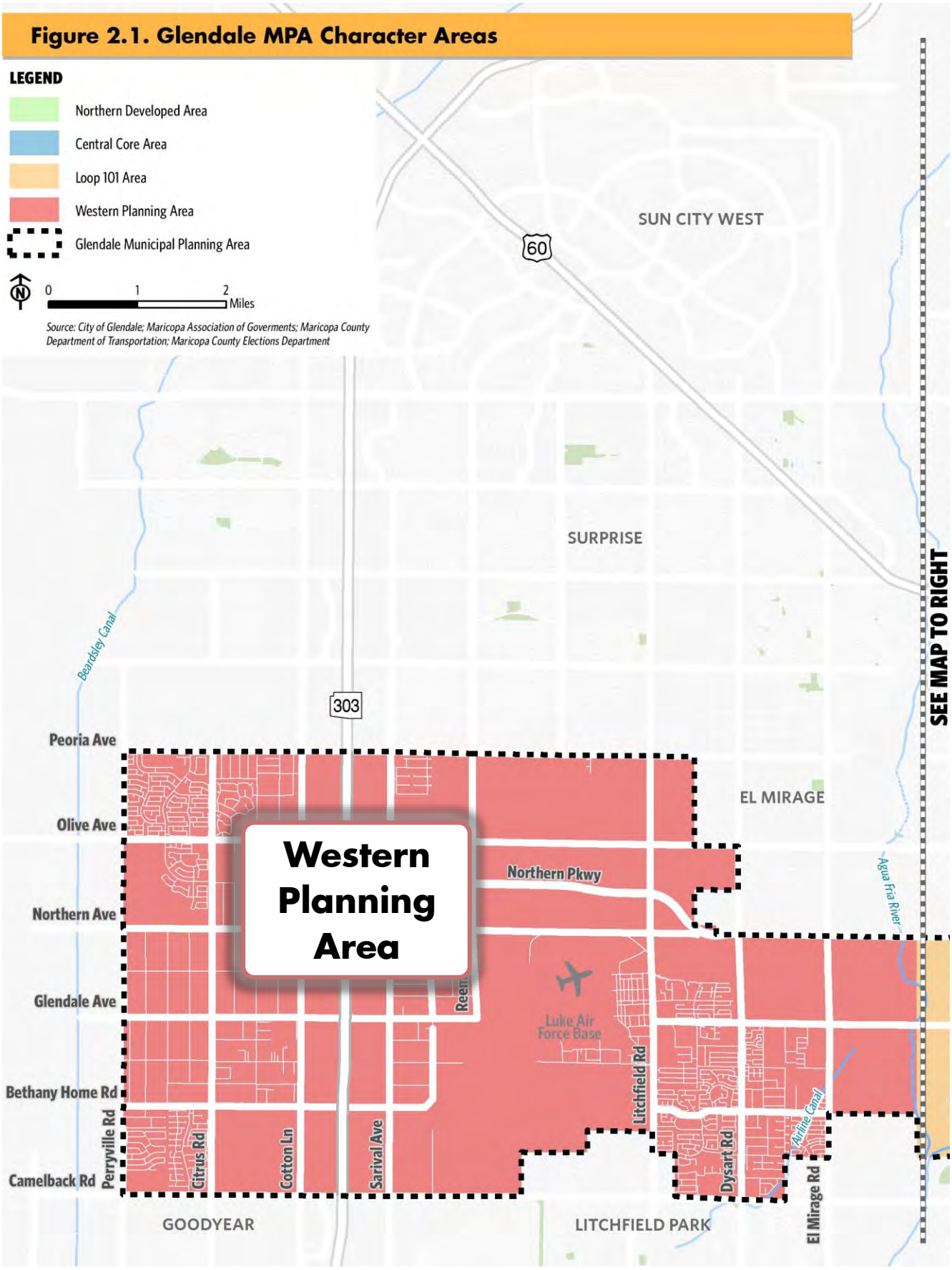

Westgate Entertainment District

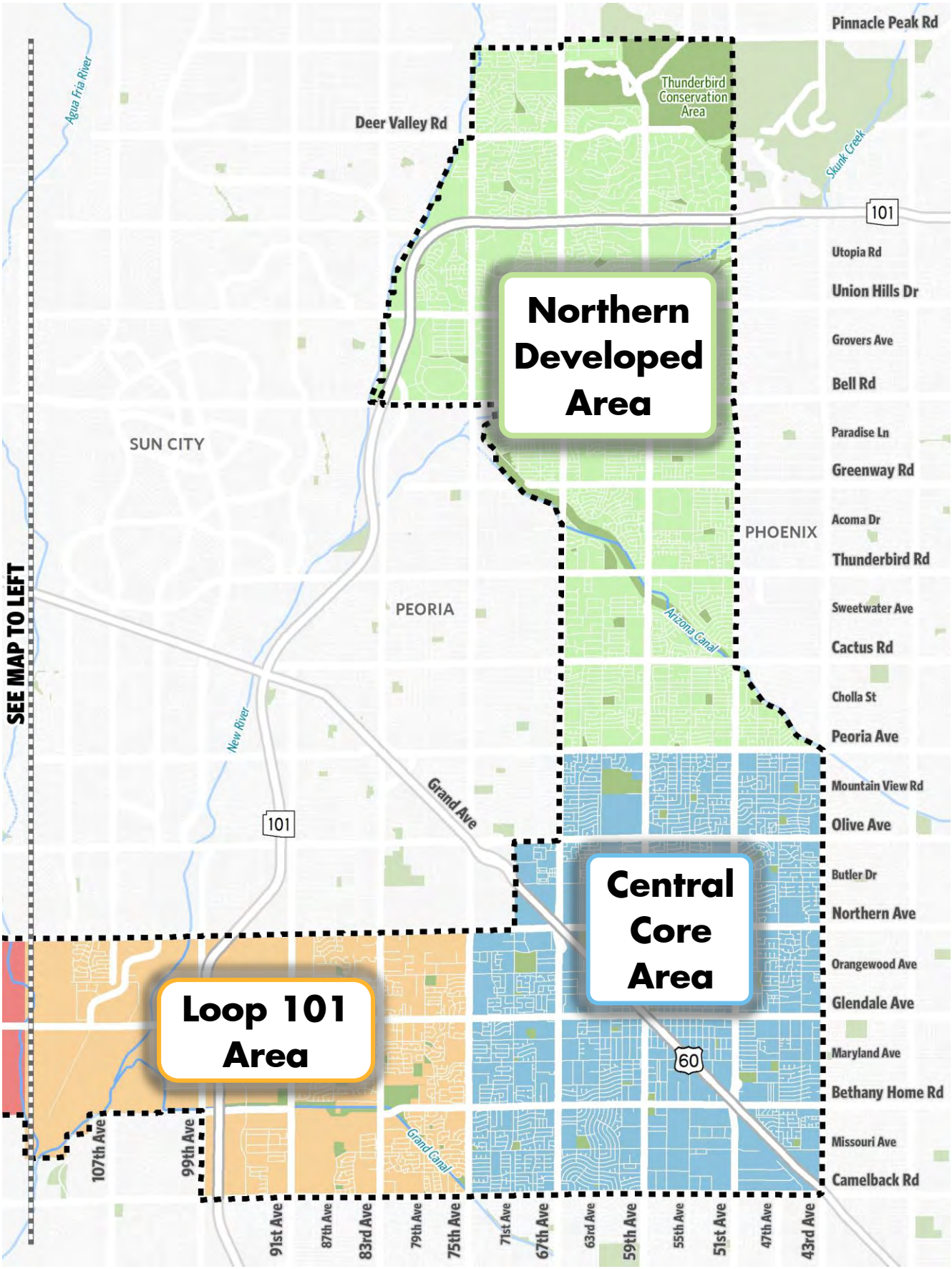
Western Planning Area

General Location: MPA area west of the Agua Fria River
Community Character: Mixture of large master-planned residential communities, large-lot single family properties, developed land surrounding Luke Air Force Base, and agriculture lands.
Key Activity Centers: Luke Air Force Base, Wildlife World Zoo


Luke Air Force Base

Figure 2.1. Glendale MPA Character Areas




Northern Developed Area
Central Core Area
Loop 101 Area
SEE MAP TO LEFT
Pinnacle Peak Rd
Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
Skunk Creek
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
Acoma Dr
Thunderbird Rd
Sweetwater Ave
Cactus Rd
Cholla St
Peoria Ave
Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
Maryland Ave
Bethany Home Rd
Missouri Ave
Camelback Rd
SUN CITY
PEORIA
PHOENIX
Arizona Canal
New River
Grand Ave
Grand Canal
60
107th Ave
99th Ave
91st Ave
87th Ave
83rd Ave
79th Ave
75th Ave
71st Ave
67th Ave
63rd Ave
59th Ave
55th Ave
51st Ave
47th Ave
43rd Ave

Population Overview

Population characteristics such as age, socioeconomic status, disability status, etc., contribute to an understanding of the needs of potential active transportation users. To determine current population characteristics, socioeconomic data was compiled from the US Census Bureau's 2010 Census and 2012-2016 American Community Survey (ACS).

Table 2.1 summarizes the study area's current population information and Figure 2.2 illustrates areas with the highest concentration of population. The highest concentrations of population occur along Bethany Home Road, Glendale Avenue, 59th Avenue, and 43rd Avenue in Glendale's Central Core area. These areas are largely zoned as high density residential, which includes a mix of apartments, condominiums, and single-family dwelling units.

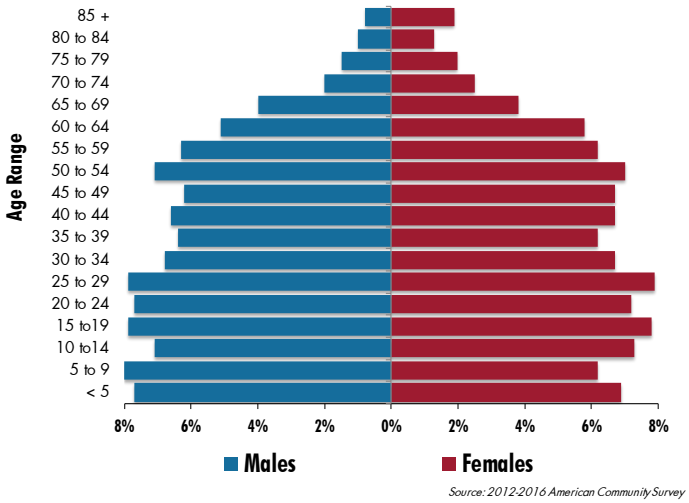
Table 2.1 Glendale MPA Population and Housing Units

Demographics	Total MPA	Northern Developed Area	Central Core Area	Loop 101 Area	Western Planning Area
Total Population	**271,142**	89,337	123,767	25,986	32,052
Occupied Housing Units	**90,600**	32,108	40,887	8,290	9,315
Average Household Size	**2.97**	2.75	3.08	3.13	3.09
Total Employed Civilian Population	**121,251**	44,458	49,753	12,912	14,128

Source: 2012-2016 American Community Survey

Age Composition

Analyzing an area's age composition helps decision-makers understand the potential need for increased walking and biking options. As people age, they may begin to drive less and require alternative modes of transportation for medical appointments, shopping, and visiting family and friends. Younger people are frequent active transportation users, who often walk and bike to school, to visit friends, and for recreational purposes. According to the ACS, older adults (persons aged 65 and older) account for approximately 10 percent of the study area's population, and children (persons aged of 18 and younger) represent approximately 26.8 percent.



Land Use and Major Destinations

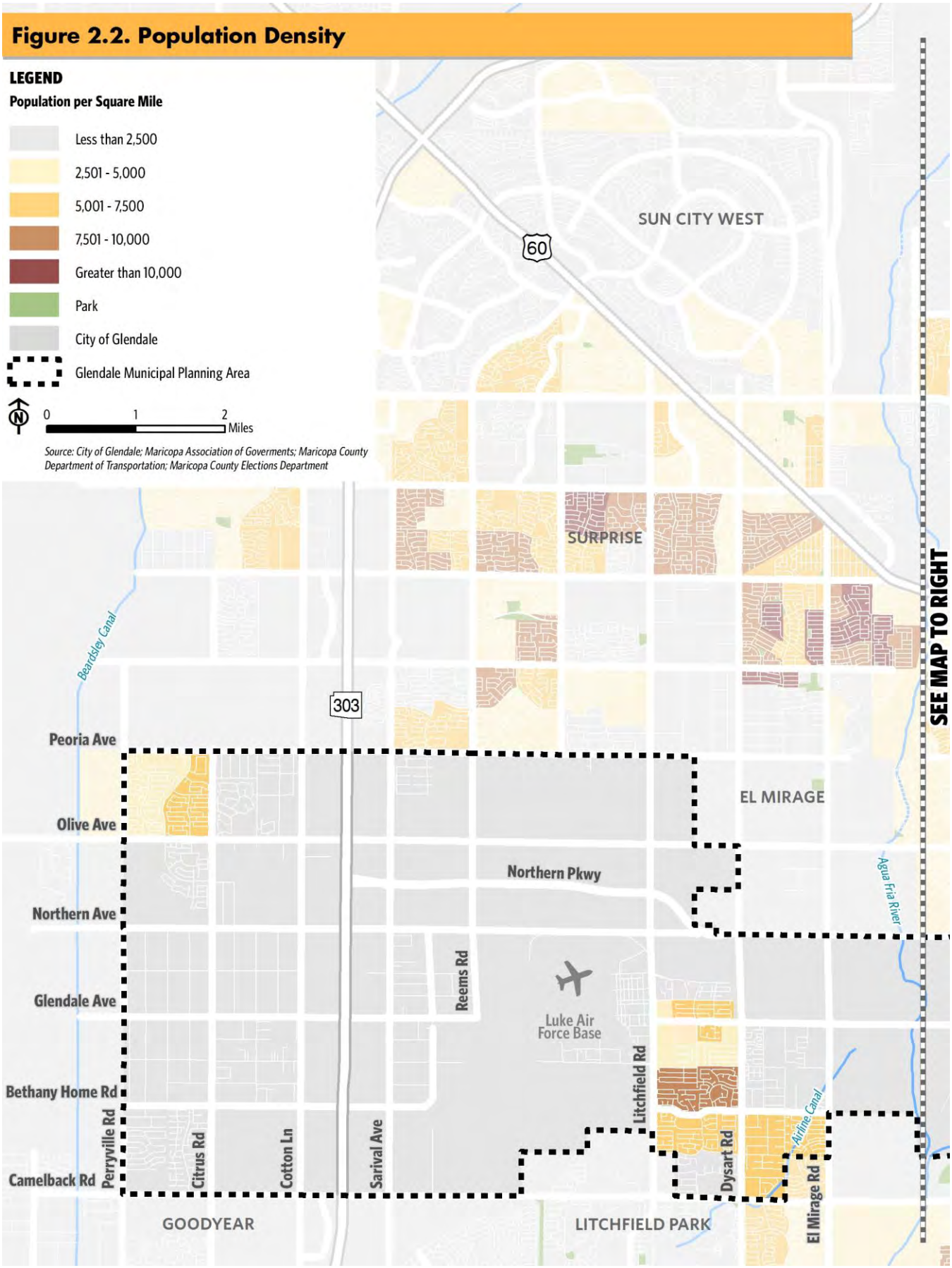
Land use and active transportation are directly connected. Dense and mixed-use areas with major destinations and services typically have more convenient biking and walking distances than single-use suburban and rural development areas. Figure 2.3 illustrates major active transportation generators in the Glendale MPA, including:

- **Major Employers.** Glendale is a growing employment market in the Phoenix metropolitan area's West Valley and home to several major employers. Major employment sectors include healthcare, entertainment, retail, and military. Luke Air Force Base is the largest employer in the study area, employing over 3,400 persons. Banner Health Thunderbird Medical Center is the second largest employer, with over 2,800 employees.
- **K-12 Schools.** In total, there are over 54 K-12 schools in the Glendale MPA with over 50,000 students. The largest schools in the study area include Copper Canyon High School, Mountain Ridge High School, and Ironwood High School. Providing active transportation facilities is critical since many children are likely to have short commutes that could be converted to walking or biking trips.
- **Higher Education.** Glendale has several private and public higher education, including the Glendale Community College, Midwestern University, and the Arizona Christian University. These educational facilities have a high demand for active transportation facilities. Glendale Community College, alone, has over 31,000 enrolled students that frequently walk, bike, and use transit to attend school.
- **Health Care Facilities.** There are several major medical/healthcare facilities located in the study area. Major healthcare facilities include Banner Health Systems, Abrazo Arrowhead Hospital, Humana Health Care, Midwestern University, and St. Joseph's Westgate.
- **Entertainment Districts.** Glendale is home to some of the Nation's finest sports and entertainment areas. Along the Loop 101, the Westgate Entertainment District hosts major sport venues, shopping outlets, and dining facilities. In the northern portion of the MPA, there are numerous golf courses and the Arrowhead Towne Center shopping area.
- **Commercial Centers.** Major retail centers are located near most major subdivisions and along major roadways. Grocery stores or big-box retailers are typically the anchor store for these centers. Commercial centers have the potential to generate significant pedestrian and bicycle traffic.
- **Retirement Communities.** As a person ages, their reliance on alternative transportation increases. Active transportation can provide elderly persons a way of accessing jobs, services, and goods, as well as facilitate healthy aging through physical activity.

Major Residential Areas

Concentrations of housing developments may be indicative of a demand for active transportation facilities. Providing direct and convenient active transportation connections between major residential communities and key activity centers creates opportunities to benefit a large number of residents. In the Glendale MPA, over 35 percent of the area is zoned as single-family or multifamily residential. Large residential developments in the far Western Planning Area, such as Cortessa and White Tank Foothills, create obstacles to active transportation as their locations are far removed from existing facilities.

Figure 2.2. Population Density



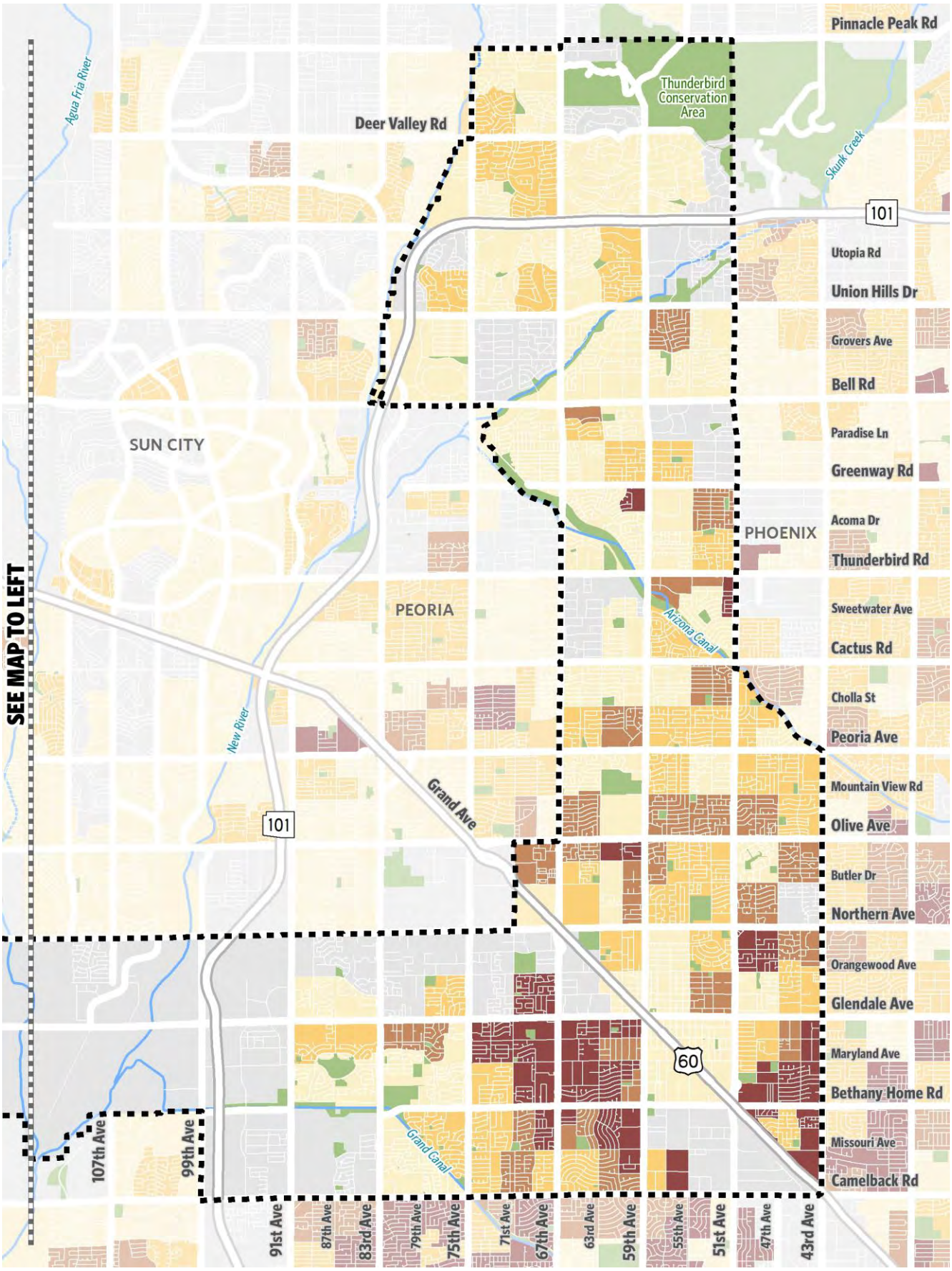
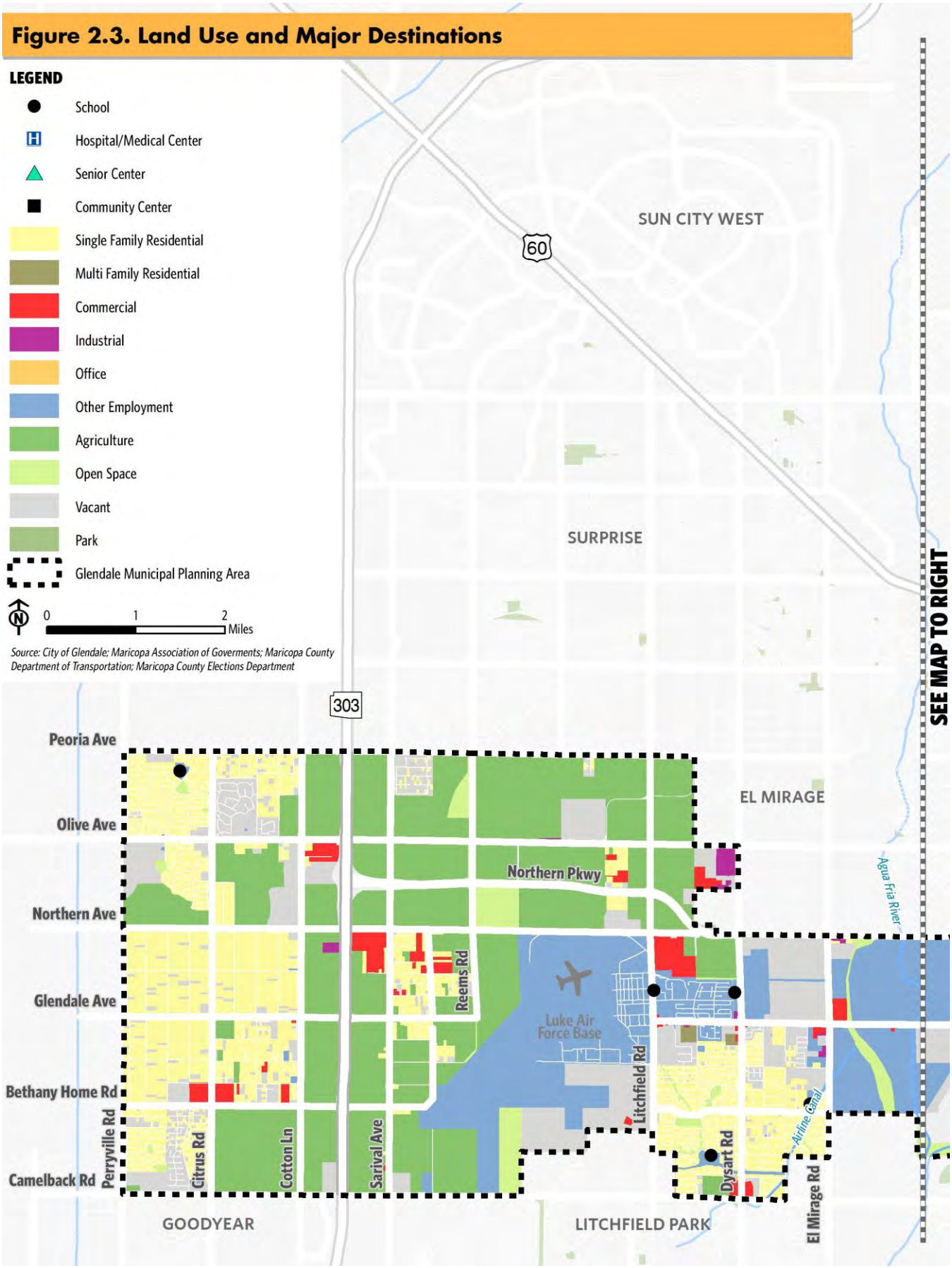
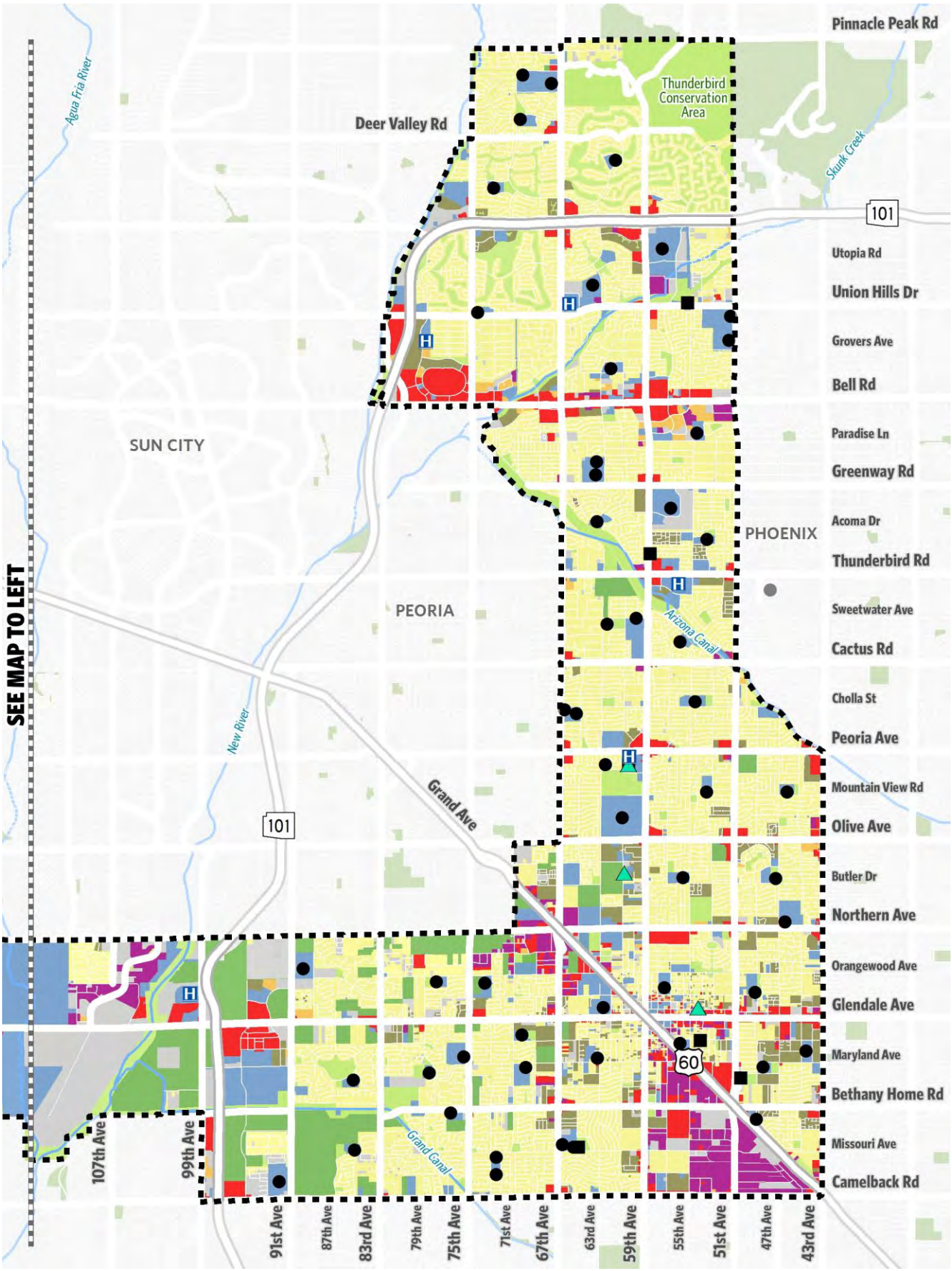

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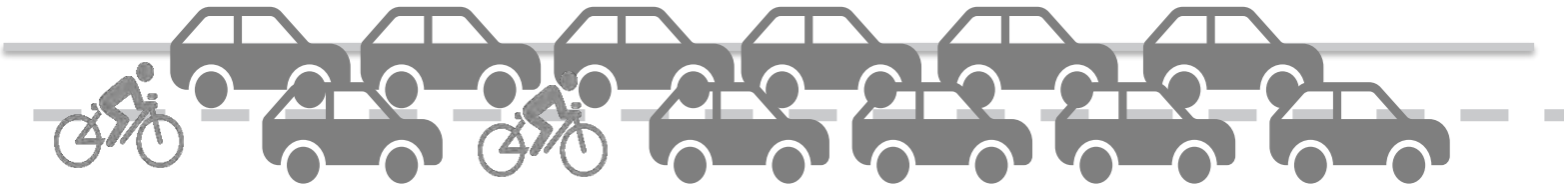
Figure 2.3. Land Use and Major Destinations




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Commute Travel Behavior

Knowing where people walk or bike, and for what purposes, informs Glendale how to effectively implement projects and programs that better serve residents. Glendale is a car-oriented community. ACS data shows that in 2016, 2.1% of workers commuted to work daily by walking or biking, while 2.7% used public transit. Although these statistics are important to the overall discussion, they only include a portion of active transportation commuters and fail to measure non-commute activities such as trips to stores, to schools, or for recreation.



Homearea.com showed that Glendale has the highest commute times in the Phoenix metropolitan area, increasing by 6.4 percent from 26.5 in 2015 to 28.2 in 2018!

Trip Reduction Program

The Maricopa Regional Travel Reduction Program (TRP), which surveys employers and students to determine commuting habits, provides a more accurate assessment of the total number of employees/students traveling by alternative modes. According to the 2015 TRP survey, commuters in Maricopa County walking and biking account for nearly 5 percent of all commuters that use alternative modes. Of students that use alternative modes, over 8.6 percent walk, which is an increase of 0.9 percent since the 2014 survey. Results of the TRP survey show employees commute 15.0 miles to work on average, while students travel an average of 7.1 miles to school. Overall, 27.6% of all TRP participants drive less than 5 miles to work/school. These 5-mile trips have the potential to be converted to walking and biking trips through a connected, efficient active transportation network.

City of Glendale Bicycle and Pedestrian Counts

In 2016, Glendale conducted peak hour traffic counts at arterial-to-arterial intersections that captured bicycle and pedestrian traffic.

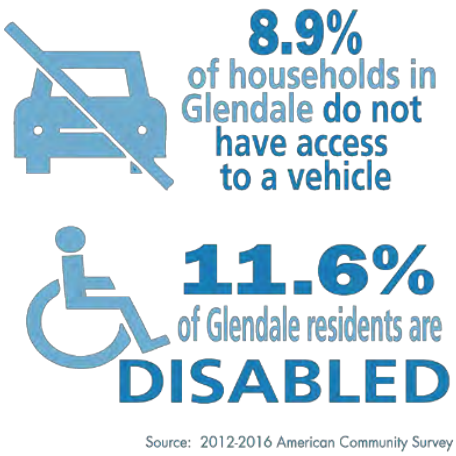
Table 2.2. Intersections with Highest Pedestrian and Bicycle Counts

Intersection	Area Characteristics/Attractions	AM	PM	Total Count
Top Three Pedestrian Count Locations				
91st Avenue/Camelback Road	Copper Canyon H.S., Walmart	471	123	**594**
43rd Avenue/Glendale Avenue	Multifamily housing, Commercial	218	205	**423**
43rd Avenue/Bethany Home Road	Multifamily housing, Fry's Grocery	187	196	**383**
Top Three Bicycle Count Locations				
43rd Avenue/Bethany Home Road	Multifamily housing, Fry's Grocery	22	52	**74**
43rd Avenue/Northern Avenue	Multifamily housing, Fry's Grocery	31	25	**56**
59th Avenue/Northern Avenue	Walmart, Single-family housing	16	34	**50**

Source: 2016 City of Glendale Count Program

Socioeconomic Equity Analysis

A socioeconomic equity analysis model was developed for the ATP to identify areas with a high percentage of population groups that traditionally rely on active transportation as their primary means of transportation. Providing active transportation connections to areas with a high concentration of disadvantaged population groups may help alleviate wider social issues such as access to jobs, healthy food, education, and healthcare.

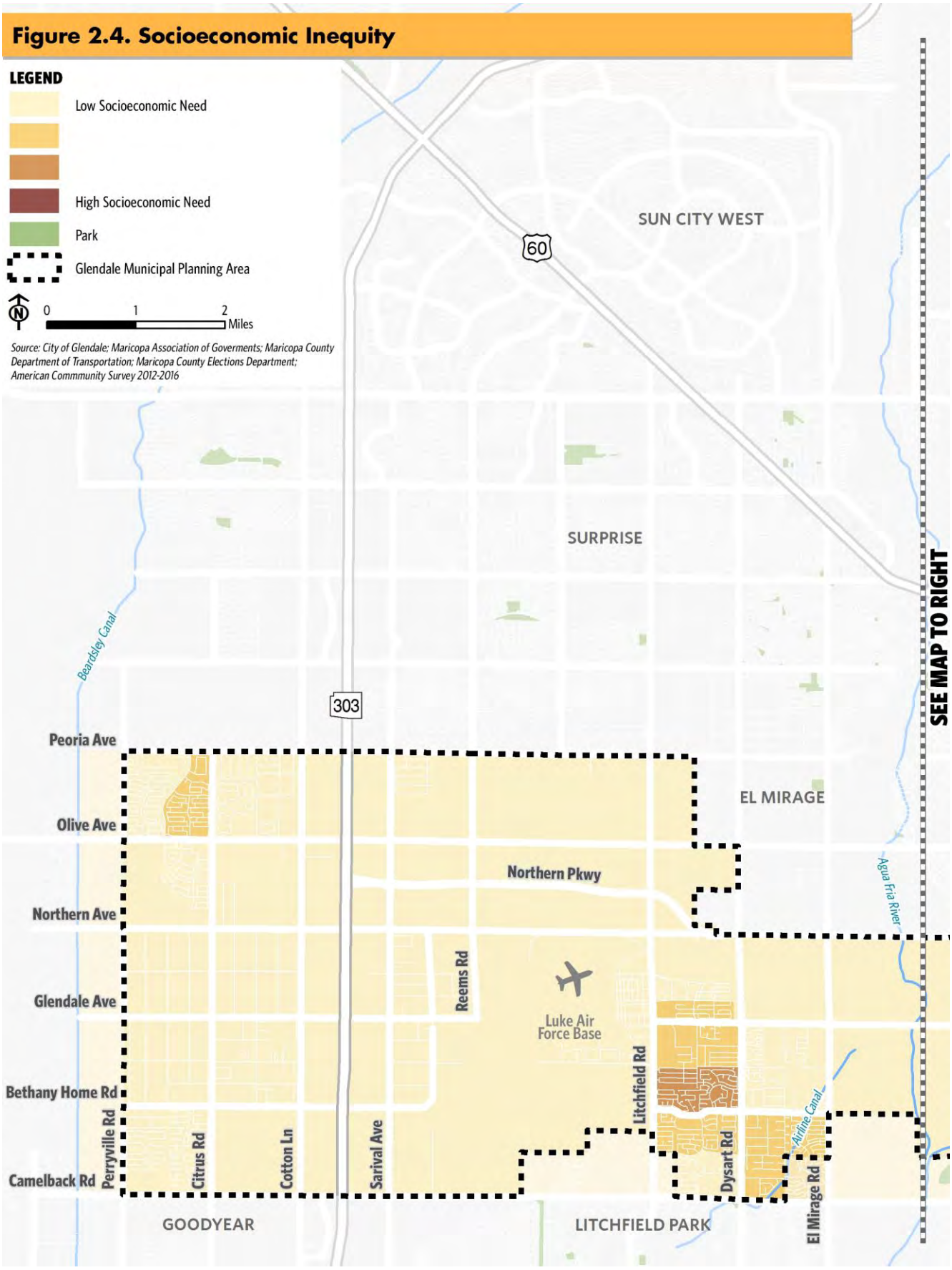


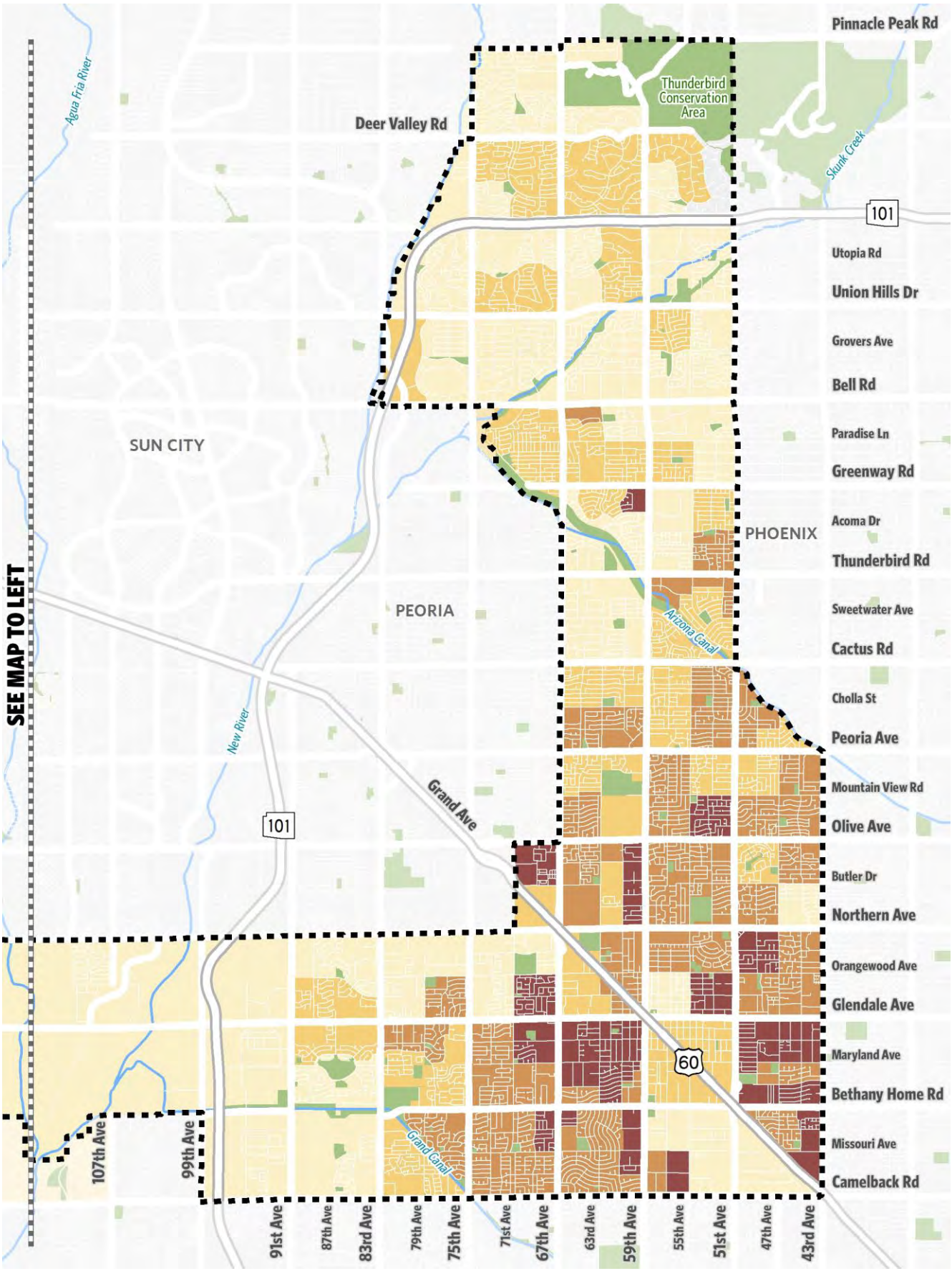
The socioeconomic equity analysis model identified levels of socioeconomic need using an index of the following indicators:

- Age – children and elderly populations
- Ethnicity – minority populations
- Disabled Populations – persons that have cognitive, visual and physical disabilities
- Low-Income – households that are financially less likely to own a vehicle
- Vehicle Ownership – households with limited or no access to a vehicle

Results of this model are displayed in Figure 2.4. Areas with the highest percentage of population groups that traditionally rely more on walking, bicycling, or transit as their primary form of transportation are depicted as having the higher socioeconomic need. As stated by the goals of the ATP, Glendale expressed its commitment to providing active transportation options to vulnerable users and traditionally underserved communities. As the ATP is implemented, additional social equity impacts, such as burden of construction on disadvantaged population groups, should be considered beyond those included in the prioritization process

Figure 2.4. Socioeconomic Inequity

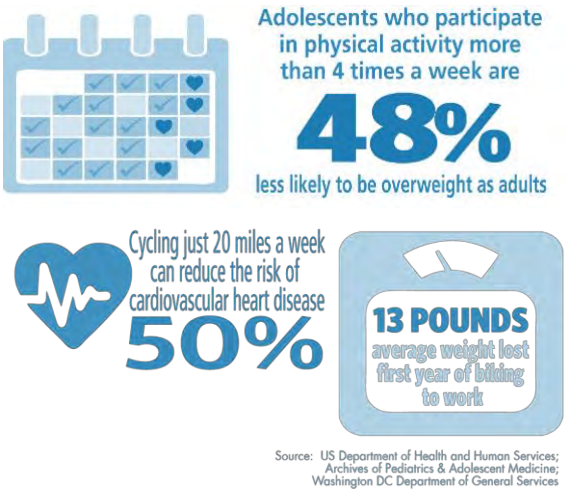



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Public Health Equity Analysis

As part of the ATP process, a health equity model was developed to identify areas with disproportionately high rates of chronic diseases that may be prevented or controlled through physical activity. Studies show that physical inactivity is linked to higher rates of heart disease and stroke. Walking and biking are basic forms of physical activity and can potentially prevent or control chronic diseases such as heart disease, stroke, diabetes, and high cholesterol and help to improve mood and lower stress levels.



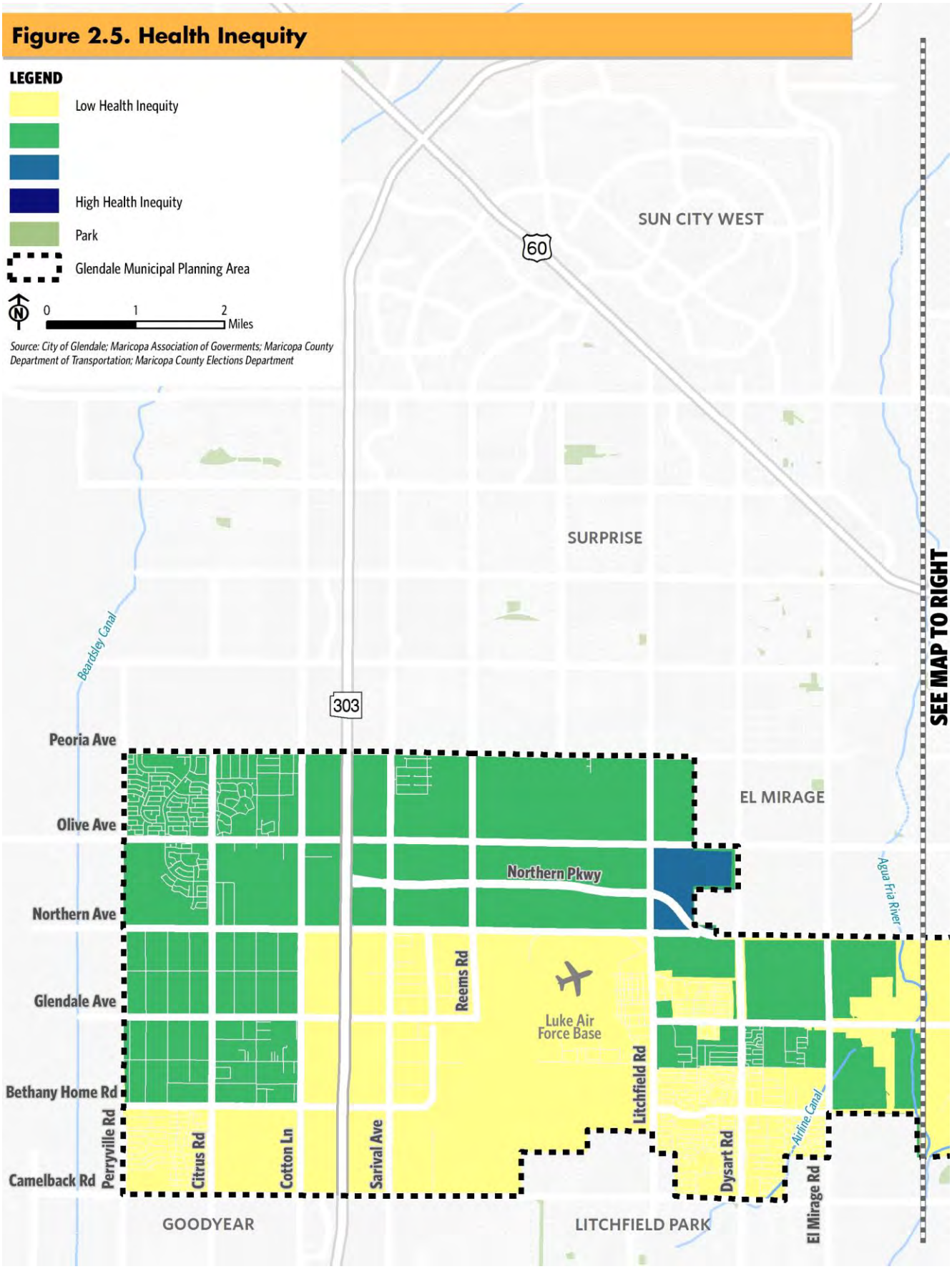
According to the Maricopa County Community Health Improvement Plan, cardiovascular disease is the second leading cause of death in Maricopa County. Furthermore, approximately 29 percent of Glendale residents are obese. Providing pedestrian and bicycle facilities that are accessible and convenient may encourage activity limited, overweight, and obese population groups to incorporate physical activity into their lifestyle.

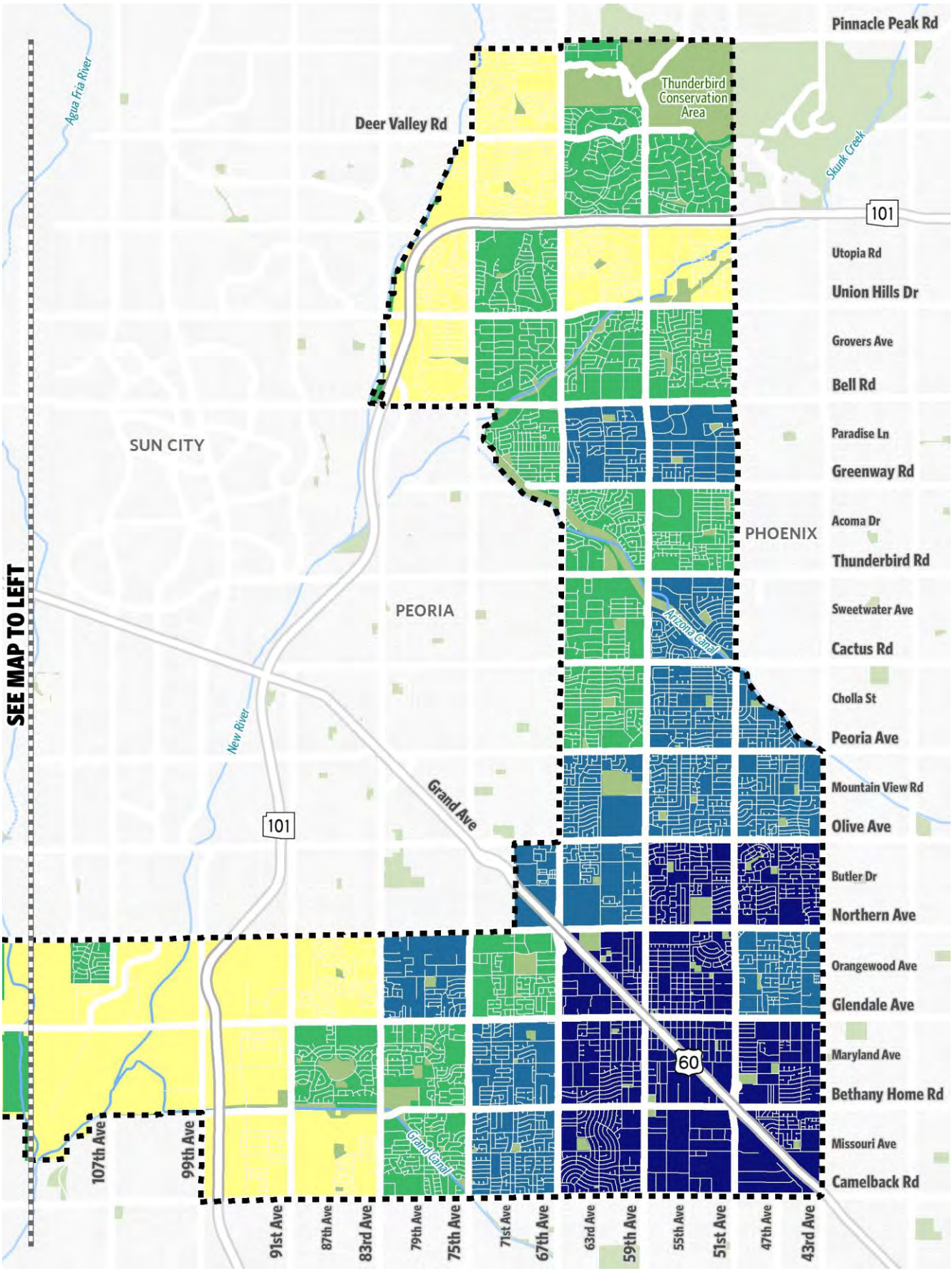
The health equity analysis model identified the rate of the following health indicators:

- Stroke
- Coronary Heart Disease (CHD)
- Diabetes
- Asthma
- High Cholesterol
- No Leisure-time Physical Activity
- Obesity

Results of the model are displayed in Figure 2.5. Utilizing the model, identifying the presence and completeness of active transportation facilities in areas with the highest rates of chronic health diseases can help Glendale align with its goals of fostering an active-living culture. By providing convenient active transportation facilities, people may be more inclined to walk or bike, and in turn help control chronic diseases while also improving mood and lowering stress levels. To aid the City in determining potential improvements that may help reduce public health issues, the project prioritization process, developed for the ATP, included health equity model results.

Figure 2.5. Health Inequity




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3. OUR CURRENT TRANSPORTATION NETWORK

WALK. BIKE. CONNECT
glendale active transportation plan

This chapter presents a synopsis of key roadway and active transportation conditions within the Glendale MPA. The purpose of this chapter is to create an overall understanding of the City as it is now, what it can become, and to develop a data-driven foundation upon which future recommendations can be built.

Existing Roadway Network

The following section summarizes typical characteristics of existing roadway conditions and characteristics in the Glendale MPA.

Major Travel Corridors

Travel corridors connect communities, land uses, employment centers, and link people to goods and services. Traditionally, roadways are grouped into a hierarchical classification, which helps identify the roadway's function, design, speed limits, access control, and adjacent land use development. Understanding roadway classification is imperative when planning an active transportation network. Vehicle volumes, number of lanes, lane width, road condition, and speed limits impact a pedestrian and bicyclist's level of comfort. As illustrated in Figure 3.1, within the Glendale MPA there is a mixture of roadways, including:

Freeway/Expressway
Controlled access roadway that provide regional connections. Typically has high speeds and high traffic volumes that make it unfavorable for pedestrian and bicycle usage.



Parkway
Enhanced arterial that eliminates left-turn movements at intersections. Connects regional destinations and communities.



Arterial
Major roadway with multiple travel lanes and higher traffic volumes and speeds. Typically, is lined with dense commercial and retail land uses and major destinations. Connects regional destinations and communities. Bike lanes may be applicable for experienced bicyclists only.



Collector
Larger corridor that has moderate traffic volumes and speeds. Distributes traffic from local roads and neighborhoods to arterials. With proper facilities, provides low stress facility for pedestrian and bicycle usage.



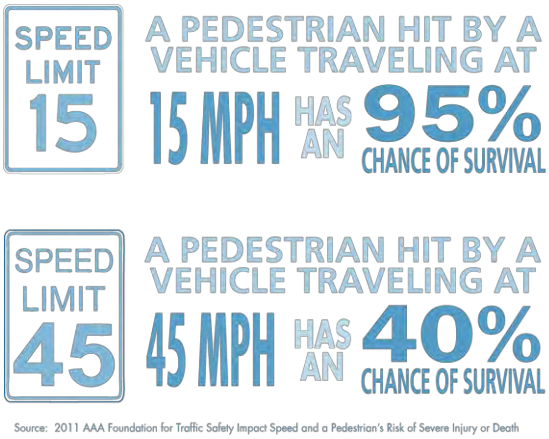
Local
Minor roadway with lower traffic volumes and speeds. Provides direct access within a neighborhood. Provides a low-stress facility for all users to walk and bike.



Due to high traffic volumes, arterials traditionally have numerous businesses, commercial services, transit stops, and other major destinations that attract pedestrians and bicyclists and, in turn, potentially create conflicts with motorists, particularly at intersections. Greater separation between the vehicle lanes and active transportation infrastructure is desirable along arterials. On lower classification roads such as collectors and local streets, bicyclists feel comfortable sharing lanes (on roads with less traffic) and pedestrians feel safer because of the lower traffic volumes and more frequent crossing opportunities.

Roadway Speed Limits

Speed is a significant factor in determining the potential use of pedestrian and bicycle facilities. The National Highway Traffic Safety Administration (NHTSA) reports that the occurrence of pedestrian crashes and risk of severe injury or death are both strongly associated with the travel speed of the motor vehicle at the time of the crash. AAA Foundation for Traffic Safety's *2011 Impact Speed and a Pedestrian's Risk of Severe Injury or Death* report found that at 15 MPH, pedestrians had a 95 percent survival rate, compared to 40 percent at 45 MPH. Risk of injury also increases significantly with age. For example, a 70-year old pedestrian struck by a vehicle traveling at 25 MPH has a similar risk of severe injury or death as a 30-year-old struck at 35 MPH. Figure 3.2 illustrates posted speed limits within the Glendale MPA.



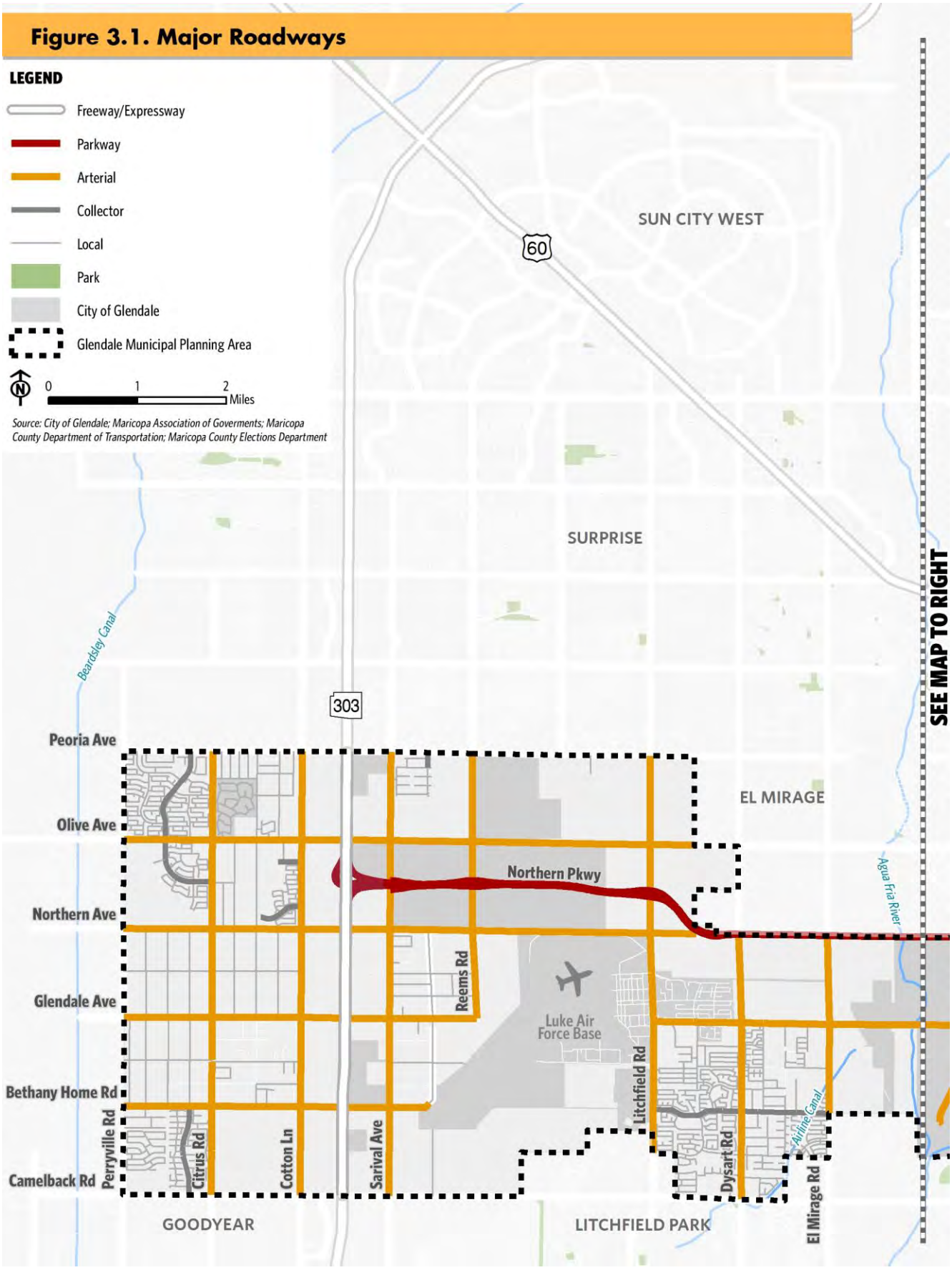
Vehicle Volumes

Very similar to speed characteristics, traffic volume is a significant factor in determining the use potential of a pedestrian or bicycle facility. Corridors with high traffic volumes and high vehicle speeds may reduce a bicyclist or pedestrians level of comfort, discouraging them from using the corridor. Figure 3.3 illustrates current traffic volumes.

Number of Lanes, Widths, and Right-of-Way

Travel lane characteristics, in conjunction with available right-of-way, play a key role in the expansion potential of pedestrian and bicycle facilities. Number of lanes and lane widths are integral in determining the stress level for bicyclists. Opportunities to increase the active transportation network within a roadway's existing right-of-way or roadway width include adding a paved shoulder or bike lane, narrowing a wide lane and installing a bike lane, constructing an off-set shared use path within a wide right-of-way, or constructing a sidewalk along a narrow roadway. In urban areas, however, right-of-way often is limited, making pedestrian and bicycle improvements challenging. Typically, arterial roadways have a lane width of 12 FT. Restriping an urban arterial road to an 11FT travel lane can provide opportunities to install sidewalks, wider bike lanes, wider curb lanes to accommodate bikes, or to provide roadside landscaped buffers. Figure 3.4 illustrates current number of travel lanes.

Figure 3.1. Major Roadways



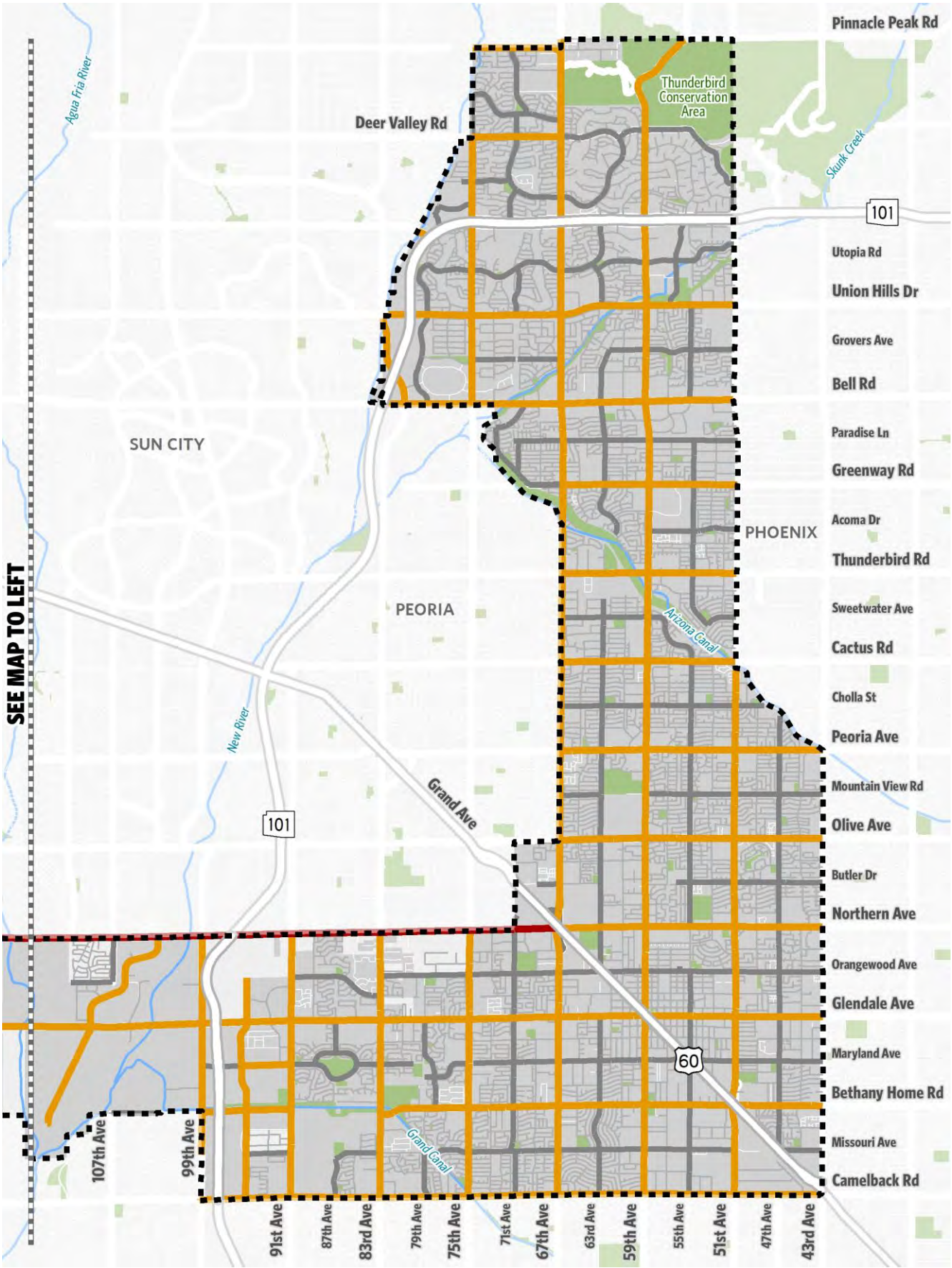
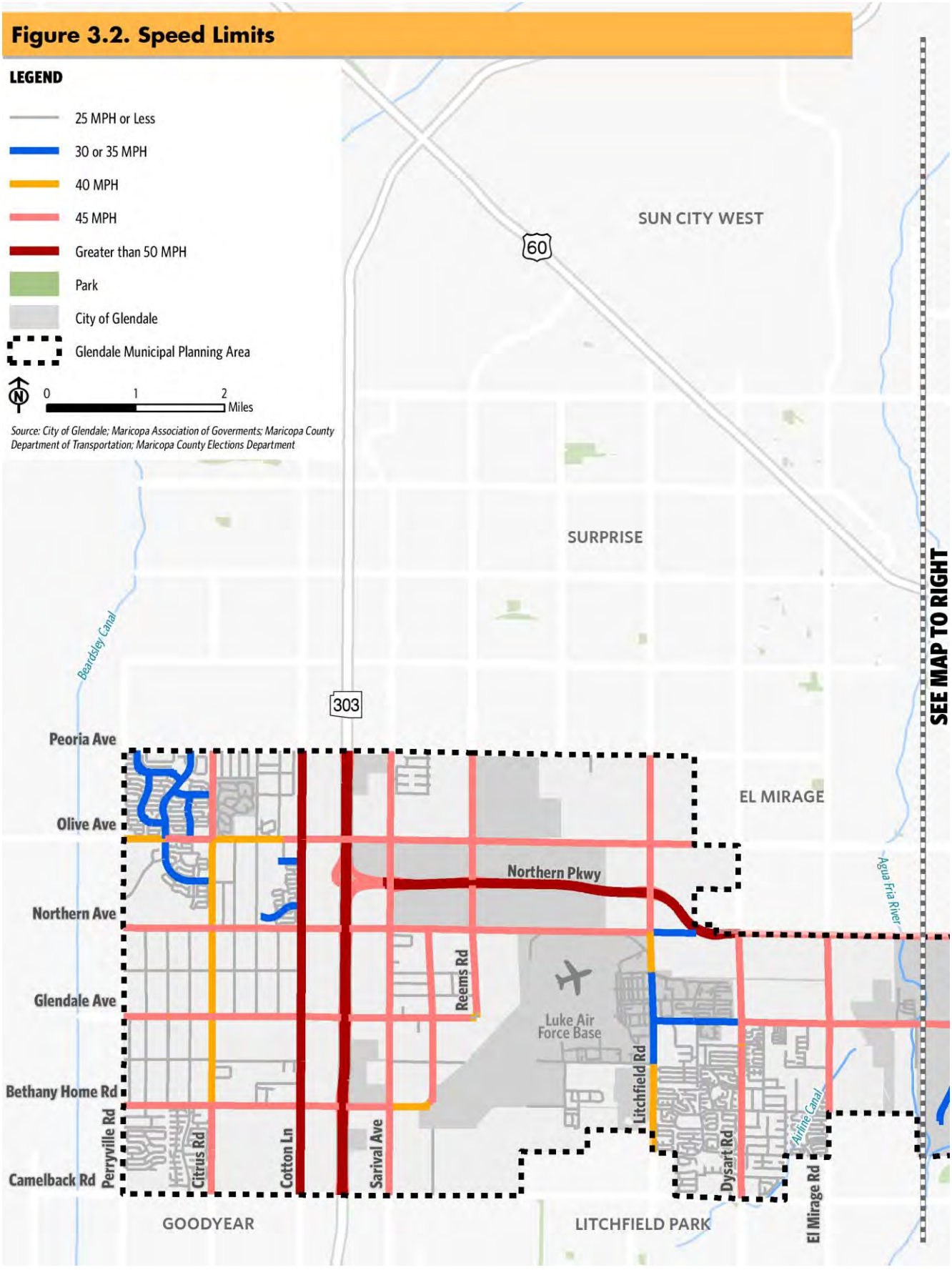

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Figure 3.2. Speed Limits



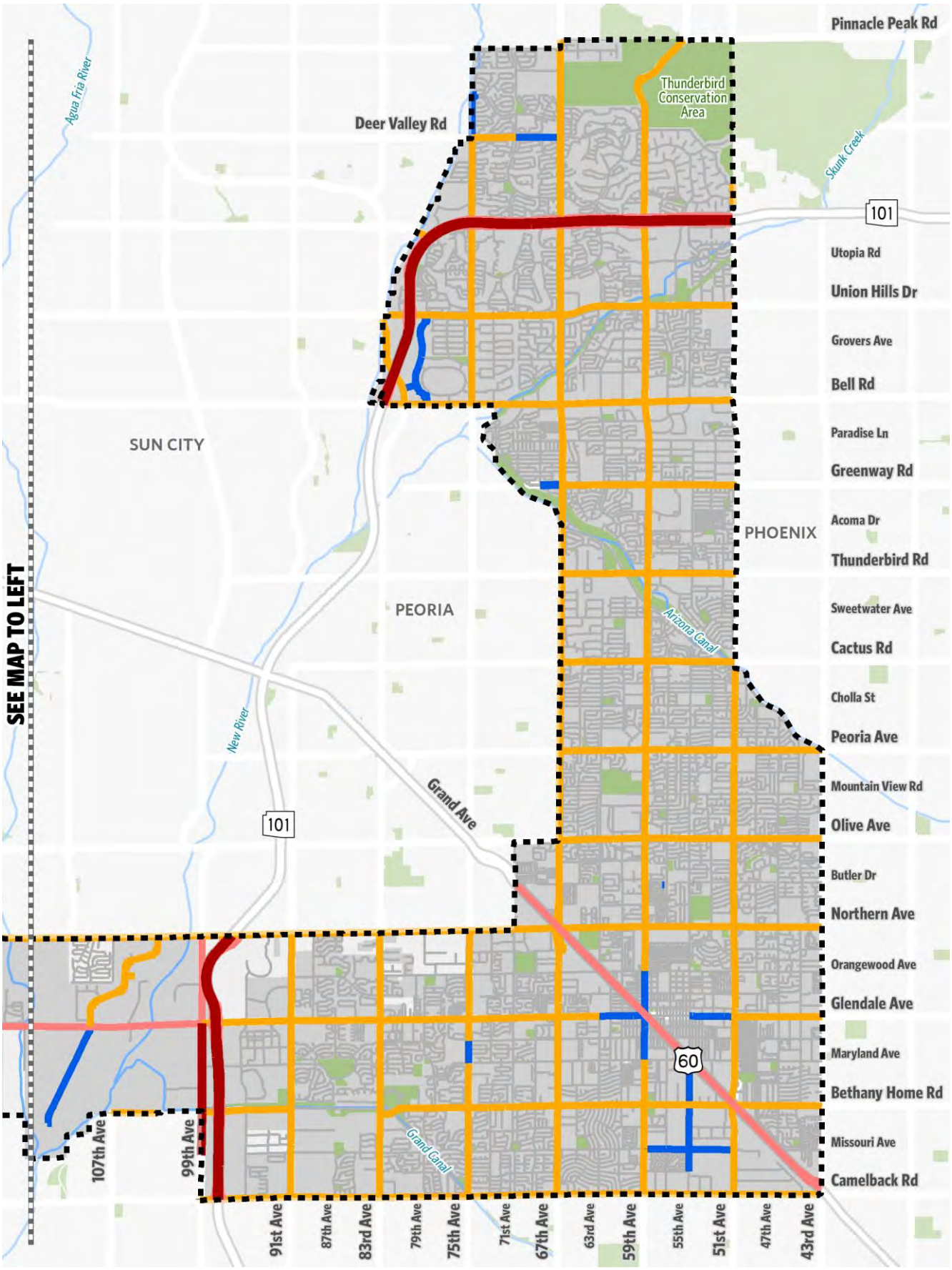
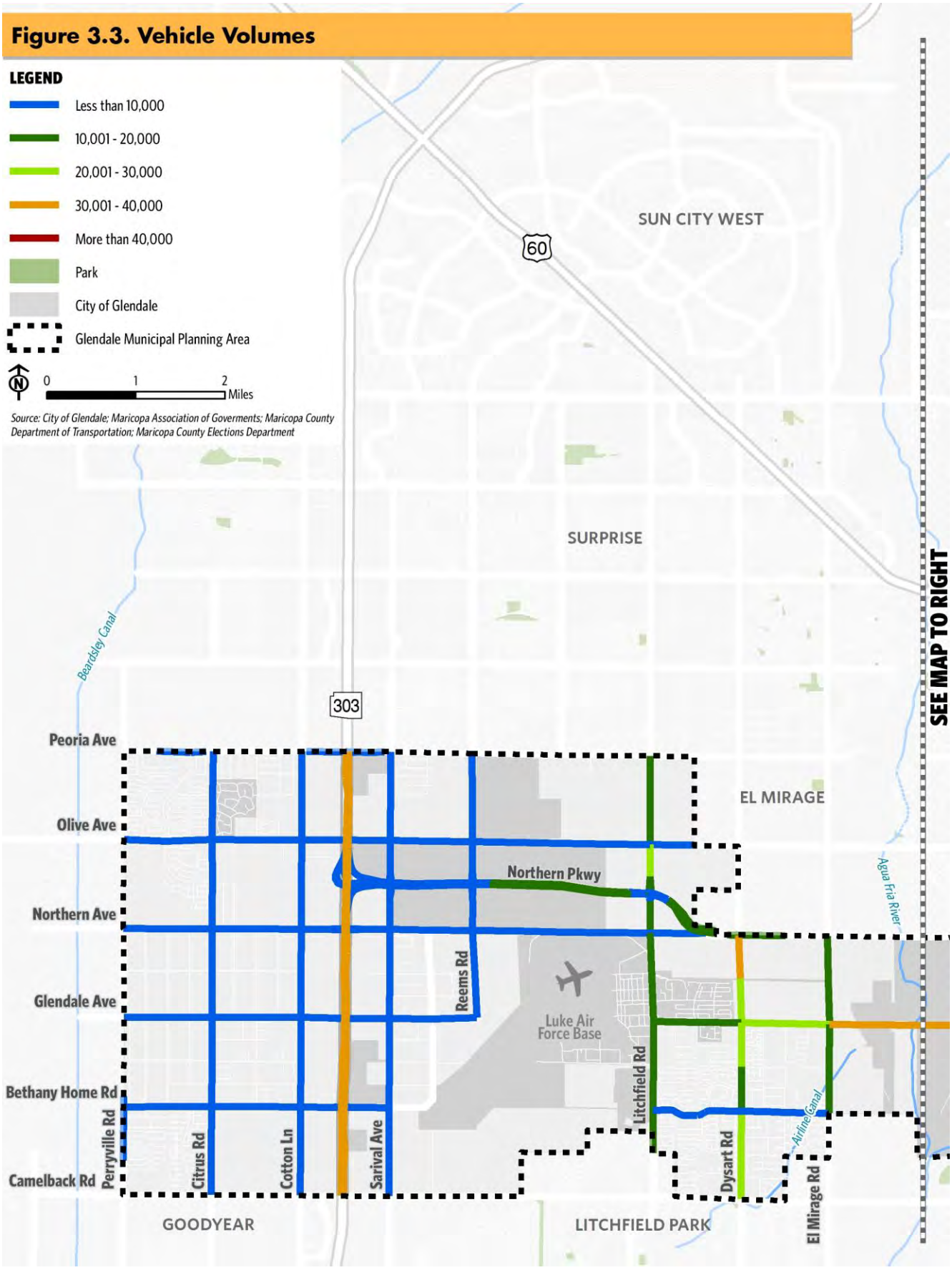

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Figure 3.3. Vehicle Volumes



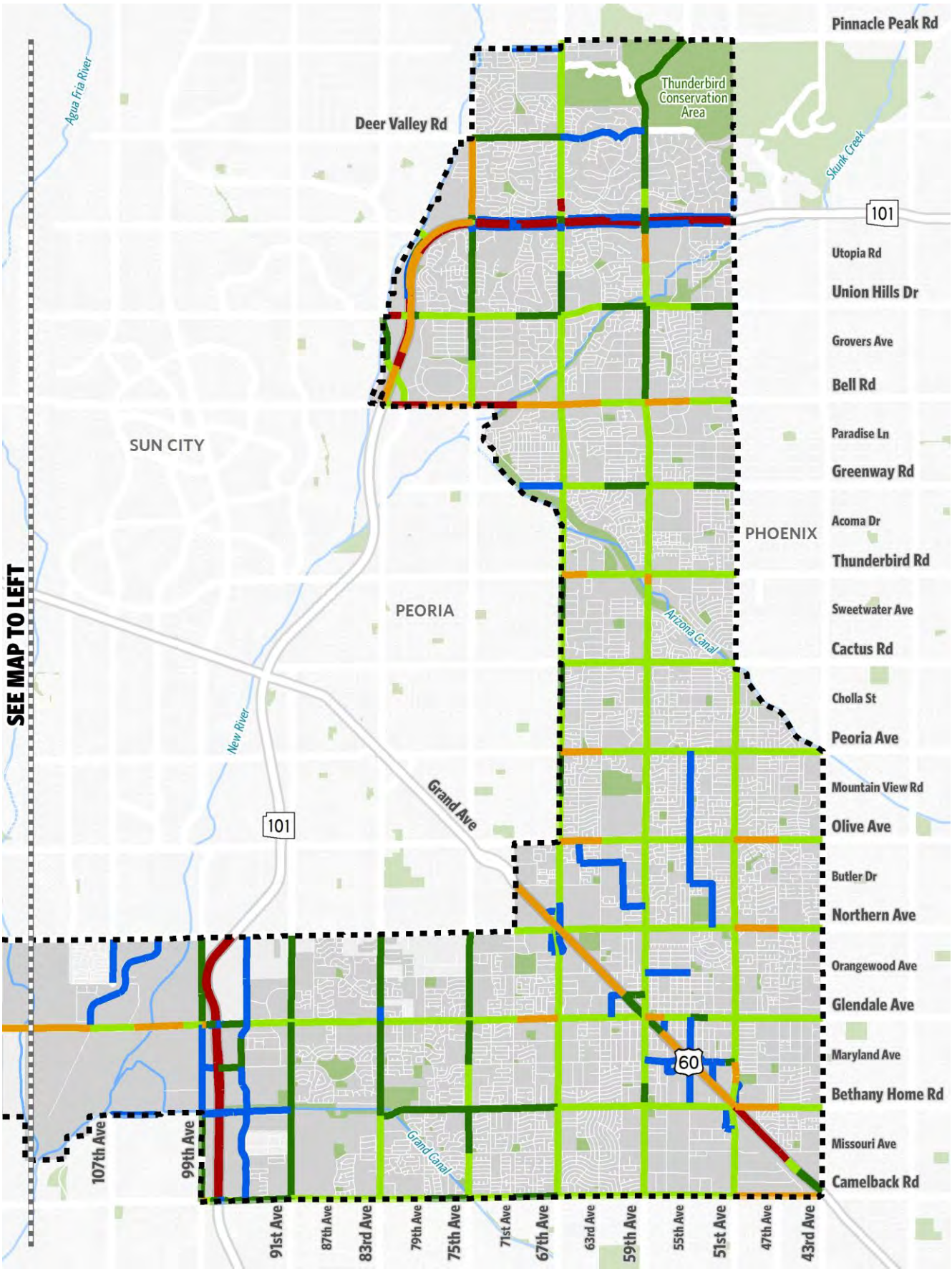
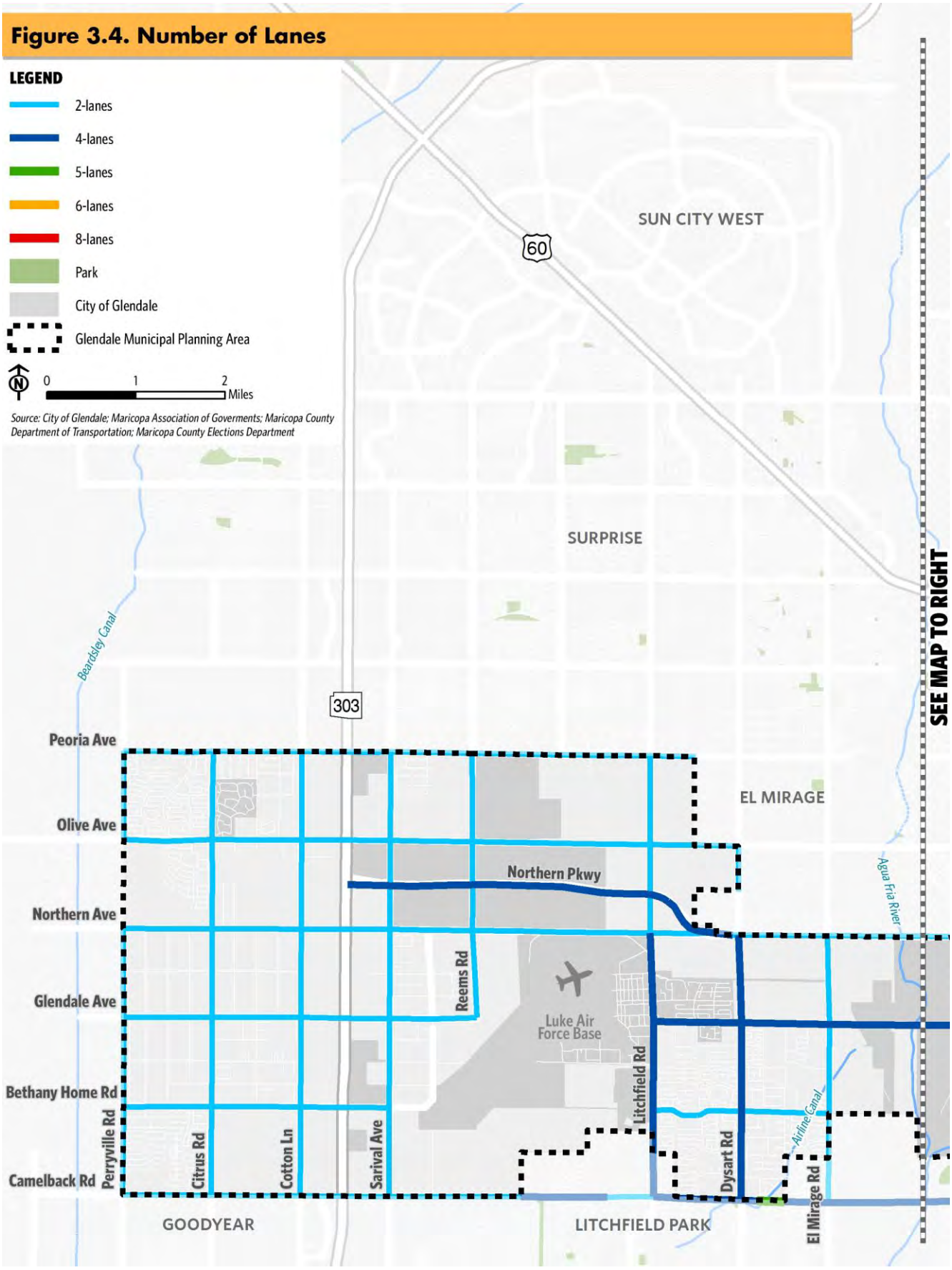
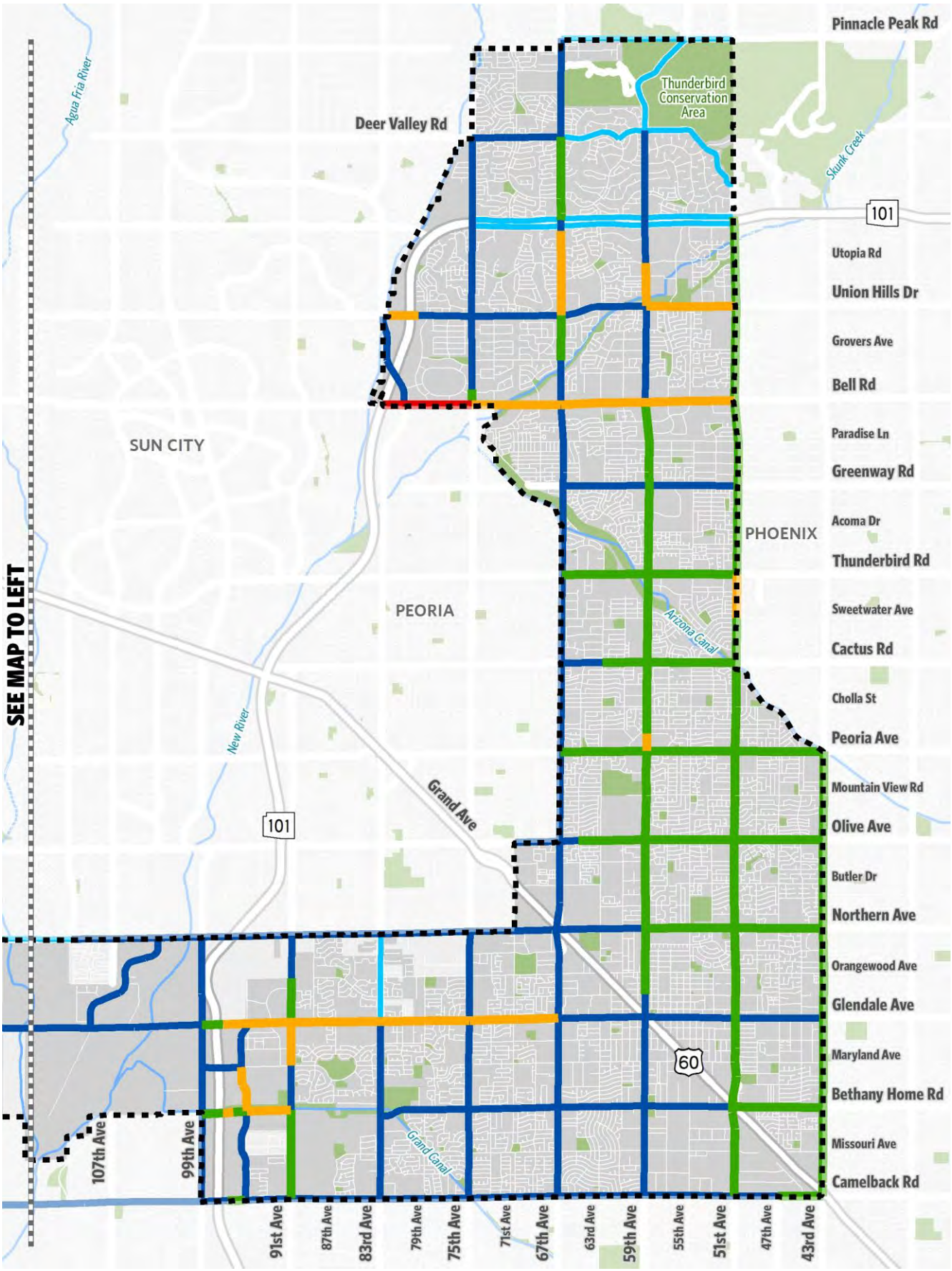

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Figure 3.4. Number of Lanes




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Safety Analysis

Analysis of pedestrian- and bicycle-related crash data provides Glendale with important safety information to help make informed decisions on improvements. A crash data analysis of pedestrian- and bicycle-related crashes over a five-year period (2012 to 2017) was conducted. Figure 3.5 and Figure 3.6 illustrate the locations of pedestrian- and bicycle-related crashes that have occurred in Glendale, respectively. Crash information is based on historical crash records reported by MAG and the Arizona Safety Data Mart and may include crashes adjacent, but not located on, Glendale roadways. As shown in Table 3.1, there were 285 pedestrian-related crashes and 312 bicycle-related crashes located on or adjacent to Glendale roadways.

Table 3.1: 5-Year Total Pedestrian and Bicycle Related Crashes

Location	Pedestrian Related	Bicycle Related	Total Fatal Crashes	Total Serious Injury Crashes
City of Glendale	285	297	45	51
Glendale MPA*	312	304	49	55

*Data includes crashes that occurred along the Glendale MPA planning boundary, which may include portions of adjacent communities. Total number of crashes includes crashes that occurred in the City of Glendale

Source: MAG and ADOT Safety Data Mart (9/1/2012 – 9/31/2012)

Driver Action

Over 25 percent of pedestrian-related crashes and nearly 44 percent of bicycle-related crashes occurred when a motor vehicle was turning right. This maneuver, often called a "right hook," takes place when a driver cuts off a bicyclist or pedestrian by turning right across their path onto a road or a driveway. In addition, over 17 percent of pedestrian-related crashes and 15 percent of bicycle-related crashes occurred when motorists made a left turn. Left turn crashes are commonly due to a motorist not seeing a pedestrian or bicyclist when turning left onto a road or intersection

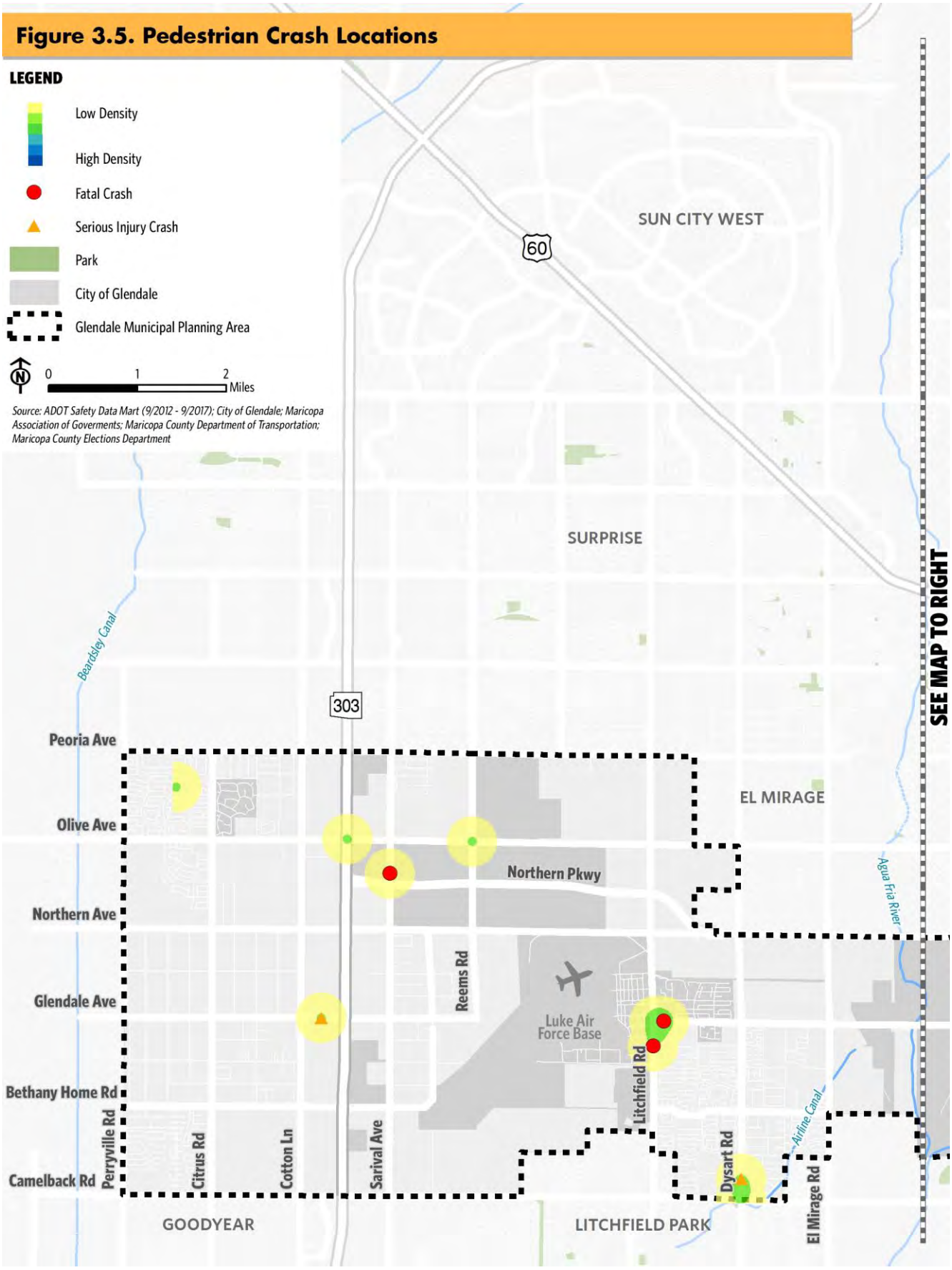


Location

Figure 3.5 illustrates the location of pedestrian- and bicycle-related crashes that have occurred in Glendale. Over 66 percent of the 616 pedestrian- and bicycle-related crashes in the Glendale MPA were intersection related. The highest number of fatal crashes occurred on Bethany Home Road from 67th Avenue to 43rd Avenue. Corridors and intersections with a significant number of pedestrian- and bicycle-related crashes include:

- Bethany Home Road: Grand Avenue to 43rd Avenue
- Glendale Avenue and 67th Avenue intersection
- Northern Avenue and 59th Avenue intersection
- Bethany Home Road and 59th Avenue intersection
- Bethany Home Road: 67th Avenue to 59th Avenue
- Glendale Avenue: 51st Avenue to 43rd Avenue

Figure 3.5. Pedestrian Crash Locations



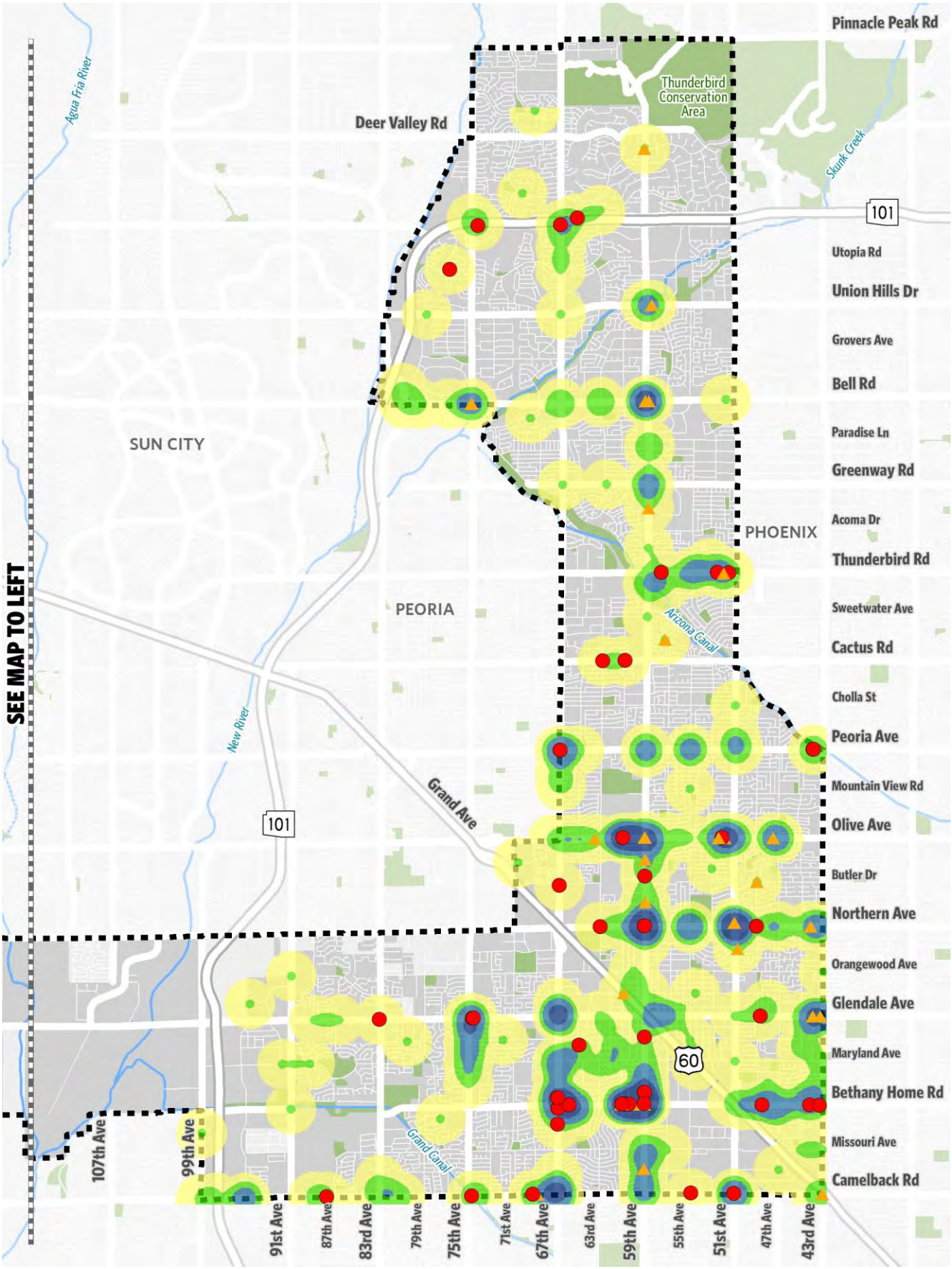
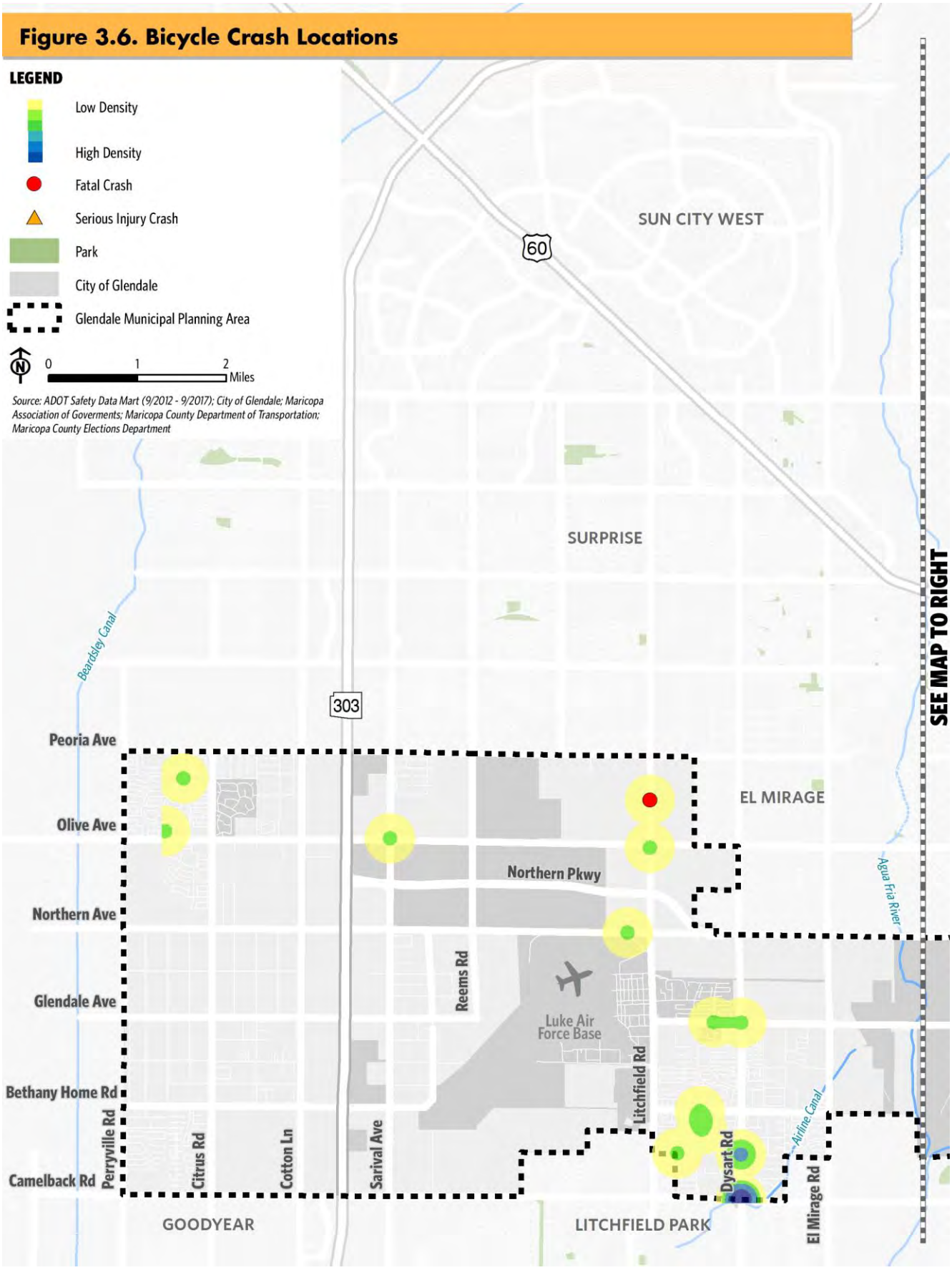
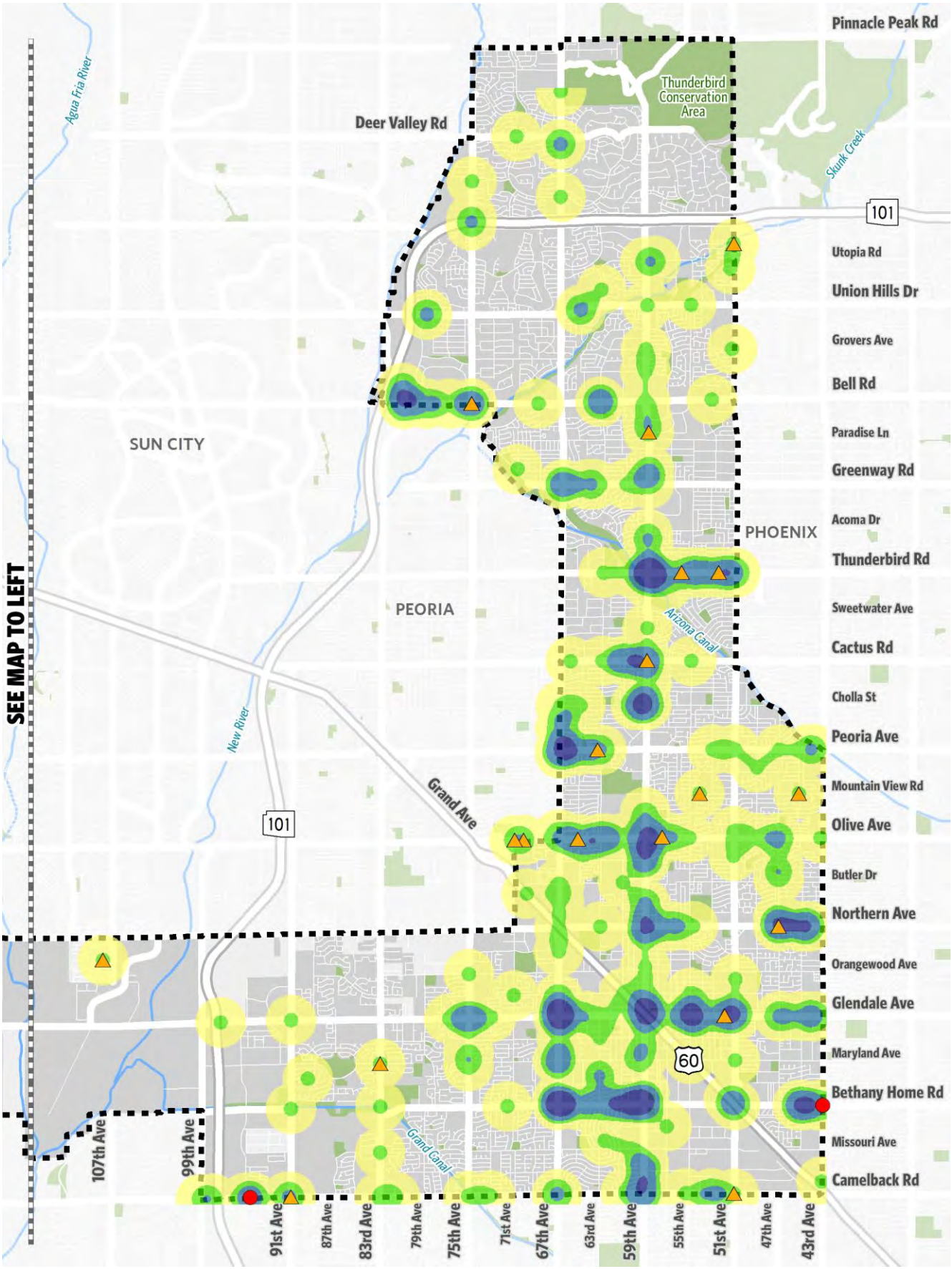

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Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
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Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
Acoma Dr
Thunderbird Rd
Sweetwater Ave
Cactus Rd
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Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
Maryland Ave
Bethany Home Rd
Missouri Ave
Camelback Rd
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43rd Ave

Figure 3.6. Bicycle Crash Locations




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Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
Skunk Creek
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Utopia Rd
Union Hills Dr
Grovers Ave
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Greenway Rd
Acoma Dr
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Existing Pedestrian Facilities and Conditions

Sidewalks provide a place for people to walk for commuting or recreational purposes; they are especially important for providing independence to the mobility impaired or persons without access to a vehicle. In addition to sidewalks, pedestrian facilities such as crossings, curb ramps, curb extensions, traffic calming features, and other improvements help create a more comfortable walking environment. In order to understand the walking environment in Glendale, an extensive mapping exercise and inventory of sidewalks and curb ramps was conducted. Glendale's *2015 Engineering Design and Construction Standards* provides guidelines for pedestrian facilities, including:

- 6 FT (minimum) sidewalks for all arterial streets, while sidewalks along collector and local roadways minimum of 5 FT wide.
- Detectable warnings are required whenever the walking surface is not separated from the roadway by curbs, railings, or other approved elements

Figure 3.7 illustrates the location and width of pedestrian sidewalks. Sporadic corridor and business development has caused gaps in Glendale's existing pedestrian network, which creates barriers to pedestrian travel.

Sidewalk Conditions

Sidewalks are the backbone of any pedestrian network; their condition affects all pedestrians, particularly individuals with disabilities. Uneven surfaces, obstructions, or poor sidewalk conditions create deterrents or barriers in the pedestrian network.

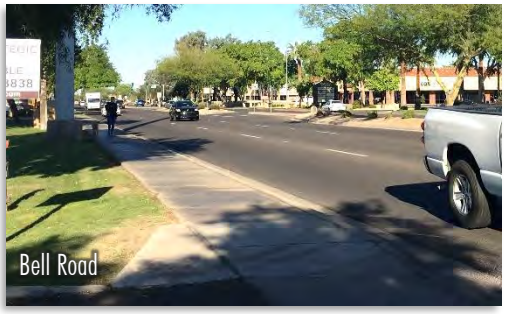
Intersection Traffic

To complete the pedestrian network, sidewalks are supported by a collection of facilities to create a more visible, navigable, and enjoyable walking experience. Facilities include signage, crosswalks, curb ramps, median refuge islands, overpasses, and signalized pedestrian crossings. At signalized intersections, it is essential that are signals phased and timed to allow pedestrians of all abilities to cross the roadway, including those who are typically slower (children, senior citizens, people with limited mobility). At the same time, signal delay must be minimized to reduce the amount of illegal and unsafe crossing that occurs when pedestrians get impatient waiting for the signal to change.

To provide pedestrian's with mid-block crossing locations, Glendale currently has pedestrian HAWK signals at the following locations (including the date signals went live):

- 61st Avenue and Bethany Home Road (5/21/2011)
- 63rd Avenue and Beardsley Road (8/1/2012)
- 65th Avenue and Bethany Home Road (9/10/2015)
- 65th Avenue and Glendale Avenue (1/16/2018)

Examples of Pedestrian Facilities



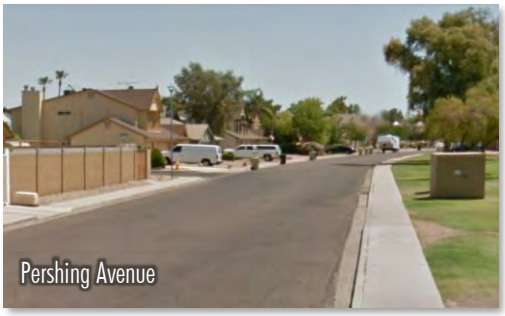
Sidewalks on Arterials

- Pathway along major travel routes with high vehicle volumes
- Intermediate and long-distance travel
- Access to many destinations



Sidewalks on Collectors

- Pathway that connects residential and commercial areas to arterials with medium to low vehicle volumes



Sidewalks on Local Roads

- Pathways on low volume roads that provide local access to homes, business, schools, etc.
- Low vehicle speeds



Buffered Sidewalks

- Pedestrian pathway offset from a roadway by a landscape, rock, or natural buffer



Curb Ramps and Crosswalks

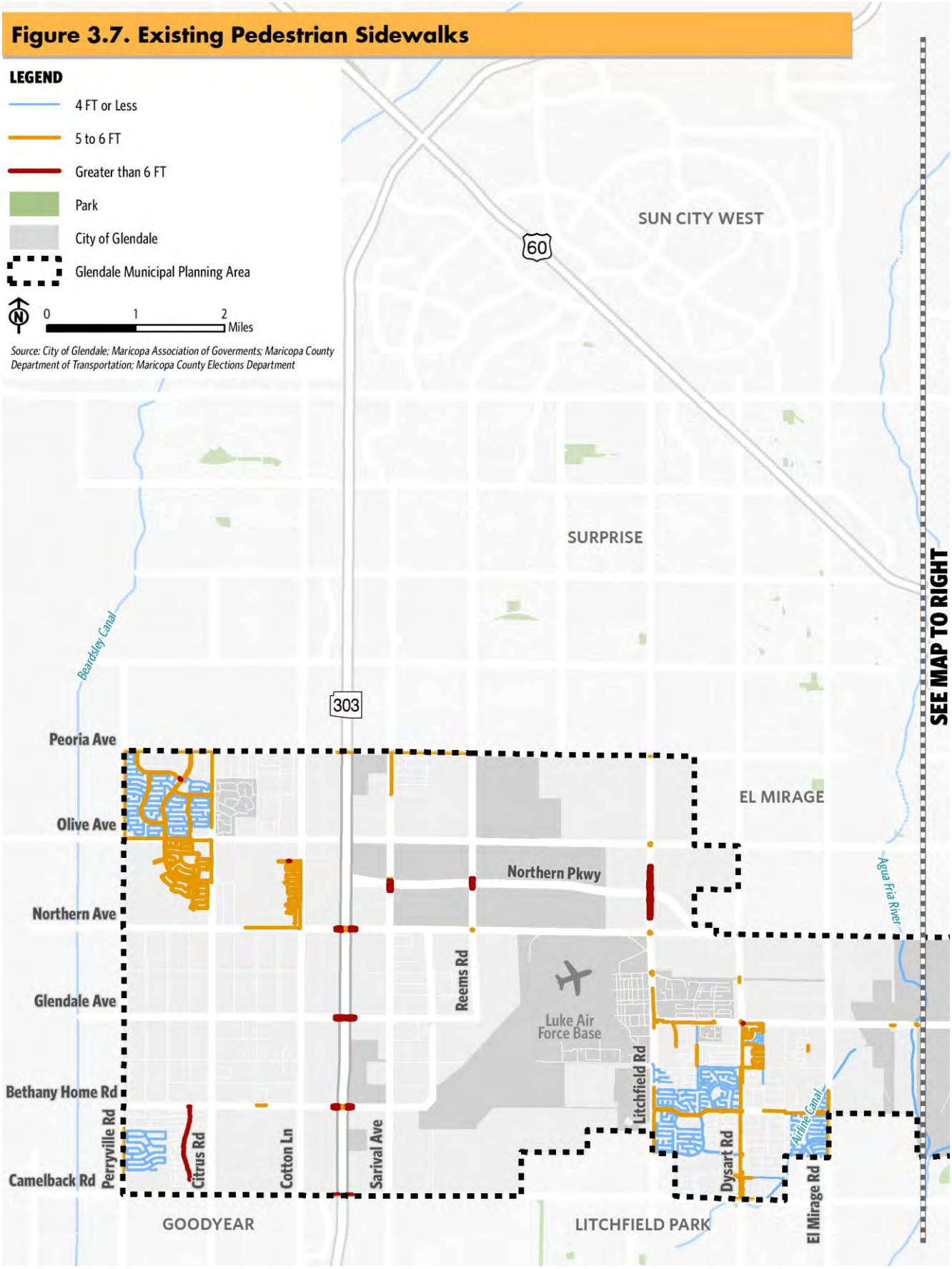
- Curb ramps provide a detectable warning for physically impaired pedestrians
- Crosswalks improve the visibility of pedestrians walking across a road

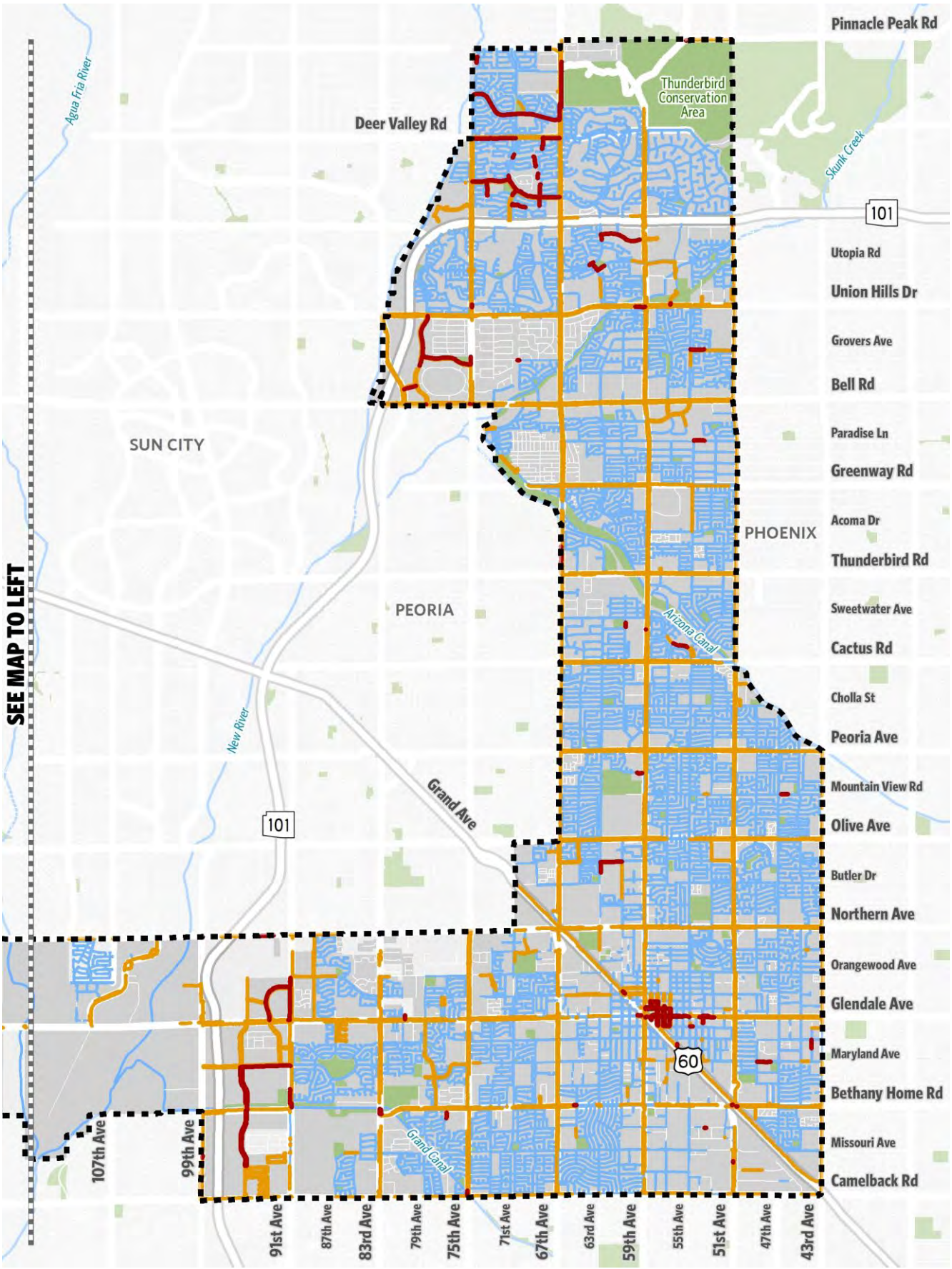


High-Intensity Activated crossWalK (HAWK)

- Traffic signal activated by a pedestrian to stop traffic and allow for the safe crossing of a road
- See page 81 for additional information

Figure 3.7. Existing Pedestrian Sidewalks




SEE MAP TO LEFT
Agua Fria River
Deer Valley Rd
Thunderbird Conservation Area
Skunk Creek
Pinnacle Peak Rd
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Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
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Bethany Home Rd
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Existing Bicycle Facilities and Conditions

Bicycling is an essential component of any transportation system that benefits numerous areas including health, economics, environment, and equity. Since the *1990 Bicycle Plan*, Glendale has made progress in providing bicycle facilities along its maintained roadways.

Location and Types of Bicycle Facilities

In total, there are 42.5 centerline miles of bike lanes and routes, 7.9 miles of paved shoulders, and 30.6 miles of shoulder stripe with parking within the Glendale MPA. Of the roadways with bike facilities, 40% have on-street parking. In addition, there are 72.4 miles of shared use paths. As illustrated in Figure 3.8, bicycle facilities are primarily located on collector streets. Shared use paths are largely located around master planned communities, which may be attributed to developers constructing the paths during development. Currently the City does not have specific design guidelines for bicycle facilities. For the purposes of the active transportation inventory, typical bicycle facilities available in the Glendale MPA include:

- Bike Lane – striped lane (4 FT or wider) may include pavement marking and a designated buffer space separating the bike lane from motor vehicle travel lanes. Facilities may have curb and gutter present or a flush shoulder.
- Bike Route – signed shared roadway that provides shared use for bicyclist and vehicles.
- Paved Shoulder – wide shoulder adjacent to the roadway that provides a functional space for bicyclists to travel. Typically, does not have curb and gutter.
- Shoulder Stripe with Parking – 5 to 6 FT wide pavement striping where parking is not prohibited.
- Shared Use Path - bidirectional pathway that is physically separated from motorized vehicular traffic by an open space or barrier. Shared use paths may be used by pedestrians, bicyclists, skaters, wheelchair users, joggers and other nonmotorized users.

Regional Bikeway Networks

Regional bicycle networks help to resolve complex, interrelated issues concerning traffic congestion, air quality, public health, and livability. Glendale's bicycle network is supported by a larger, regionwide network of bicycle facilities that has been developed and is maintained by the numerous municipalities and agencies in the region. Collectively, this regional bicycle network consists of a combination of bike lanes, bike routes, paved shoulders, and shared use paths. While the network is vast and far-reaching, Glendale has a tremendous opportunity to collaborate with neighboring jurisdictions to expand the region's bicycle network.

Off Street Paths and Trails

When bicycle and pedestrian facilities are connected to recreational areas they act as an extension of the transportation system. Connecting parks and other recreational facilities via bicycle and pedestrian facilities is a way to make parks more accessible and provide a safe and convenient means for residents to explore the recreational system. Off-street paths and trails are open to bikers, walkers, hikers, runners, and often equestrians. Figure 3.8 illustrates the locations of major local and regional trails.

Examples of Bicycle Facilities



Bike Lane

- Striped lane with pavement markings and signs that designate an exclusive lane for bicycle use
- 48.4 centerline miles



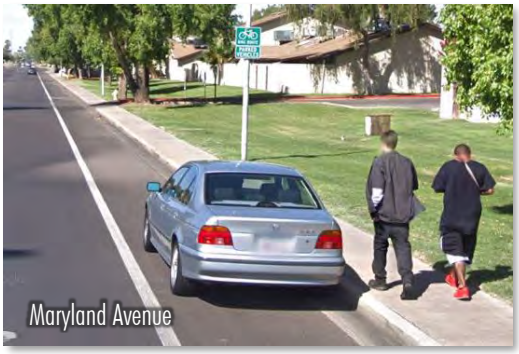
Buffered Bike Lane

- Increases riding space and comfort by adding a painted buffer to a standard bike lane
- 3.2 centerline miles



Bike Route

- Route where the travel lane is shared by drivers and bicyclists.
- 3.0 centerline miles



Shoulder Stripe with Parking

- Signed as "Bike Route Parked Vehicles". Striped shoulder typically 5-6 FT wide that is shared with parked vehicles
- 92.3 centerline miles



Paved Shoulder

- Paved shoulder on the edge of pavement can be widened and enhanced to allow bicyclists more separation from vehicles
- 15.6 centerline miles

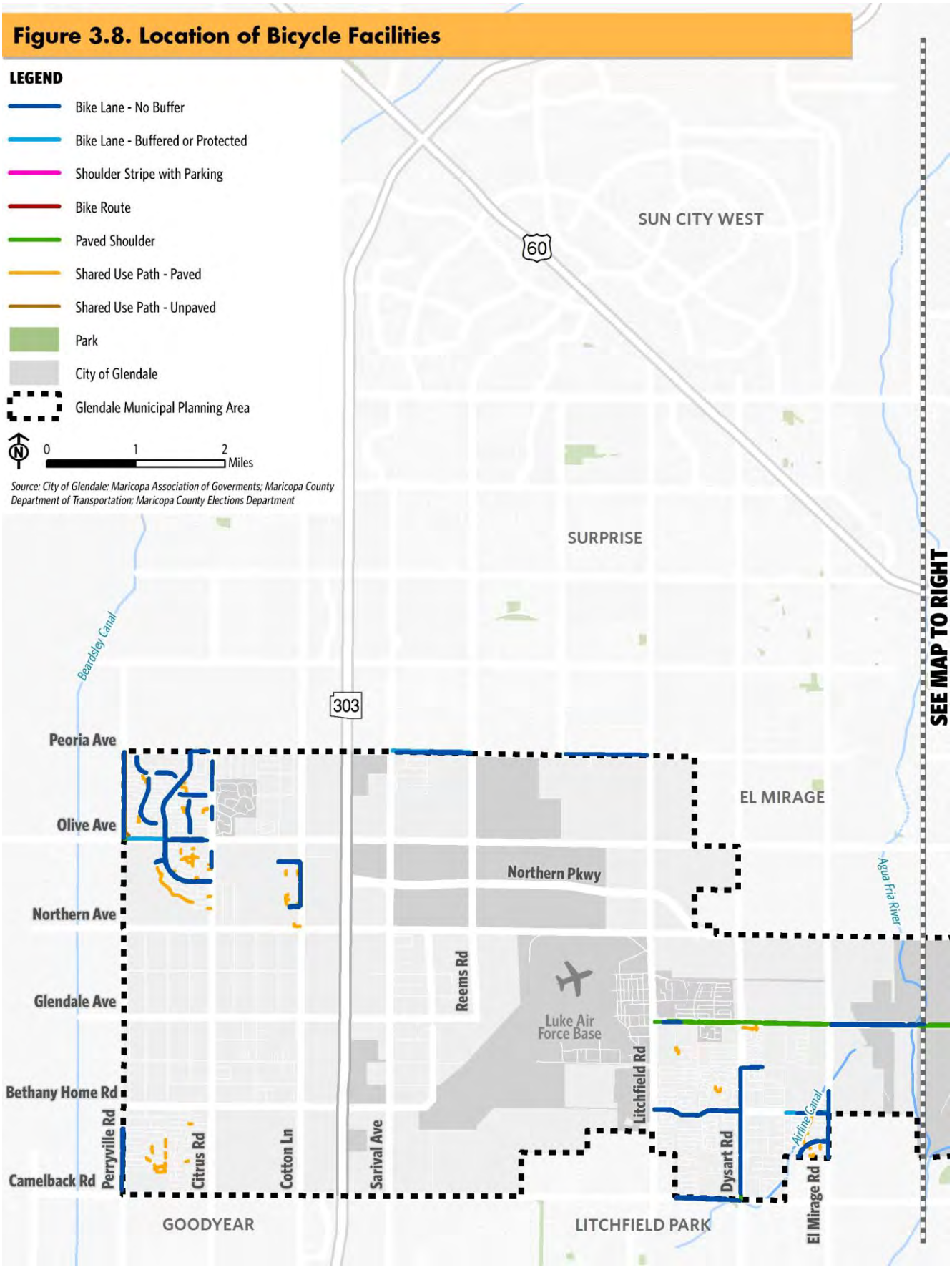


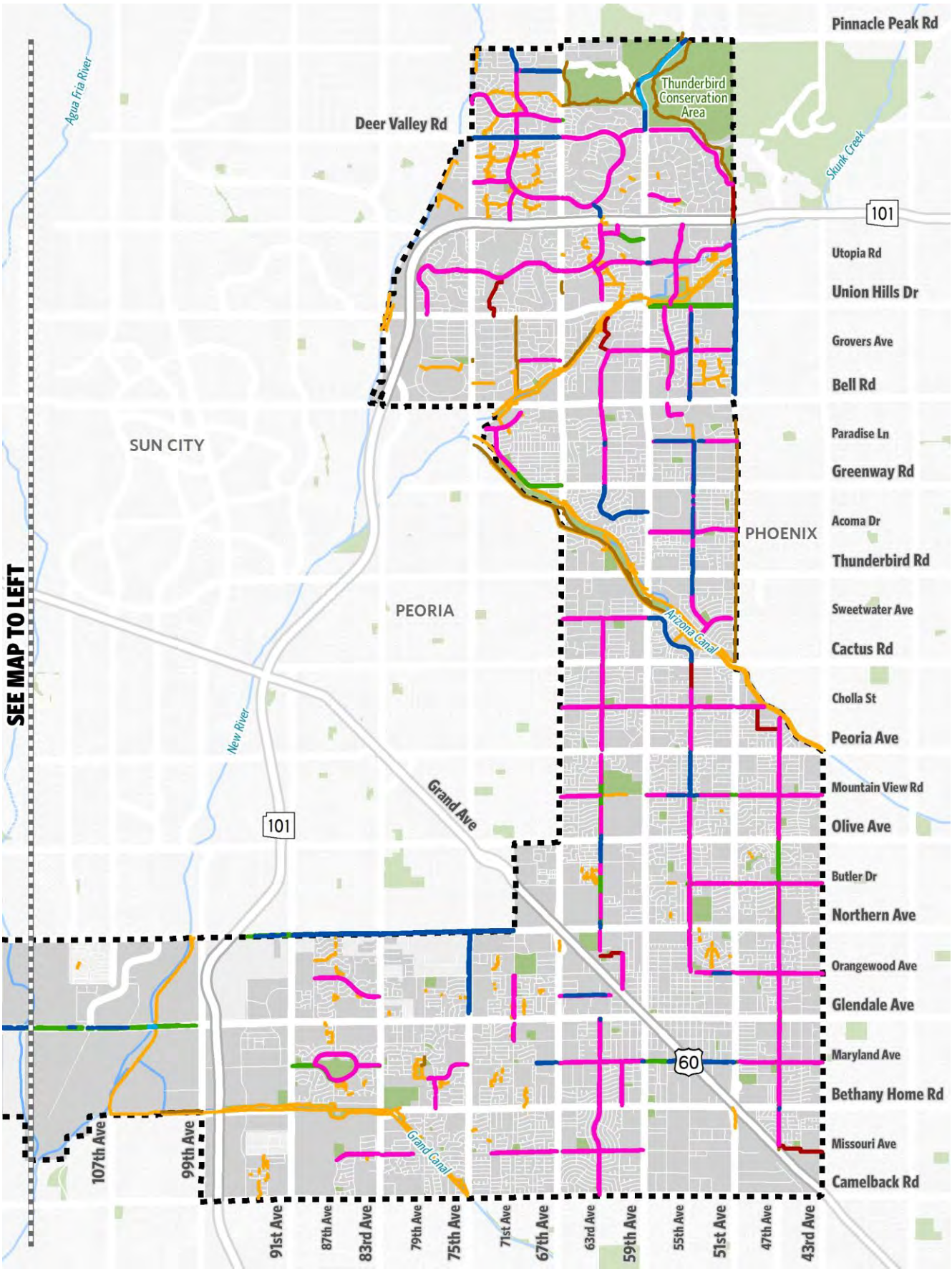
Shared Use Path

- Off-street facility dedicated exclusively for non-motorized travel
- 72.4 miles in Glendale (includes non-Glendale maintained shared use paths)

Note: Mileage includes all facilities located within the Glendale MPA and may include facilities not owned or maintained by the City.

Figure 3.8. Location of Bicycle Facilities




SEE MAP TO LEFT
Agua Fria River
Deer Valley Rd
Thunderbird Conservation Area
Skunk Creek
Pinnacle Peak Rd
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
Acoma Dr
Thunderbird Rd
Sweetwater Ave
Cactus Rd
Cholla St
Peoria Ave
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43rd Ave

Existing Transit Services

Understanding the location of transit routes and stops is critical when developing a walking and biking network, because almost every transit trip begins or ends with walking and/or biking. Often people who could potentially utilize transit choose to drive because no transit stops are conveniently located near their starting points or final destinations. Placing walking and biking facilities along "first and last mile" paths expands a person's transportation choices by making transit more accessible. Integrating active transportation networks and transit helps to create a balanced and efficient multimodal transportation network that makes transportation affordable, convenient, and flexible for all users regardless of their age, ability, or socioeconomic status.

In the study area, the City of Glendale, Valley Metro, and the City of Phoenix provide public transportation services, including:

- Glendale Urban Shuttle (GUS) – free, local neighborhood circulator system that connect residential areas to major destinations and the regional transit network. All GUS vehicles are wheelchair accessible and include bicycle racks.
- Fixed-Route – 13 fixed-routes operate in Glendale, primarily along arterial roads. Service hours and frequency vary by route, as well as the condition of bus stops.
- Express Bus – two express bus routes provide peak hour service to and from downtown Phoenix.



Picture of the Glendale Urban Shuttle (GUS) free, local neighborhood circulator shuttle

Existing transit service routes within the study area are shown in Figure 3.9.

Ridership

Ridership information provides important information on where people are accessing transit. Table 3.2 outlines the bus stops with the highest average daily ridership within the Glendale MPA.

Table 3.2. 2017 Average Daily Ridership in Glendale MPA

Intersection*	Route Served	Weekday	Weekend	Total
Glendale Avenue/67th Avenue	GUS 1 | 67 | 70	1,327	526	1,853
Bethany Home Road/ 59th Avenue	59 | 60	1,087	566	1,653
Glendale Avenue/43rd Avenue	GUS 2 |43 | 70	905	571	1,476
Northern Avenue/51st Avenue	GUS 2 |51 | 80	832	458	1,290
Bethany Home Road/43rd Avenue	GUS 2 |43 | 70	905	371	1,276
Olive Avenue/59th Avenue	GUS 3 | 59 | 90	761	307	1,068
Arrowhead Mall Transit Center	67 | 83 | 170 | 186	476	427	903
Northern Avenue/59th Avenue	GUS 1, 3 | 59 | 80	647	327	974

**Only includes bus stops located within the Glendale MPA. Source: Valley Metro and City of Glendale*

Bicycle and Wheelchair Boardings

Valley Metro and the City of Glendale track the number of bicycle and wheelchair boardings, which provides vital information on where active transportation facilities are most needed. Table 3.3 outlines the routes with the highest annual bicycle and wheelchair boardings in Glendale.

Table 3.3. Bicycle and Wheelchair Boardings (FY 2018)

Highest Annual Bicycle Boardings				Highest Annual Wheelchair Boardings			
Route	Boardings	Bikes	%	Route	Boardings	Wheelchair	%
70: Glendale Ave.	482,035	1,091	0.2%	GUS	104,385	316	0.3%
59: 59th Avenue	428,842	727	0.2%	59: 59th Avenue	428,842	279	0.1%
51: 51st Avenue	226,983	625	0.3%	70: Glendale Ave.	482,035	257	0.1%
67: 67th Avenue	299,522	605	0.2%	67: 67th Avenue	299,522	161	0.1%
60: Bethany Home Rd.	194,749	459	0.2%	51: 51st Avenue	226,983	135	0.1%
170: Bell Rd.	122,571	306	0.2%	60: Bethany Home Rd.	194,749	118	0.1%
90: Olive Ave.	146,197	258	0.2%	90: Olive Ave.	146,197	94	0.1%
80: Northern Ave.	119,444	245	0.2%	170: Bell Rd.	122,571	83	0.1%
106: Peoria Ave.	91,261	212	0.2%	80: Northern Ave.	119,444	77	0.1%
186: Union Hills Dr.	34,885	184	0.5%	106: Peoria Ave.	91,261	76	0.1%
83: 83rd Avenue	43,299	135	0.3%	138: Thunderbird Rd.	62,278	43	0.1%
138: Thunderbird Rd.	62,278	132	0.2%	83: 83rd Avenue	43,299	30	0.1%
GUS	104,385	83	0.1%	186: Union Hills Dr.	34,885	28	0.1%
Total	**2,356,451**	**5,062**	**0.2%**	**Total**	**2,356,451**	**1,697**	**0.1%**

Bus Stops

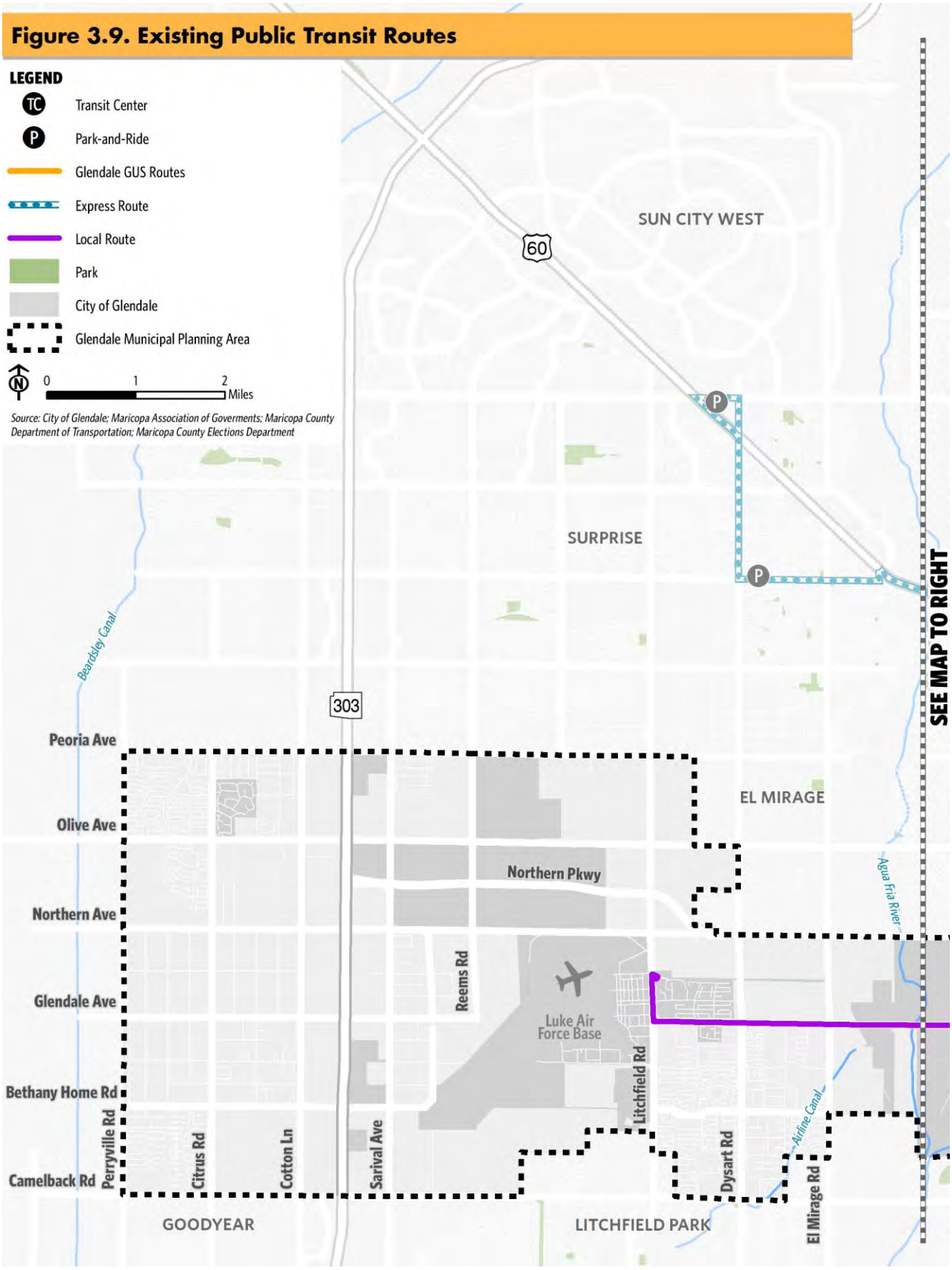
Bus stops provide a safe boarding location for passengers while each route provides options for Glendale residents to access nearby and regional destinations. Typically bus stops are located at one-quarter mile intervals for convenience. In addition to Valley Metro routes, Glendale operates three circulator services called Glendale Urban Shuttle (GUS). GUS 1 is a downtown circulator that provides daily services in a 30-minute loop. GUS 2 and 3 are larger community circulator routes that provide service Monday through Friday. More information on GUS can be found at https://www.glendaleaz.com/transit/GUS.cfm.

In 2017, Valley Metro conducted a bus stop inventory to determine what types of facilities are present at each stop. Data developed by Valley Metro was verified and updated to reflect existing conditions in Glendale. Within the study area, there are 539 bus stops with varying amenities. Typical facilities found within the study area include:

- Bus Shelters – 172 bus stops are sheltered
- Bicycle Racks – 10 bus stops are equipped with bicycle racks
- Bicycle Lockers – Glendale Ave/99th Ave Park-and-Ride has four bike lockers
- Benches – 224 bus stops have benches
- Sidewalk Connectivity – 26 bus stops do not have sidewalk connections

Figure 3.10 illustrates the results of the bus stop inventory.

Figure 3.9. Existing Public Transit Routes



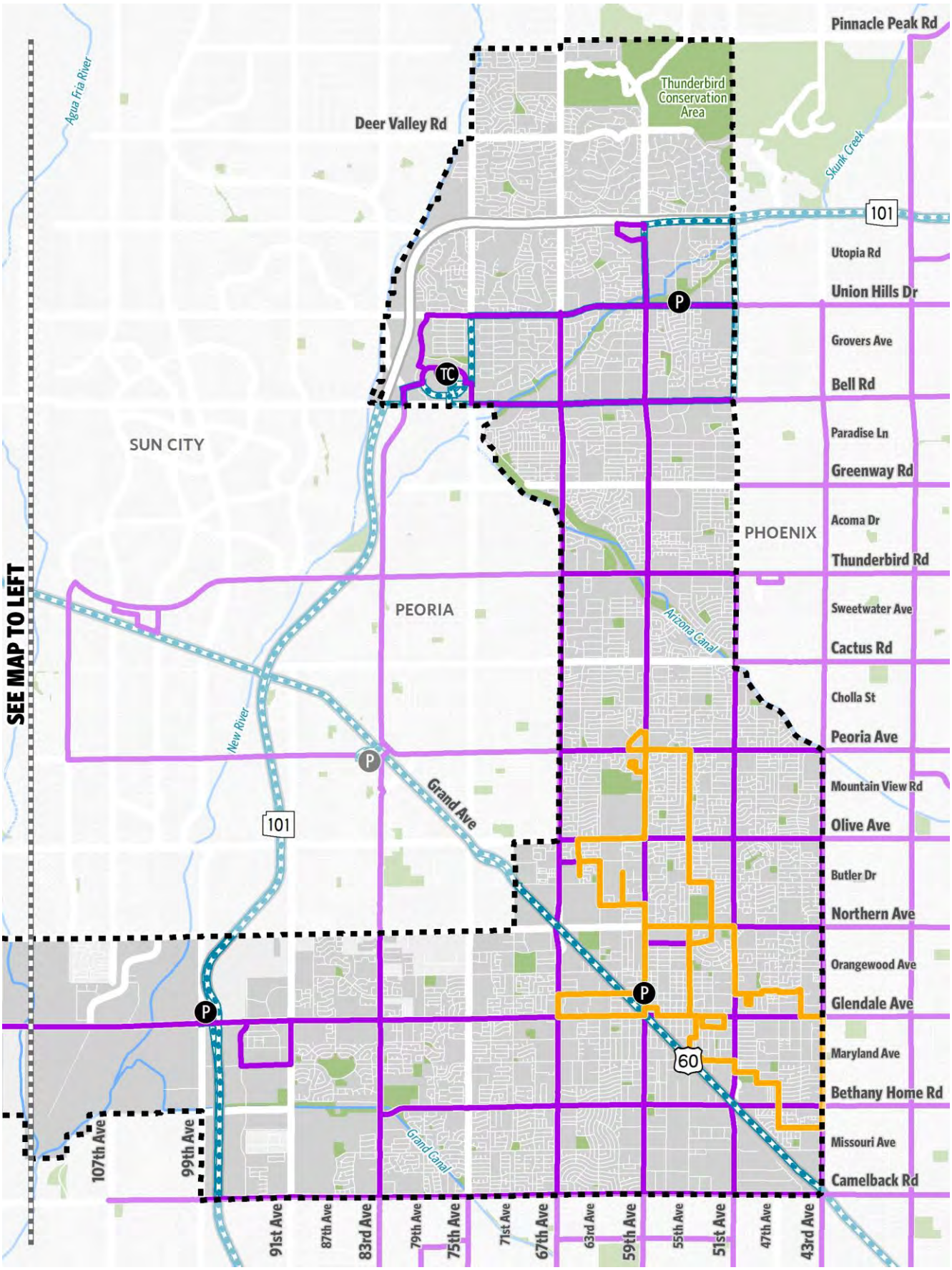
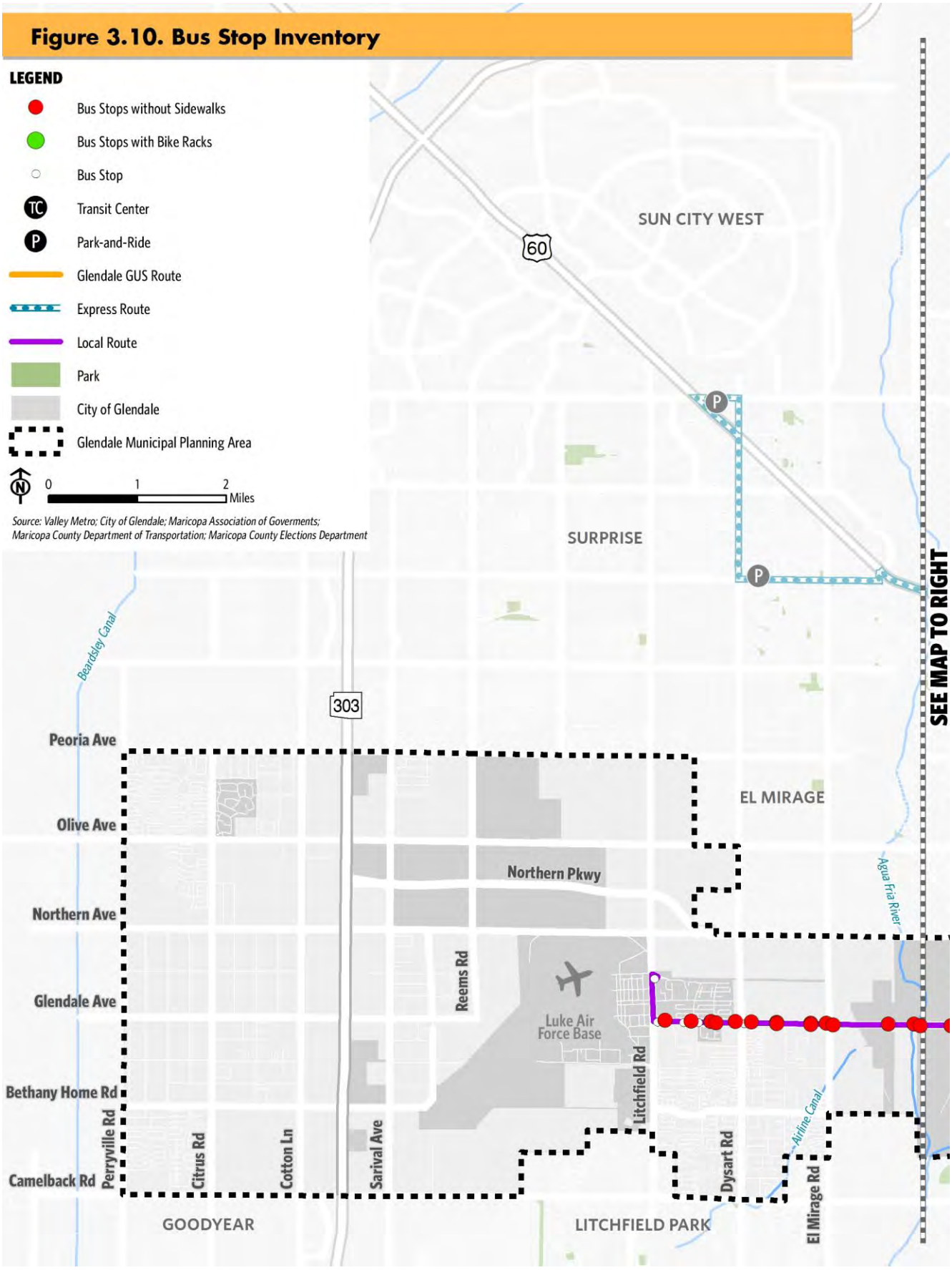
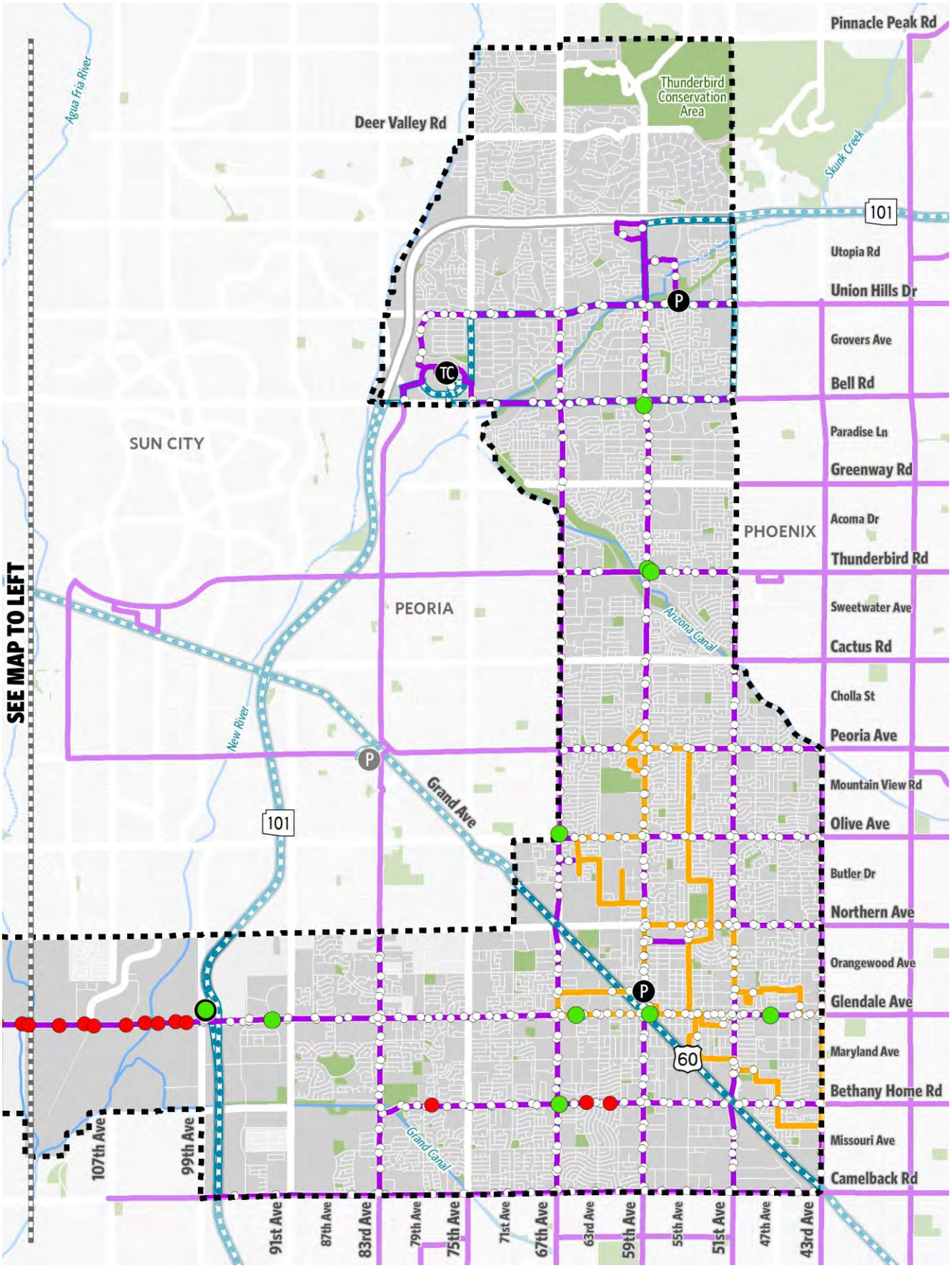

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Figure 3.10. Bus Stop Inventory




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Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
Skunk Creek
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Active Transportation Demand Analysis

The active transportation demand model estimates the level of potential walking and biking trips based on where people live, work, play, access transit, go to school, and other factors known to influence travel behavior. Often referred to as "Latent Demand", the model looks at potential trip generators (i.e., where people begin or end their trips such as home or work), as well as trip attractors (i.e., school, library, places to shop, etc.).

The results of the Active Transportation Demand Model, shown in Figure 3.11, illustrate a high demand for walking and biking. Glendale areas with the highest latent demand include:

- Areas along Bell Road, Greenway Road, and Union Hills Drive in the Northern Developed area
- Downtown Glendale
- Areas along Olive Avenue, Northern Avenue, and Bethany Home Road in the Downtown Core area
- Areas along Bethany Home Road
- Residential areas near the Luke Air Force Base

Demand results provide important information on where active transportation investments are most needed. It is important to note though, that distance is an important predictor for where active transportation investments may be the most impactful. Even if an appealing and comfortable path is available, the average person is unlikely to travel more than five miles by bike or a mile on foot to commute to work or school, grocery shopping, etc.

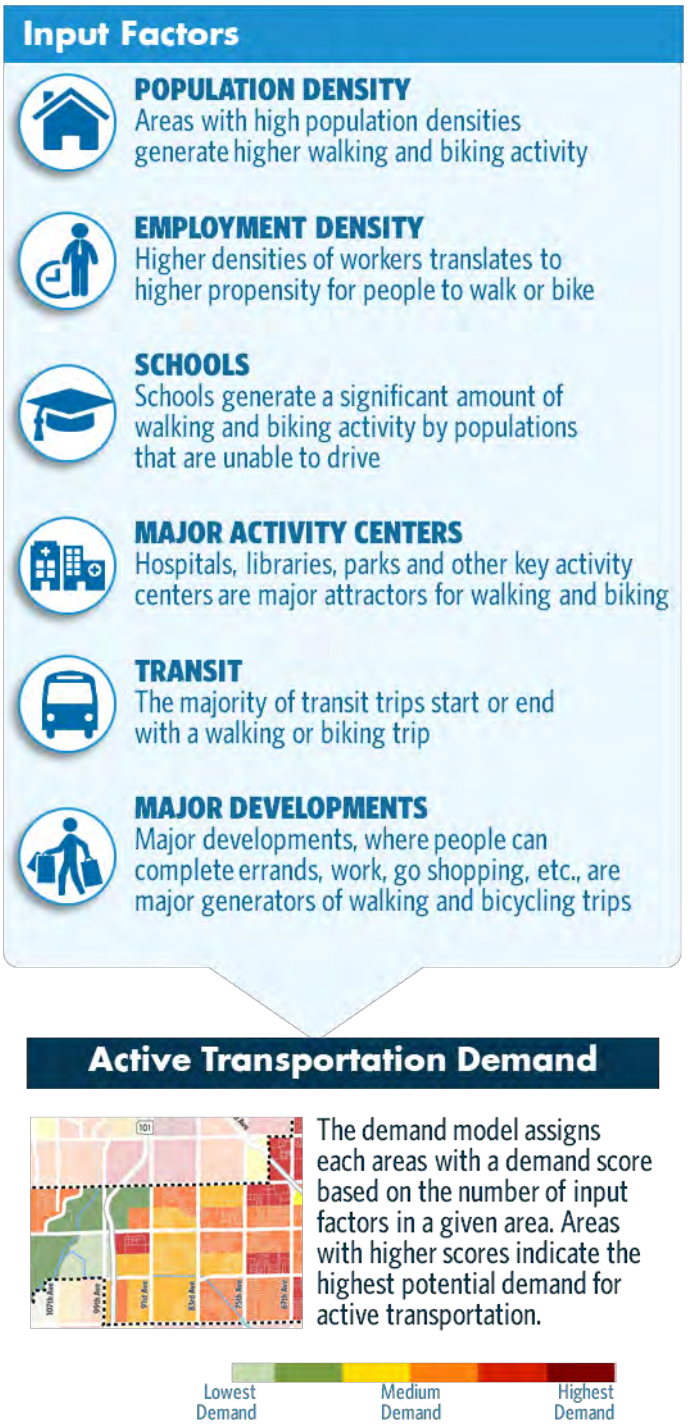
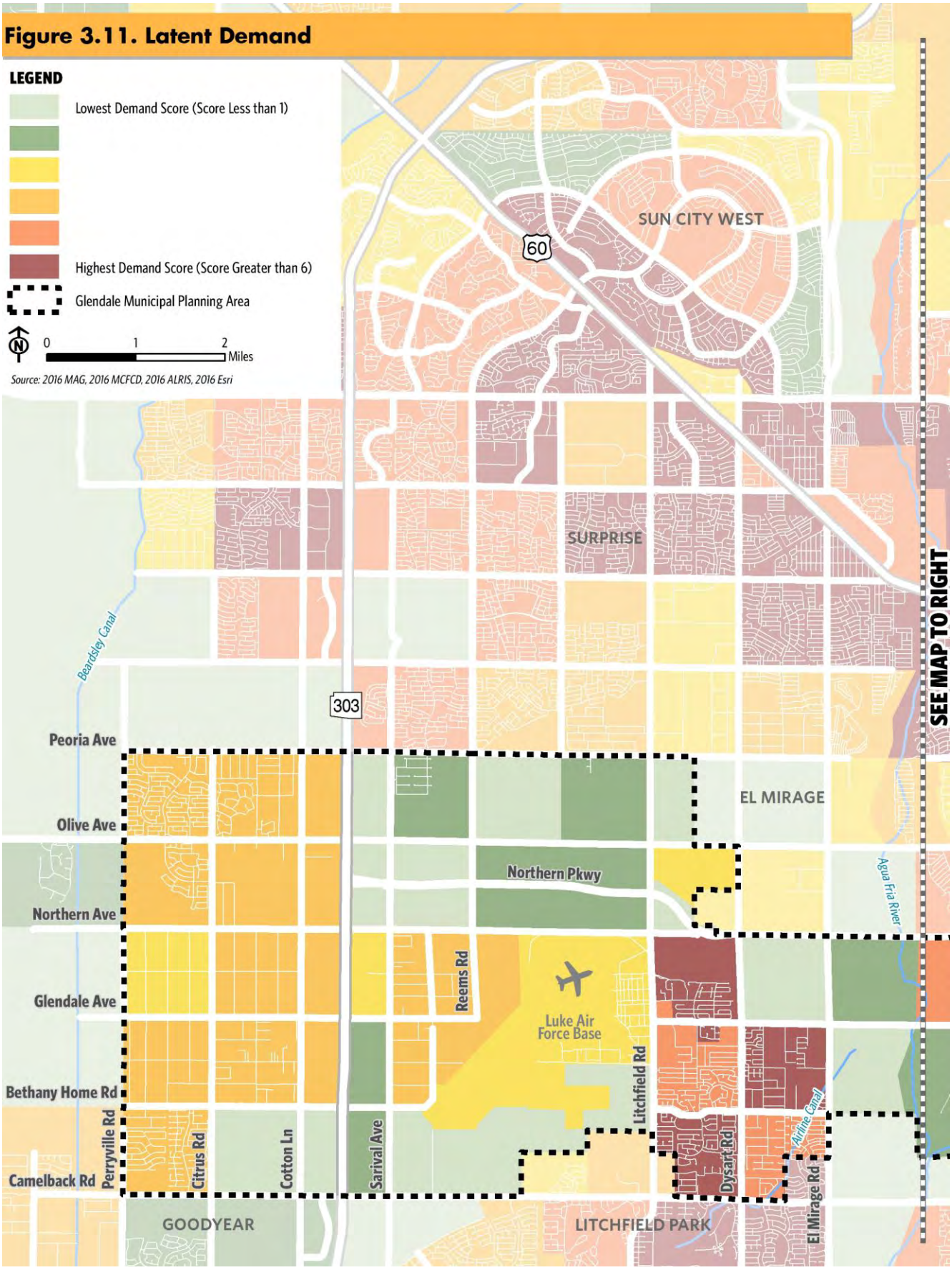
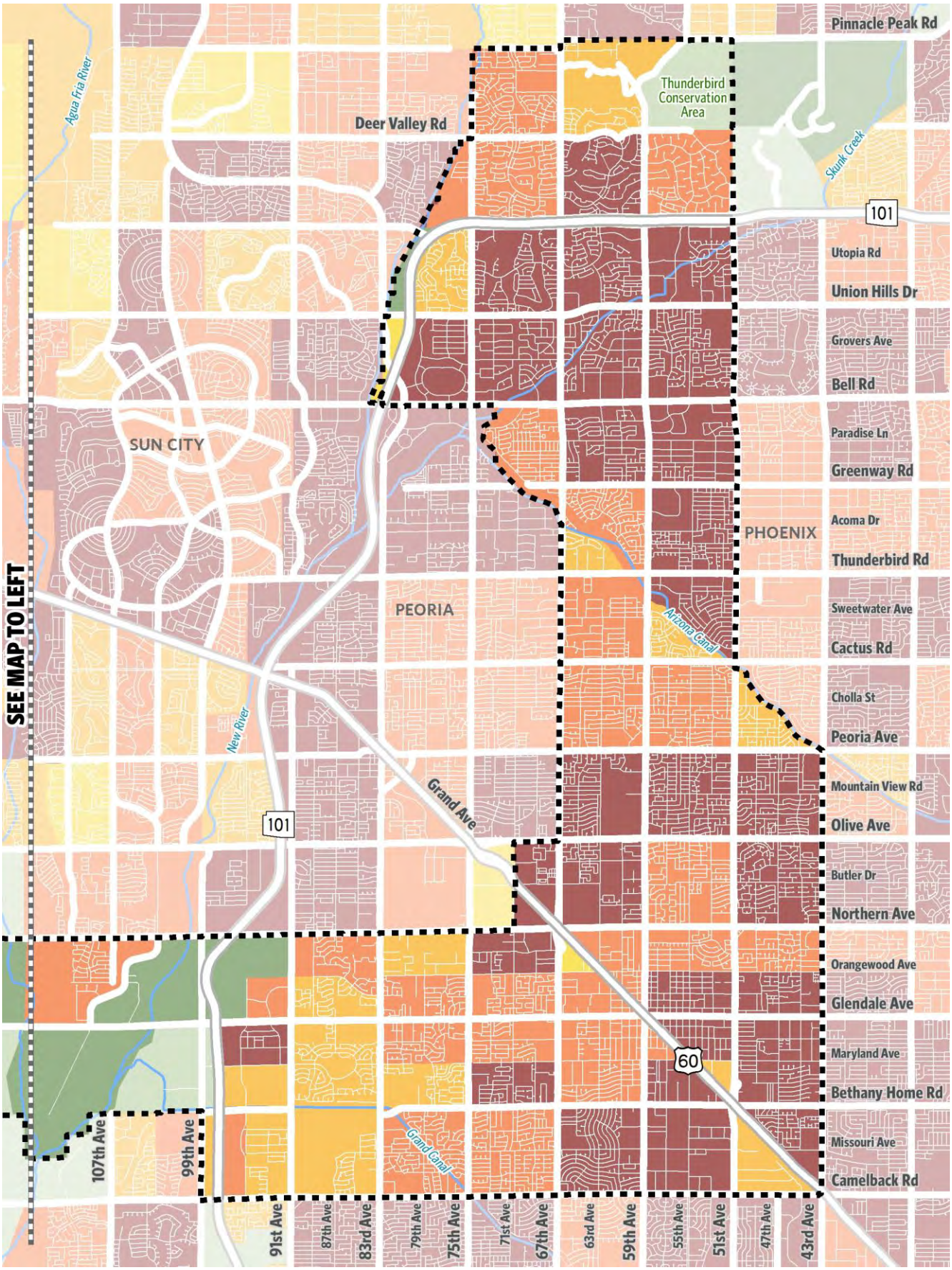


Figure 3.11. Latent Demand




Pinnacle Peak Rd
Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
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Pedestrian Level of Comfort

While each person experiences a different level of comfort when walking, there are basic roadway characteristics that affect the user experience. Nationally, there is no standard methodology to quantify a pedestrian's level of comfort; however, a pedestrian's comfort is primarily affected by factors such as:

- Sidewalks: existing/non-existing, buffer, condition
- Roadways: width, speed limit, vehicle volumes

Generally, wider pedestrian spaces, lower speed limits, and the presence of a buffer correlate to a higher comfort level. An incomplete sidewalk network, higher speeds, and a greater number of lanes correlate to a lower comfort level. Bicycle lanes or on-street parking act as buffers between pedestrians and vehicle traffic, increasing comfort. Examples of sidewalks with varying comfort levels are shown on the right.

Sidewalks in poor condition also lower a pedestrian's level of comfort, as they pose tripping hazards and make pushing a stroller, skateboarding, or using an assistive mobility device more difficult. To identify the potential comfort of Glendale's pedestrian network, a complete inventory of sidewalk locations, condition, width, and presence of curb ramps was conducted. The inventory was the basis for developing pedestrian network improvements.

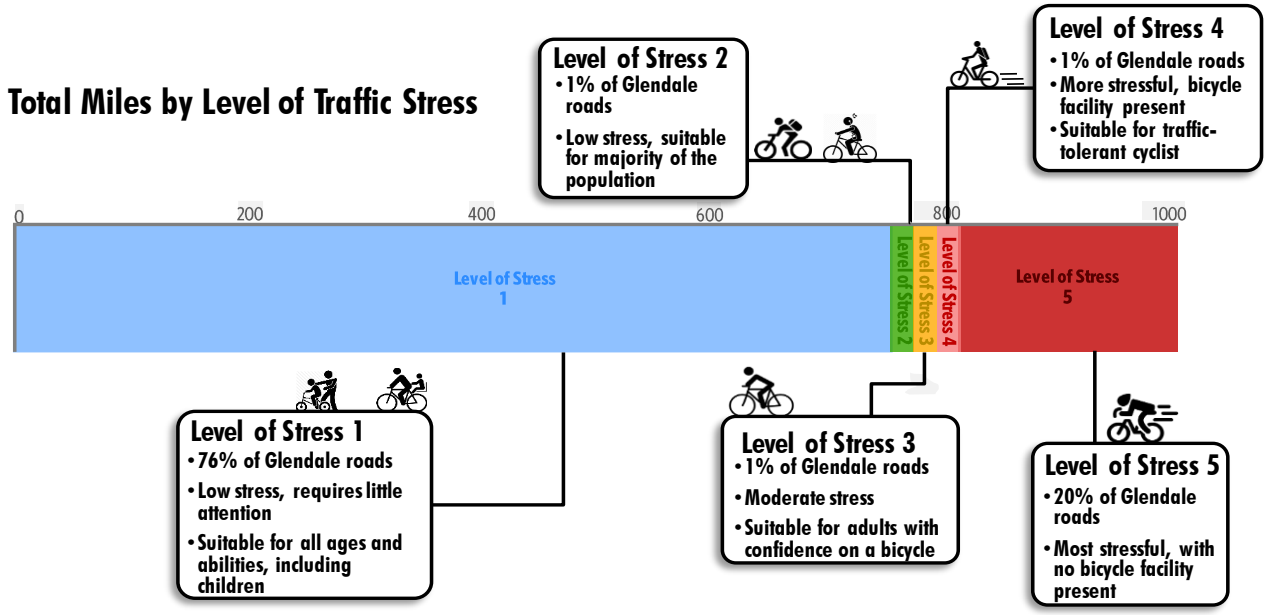


Bicycle Level of Traffic Stress (LTS)

Bicycle Level of Traffic Stress (LTS) quantifies the perceived sense of comfort associated with biking along a given roadway. Originally developed by Alex Sorton and the Northwestern University Traffic Institute, LTS has become the industry standard for assessing the comfort and connectivity of bicycle networks. For the Glendale ATP, the following roadway characteristics were used to predict the stress experience (or comfort level) of bicyclists:

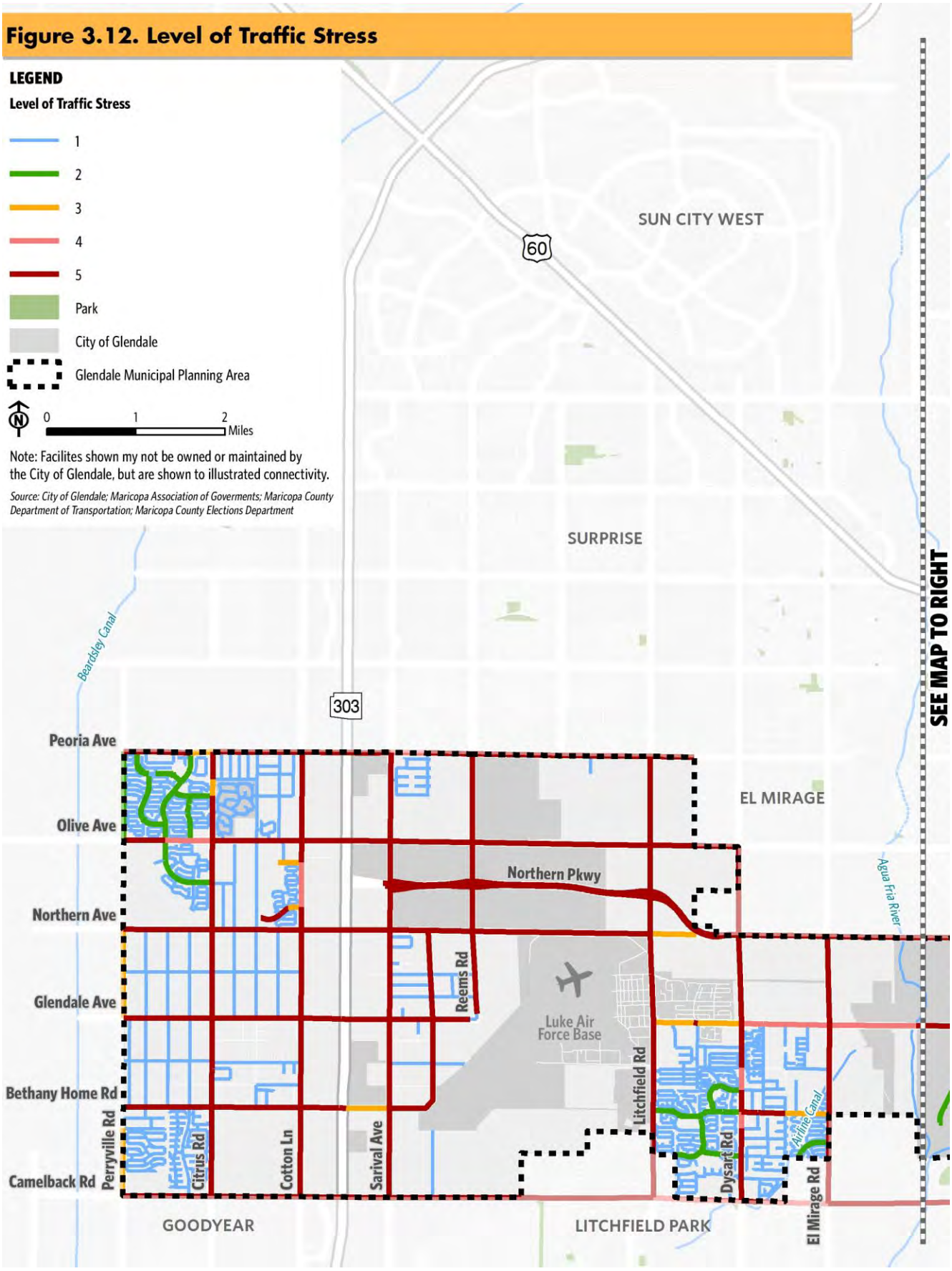
- Posted speed limit
- Number of lanes
- Annual average daily traffic volumes (AADT)
- Presence (and width) of a bicycle facility
- Presence (and width) of on-street parking

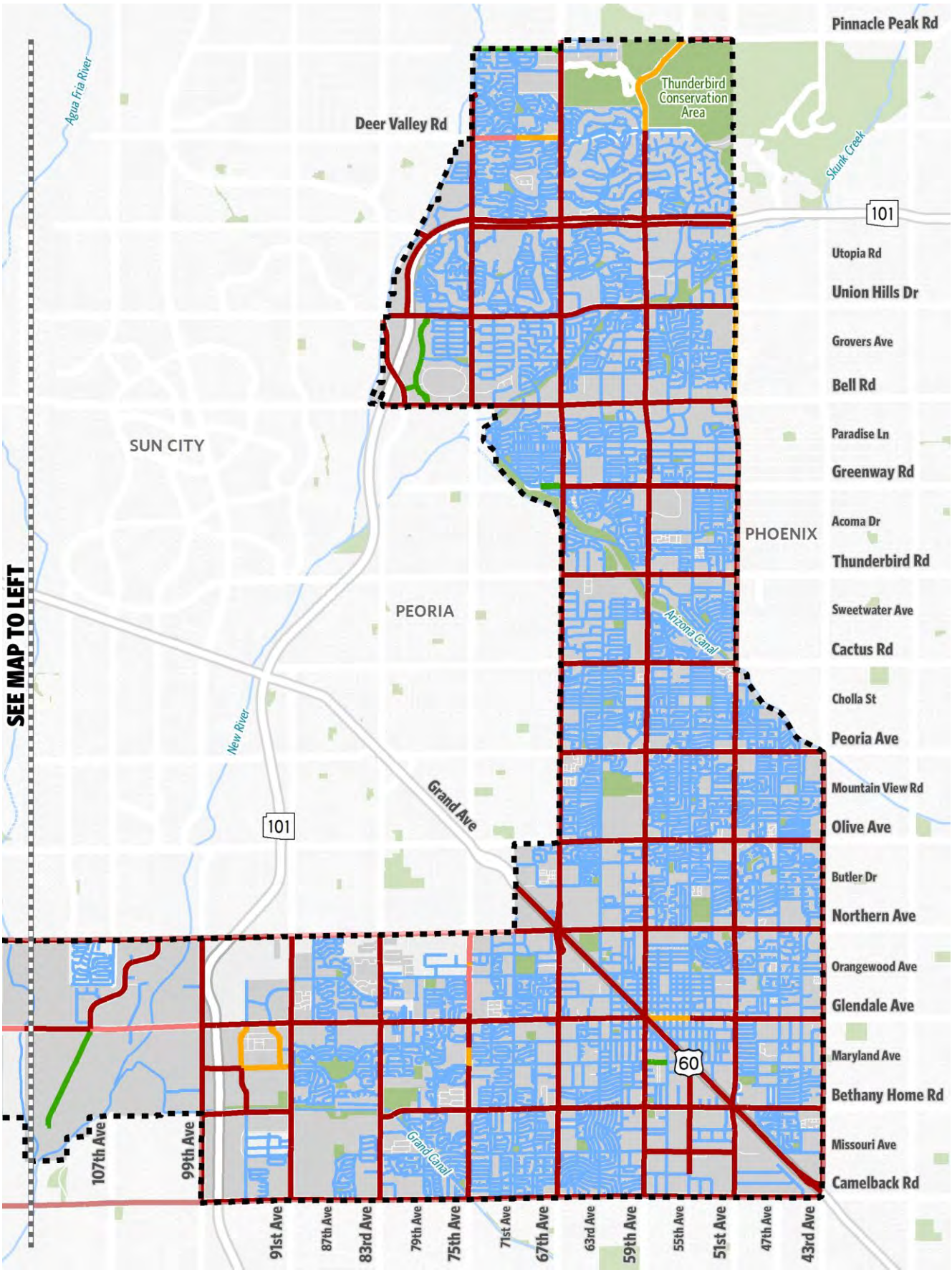
Each roadway was assigned an LTS score between 1 (low stress) and 5 (high stress). The most desirable bicycling score, LTS 1, is assigned to roads that would be suitable for most children to ride. Roadways that are scored LTS 4 or 5 are only suitable for "strong and fearless" bicyclists who tolerate roadways with high vehicle speeds and volumes.



As shown in Figure 3.12, the majority of the roadway network within the Glendale MPA is LTS 1 (shown in light blue). These LTS 1 roadways are primarily local, internal neighborhood roads and not necessarily a part of the regional bicycle network. These low-stress facilities are primarily smaller, local roadways within residential areas. In total, nearly 79 percent of roadways within the study area are classified as low to moderate stress (LTS 1 – LTS 3). Moderate- to high-stress roadways are primarily arterials that provide regional connections. It is important to note that this analysis represents existing stress level conditions; therefore, the addition or widening of bicycle facilities may improve a user's perceived level of comfort and in turn improve a roadway's LTS score. The study area has a large network of roadways that can be utilized to develop a complete and connected bicycle network to encourage "No Way, Now How" and "Interested but Concerned" riders to get on their bikes.

Figure 3.12. Level of Traffic Stress




Pinnacle Peak Rd
Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
Skunk Creek
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
SUN CITY
Paradise Ln
Greenway Rd
Acoma Dr
PHOENIX
Thunderbird Rd
PEORIA
Sweetwater Ave
Arizona Canal
Cactus Rd
Cholla St
Peoria Ave
New River
Grand Ave
101
Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
Maryland Ave
60
Bethany Home Rd
Missouri Ave
Camelback Rd
Grand Canal
107th Ave
99th Ave
91st Ave
87th Ave
83rd Ave
79th Ave
75th Ave
71st Ave
67th Ave
63rd Ave
59th Ave
55th Ave
51st Ave
47th Ave
43rd Ave
SEE MAP TO LEFT

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4. COMMUNITY OUTREACH

WALK. BIKE. CONNECT
glendale active transportation plan

The Glendale ATP incorporated a robust community outreach process that solicited input about existing walking and biking conditions throughout the Glendale MPA, and where the public thought improvements were needed. To garner input from a wide cross-section of the public, multiple outreach methods were utilized. Comments received supported the fact that people who live and work in Glendale care about safety and convenience when traveling. Priorities centered on closing gaps in the local and regional network, educating motorists on pedestrian and bicycle safety, improving crossings, and preserving and maintaining existing facilities.

Outreach Methods

A project website was developed and launched in April 2018, allowing the public easy access to important information about the ATP. The website GlendaleATP.com includes outreach event dates, project document links, and survey links. A comment form is also provided to allow the public to submit a question or concern directly to the study team and the City of Glendale.



Comments Received Via Project Website

To date, comments received directly via the Project Website include the following examples:

"Pedestrian traffic on 69th Avenue during the school year is higher than would be expected as: 1) a great deal of Challenger Middle School students that live in the Desert Eagle Apartments jump the north fence around the park and travel 69th Ave. 2) Challenger Middle School students that live south of Maryland & have family members that attend Desert Garden (Kindergarten-3rd grade) walk to Desert Garden to drop off younger students then walk back to Challenger for classes in the morning and again in the afternoon walk to Desert Garden to pick up younger students and travel back to their homes."

"I would like to see the bike route lanes in our neighborhood Arrowhead Ranch Phase 2 changed so there is no parking of vehicles allowed for the purpose of safety and enjoyment of all cyclists using them. I am glad to see that the City of Glendale is interested in improving the environment of the cycling community. Please take this into consideration."

"Would like to see more pedestrian bridges to the parks and bike trails so you do not have to cross busy traffic lanes on 75th Ave in south Glendale."

"Please consider that Poor road conditions do damage to bikes. I could not find the survey to take."

"Litchfield Road between Missouri and Glendale is in serious need of a bike lane. There is relatively high speed traffic and 4 lanes with no shoulder. There are also many bike riders due to Dreaming Summit development and Luke Air Force Base."

"There needs to be a buffer zone between the street and the sidewalks. I will never walk or allow my kids to walk on our sidewalks when they are right up against the street and flow of traffic on busy roads. Look at how things are designed in midwest and east coast."

Advertisement and Promotion

To make the public aware of the purpose of the Glendale ATP and to invite them to participate in an online survey and mapping exercise, the study team advertised and promoted the ATP utilizing various platforms. Advertisements and promotions completed to date include:



- Social media posts on the City of Glendale Transportation Department's Facebook, Twitter, and Instagram feeds
- E-mail information and link to website via the City of Glendale's Parks and Recreation and Homeowner's Association email list
- Distribution of project information on NextDoor, which reaches 30,000 residents
- KJZZ press release and interview with City of Glendale project team member

Additional advertisements and promotions will be conducted during future phases of the ATP.

Community Events

At community events, study team members hosted booths and asked participants to fill out a brief survey on their active transportation concerns, issues, and needs. Events attended to date include:



- Glendale Bike to Work Day – April 4, 2018
- Cactus District Meeting – April 11, 2018
- Glendale Family Bike Ride – April 22, 2018
- Glendale Farmer's Market – May 12, 2018
- Design Assist Project Meeting – September 11, 2018
- Touch-a-Truck – October 20, 2018
- Glendale Glitters – December 14-15, 2018
- Pop-Up Events – March 2019

Touch-a-Truck Community Event

Team members hosted a booth at the Walmart Transportation Touch-a-Truck event on October 20, 2018 at Westgate Entertainment District from 10:00 am to 1:00 pm to gather public input regarding walking and bicycling in Glendale. The team's efforts included:

- An interactive survey exercise, in which adults and children were able to place a ping pong ball in the bucket that corresponded with the improvement they would most like to see in Glendale. There were ten buckets in all, with each one representing one of the following improvement types:
  - On-street bike lanes
  - Buffered bike lanes
  - Painted bike lanes
  - Off-street walking and biking paths
  - Fill-in sidewalk gaps
  - Enhanced street crossings
  - Improved lighting
  - Better access to bus stops
  - High visibility crosswalks
  - Shade





- A map on which residents could place a sticker at locations they felt were important to this plan (bicycle facility/sidewalk improvements, transit connections, need for crossing, etc.)



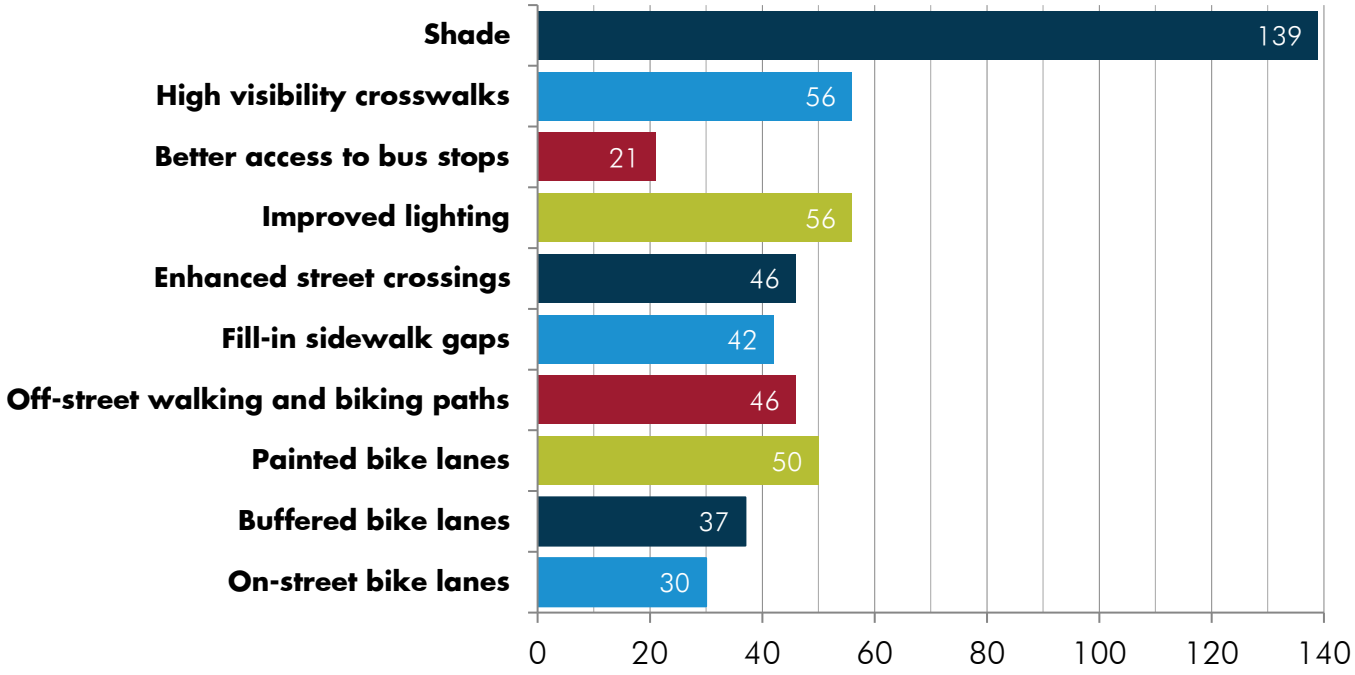




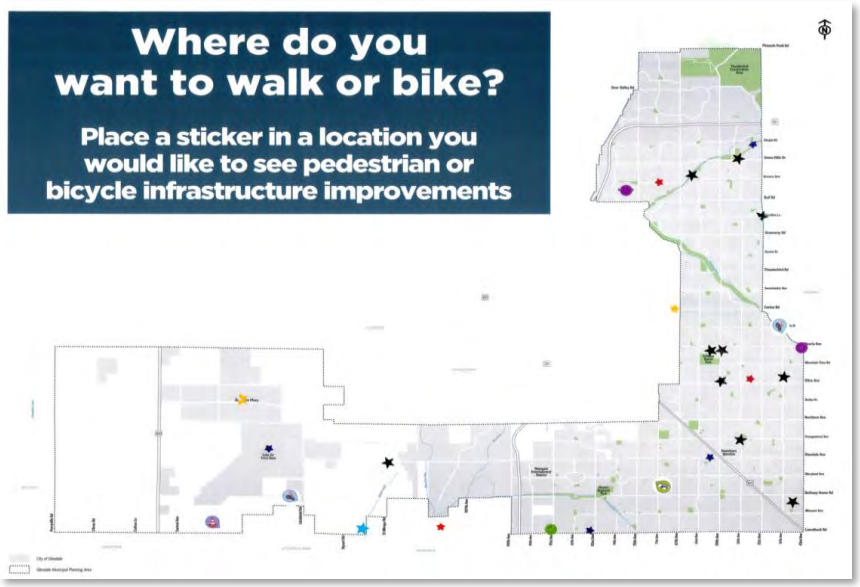
Touch-a-Truck Community Event – Survey Results

In total, 529 ping pong balls were counted in the ten buckets. The graph below illustrates results of the survey. Findings from the survey included:

- More shade had the highest number of votes with 139 votes or 27 percent of responses
- Safety improvements, such as high visibility crosswalks, enhanced street crossings, and improved lighting, accounted for 30 percent of responses (with 158 votes)
- Painted bike lanes, buffered bike lanes, and on-street bike lanes accounted for 22 percent of responses (with 117 total votes)
- "Better access to bus stops" received the fewest votes (21 votes or 4 percent responses)



A total of 23 stickers were placed on the map near participants' homes and neighborhoods where they would like to see better bicycle facilities. Increased access to parks, libraries, schools and other amenities was important to many event attendees, and Skunk Creek Trail was identified as an important bicycle corridor.



Glendale Glitters

Team members participated in the Glendale Glitters event in Downtown Glendale on Friday, December 14th and Saturday, December 15th, 2018. The Glendale ATP booth included three boards – two boards on which residents could select the roadway cross section they preferred and one board depicting the future bike network, on which residents could include their comments and suggestions.



Table 4.1provides a summary of the results of the events surveys.

Table 4.1: "Pick the Road you Like the Most" - Results

Cross Section	Number of "Likes"	Percentage of Total
Shared lanes (parking on both sides)	1	2%
Bike lanes with parking on one side	15	33%
Bike lanes with parking on both sides	6	13%
Bike lanes (no parking)	1	2%
Buffered bike lanes	22	49%

The large majority of respondents preferred either buffered bike lanes (49 percent) or bike lanes with parking on one side (33 percent). Safety was the most common concern – respondents felt the buffer added much needed space between vehicles and bicycles. When parking was important to the respondent, most were willing to give up parking on one side of the street to accommodate a buffer for the bike lane adjacent to the parking lane. There was some concern that on-street parking on local and collector streets is unsafe for bicyclists, particularly for young children. Only one respondent preferred a cross section with no dedicated bicycle lanes.



The project team also asked participants to identify any areas throughout the City where more bicycle facilities were needed. Two locations were identified:

- 59th Avenue between Loop 101 and Deer Valley Road
- The intersection of Bethany Home Road and 107th Avenue

Pop-Up Outreach Events

To further increase participation from residents that traditionally don't participate in the planning process, the Glendale ATP study team conducted sex pop-up outreach events in early 2019. The purpose of the pop-up events was to talk one-on-one with community members to gauge their support for recommended improvements and to understand additional issues and concerns they may have.



Pop-up events were conducted at:

- Pizza Old Roma in Downtown Glendale – January 31, 2019
- Bonsai Park – February 20, 2019
- Glendale Adult Center – February 27, 2019
- Maricopa County Women, Infant, Children (WIC) Clinic – March 3, 2019
- Glendale Community Action Program Office – March 6, 2019
- Thunderbird Paseo Park – March 9, 2019
- Glendale Community Center – March 25, 2019



At the events, a project team member asked participates to review a poster board displaying recommended active transportation improvements in the general area of the pop-up. Participants were asked to place stickers and write comments on improvements they agreed with or additional improvement recommendations. In total, there were 43 participates that the pop-up events. In general, most participants agreed with recommended improvements. Comments received during the events included:



- The crosswalk on Glendale Ave at City Hall was an issue for several respondents who agreed that it should be enhance (i.e., increase visibility) or be eliminated.
- The lack of walking accessibility hurts local businesses and suggested more space be allocated to foot traffic, so business can stay afloat and emptied storefronts can be used.
- There is a need for new bike lanes that are clearly marked/ not mistaken for parking spots.

Public/Stakeholder Open House



Public/Stakeholder Open House #2

The purpose of the first phase of stakeholder/public outreach was to seek input from the public regarding the existing and future deficiencies and needs. To provide ample opportunities for residents to participate, two separate meetings at the Glendale Public Library from 11:00 am to 1:00 pm and 5:00 pm to 7:00 pm. The meetings were advertised to City residents through social media, radio spots, fliers, email, and NextDoor. Direct email invitations were also sent to representatives from local community groups, bike shops, colleges, and area stakeholders.

The meetings were held in open house style, with attendees invited to casually mingle with the study team. A series of boards were provided that outlined identified conditions, issues, and concerns in the City. Attendees were asked to identify additional issues, needs, and improvement ideas utilizing stickers or in writing. Comment forms were also distributed to gain additional input from attendees. A Spanish translator was also in attendance to accommodate Spanish speaking participants. In total, there were 24 members of the community in attendance, not including study team members. Key comments received during the meetings included:





- Lack of connectivity was repeatedly mentioned as a major impedance.
- Multiple residents cited concerns regarding safe access to school for children using various modes.
- Bicyclists and pedestrians have a difficult time crossing Grand Avenue.
- Narrow bike lanes on arterial streets make bicycling dangerous.
- Paved shoulders that allow for automobile parking are a nuisance for bicyclists.
- Connections between multi-use paths and streets need to be enhanced.
- Many neighborhoods lack ADA-compliant sidewalks and driveways.
- Bicycle facilities abruptly end at river crossings.
- Rideshare services (scooters) are beginning to be a nuisance, as the users are riding on the sidewalks and parking on private property.

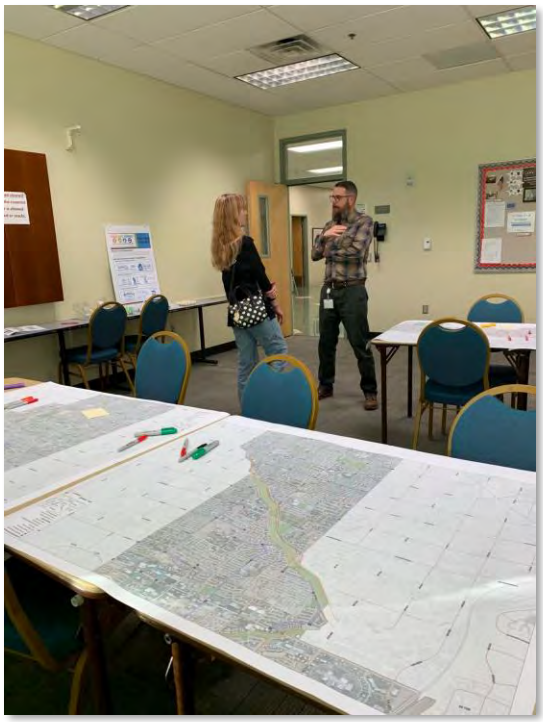
Public/Stakeholder Open House #2

The purpose of the second phase of stakeholder/public outreach was to seek input from the public regarding the recommended improvements. The open house was held at the Glendale Adult Center on February 27th from 5:00 pm to 7:00 pm. The meeting was held in open house style, with attendees invited to talk one-on-one with the study team. Boards illustrating recommended pedestrian and bicycle improvements, as well as potential cross-section were on display. Additionally, large area plots displaying recommended improvements were provided for participatnts to mark additional comments or concerns. Attendees were asked to identify additional issues, needs, and improvement ideas utilizing stickers or in writing.



Key comments received during the meetings included:

- Evaluate the need for additional equestrian staging areas and crossings to improve access to trails.
- Parks should have bicycle infrastructure access.

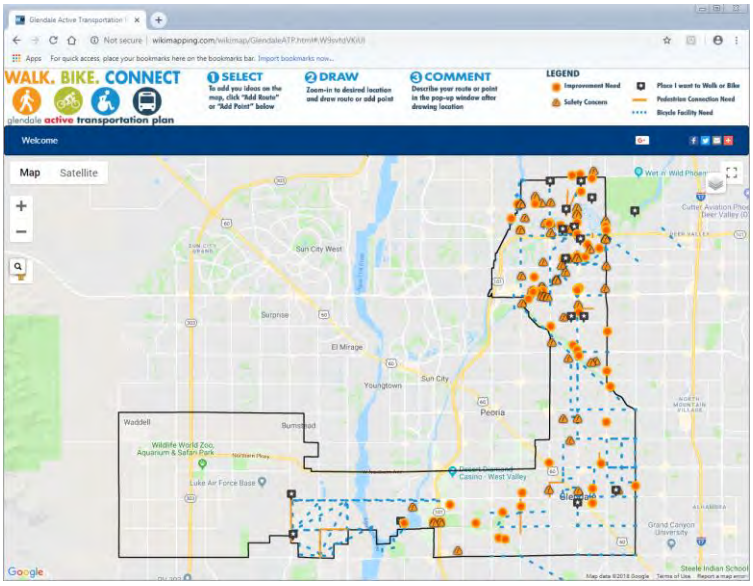


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Online Mapping Tool

An online public survey tool was developed to gather input on existing active transportation conditions, issues, and needs. The tool allows users to enter improvements they would like to see for each of the following:

- Routes where pedestrian or bicycle improvements are needed
- Spot locations where pedestrian and facility improvements are needed
- Trail connection needs
- Pavement or sidewalk maintenance/repair
- Locations with safety concerns
- Locations where they would like to walk or bike



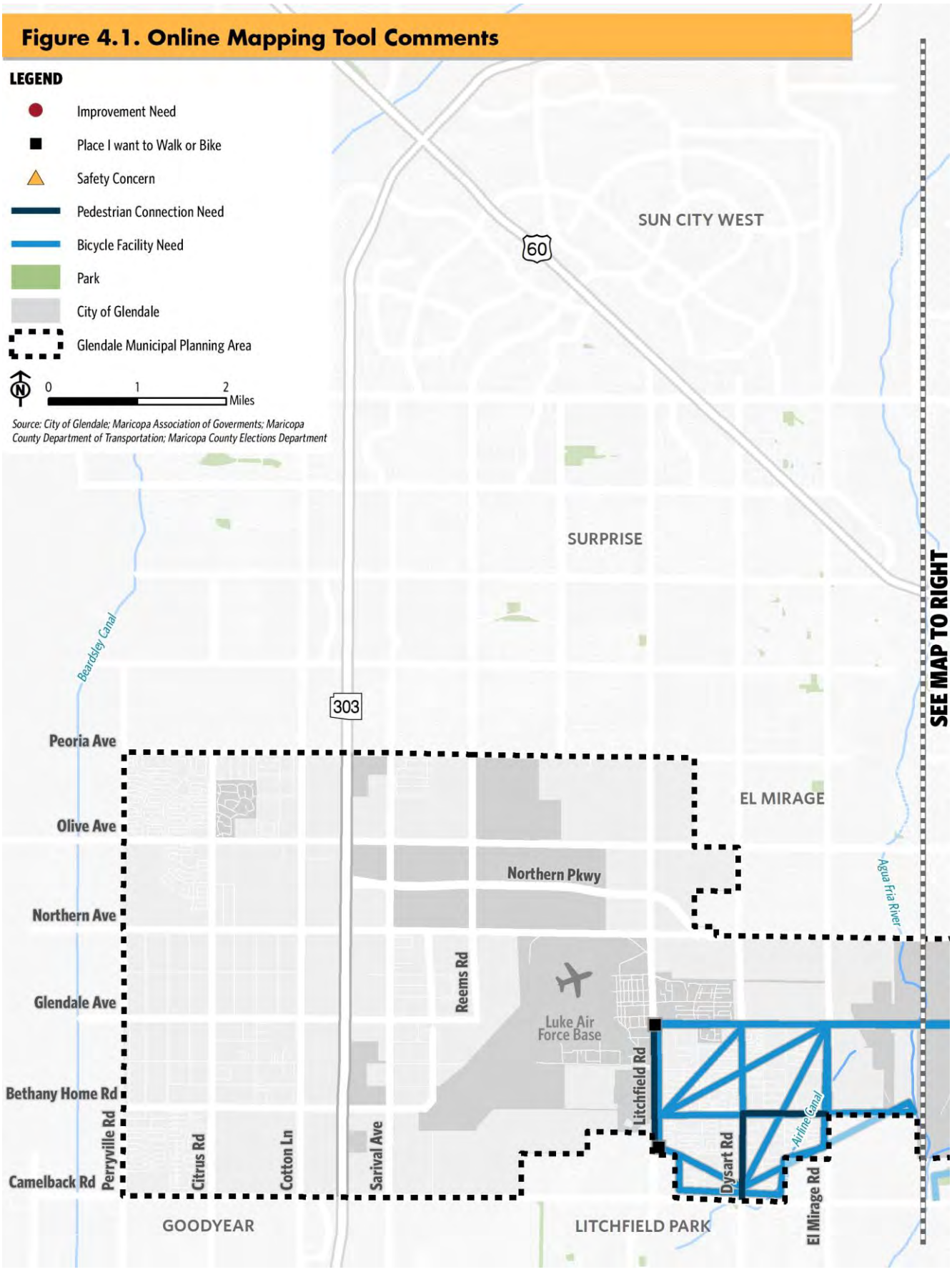
www.GlendaleATP.com

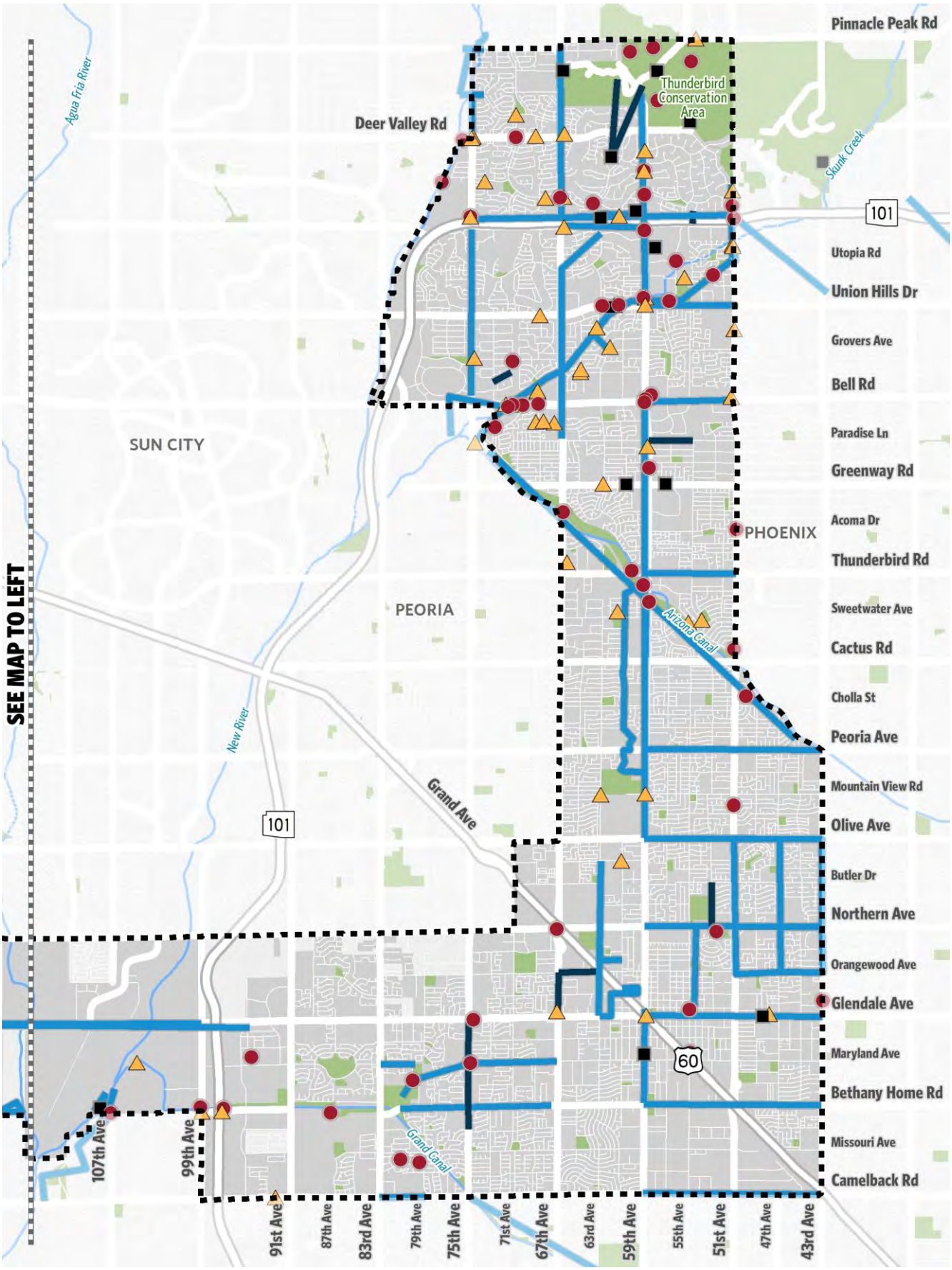
In addition, users are able to agree, disagree, or comment on previously submitted comments.

The web map was launched in April 2018 and will remain open for the duration of the ATP. As of November 31, 2018, a total of 221 points and lines have been drawn and 51 additional comments have been received. Figure 4.1 illustrates comments received as of November 31, 2018. In addition, comments collected during the MCDOT Active Transportation Plan are displayed. As illustrated in the Figure, a high number of comments were received in the Northern Developed and Central Core area. Highlights of the comments received, include:

- Several comments were made regarding using Skunk Creek Trail as a bike & pedestrian connection to Peoria and Phoenix and updating lighting and pedestrian amenities such as benches and shade.
- Four participants commented on the need for better and safer bike facilities on Glendale west of the Loop 101.
- Four participants agreed that the bike lane on Deer Valley Road at 75th Avenue needs to be extended/completed, as it currently ends abruptly, forcing bicyclists into dangerous situations.
- Several comments were received on the lack of bicycle infrastructure along arterials.
- Five participants identified the need to connect existing active transportation facilities to the Loop 101 pedestrian bridge at 63rd Avenue.

Figure 4.1. Online Mapping Tool Comments




SEE MAP TO LEFT
Agua Fria River
Deer Valley Rd
Thunderbird Conservation Area
Skunk Creek
Pinnacle Peak Rd
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
Acoma Dr
Thunderbird Rd
Sweetwater Ave
Cactus Rd
Cholla St
Peoria Ave
Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
Maryland Ave
Bethany Home Rd
Missouri Ave
Camelback Rd
SUN CITY
PEORIA
PHOENIX
New River
Grand Ave
Arizona Canal
Grand Canal
60
107th Ave
99th Ave
91st Ave
87th Ave
83rd Ave
79th Ave
75th Ave
71st Ave
67th Ave
63rd Ave
59th Ave
55th Ave
51st Ave
47th Ave
43rd Ave

Community Survey

To gain additional insight in the type of active transportation users in the study area, as well as their opinion of active transportation and their travel behaviors, a survey was developed. This survey is available online at GlendaleATP.com and was distributed at community events and other outreach activities. It includes 15 questions and three main sections:

- Perceptions of walking and biking
- Active transportation issues and improvement ideas
- Demographic information



As of November 31, 2018, a total of 329 surveys were completed. Respondents primarily resided in zip codes 85308 (37 percent), 85302 (11 percent), and 85301 (8 percent).

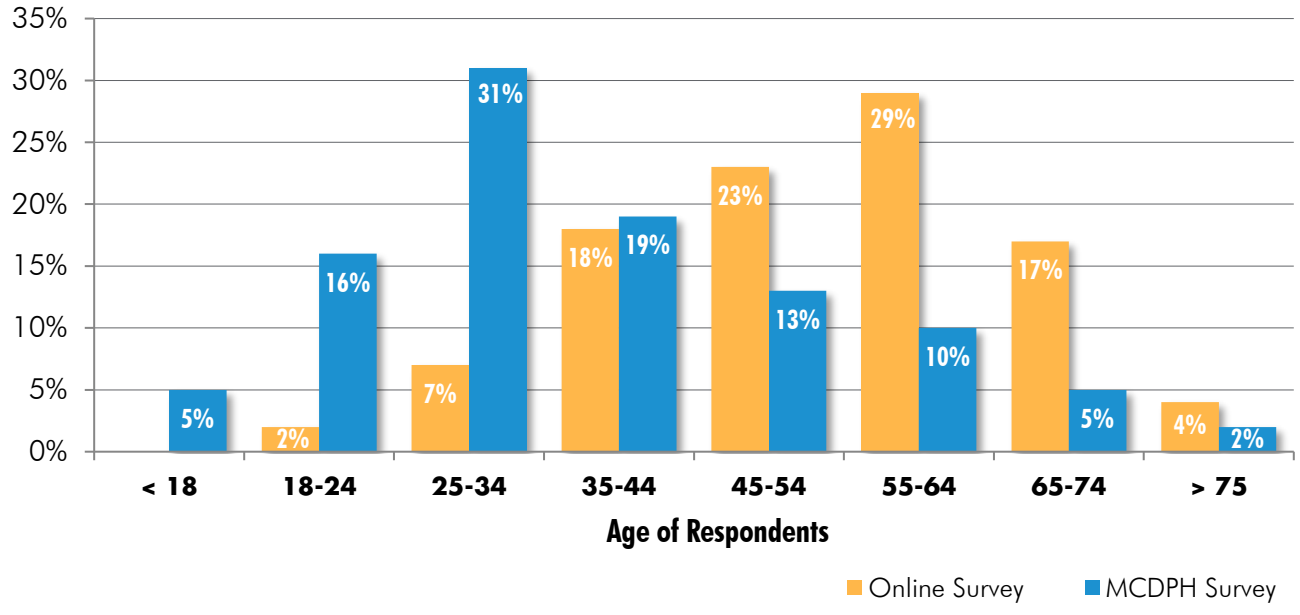
Maricopa County Department of Public Health Survey

To supplement the efforts of the online outreach, the Maricopa County Department of Public Health (MCDPH) administered 199 one-on-one community surveys to residents that don't traditionally participate in the public outreach process. The survey was available in both English and Spanish and included the same questions as the online survey. Surveys were collected at the Arrowhead Transit Center, at key stops along GUS route, and at the Glendale Public Library. Respondents primarily resided in zip codes 85308 (20 percent) and 85301 (17 percent) and were markedly younger than those who took the online survey.

Maricopa County Public Health
WeArePublicHealth.org

Figure 4.2 illustrates the overall age range of respondents from both the online and MCPHD survey. As shown in the Figure, MCPHD survey primarily consisted of residents aged 25 to 34 years old, while the online survey was primarily completed by persons ages 55 to 64 years old.

Figure 4.2: Survey Demographics

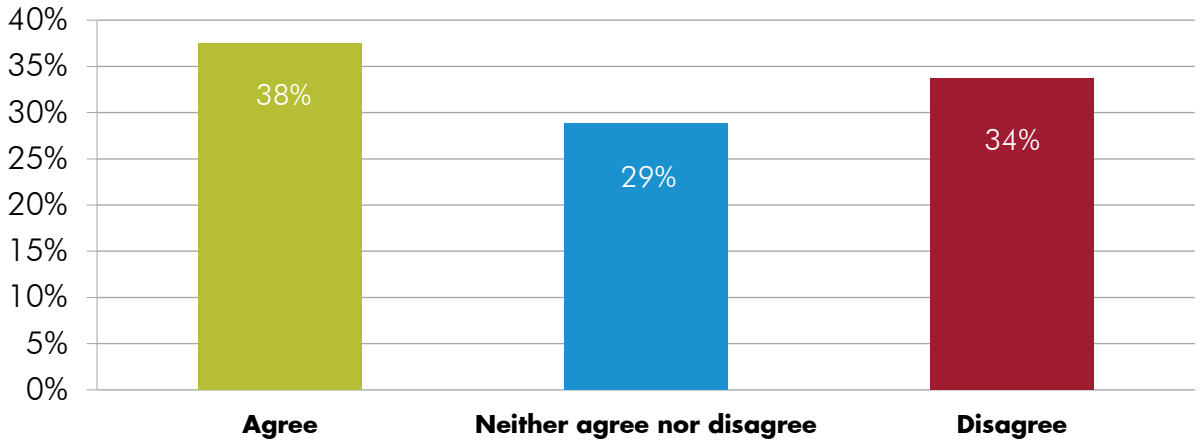


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Summary of Survey Results

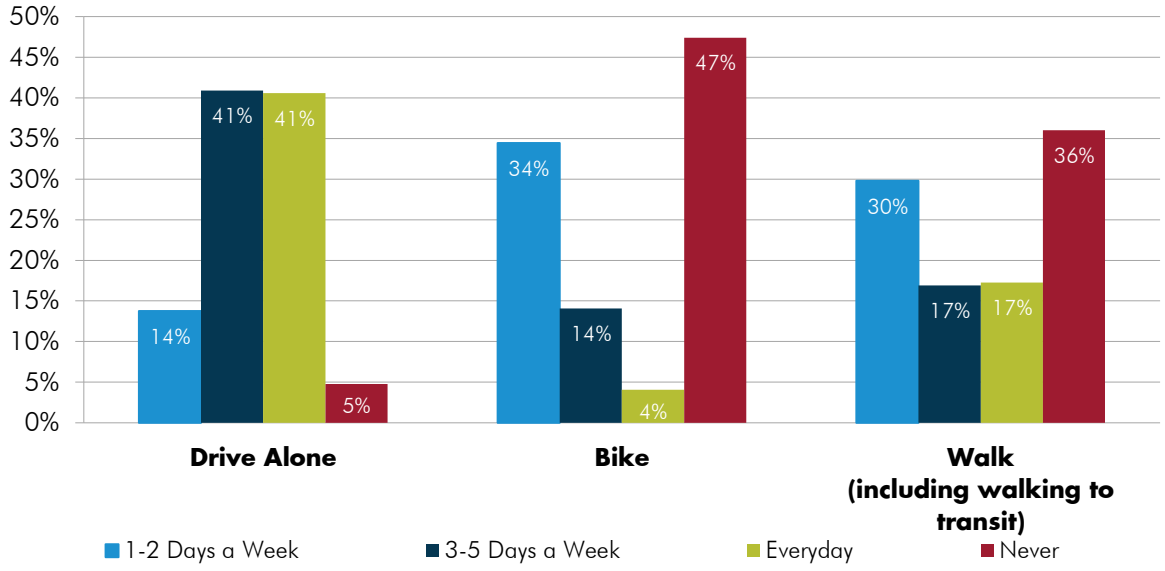
The following section summarizes the combined online and MCPHD survey results (528 total surveys) noting interesting distinctions between the surveys when applicable.

Q. 1: Walking and biking in Glendale is a safe, convenient, and practical way to get from one place to another.



The first survey question asked respondents to share their perception of walking and biking in Glendale today. Slightly more respondents agreed with the statement than disagreed. Notably, younger respondents (younger than 45) were more likely to agree with the statement (43 percent) than older respondents (34 percent).

Q. 2: In an average week, how many days a week do you use the following modes of transportation (from online survey respondents only)?



The second survey question asked respondents to share their current travel behaviors. Results show that driving alone was respondents' primary mode of transportation. However, 52 percent of respondents stated they bike at least 1 day per week, which may represent their recreational weekend rides.

Q. 3 to Q. 5: Would you ride your bike on this road (online survey only)?

Respondents were asked about their level of comfort riding a bike on various corridors. The survey included images of corridors and noted example design components (such as type of bike facility, speed, and number of lanes). The summarized percentages represent the most popular responses by corridor. Respondents also had the choices "I don't ride a bike," or "Other." Because only the most popular choices are summarized, the totals do not equal to 100%.



Designated bike lane, 40 MPH speed limit, two traffic lanes in each direction

- Yes, I am comfortable riding on roads with traffic – 32 percent
- Maybe, it would depend on how much traffic is on the road - 29 Percent
- No, I am not comfortable riding on roads in traffic – 18 percent



Designated bike lane, commercial corridor, 45 MPH speed limit, three traffic lanes in each direction

- No, I am not comfortable riding on roads in traffic – 47 percent
- Maybe, it would depend on how much traffic is on the road - 22 Percent



Low speed, low traffic volumes, one traffic lane in each direction, shared bike lane/parking lane

- Yes, I am comfortable riding on roads with traffic – 40 percent
- Maybe, it would depend on how much traffic is on the road - 33 Percent

Q. 6: Why do you walk or bike?

Understanding why users walk or bike informs decisions on future facilities. Key findings show:

- 64 percent of respondents shared that they walk or bike for fun, recreation, or exercise.
- MCPHD survey respondents ride their bikes to work or school or to run errands (48 percent to go to work or school; 34 percent to run errands) more than online survey respondents (21 percent to go to work or school; 19 percent to run errands).
- Lack of car ownership may account for this difference as a much higher percentage of MCPHD respondents (50 percent) do not own a vehicle compared to the online respondents (3 percent).

Q. 7: What prevents you from walking and biking more?

Respondents were asked to choose their top barriers to walking and biking from a list. Feeling unsafe, not having enough time/destinations too far, and lack of dedicated bike space were the most commonly reported barriers. Other barriers cited included: lack of driver awareness and education, weather/heat, and gaps in the network.

Answer Choices	Number of Responses
Lack of sidewalks	89
Lack of dedicated bike space	177
Do not feel safe (i.e., too much traffic, inadequate lighting, etc.)	254
Sidewalks or bike paths in poor condition	133
Not enough time/destinations too far	189
There are too many physical barriers along my travel route (i.e. highways, canals, railroads, etc.)	106
Lack of sidewalks connecting to bus stops	42
I don't own a bike	72
Other	108

Q. 8: What improvements would you like to see Glendale invest in? Rank 1 (highest) to 7 (lowest priority).

This question allowed respondents to provide their input on where they would like to see financial investments. The results highlight the types of improvements that may have the most public support. Because respondents ranked their choices, the totals do not equal to 100%.

- More than half of all respondents shared that they believe new off-street walking and bicycling paths are the highest investment priority in Glendale.
- Filling in sidewalk gaps, enhanced street crossings, and improved lighting were also identified as high priority improvements.

Answer Choices	Number of "Highest Priority" Rankings
New on-street bike lanes	157
New off-street walking and bicycling paths	234
Fill-in gaps in existing sidewalk network	177
Enhanced street crossings	172
Improved lighting	165
Better access to bus stops	90
Additional recreational trails (including equestrian)	159

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5. ACTIVE TRANSPORTATION RECOMMENDATIONS

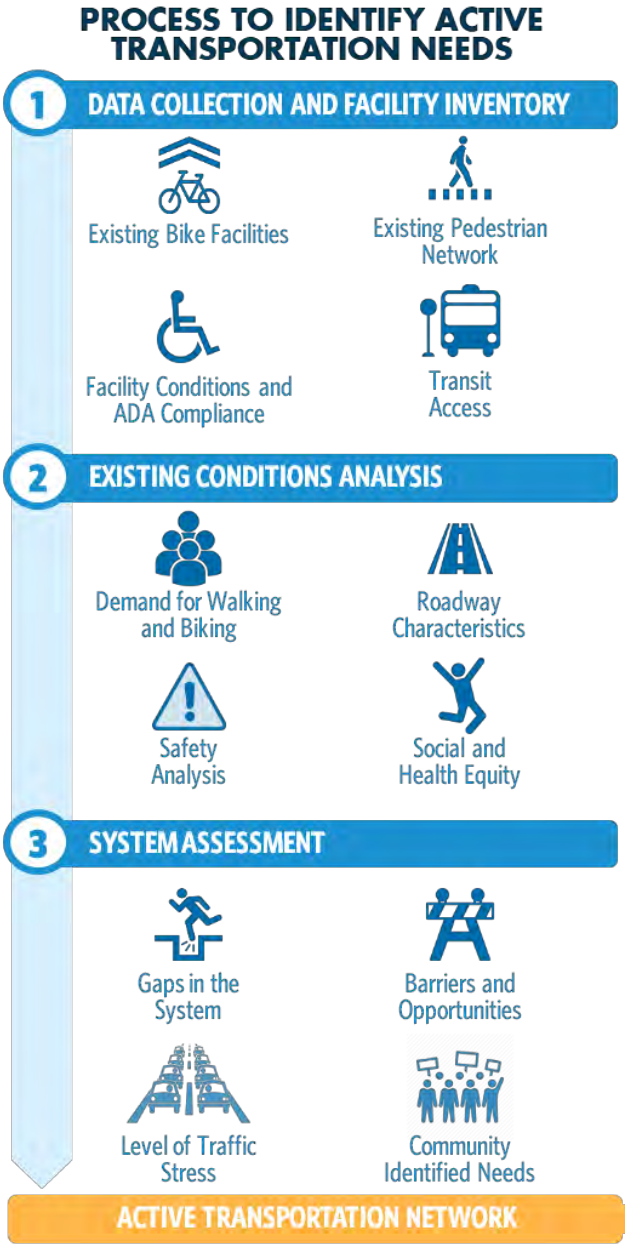
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The need and recommendations identified in the Glendale ATP address gaps in the active transportation network that create barriers and may discourage people from choosing to walk, ride a bike, or access transit.

Identification of Needs

Identification of active transportation needs and potential projects is the first step in developing the future active transportation network. To begin, a comprehensive listing of all locations that may benefit from pedestrian and/or bicycle facilities improvements was developed. Developing a list of potential improvements included a multi-step process involving complex data analysis and ongoing dialogue with Glendale staff, stakeholders, and the public. These improvements included:

- Linear improvement locations that lack pedestrian and/or bicycle facilities, creating a gap in the local and regional system;
- Spot improvement locations, including intersections lacking sidewalks or bicycle facilities, inadequate railroad crossings, and vertical gaps;
- First and last-mile connections to transit for people walking and biking;
- Potential freeway, canal, railroad or river crossing locations;
- Locations where a low-stress bicycle network can be developed;
- Locations where strategic improvements can aid in expanding the regional active transportation network, such as additional access to regional parks and trails, or new routes to create new long-distance cycling routes; and
- Locations identified by the public, local agencies, and Glendale staff that would benefit from improved pedestrian and/or bicycle facilities.



Gaps in the network may be due to many factors, including but not limited to; inconsistent corridor development, physical constraints, and right-of-way issues. Filling-in these gaps has the potential to link thousands of people to jobs and provide choices for convenient travel by foot or bicycle. Figure 5.1. illustrates gaps and needs identified in the Glendale MPA.

Examples of Gaps and Needs in Glendale's Active Transportation Network



Intersection Improvements

Missing sidewalks, bike lane striping that ends prior to the stop bar, and other at intersection improvements that create barriers to active transportation



Crossing Needs

Potential need for crosswalk at an intersection or where a trail intersects with a busy roadway



Spot Improvements

Railroad crossings that are not ADA compliant, transit stops without pedestrian access, and other spot needs.



In-fill Improvements

Includes small breaks in the active transportation network that may be due to sporadic corridor development



Expansion Opportunities

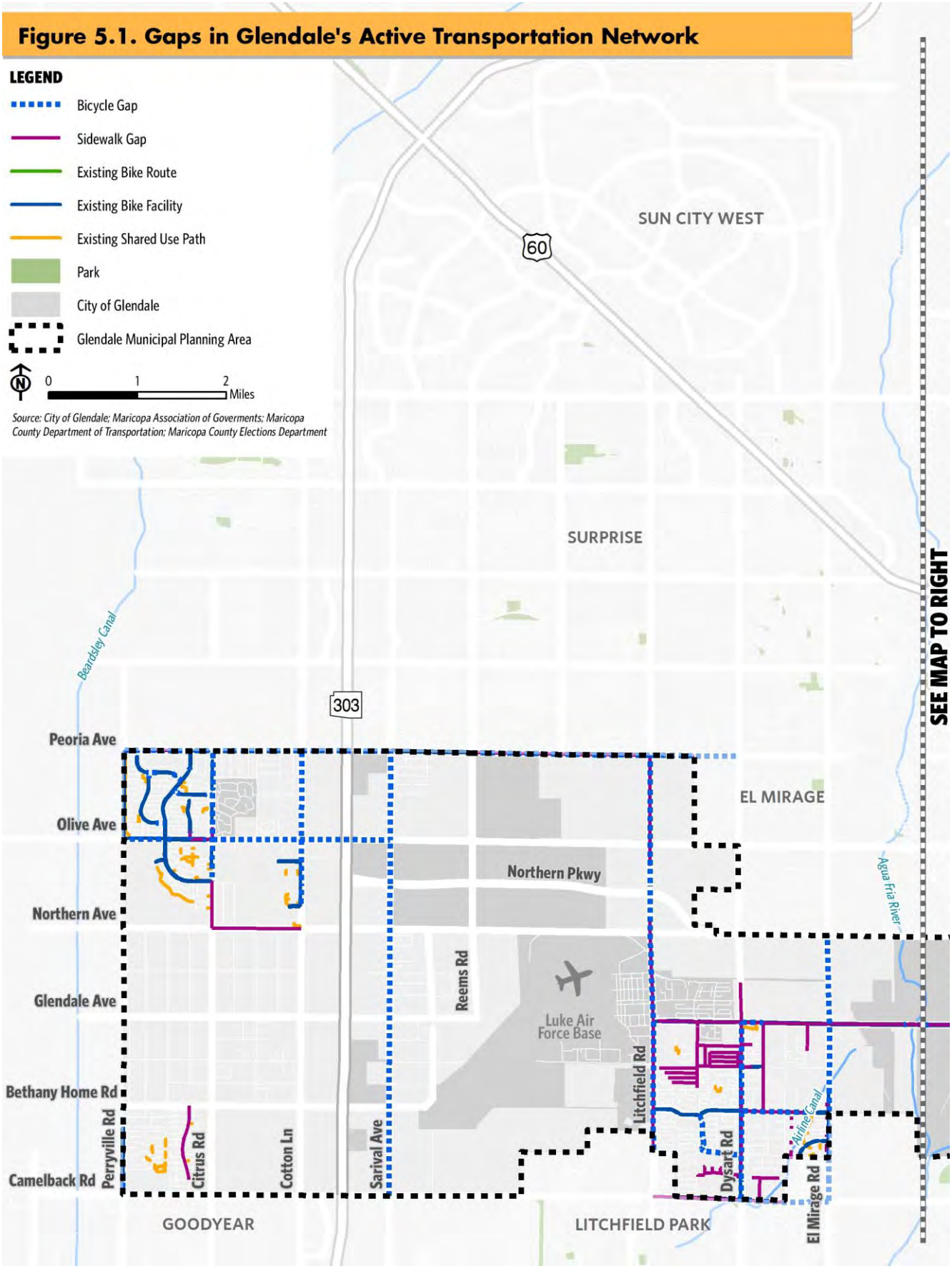
Key corridors that would provide low-stress connections or vital links to major activity centers.

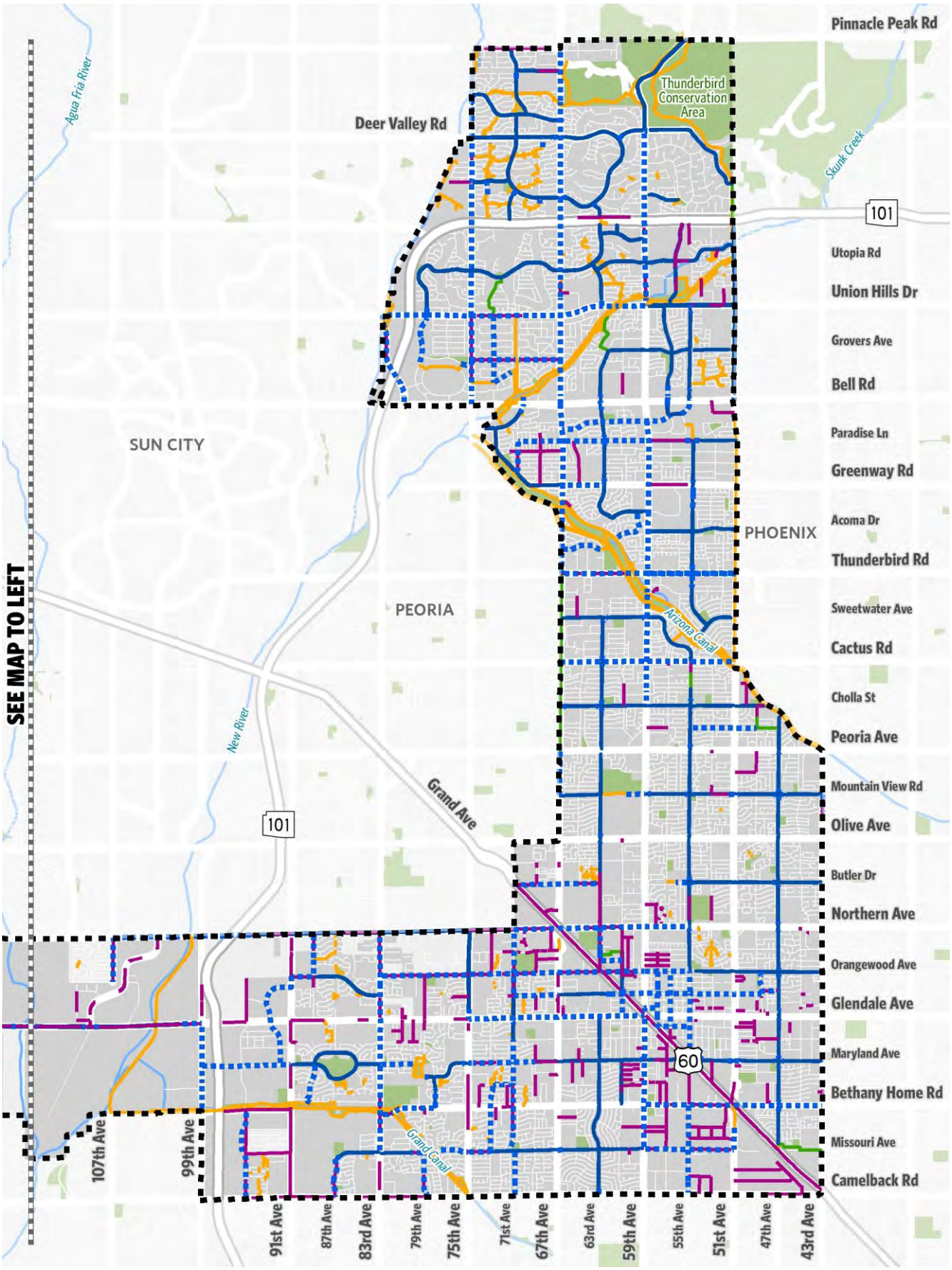


Regional Connections

Gaps in the active transportation network between existing facilities in neighboring jurisdictions, park entrances, canal trails, and other regional attractions.

Figure 5.1. Gaps in Glendale's Active Transportation Network




Pinnacle Peak Rd
Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
Skunk Creek
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
Acoma Dr
Thunderbird Rd
Sweetwater Ave
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Prioritization of Improvement Needs

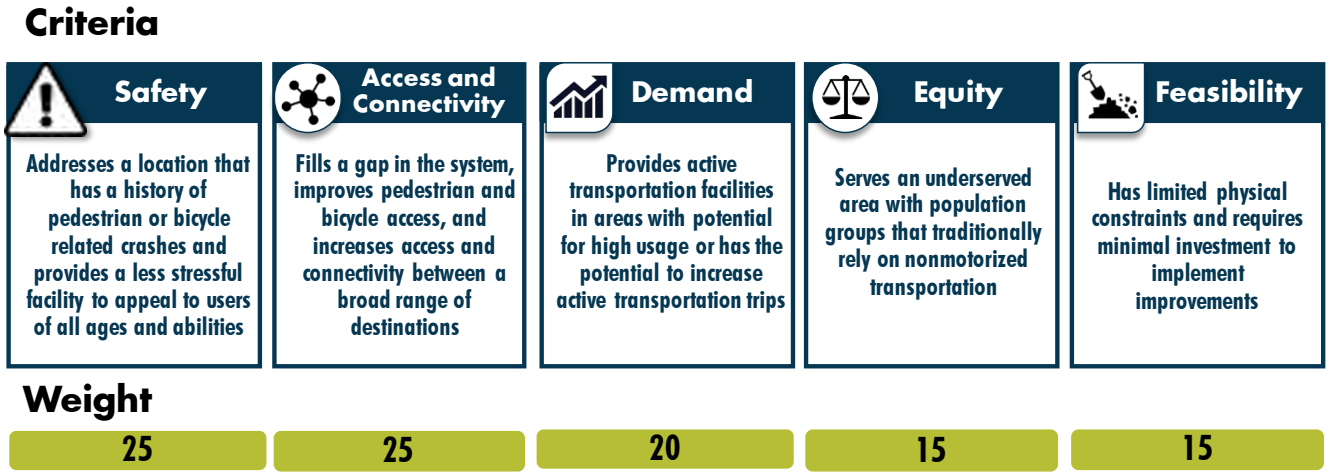
Once a complete set of infrastructure improvement needs were identified, a prioritization tool was developed to quantify the magnitude of each project's contribution to the overall vision and goals of the ATP. The prioritization tool is a flexible approach, intended to provide clear direction for proactively seeking project funds and completing design and engineering of the most critical projects, while still allowing for opportunistic implementation of the entire network.

Project Prioritization Criteria

Five key prioritization categories were identified and weights were assigned based on their overall contribution to the goals of the ATP Key prioritization criteria included:

- Safety: project addresses a location that has a history of pedestrian- or bicycle-related crashes and provides a less stressful facility to users of all ages and abilities;
- Access and Connectivity: project fills a gap in the system and improves pedestrian and bicycle access to key destinations, including critical facilities and transit;
- Demand: project likely addresses unmet nonmotorized travel demand or has the potential to increase active transportation trips;
- Equity: project serves an underserved area with population groups that traditionally rely on nonmotorized transportation; and
- Feasibility: project has few physical constraints and is a modest investment.

The results of the prioritization process are not intended to preclude projects from receiving funding or from being incorporated into a new development or other roadway project. All recommended improvement projects have merit and should be implemented as timely and cost-effective opportunities arise.



Project Prioritization Criteria Scoring

For each of the criteria presented in Figure 5.2, measurable factors (referred to as sub criteria) were first identified and an objective scoring system was created that rates each project on a scale of 0 to 100. Points were allocated to each project based on GIS data analysis, creating a transparent, data driven process that identifies the projects that are most consistent with the goals of this plan.

Table 5.1 outlines the project prioritization sub criteria scoring utilized. Ultimately, the sub criteria scores for each project are combined to generate an overall project score. A high priority project (a project with a higher overall score) provides the greatest benefit for the criteria identified. Glendale staff should regularly revisit the project list and reapply the prioritization methodology to help schedule short-term improvements.

Table 5.1. Project Prioritization Scoring Criteria

Safety

Improves safety and provides a comfortable facility for people of all ages and abilities

Criteria Weight	Sub Criteria	Description	How is it Measured	Sub Criteria Score	Scoring
25	Addresses pedestrian/ bicycle safety issues	Project may address corridors/ intersections with a high number of pedestrian/bicycle related crashes	Total number of pedestrian and or bicycle related crashes within 100 FT of project	12	12 = High number of ped/bike crashes 6 = Low number of ped/bike crashes 0 = No ped/bike crashes
	Provides a less stressful facility	Project provides a facility that appeals to users of all ages and abilities	Average of current Level of Traffic Stress (LTS)	8	0 = LTS 1 2 = LTS 2 4 = LTS 3 6 = LTS 4 8 = LTS 5
	May address related safety issues at project location	Project may contribute to reduction in overall number of crashes	Total number of crashes within 100 FT of project	5	5 = High number of total crashes 3 = Low number of total crashes 0 = No crashes

Table 5.1. Project Prioritization Scoring Criteria (Continued)

Access and Connectivity

Increases access and connectivity between a broad range of destinations

Criteria Weight	Sub Criteria	Description	How is it Measured	Sub Criteria Score	Scoring
25	Fills a gap in the existing system	Project addresses system gap to create a continuous and interconnected active transportation network	Spatial overlay, if it connects to existing/planned network	8	8 = Addresses connection gap in the existing and planned network 0 = Does not provide connection in the existing and planned network
	Connects neighborhoods /parks/multiple jurisdictions	Project connects people within neighborhoods, to recreational facilities, or across multiple jurisdictions	Total number of activity centers within 1/4 mile of project	6	Scale Range of 0 - 6 points 6 = High number of major activity centers within 1/4 mile 0 = Low number of major activity centers within 1/4 mile
	Provides access to critical facilities	Project improves direct access to critical facilities such as schools and medical services	Total number of critical facilities within 1/4 mile of project	6	Scale Range of 0 - 6 points 6 = School(s) or hospital(s) located within 1/2 mile 0 = No school or hospital located within 1/2 mile
	Addresses first mile/last mile connection	Project improves first/last mile connections between transit stops and surrounding destinations	Bus stop located with 1/8 mile of project	5	Scale Range of 0 - 5 points 5 = Connects to transit route(s)/stop(s) 0 = Does not connect to transit route/stop

Table 5.1. Project Prioritization Scoring Criteria (Continued)

Demand

Provides active transportation facilities in areas with potential for high usage

Criteria Weight	Sub Criteria	Description	How is it Measured	Sub Criteria Score	Scoring
20	Serves an area with a high demand for active transportation	Project located in a high activity area	Average latent demand within 100 FT of project	9	Scale Range 9 = Highest Latent Demand Score 0 = Lowest Latent Demand Score
	Has public/agency support	Project received public or agency support during public/ stakeholder outreach	Total number of comments on facility	7	Scale Range 7 = Public and agency comment(s) received on corridor/intersection 0 = No public input received
	Provides facilities in an area with high density of access points	Project located in an area with multiple activity centers	Total number of driveways and intersections within 1/8 mile	4	Scale Range 4 = High density of activity centers along corridor 0 = Low density of activity along corridor

Table 5.1. Project Prioritization Scoring Criteria (Continued)

Equity

Provides active transportation facilities for everyone regardless of physical, social, or economic abilities

Criteria Weight	Sub Criteria	Description	How is it Measured	Sub Criteria Score	Scoring
15	Serves an area with high number of low income and zero car household populations	Project provides transportation options for persons below the poverty level or without access to an automobile	Density of below poverty and households with no access to an automobile immediately surrounding the project	5	Scale Range 8 = Highest percentage range of populations living below poverty level or without access to an automobile 0 = Lowest percentage range of populations living below poverty level or without access to an automobile
	Serves an area with high number of individuals with disabilities	Project provides transportation options for persons living with limited mobility	Density of persons with a disability immediately surrounding the project	4	Scale Range 5 = Highest percentage range of households with a person living with limited mobility 0 = Lowest percentage range of households with a person living with limited mobility
	Serves areas with disproportionate rate of vulnerable users	Project provides transportation options for children, elderly, minorities, and other vulnerable population groups	Density of underserved populations immediately surrounding the project	4	Scale Range 5 = Highest composite score of children, elderly, and minority populations 0 = Lowest composite score of children, elderly, and minority populations
	Provides active transportation facilities in an area with high levels of chronic health issues	Project provides opportunities for areas with high rates of heart disease, diabetes, and stroke to engage in physical activity	Average of Health Inequity composite score immediately surrounding the project	2	Scale Range 2 = Highest Health Inequity composite score 0 = Lowest Health Inequity composite score

Table 5.1. Project Prioritization Scoring Criteria (Continued)

Feasibility

Project has limited constraints for implementation

Criteria Weight	Sub Criteria	Description	How is it Measured	Sub Criteria Score	Scoring
15	Quick Win	Project requires a modest investment, has few constraints, and can be constructed in a short time frame	Data analysis of right-of-way, potential constraints, project length, and complexity of potential improvement	5	Scale Range 5 = Low cost, few barriers, project can be constructed in short period 0 = High costs, barriers, project has a longer construction period
	Project cost	Planning level cost estimate for construction	Estimate of cost to construct project	4	4 = Project has a low planning level cost estimate 2 = Project has a medium planning level cost estimate 0 = Project has a high planning level cost estimate
	Maintenance Costs	Planning level cost estimate for ongoing maintenance	Estimate of cost to maintain project	3	3 = Project has low ongoing maintenance 2 = Project has a medium ongoing maintenance 0 = Project has a high ongoing maintenance
	Physical constraints	Major right-of-way or other physical constraints	Overlay analysis of right-of-way, location of utilities, drainage, and other potential constraints	3	3 = Project is feasible (no major right-of-way or other physical constraints) 0 = Major right-of-way or other physical constraints possible

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Pedestrian Network Improvement Recommendations

The approach to developing the pedestrian network intends to concentrate resources in areas where improvements are most needed and where people are most likely to walk. Proposed pedestrian improvements aim to close sidewalk gaps and provide a safe and comfortable experience for users of all ages and abilities. Combined with Glendale's extensive existing pedestrian network, the identified pedestrian needs create a more robust, connected, and comfortable walking network throughout the study area. This section describes the recommended pedestrian network improvements. Network improvements are categorized as follows:

- Sidewalk Repairs: sidewalks identified as being in poor condition at the time of the inventory.
- Sidewalk Infill Improvements: short-trip pedestrian facility improvements at specific locations where gaps occur in the overall pedestrian network.
- Intersection Improvements: improve pedestrian safety and comfort when crossing the road as well as minimize any potential conflicts with vehicles.
- Transit Connections: addresses first/last mile issues by providing needed pedestrian connections to and from the public transportation network.
- Accessibility for People with Disabilities: sidewalk and other pedestrian facility improvements to accommodate all persons with cognitive, visual, or physical disabilities.

Appendix A provides a full listing of improvement recommendations, their priority scoring, and a planning level cost estimate.

Sidewalk Repair

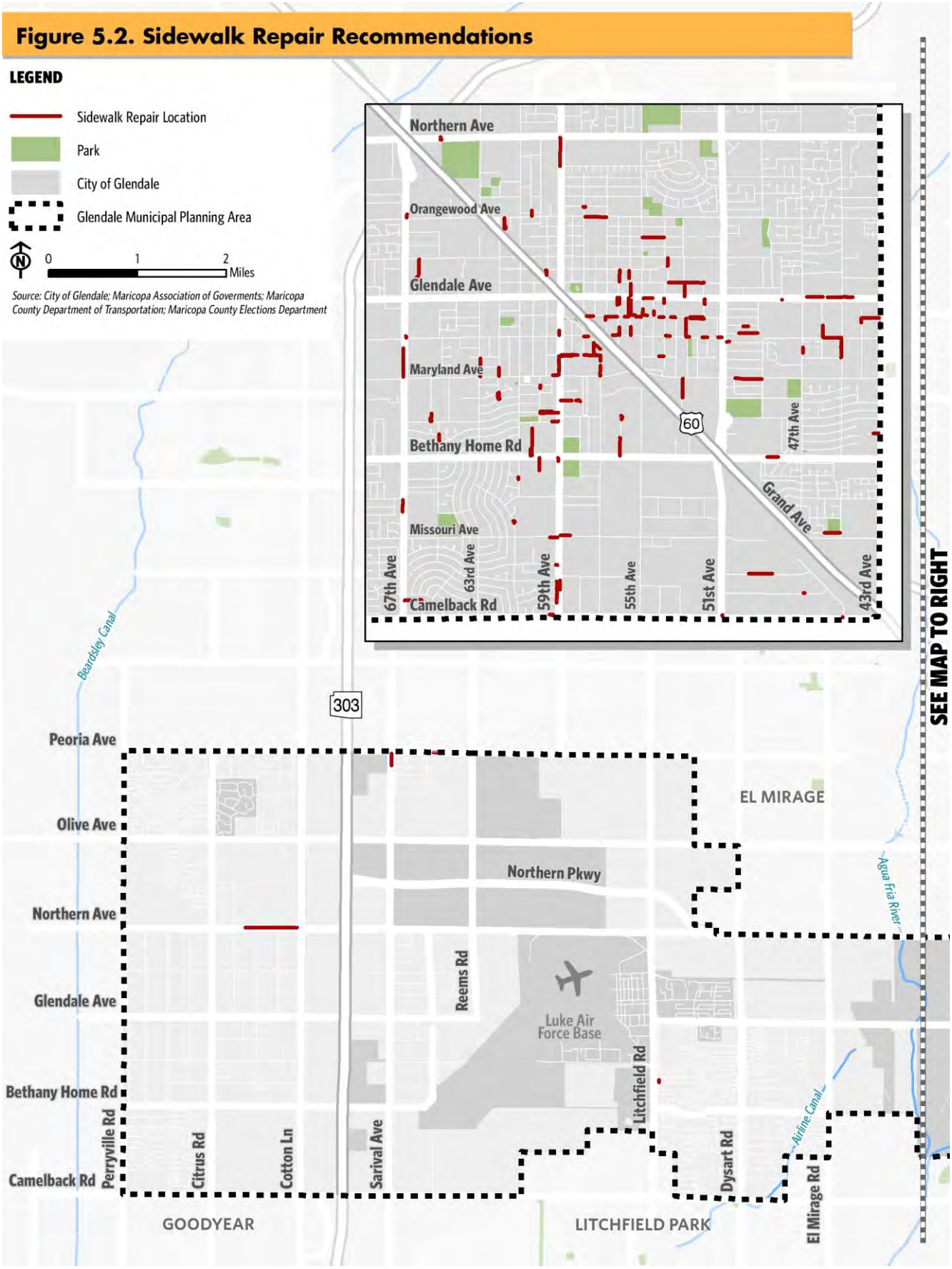
Sidewalk repair projects are proposed for sidewalks identified as being in poor condition at the time of inventory. Sidewalk issues may include obstructions, large cracks, uneven surfaces, and damaged segments. Sidewalks in need of repair can limit pedestrian access and pose safety concerns for pedestrians. Figure 5.2 illustrates the location of sidewalks identified as in poor condition and Table 5.2 provides examples of repair needs.

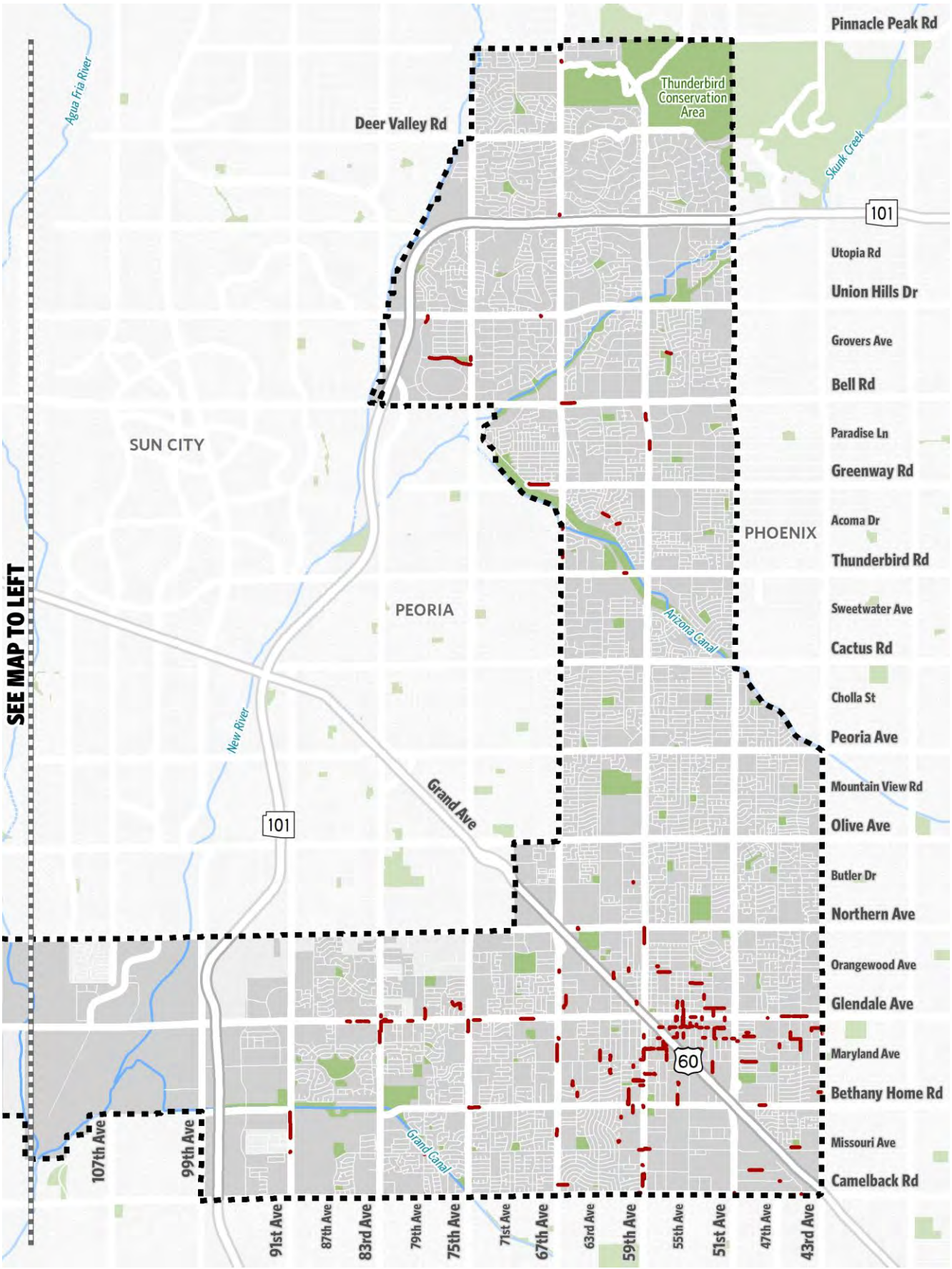


Table 5.2. Examples of Sidewalk Repair Needs

On Road	Location	Miles	Need
55th Dr	Glendale Ave to Glenn Dr	0.08	Repair sidewalk in poor condition
Lamar Rd	58th Ave to 57th Dr	0.02	Replace asphalt sidewalk in poor condition
59th Ave	Monte Cristo Ave to Paradise Ln	0.09	Replace uplifted sidewalk segments

Figure 5.2. Sidewalk Repair Recommendations




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Agua Fria River
Deer Valley Rd
Thunderbird Conservation Area
Skunk Creek
Pinnacle Peak Rd
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
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47th Ave
43rd Ave

Intersection Improvements

Intersections are potential conflict points between pedestrians and vehicles and thus require design treatments that set clear expectations for all roadway users. Every intersection in Glendale should be designed for pedestrian safety and comfort, with pedestrian enhancements appropriate for the roadway conditions. Sidewalks at intersections provide pedestrians with a refuge area while waiting to cross the street. Additionally, enhancements such as adding marked crosswalks, installing enhanced crossing treatments, or signalizing an intersection help to create more pedestrian-friendly intersections. The specific type of treatment at a crossing may range from simple, marked crosswalks to full traffic signals or grade-separated crossings. Appendix B provides examples and design guidance of different crossing treatments Glendale should consider.

Figure 5.3 illustrates the locations in need of pedestrian intersection improvements and Table 5.3 provides examples of these intersection needs. Appendix A provides a full listing of improvement recommendations, their priority scoring, and a planning level cost estimate.

Table 5.3. Examples of Intersection Improvement Recommendations

On Road	Location	Project Description	Comments	Potential Constraints	Prioritization Score
Peoria Ave	53rd Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops.	No	51.5
Solano Dr	65th Ave	Install sidewalk and ADA ramp at NW corner		No	51
Peoria Ave	49th Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops. If recommended bicycle facilities installed, provides bicycle connection across arterial.	No	50
Peoria Ave	60th Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops.	No	49
Thunderbird Rd	56th Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods, to hospital, and to transit stops.	No	47.5

Sidewalk Infill Improvements

A pedestrian network is only as strong as its weakest link. Sporadic corridor and business development have caused small gaps in the existing pedestrian network, which creates barriers to pedestrian travel. Strategically filling these gaps can link people to jobs, shopping, recreational opportunities, and provide choices for convenient travel by foot. Figure 5.4 illustrates the locations sidewalk infill improvements by their priority scoring and Table 5.4 provides examples of these infill needs. Appendix A provides a full listing of improvement recommendations, their priority scoring, and a planning level cost estimate.

Table 5.4. Examples of Sidewalk Infill Improvement Recommendations

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score
Bethany Home Rd	52nd Ave	51st Ave - W of	Left	0.04	No	73.5
Bethany Home Rd	57th Ave - E of	56th Ave - W of	Left	0.07	No	73
Bethany Home Rd	55th Ave - E of	52nd Dr - W of	Both	0.03	No	71.5
75th Ave	Bethany Home Rd	Griffin Ave	Right	0.38	No	71.5
Greenway Rd	59th Ave	57th Ave	Right	0.30	No	71.5

Railroad Crossings

Railroads often act as barriers in the active transportation network by limiting the pedestrian and cyclist's ability to access a location. Providing sidewalk, bicycle, and ADA facilities at railroad crossings aids in connecting pedestrians and cyclists to destinations. Figure 5.5 presents the priority scoring for locations that may benefit from pedestrian crossing enhancements. It's important to note that any improvements to railroad crossings requires close coordination, review, assessment, and buy-in from the railroad operator. Warrants for at-grade crossing infrastructure also requires an assessment of train frequency, pedestrian and bicycle level, and on-the-ground conditions to determine if additional active warning devices or grade separation is required. The *MAG Bicycle and Pedestrian Pathway/Railroad Crossing Recommendations* report provides additional information in regard to potential treatments and design guidelines for at-grade railroad crossings.

Transit Connections

Connecting bus stops to nearby active transportation facilities addresses first/last mile issues by facilitating walking and bicycling to and from the public transportation network. Figure 5.5 present the priority scoring for transit stops that require pedestrian facility connections. Partnerships can be made with Valley Metro to enhance and expand accessibility to bus stops located along Glendale-maintained roadways. Additionally, evaluation and need for mid-block crossing and transit stops, particularly high ridership stops, should be considered.

Accessibility for People with Disabilities

A facility's compliance to the ADA not only affects individuals with disabilities, but all pedestrians. Maintaining a smooth sidewalk surface free of obstructions helps to eliminate tripping hazards, especially for visually impaired pedestrians, and makes the path easier to traverse, particularly for persons using a wheelchair, cane, or other mobility assistance device. Furthermore, curb ramps, and particularly detectable warning surfaces, create a safe transition for blind and low-vision persons and provide a vital safety cue that you are entering a roadway. Currently Glendale adheres to ADA standards for all new construction and alterations to existing public rights-of-way, as outlined in Glendale Engineering Standard Details. ADA standards include, but are not limited to:

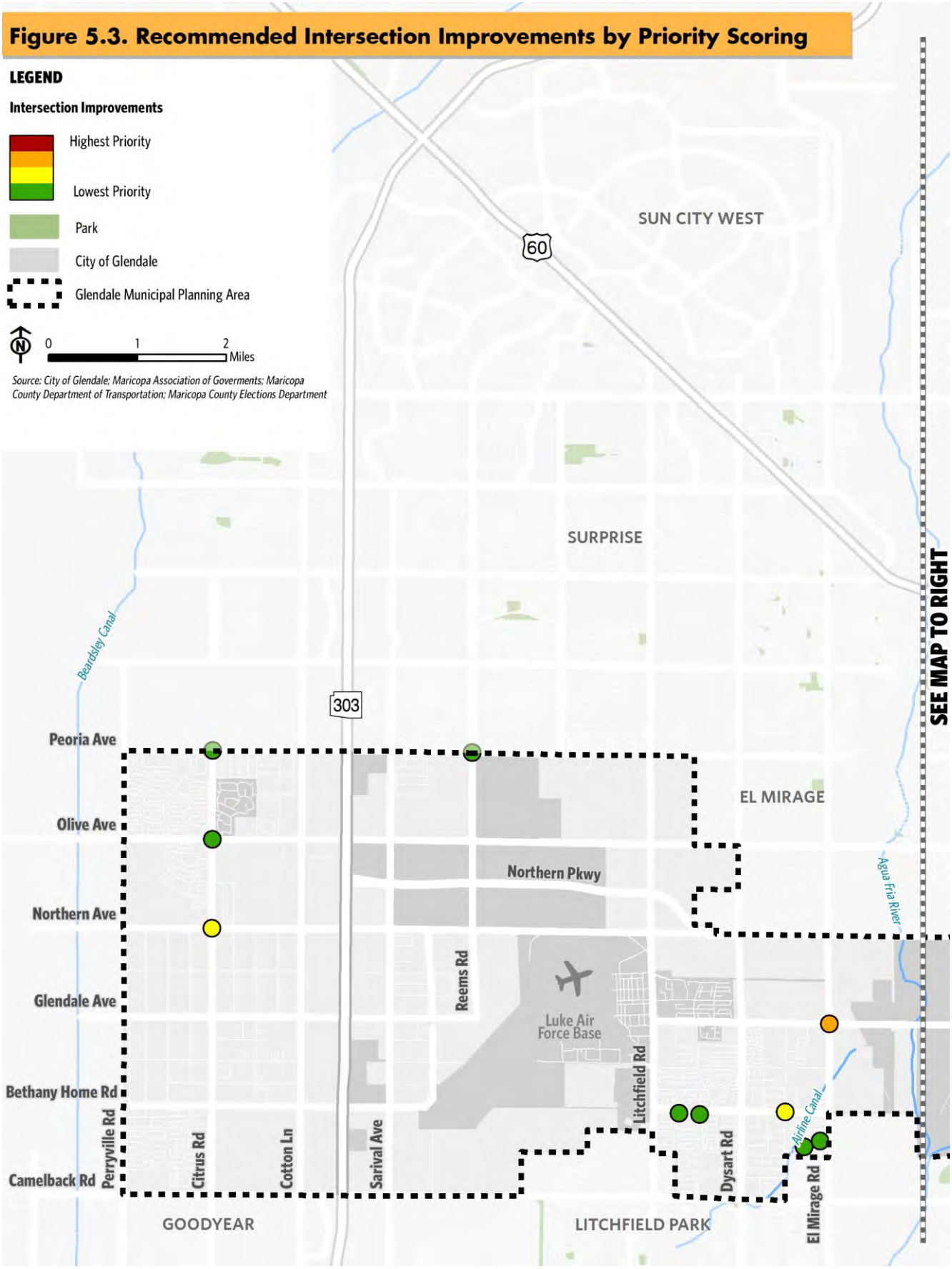
- A 5 FT x 5 FT passing zone is required every 200 FT on sidewalks with widths less than 5 FT; and
- Curb ramps must be 36 inches in width or greater, meet cross and running slope standards, and have a high-visibility truncated dome present.

As part of the ATP process, an aerial assessment of sidewalk conditions, as well as the location and conditions of curb ramps, was conducted to assist Glendale in identifying locations where sidewalk and curb ramp upgrades may be needed for ADA compliance. The results of the inventory found:

- Sidewalks were primarily less than 5 FT in width;
- 9,176 curb ramps do not have a detectable warning or are in poor condition; and
- 10,077 driveways had potential ADA compliance issues.

The findings from the ATP should be utilized as a base for implementing citywide ADA improvements. Detailed, on the ground assessments are needed to confirm ADA compliance as additional factors impact a facility's compliance. Where possible, upgrades to facilities to conform with ADA standards should be incorporated into maintenance programs and programmed improvements. For all future improvements, facilities should be constructed in compliance with ADA standards.

Figure 5.3. Recommended Intersection Improvements by Priority Scoring



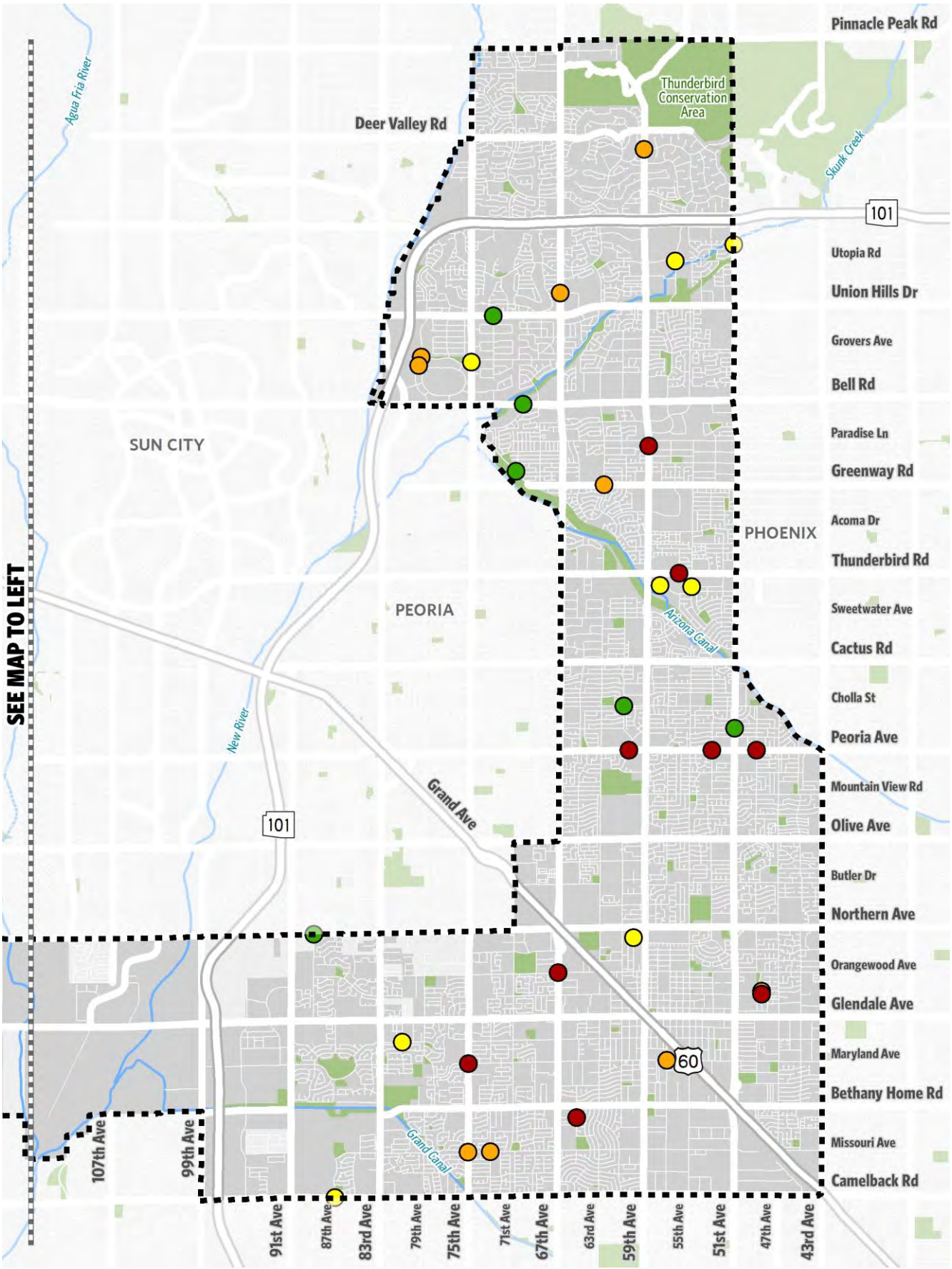
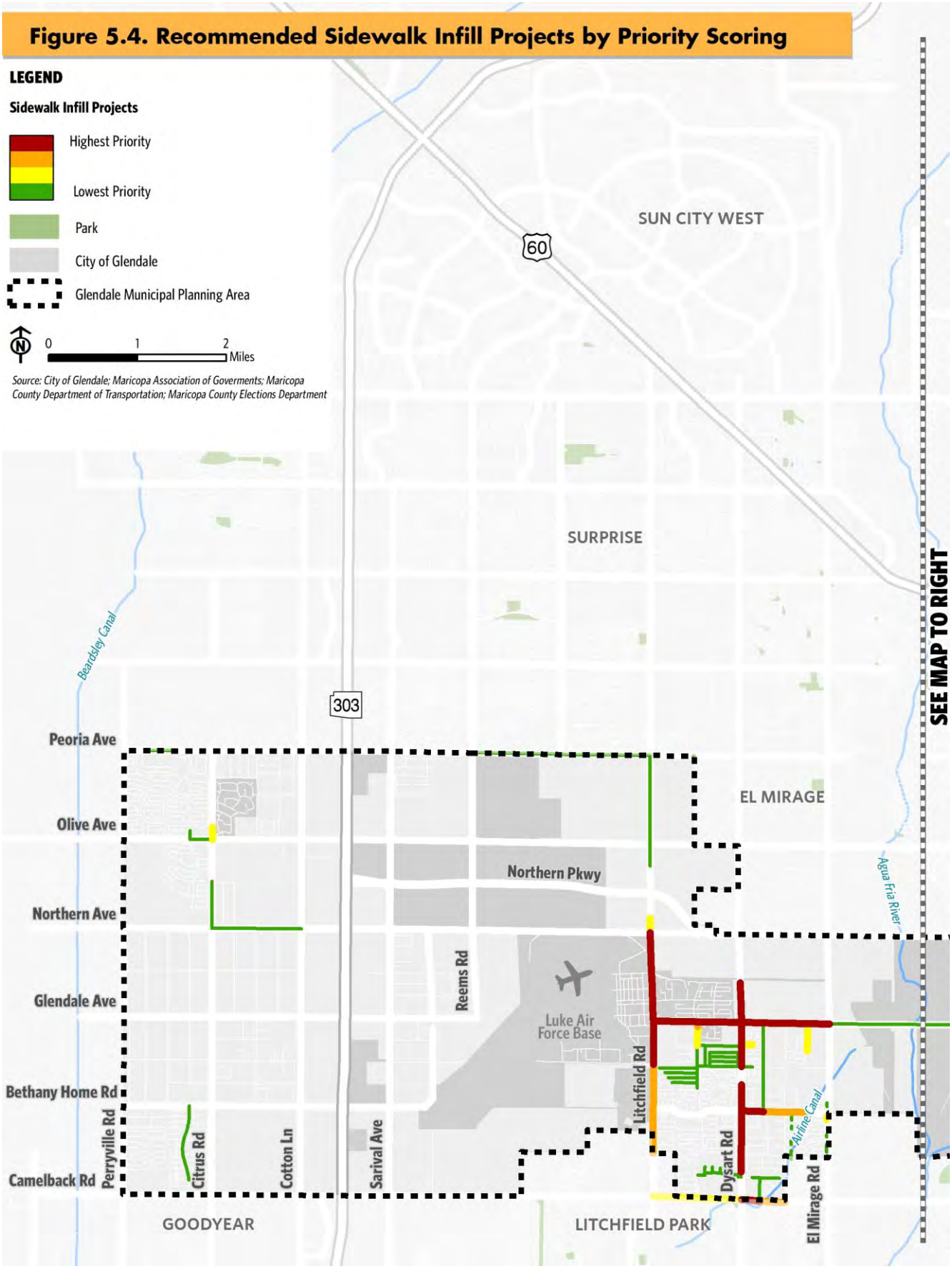

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Figure 5.4. Recommended Sidewalk Infill Projects by Priority Scoring



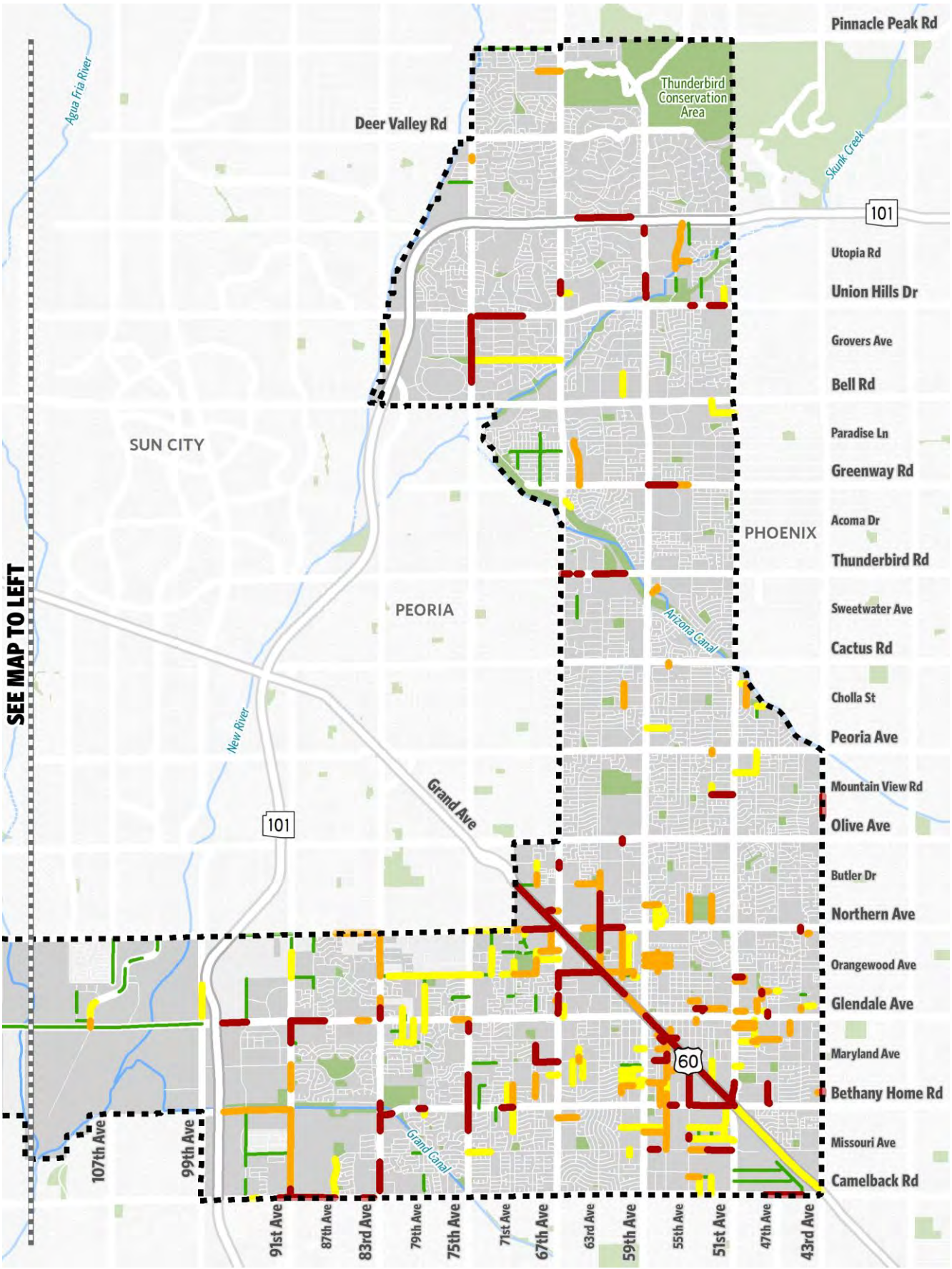
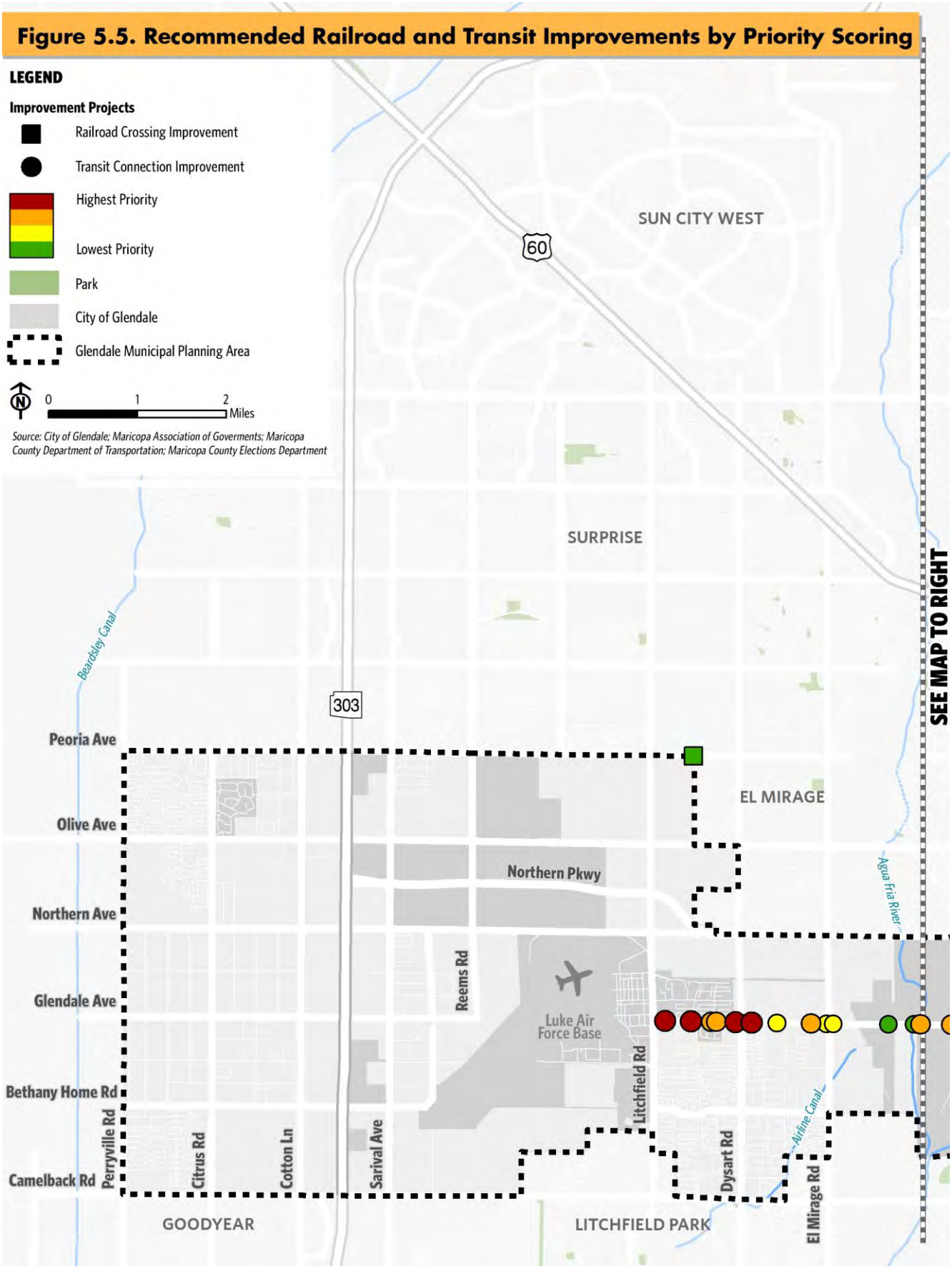
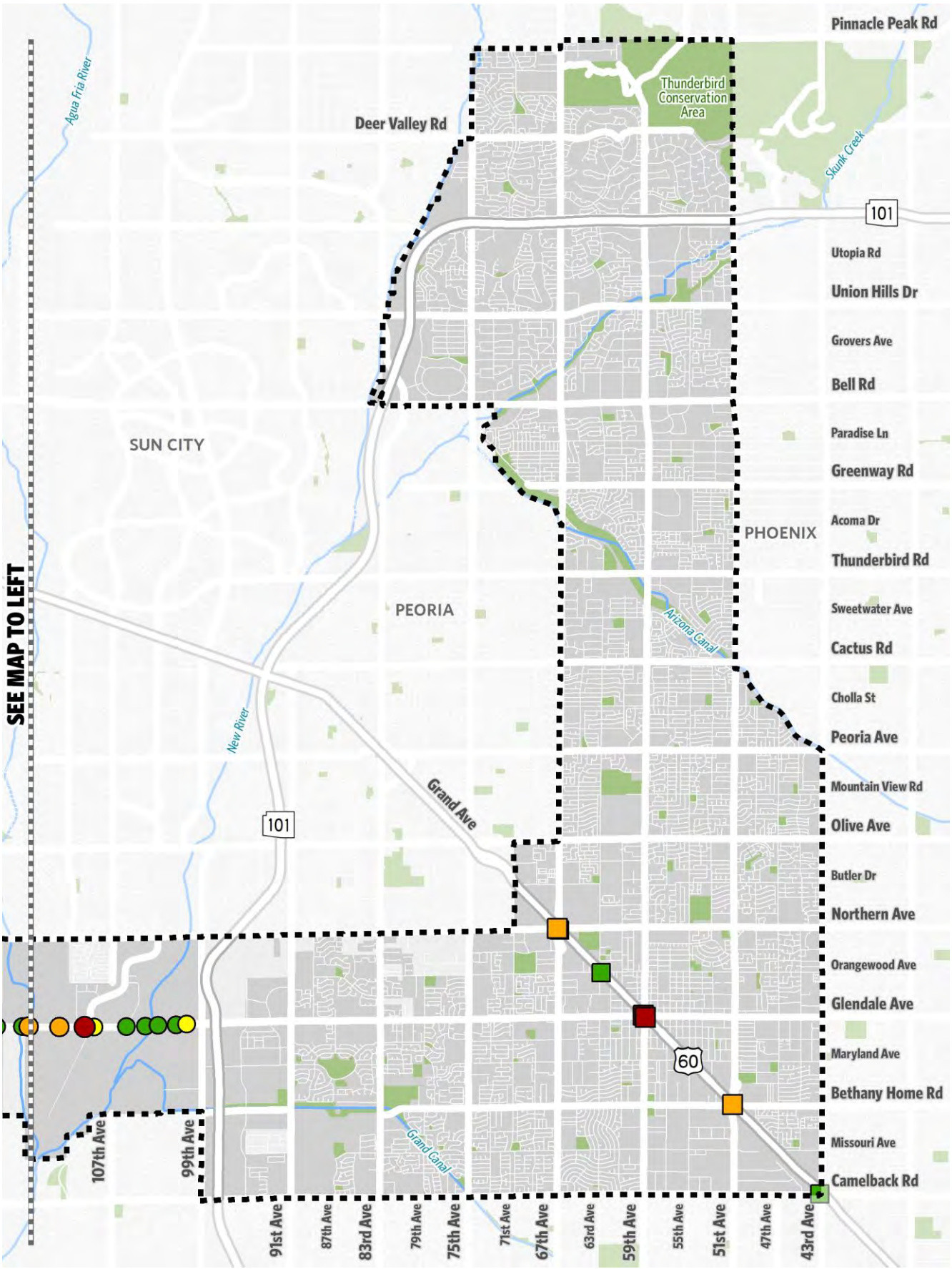

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Figure 5.5. Recommended Railroad and Transit Improvements by Priority Scoring




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Bicycle Network Improvement Recommendations

A comprehensive bikeway network improves bicyclists' level of safety, convenience, and access to key destinations. Planning a bikeway network enables Glendale to prioritize and seek funding to construct bicycle facilities where they will provide the greatest benefit to bicyclists and the community-at-large. The recommended bicycle network presented in this section was developed with five main objectives:

- Define bicycle facilities and create consistent use of treatments and application so users can plan trips based on their comfort level;
- Logically connect existing facilities to improve local and regional nonmotorized mobility;
- Identify low-cost solutions that fit within the existing roadway footprint;
- Determine upgrades to accommodate all ages and abilities; and
- Identify long-term, regional needs.

Addressing all identified recommendations creates a robust regional bicycle network that provides a more comfortable riding experience for experienced bicyclists, and low-stress options for children and those not as confident riding a bike.

Upgrades to Current Facilities

Table 5.5 provides examples of recommended upgrades to current on-street bicycle facilities to further improve a bicyclists level of comfort and improve messaging. Figure 5.6 illustrates the locations of upgrade recommendations. Appendix A provides a full listing of improvement recommendations, their priority scoring, and a planning level cost estimate, while Appendix B provides examples and design guidance of different bicycle facilities Glendale should consider.

Table 5.5. Examples of Upgrades to Current Facilities

On Road	From	To	Miles	Recommended Bike Facility	Comments	~ Pavement Width (FT)	Fits in ROW	Prioritization Score
47th Ave	Northern Ave	Butler Dr	0.48	Bike Lanes	On-street parking must be removed to accommodate bike lanes	53	Yes	62.5
47th Ave	Butler Dr	Olive Ave	0.47	Bike Lanes		40	Yes	61.5
61st Ave	Bethany Home Rd	Glendale Ave	0.99	Signed Bike Route		42	Yes	57
55th Ave	Orangewood Ave	Northern Ave	0.49	Bike Lanes with Parking on Both Sides		50	Yes	56
Missouri Ave	67th Ave	59th Ave	0.95	Bike Lanes with Parking on Both Sides		45	Yes	55.5

New Bicycle Facilities

People who ride bicycles vary in their physical abilities, experience levels, and comfort level riding adjacent to motor vehicles. The identified new bicycle facility recommendations include gaps in the current network that limit a person's ability to bike to their destination. In addition, opportunities to expand Glendale's bicycle network and create regional recreation routes were also identified. Filling in these network gaps links hundreds of thousands of residents to jobs and provides choices for convenient travel by bicycle.

Table 5.6 provides examples of recommended new bicycle facilities to expand Glendale's bicycle network and to provide a more comfortable riding experience for bicyclists. Figure 5.7 illustrates the locations of new bicycle facilities. Appendix A provides a full listing of improvement recommendations, their priority scoring, and a planning level cost estimate.

Table 5.6. Examples Recommended New Bicycle Facilities

On Road	From	To	Miles	Recommended Bike Facility	Comments	~ Pavement Width (FT)	Fits in ROW	Prioritization Score
67th Ave	Camelback Rd	Glendale Ave	2.01	Bike Lanes	Reduce travel lanes and median	65	No	76.5
Glendale Ave	67th Ave	59th Ave	0.97	Protected Bike Lane	Must reduce median to accommodate buffered bike lanes	65	No	75.5
59th Ave	Bethany Home Rd	Maryland Ave	0.50	Bike Lanes	Narrow or reduce center turn lane	65	No	74
59th Ave	Maryland Ave	Glendale Ave	0.50	Bike Lanes	Narrow or reduce center turn lane	65	No	70.5
Ocotillo Rd	75th Ave	59th Ave	1.97	Shared Lanes			Yes	64

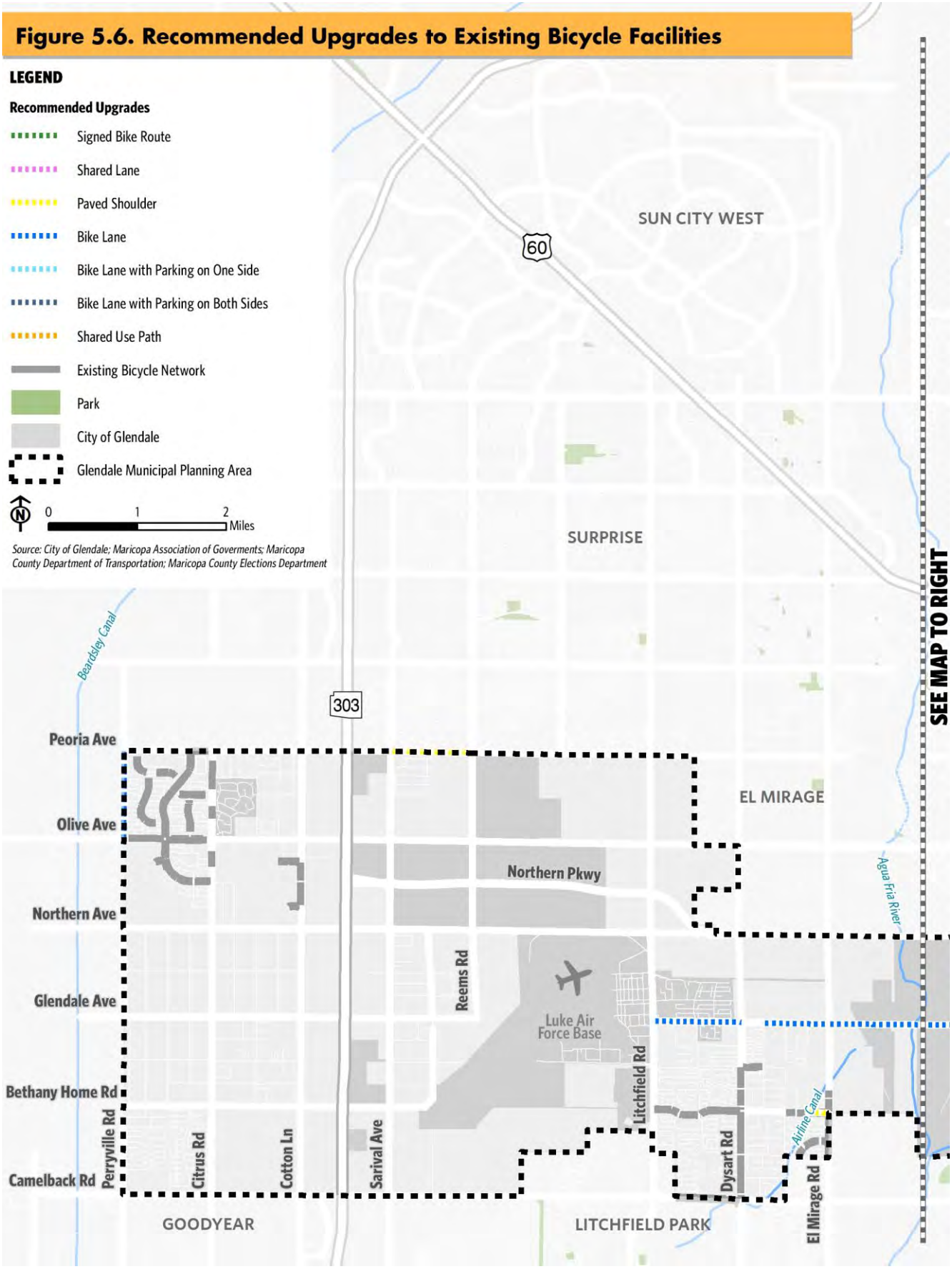
Bicycle Facilities on Arterials

Arterials are the backbone of the Phoenix metropolitan's regional transportation network. In most of the region, arterials provide the only through-route between cities and across rivers, canals, mountains, and freeways. While the Glendale ATP focuses on creating a network for persons of all ages and abilities, accommodating bicycle travel along arterials not only provides regional access, but also provides direct connections to the places people want to travel – shopping centers, businesses, jobs, grocery stores, medical facilities, and transit stops. Specific bike treatments and the feasibility of installing bicycle facilities should be considered to accommodate both the experienced, commuter bicyclist and the less experienced rider. Key arterials that should be evaluated for potential bicycle accommodation include:

- Olive Ave: 67th Ave to Glendale Community College. Key transit corridor that connects the popular 67th Avenue transit route to Glendale Community College.
- 67th Ave, 59th Ave, 43rd Ave (Phoenix maintained), Bethany Home Rd, Glendale Ave, Northern Ave, Peoria Ave, Thunderbird Rd, and Bell Rd. High bicycle crash corridors and the highest ridership transit routes in the City. Arterials, such as Bell Road, pose numerous design constraints to incorporate bike facilities.

Arterials have higher traffic volumes than local streets, and typically have traffic signals at major intersections. Preferred treatments on arterials include barrier-protected bike lanes, cycle tracks, and buffered bike lanes. On many arterial streets, accommodating bicycle facilities requires major investment and reallocating space within the existing right-of-way.

Figure 5.6. Recommended Upgrades to Existing Bicycle Facilities



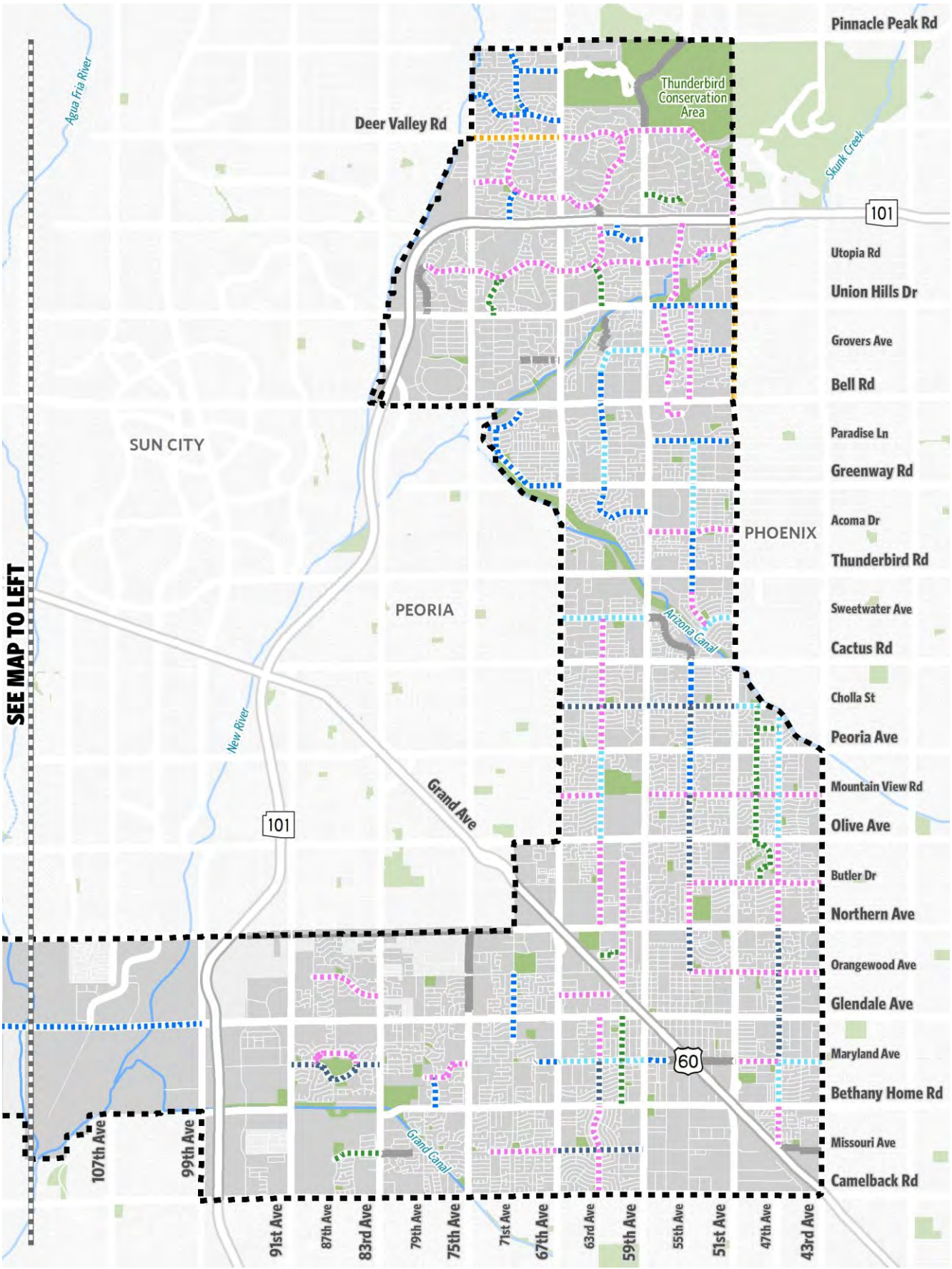
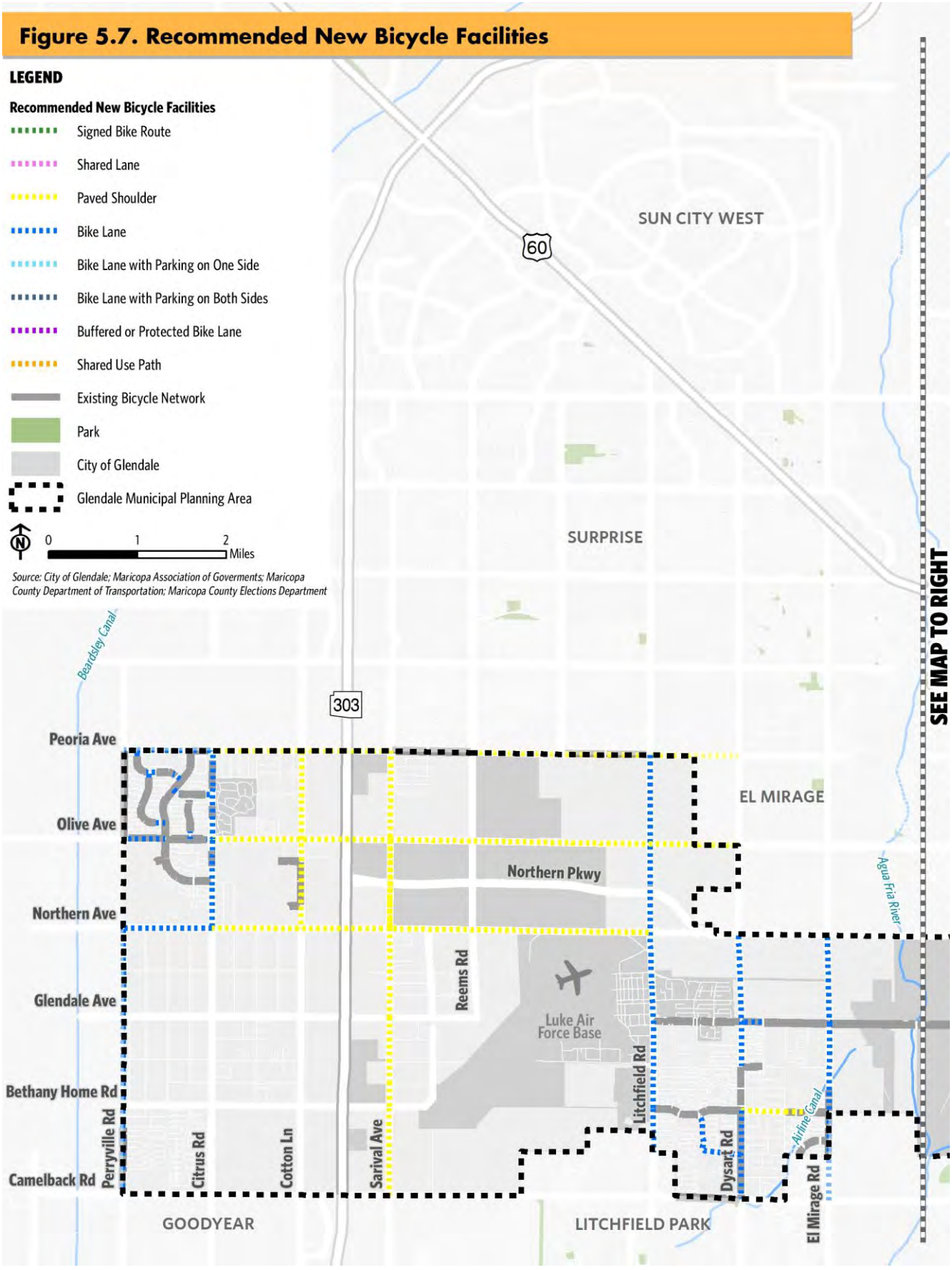
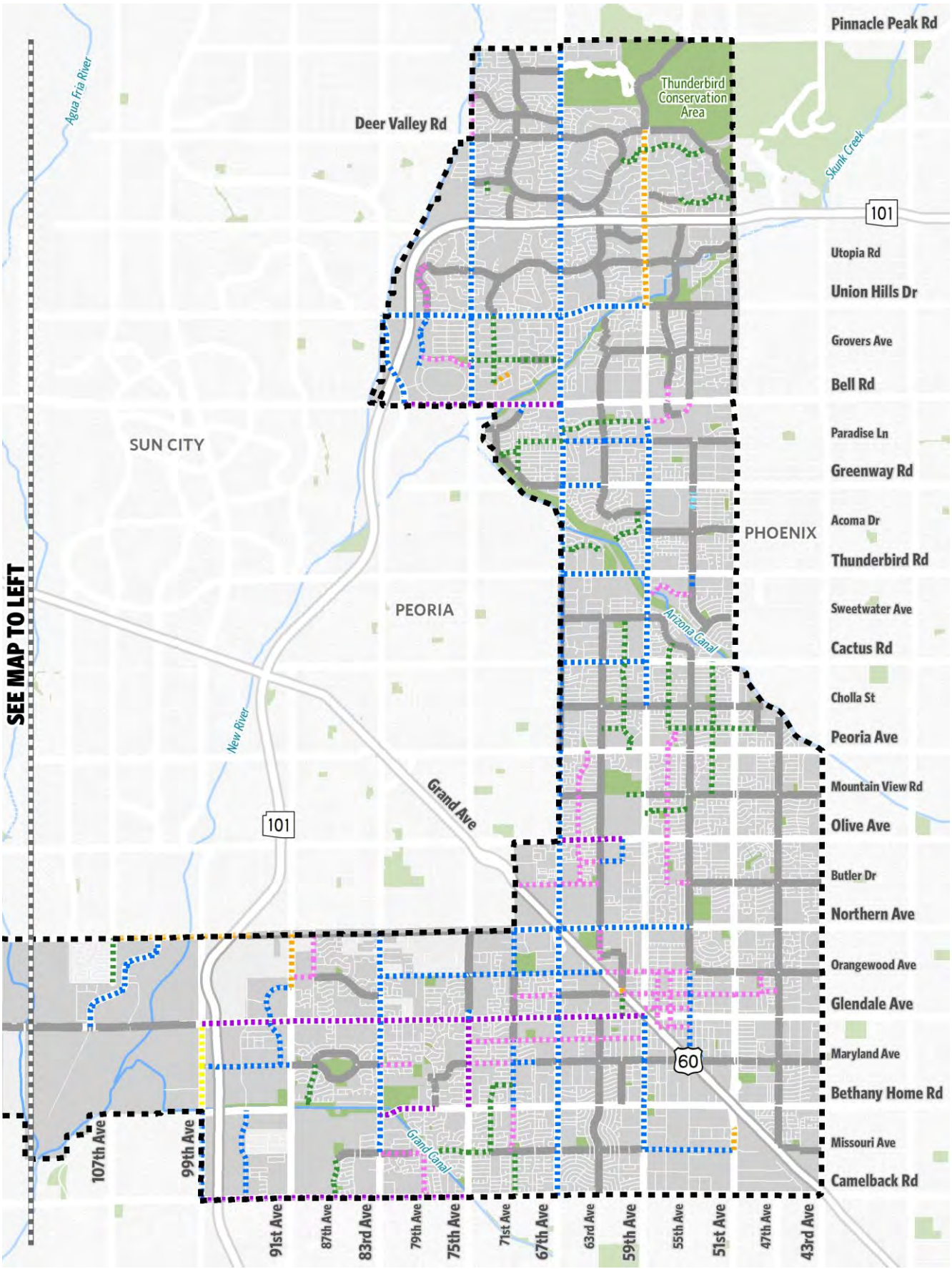

SEE MAP TO LEFT
Agua Fria River
Deer Valley Rd
Thunderbird Conservation Area
Pinnacle Peak Rd
Skunk Creek
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
SUN CITY
Paradise Ln
Greenway Rd
PHOENIX
Acoma Dr
Thunderbird Rd
PEORIA
Sweetwater Ave
Arizona Canal
Cactus Rd
Cholla St
Peoria Ave
New River
Grand Ave
101
Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
Maryland Ave
60
Bethany Home Rd
Missouri Ave
Camelback Rd
Grand Canal
107th Ave
99th Ave
91st Ave
87th Ave
83rd Ave
79th Ave
75th Ave
71st Ave
67th Ave
63rd Ave
59th Ave
55th Ave
51st Ave
47th Ave
43rd Ave

Figure 5.7. Recommended New Bicycle Facilities




SEE MAP TO LEFT
Agua Fria River
Deer Valley Rd
Thunderbird Conservation Area
Pinnacle Peak Rd
Skunk Creek
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
Acoma Dr
Thunderbird Rd
Sweetwater Ave
Cactus Rd
Cholla St
Peoria Ave
Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
Maryland Ave
Bethany Home Rd
Missouri Ave
Camelback Rd
SUN CITY
PEORIA
PHOENIX
New River
Grand Ave
Arizona Canal
Grand Canal
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107th Ave
99th Ave
91st Ave
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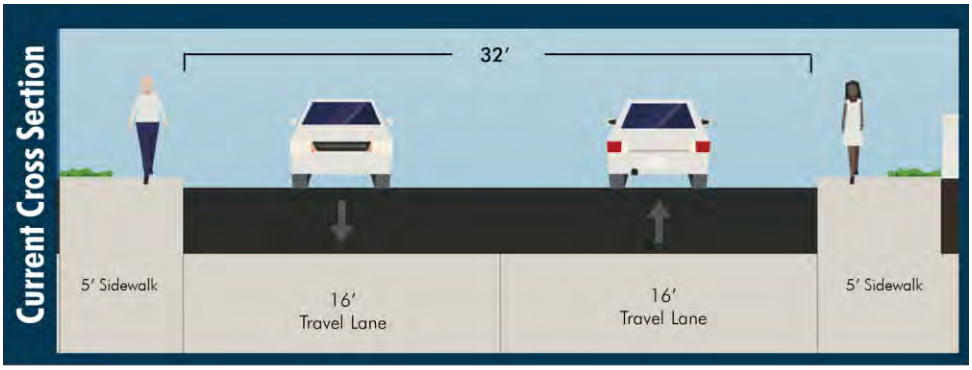
Accommodating Bicyclists Within Current Cross Sections

Glendale's existing roadway cross section options includes limited bicycle facilities. Low-cost upgrades to accommodate bicyclists within current cross-sections can be accommodated in some locations by:

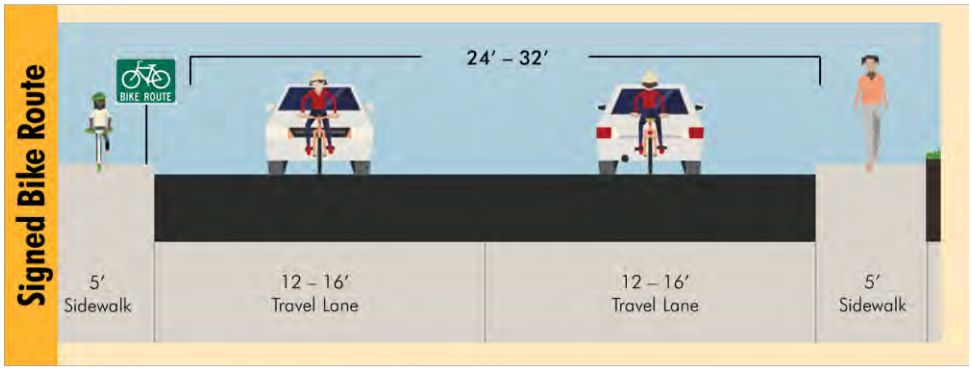
- Restriping and lane narrowing within the existing pavement width. Implementation of these improvements are best achieved during roadway resurfacing or restriping projects.
- Reallocating space for cross sections' elements within the existing right-of-way to realize higher quality active transportation facilities while considering the needs of motorized users.

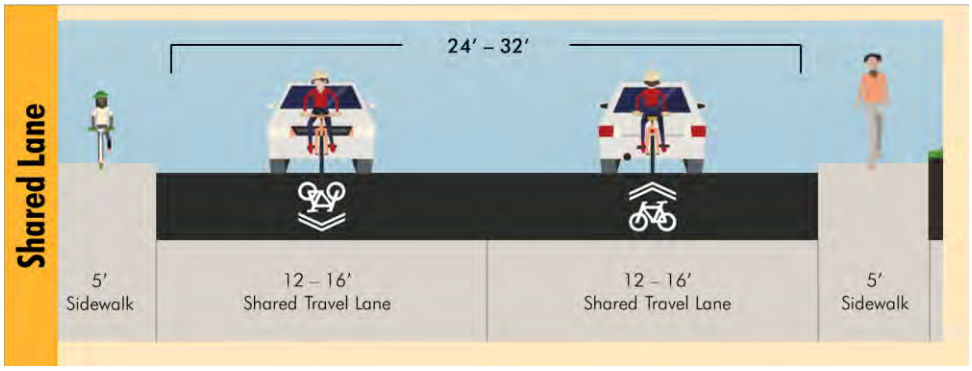
Local Cross Section

This represents a typical residential street in Glendale. Lanes and parking areas are usually not delineated. Per the City's 2015 Standard Details, reduced pavement widths may be allowed. To accommodate the needs of bicyclists, the modified cross sections below provide a range of potential configurations.



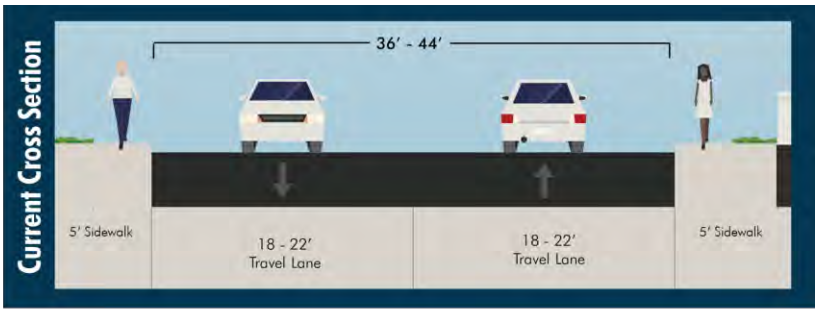
Modified Restriping



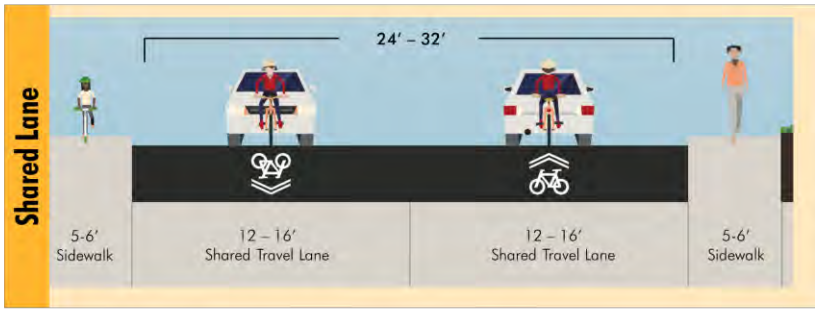


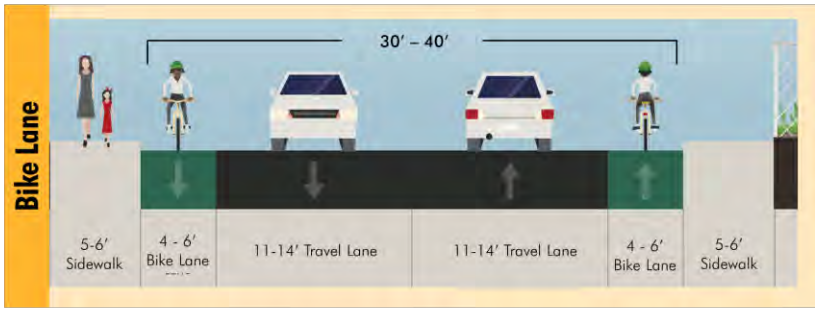
Collector Cross Section

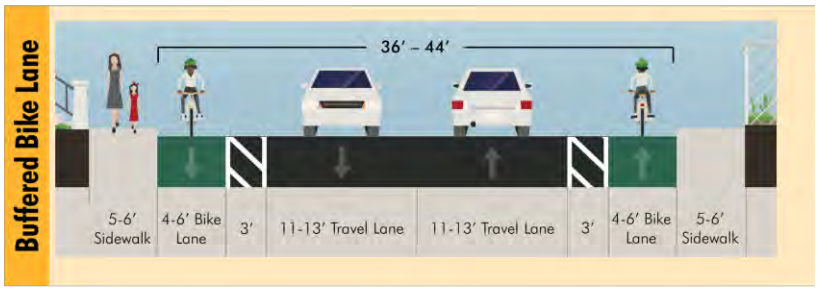
Collectors are designed to accommodate motor vehicle traffic between local, residential roads and busier, wider roads. Collectors typically create low-stress connections for bicyclists to travel. Per the City's 2015 Standard Details, reduced pavement widths may be allowed. To accommodate the needs of bicyclists, the modified cross sections below provide a range of potential configurations.

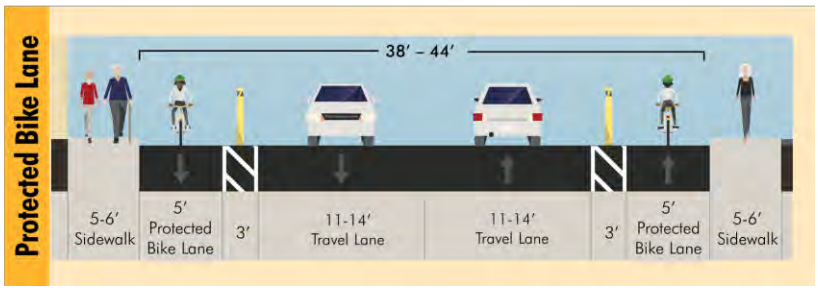


Modified Restriping



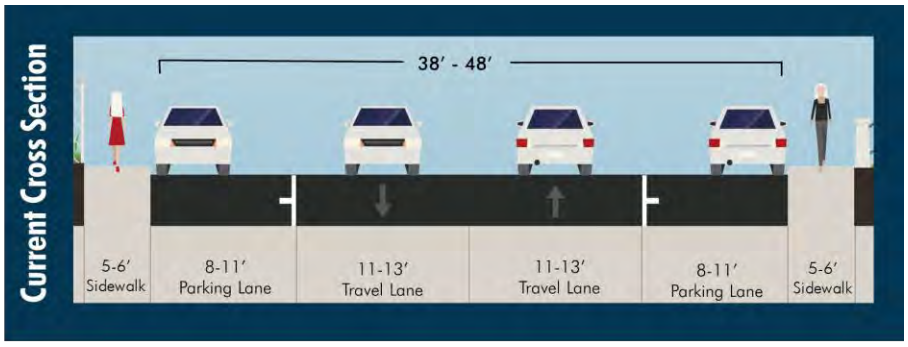




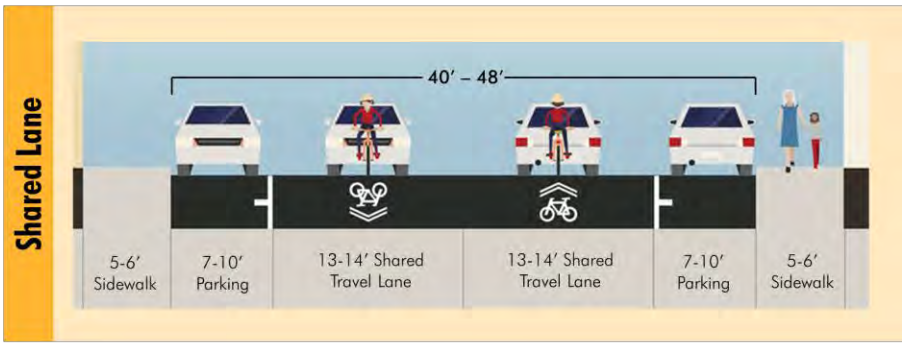


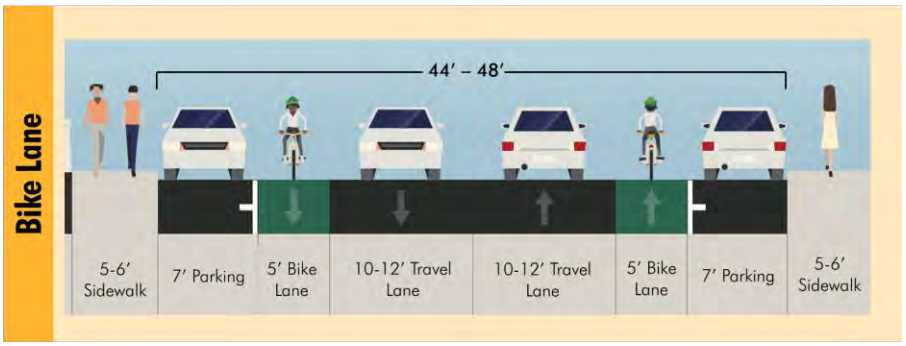
On-Street Parking Cross Section

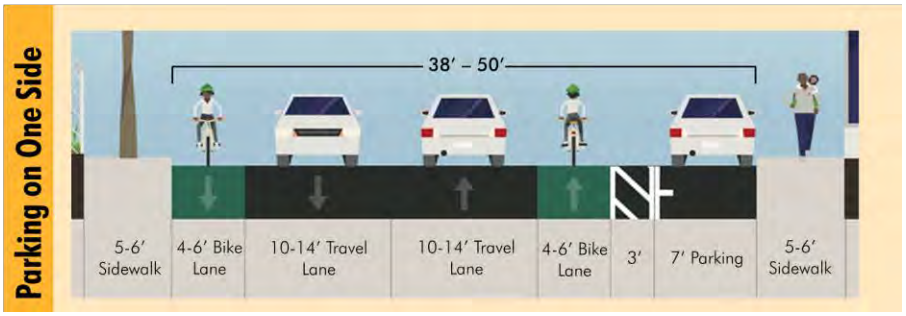
Currently, Glendale has over 90 centerline miles of roadways classified as "Shoulder Stripe with Parking". These routes are classified as and signed as "Bike Route Parked Vehicles", which creates a shared shoulder for bicyclists and parked vehicles. Sharing bicycle facilities with parked vehicles can create an unsafe and uncomfortable riding experience for most cyclists, as they may need to weave in and out of traffic to avoid parked vehicles. To accommodate bicycle facilities on streets with parked vehicles, the following cross sections can be applied. For streets with on-street parking that do not have businesses, homes, or parks located along the route, current shoulder striping could be easily signed as a bike lane and on-street parking removed.



Modified Restriping

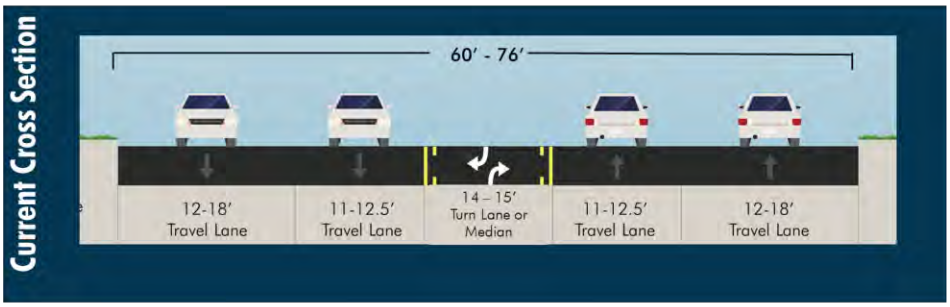




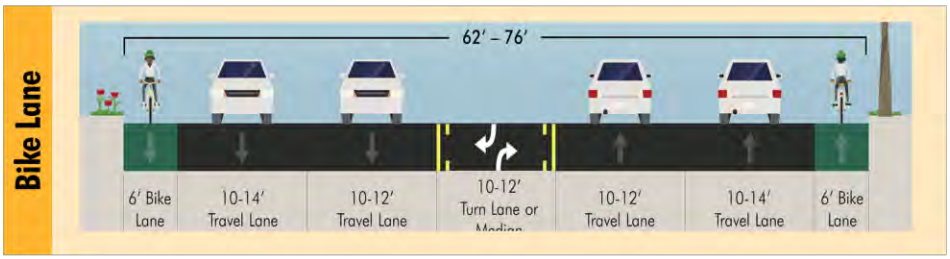


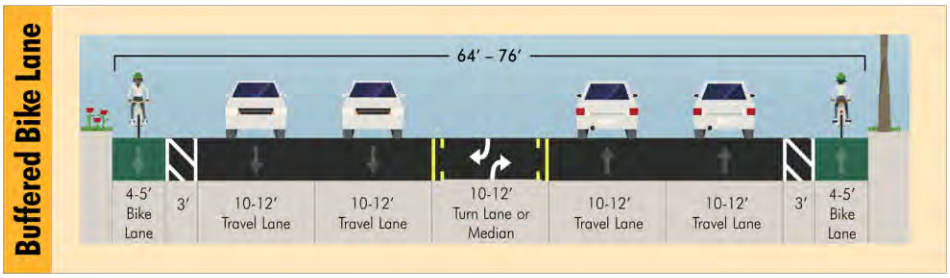
Arterial Cross Section

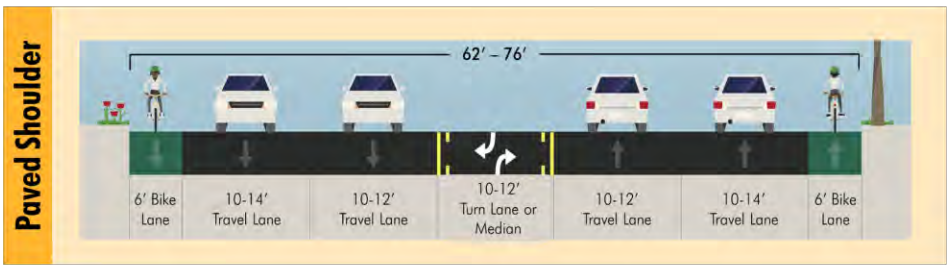
Arterials are vital to Glendale's transportation network, as they provide direct access to jobs, shopping centers, major destinations, and regional facilities and also provide the only access across canals and rivers. Retrofitting Glendale's arterial network with bicycle facilities can help expand the regional bicycle network and provide necessary commuting routes to jobs and transit stops. The cross sections presented below provide several options to fit within Glendale's many arterial cross sections.

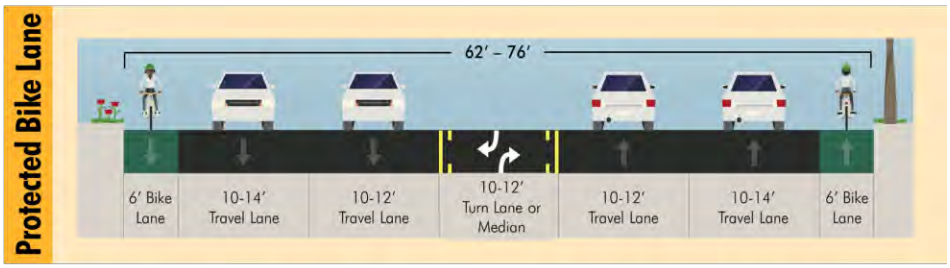


Modified Restriping









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Bicycle Treatments at Intersections

Bike lanes define a portion of the roadway that is preferential or exclusive for use by bicyclists; allowing bicyclists to travel alongside traffic. In addition, bike lanes remind motorists to look for bicyclists when turning and that bicyclists have the right to the road. A bike lane along a corridor may provide a comfortable riding experience; however, if that designated space disappears at an intersection, the entire corridor may no longer be attractive to bicyclists. As noted in Chapter 3, intersections are common locations of conflict between drivers and bicyclists. Therefore, particular attention to intersections is necessary in order to increase motorist awareness, provide bicyclists with a sense of comfort, and improve the riding experience along a corridor. Figure 5.8 illustrates locations where intersection improvements are needed to extend bicycle facilities through intersections to alert motorists of bicycle traffic. Appendix A provides a full listing of improvement recommendations, their priority scoring, and a planning level cost estimate.

River, Canal, Freeway Crossings

Rivers, canals, and freeways create barriers in the active transportation network. Crossings at these facilities are needed to provide a continuous pedestrian and bicycle network. Study roadways were evaluated to identify locations that currently lack bicycle connectivity at river, canal, and freeway crossings. In addition, locations that may warrant crossings if development occurs were identified. Figure 5.8 and Table 5.7 presents example of locations that may benefit from crossing facilities.

Table 5.7. Example of River and Canal Crossings

On Road	Crossing	Improvement Description	Comments	Prioritization Score
Bethany Home Rd	Grand Canal Path	Conduct additional engineering analysis to determine feasibility/need for crossing on west side of canal	Provides at-grade arterial crossing connecting paved canal paths	41.5
Union Hills Dr	Skunk Creek Trail	Conduct additional engineering analysis to determine feasibility/need for crossing on west side of canal	Provides arterial crossing connecting the western paved canal path. Median pedestrian refuge island would be needed.	40.5
Thunderbird Rd	Arizona Canal Trail	Conduct additional engineering analysis to determine feasibility/need for crossing on west side of canal	Provides at-grade arterial crossing connecting paved canal paths	39

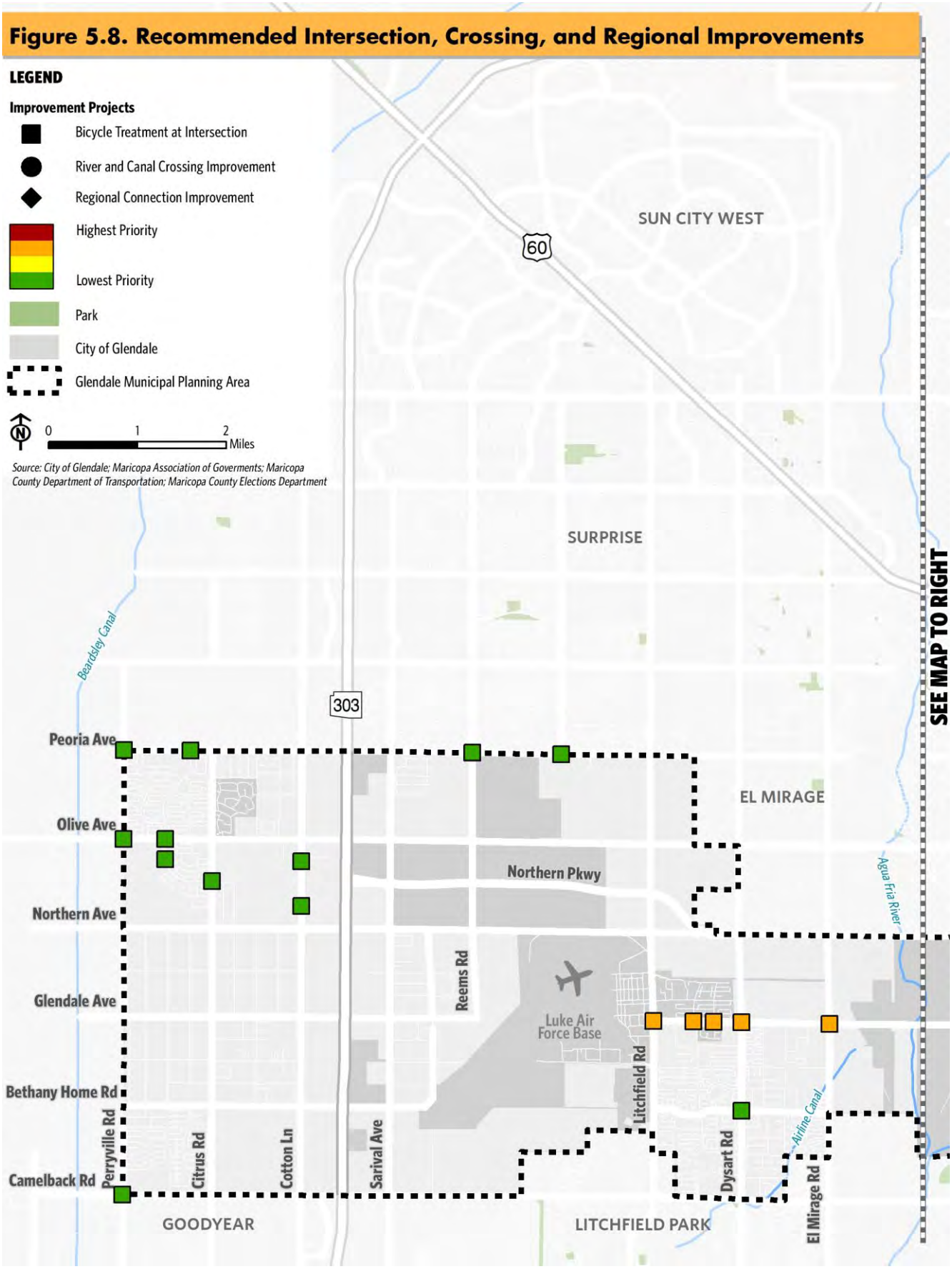
Regional Connections

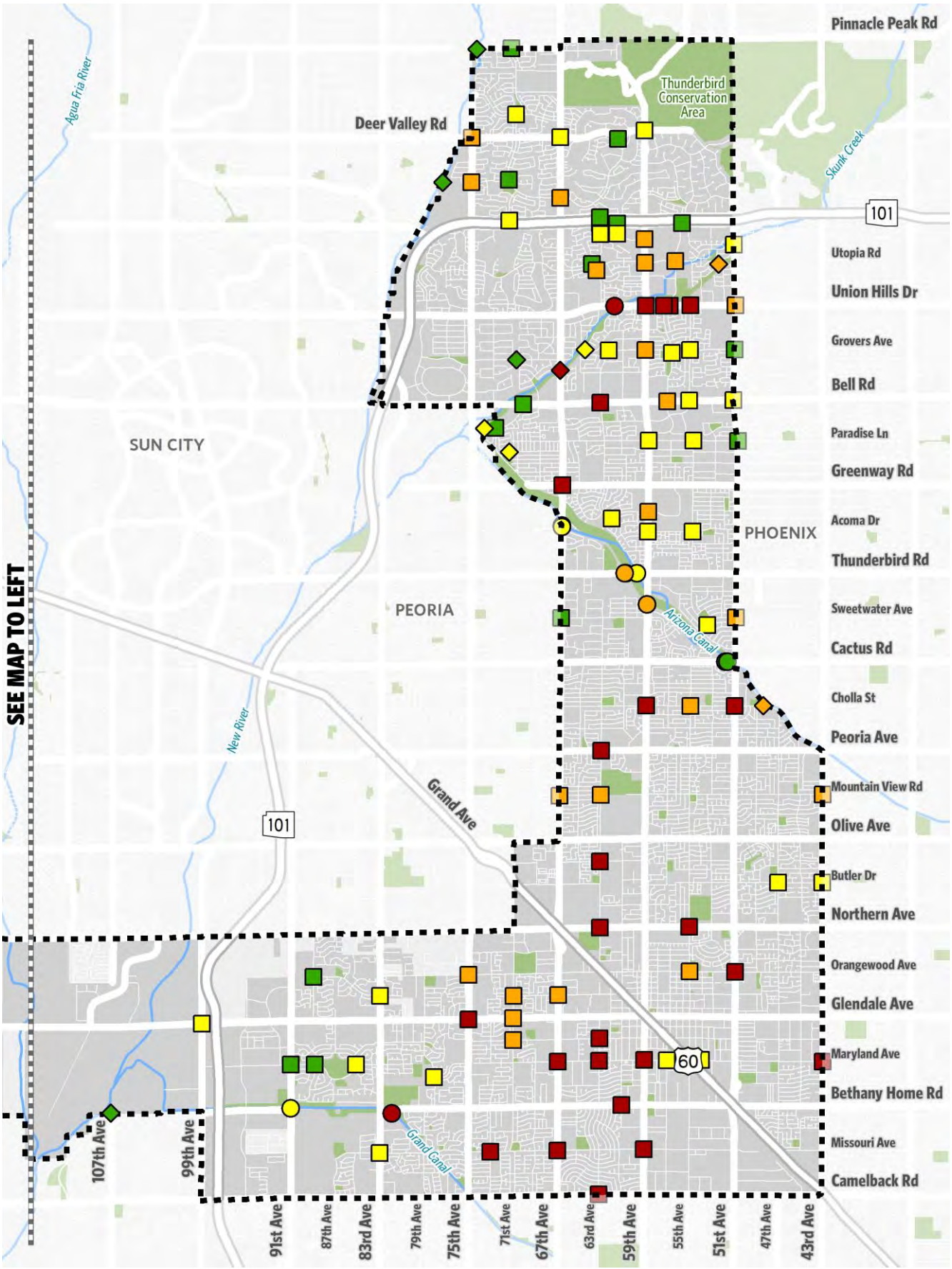
Glendale has a tremendous opportunity to collaborate with neighboring jurisdictions to expand the region's active transportation network. Regional bicycle connection and expansion opportunities address regional connectivity needs between jurisdictions and key regional destinations, as well as regional parks, and long-distance recreational opportunities. Figure 5.8 and Table 5.8 presents example of locations that may benefit from crossing facilities.

Table 5.8. Regional Connections

On Road	Crossing	Improvement Description	Comments	Prioritization Score
67th Ave	Skunk Creek Trail	Install connection from sidewalks on 67th Avenue to Skunk Creek Linear Park	No connection from on-street facilities and sidewalks present to connect to trails.	34.5
Cholla St	Arizona Canal Trail	Install connection to Arizona Canal Trail	No connection from on-street facilities and sidewalks present to connect to trail.	23.5
Taro Ln	Skunk Creek Trail	Upgrade connection to Skunk Creek Linear Park	Upgrade connection from on-street facilities and sidewalks to create a more visible, user-friendly connection to Skunk Creek Trail.	23

Figure 5.8. Recommended Intersection, Crossing, and Regional Improvements




Pinnacle Peak Rd
Thunderbird Conservation Area
Deer Valley Rd
Agua Fria River
Skunk Creek
101
Utopia Rd
Union Hills Dr
Grovers Ave
Bell Rd
Paradise Ln
Greenway Rd
Acoma Dr
Thunderbird Rd
Sweetwater Ave
Cactus Rd
Cholla St
Peoria Ave
Mountain View Rd
Olive Ave
Butler Dr
Northern Ave
Orangewood Ave
Glendale Ave
Maryland Ave
Bethany Home Rd
Missouri Ave
Camelback Rd
SUN CITY
PHOENIX
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SEE MAP TO LEFT
New River
Grand Ave
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107th Ave
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91st Ave
87th Ave
83rd Ave
79th Ave
75th Ave
71st Ave
67th Ave
63rd Ave
59th Ave
55th Ave
51st Ave
47th Ave
43rd Ave

Near-Term and Quick Win Projects

Improvements to the active transportation network may require significant investment and take years to complete. To build momentum behind the ATP, short-term actions that the City should focus on to begin implementing. These near-term/quick win projects were selected based on a variety of factors, such as priority scoring, ease of implementation, constructability, potential benefit to socioeconomically disadvantaged population groups, and strong public or stakeholder interest.

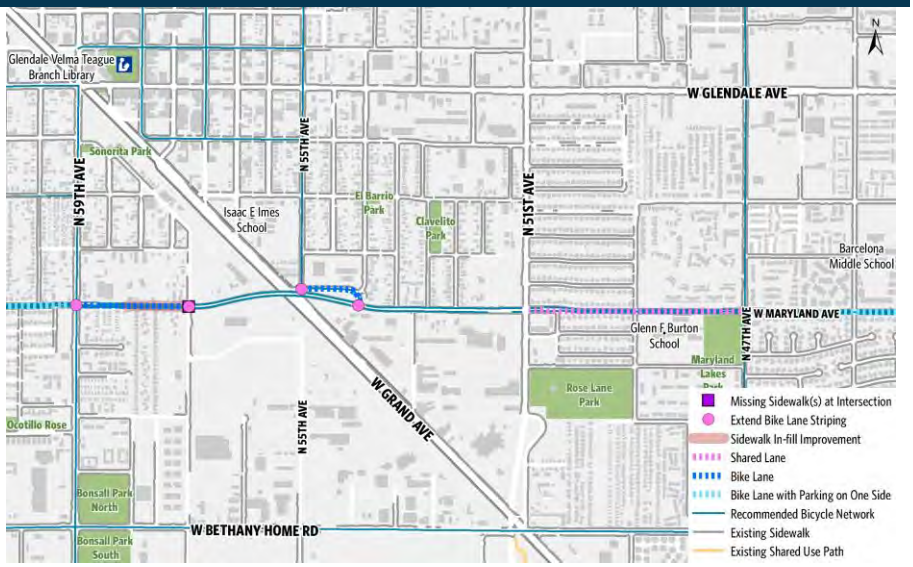
Maryland Ave: 59th Ave to 43th Ave

Maryland Ave is a two-lane, low-speed roadway that provides regional connectivity to Phoenix. Bike lanes currently are present between 57th Ave and 51st Ave. Ample right-of-way is available.

Priority Score: 62.5

Projects:

- Extend bike lane striping at intersections
- Add bike facilities throughout corridor
- Fill-in sidewalk gaps
- Connect corridor to 55th Ave



Project Benefits:

- Low-cost regional bicycle connection over Grand Avenue
- Improves walking/biking conditions in socioeconomically disadvantaged area
- Creates connectivity to Downtown Glendale

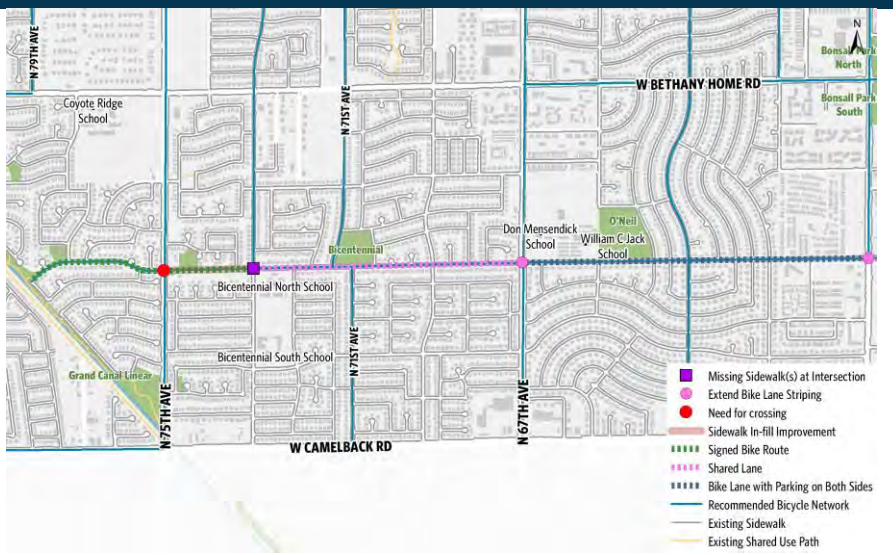
Missouri Ave: Grand Canal Linear Park to 59th Ave

Missouri Ave is a two-lane, low-speed roadway that provides direct access to numerous schools and parks. Currently the corridor has a striped shoulder with on-street parking.

Priority Score: 59

Projects:

- Extend bike lane striping at intersections
- Add bike facilities throughout corridor
- Fill-in sidewalk gaps and add sidewalks at intersections



Project Benefits:

- Provides a low-stress, regional connection from Grand Canal Linear Park to homes
- Provides a Safe Route to School for students
- Project Assessment complete for portion of corridor

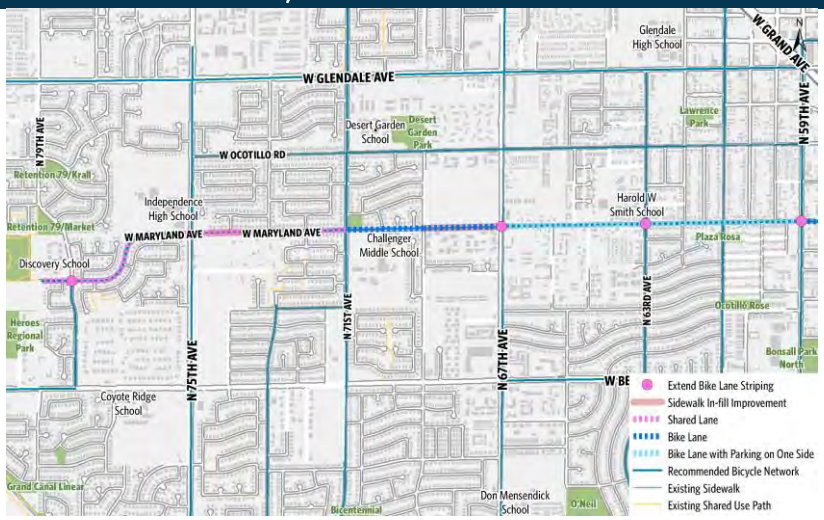
Local Connection to Discovery Park, Glendale Heros Park, and Schools

Maryland Ave is a two-lane, low-speed roadway that provides connectivity to numerous schools, homes, and recreation opportunities. Maryland Ave does not currently connect from 75th Ave to 74th Ave

Priority Score: 53.5

Projects:

- Add crosswalk at 75th Ave/Maryland Ave
- Connect 75th Ave to 74th Ave
- Extend bike lane striping at intersections
- Bike facilities throughout corridor
- Fill-in sidewalk gap



Project Benefits:

- Creates local and regional bicycle route
- Provides a Safe Route to School for students
- Improves walking/biking conditions in socioeconomically disadvantaged area

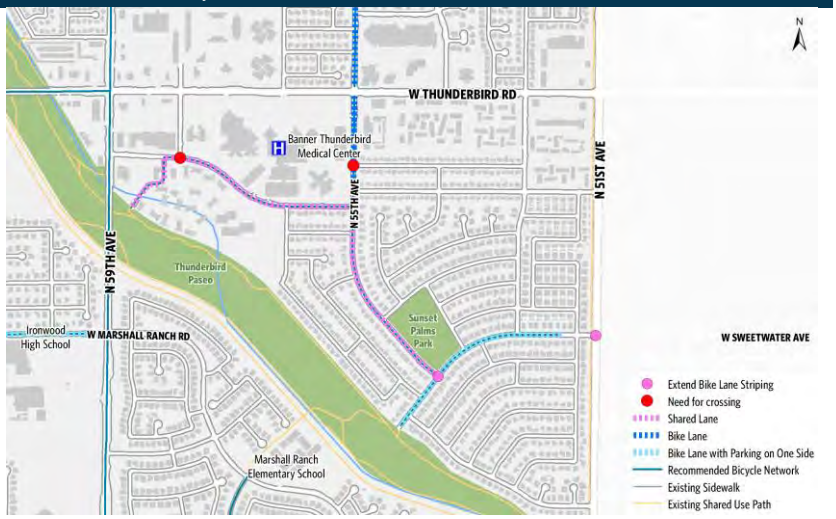
Area Improvements from Arizona Canal Trail to 51st Ave/Sweetwater Ave

Bike lanes on Sweetwater Ave east of 51st Ave provides regional connections to Arizona State University – West Campus and I-17 in Phoenix.

Priority Score: 57.5

Projects:

- Convert Sweetwater Ave to bike lanes with parking on one side
- Add shared lanes on 55th Ave and Eugie Ave
- Evaluate the need for crosswalks at Banner Thunderbird Medical Center entrances



Project Benefits:

- Low-cost solution that creates local and regional bicycle route
- Provides low-stress connection to regional medical hospital
- Improves recreation access

Union Hills Dr/Skunk Creek Park Pedestrian Improvements

Small gaps in the pedestrian network create barriers to active transportation along Union Hills Drive, the Glendale Foothills Library, and Skunk Creek Park.

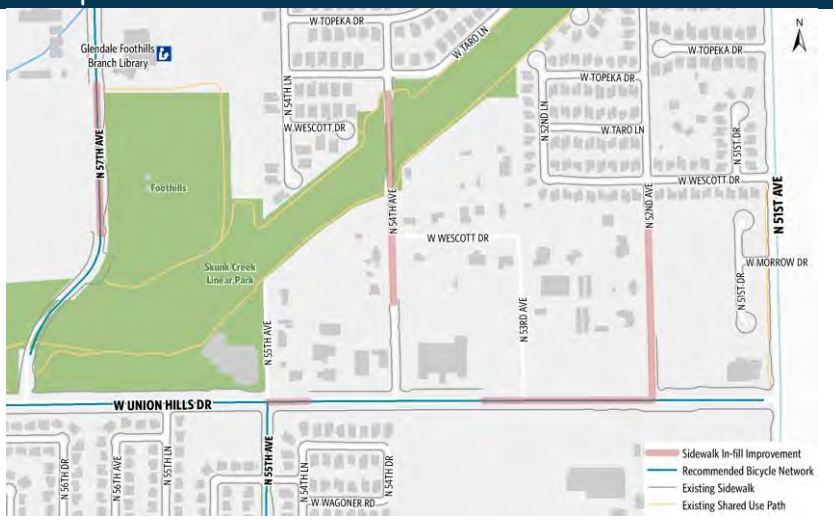
Priority Score: 45.5 to 71.5

Projects:

- Fill-in sidewalks along Union Hills Dr, on 57th Ave, 54th Ave, and 52nd Ave

Project Benefits:

- Improves pedestrian access to the Glendale Foothills Library, Skunk Creek Park, and the Foothills Recreation Center
- Provides a Safe Route to School for students to Deer Valley High School



Low-Cost Signage/Striping Improvements

Roadways that currently have a stripped on-street parking facility, but no direct business or residential driveway access could easily be converted into a bike lane with proper signage. Additionally, low stressed facilities recommended to be a bike route can easily be signed to provide bicyclist's with directional guidance of Glendale's bicycle network.

Projects Include:

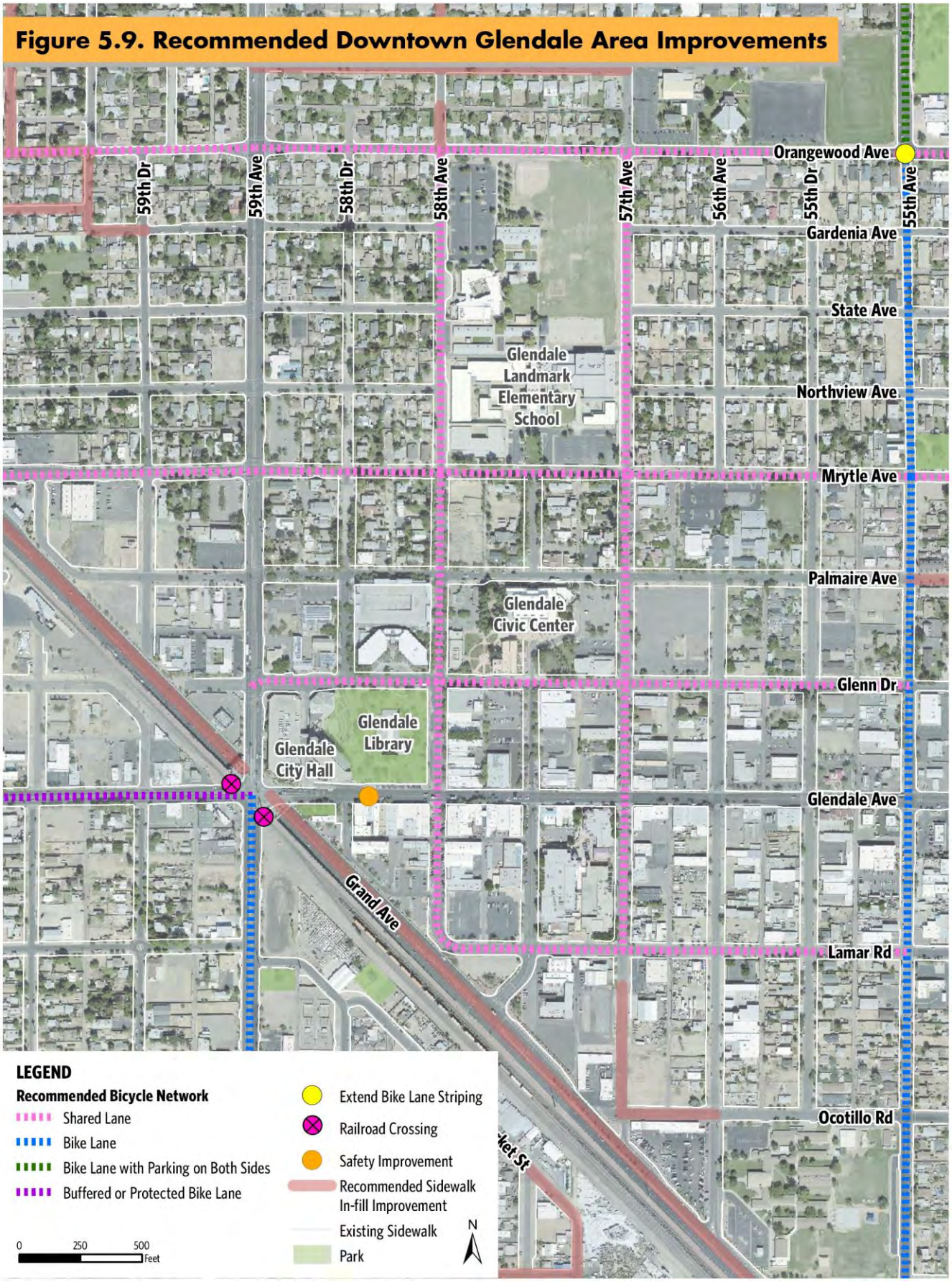
- Greenway Rd: Grandview Rd to 67th Ave
- 55th Ave: Cholla St to Cactus Ave
- 63rd Ave: Paradise Ln to Greenbrier Dr
- Grovers Ave: West of 55th Ave
- Country Gables Dr and 63rd Ave: 59th Ave to Greenway Rd

Recommended Downtown Glendale Area Improvements

As the urban core of the City, Downtown Glendale features shopping, public facilities including City Hall and the Glendale Civic Center, and parks, and has relatively low speed limits and wide sidewalks. Active transportation improvements such as providing enhanced pedestrian crossings, creating new bicycle facility connections, and placemaking can help improve the overall livability and community character of the downtown area. Figure 5.9 illustrates recommended improvements in Downtown Glendale, which includes:

Create Low-Stress Downtown Bicycle Route	Designate selected local, low-stress roadways as Shared Lanes to create a downtown bicycle route that connects Downtown Glendale to existing and recommended regional bicycle facilities and parks. Recommended Downtown Bicycle Route includes: • Lamar Rd: 58th Ave to 55th Ave • Glenn Dr: 59th Ave to 55th Ave • Myrtle Ave: Grand Ave to 48th Dr • 58th Ave: Lamar Rd to Orangewood Ave • 57th Ave: Lamar Rd to Orangewood Ave Route can be utilized for future Glendale bicycle events to further encourage and educate residents. To further improve a bicyclist's level of comfort, consider including bicycle detection at arterial intersections.
Add Bike Lanes on 55th Ave from Maryland Ave to Orangewood Ave	55th Avenue from Maryland Ave to Orangewood Ave is a low-stress, two lane roadway with sporadic on-street parallel parking. Adding bike lanes to the corridor provides an opportunity for cyclists to travel to Downtown Glendale from existing and planned bicycle facilities on Maryland Ave and Orangewood Ave.
Add Bike Lanes on 59th Ave from Maryland Ave to Glendale Ave	Conduct an engineering assessment to determine the feasibility of incorporating bike lanes onto 59th Ave to help bicyclists reach Downtown Glendale from east of Grand Avenue. To accommodate on-street bike lanes, the existing center turn lane and travel lane widths would need to be narrowed.
Upgrade City Hall Pedestrian Crossing on Glendale Ave	Upgrade the existing, unsignalized midblock pedestrian crossing on Glendale Ave directly east of 58th Ave. Recommended optional upgrades include: • **Option 1: Eliminate Crossing -** directing traffic to cross at 58th Ave signal • **Option 2: Upgrade Crossing -** include Rectangular Rapid Flash Beacon (RRFB) to increase driver yielding behavior, perform maintenance on brick crossing, and repaint crosswalk to include reflective markings.
Intersection Crossing Maintenance	Perform maintenance on existing pedestrian crossings at Glendale Ave, Glenn Dr, Palmaire Ave, 58th Dr, 57th Dr, 57th Ave, including: • Restripe textured crosswalks with reflective lines • Fix bricks and pavers that may have popped out of place creating a tripping hazard • Install ADA compliant curb ramps
Install Pedestrian and Bicycle Street Amenities	Create a better sense of place, improve walkability, and enhance bicycle circulation by installing: • Bicycle and pedestrian wayfinding system that includes signs and pavement markings • Bicycle parking racks or lockers, particularly at transit stops and at the pedestrian mall • Upgraded lighting at pedestrian crossing (additional engineering analysis is needed)

Figure 5.9. Recommended Downtown Glendale Area Improvements

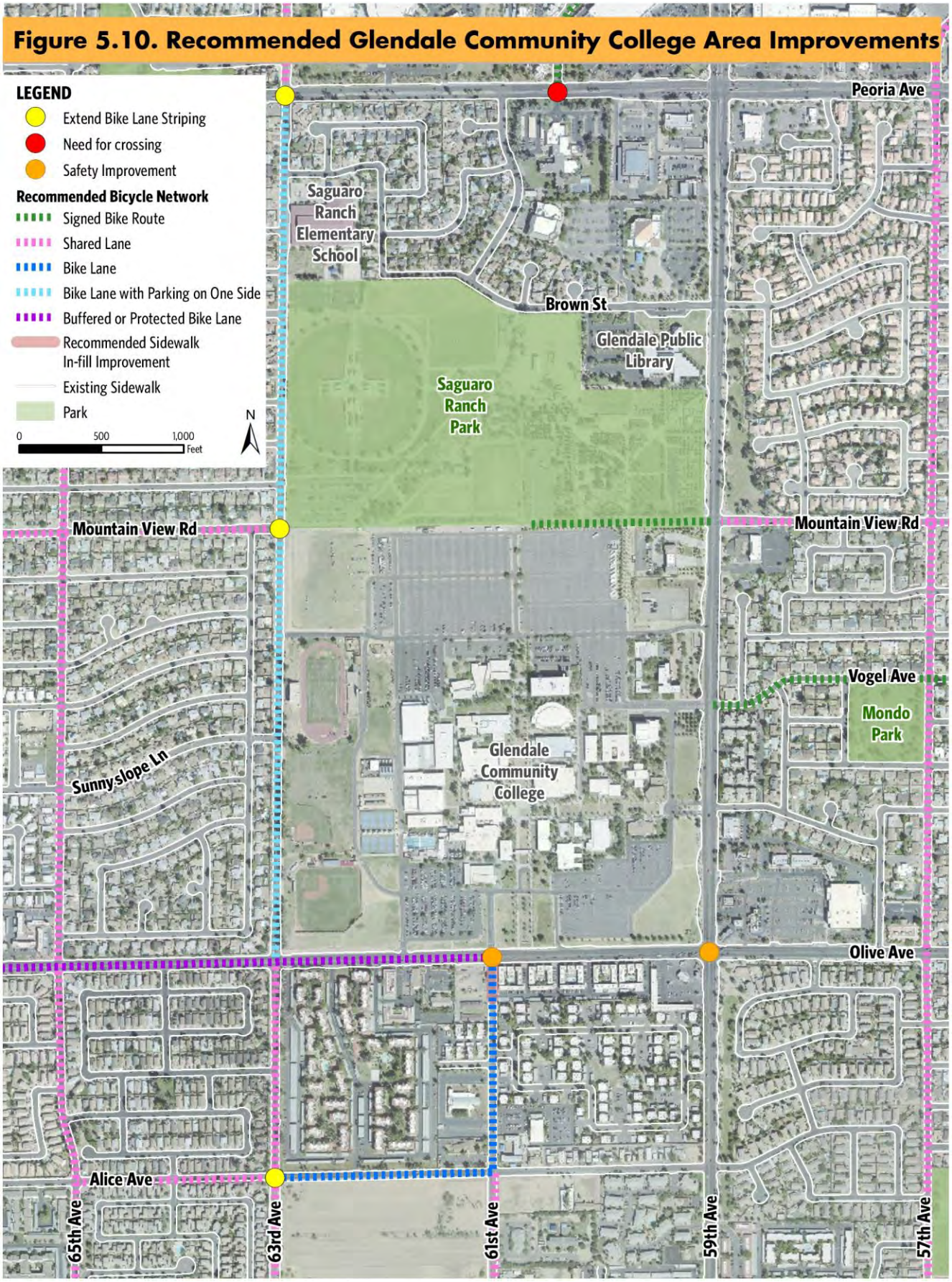


Glendale Community College Area Improvements

Glendale Community College, located at the northwest corner of Olive Ave and 59th Ave, is one of the largest public colleges in the Phoenix metropolitan area. With over 19,000 students on 28 acres, Glendale Community College is one of Glendale's largest activity centers with numerous students and employees traveling via walking, biking, transit, and vehicle to the college daily. This high activity zone has historically experienced significant pedestrian and bicycle safety issues. Active transportation improvements such as improving safety conditions at entrances and creating new bicycle facility connections can help improve safety and accessibility for students and employees. Figure 5.10 illustrates recommended improvements surrounding Glendale Community College, which include:

Olive Avenue Safety Improvements	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues along the corridor and specifically at intersections. The RSA should evaluate: • Sight distance and lighting issues at intersections that may attribute to left and right hook crashes. • Traffic and pedestrian crossing signal timing to accommodate pedestrians crossing and to better coordinate traffic flow • Need for right-turn-on-red restrictions at signalized intersections • Incorporating curb cuts on the southside of the corridor from 57th Ave to 55th Dr • Need for improved high visibility crosswalks • Signage applications to warn motorists of high pedestrian activity crossing areas • Application and feasibility of constructing a pedestrian bridge through a public-private partnership at 61st Ave • Targeted enforcement measures
Bike Facilities on Olive Ave from 67th Ave to 59th Ave	Conduct an engineering assessment to determine the feasibility of reallocating space along the corridor to include an on-street bicycle facility to address first/last-mile transit access at bus stops on Olive Ave, 67th Ave, and 59th Ave, which have some of the highest daily ridership in the City. To offer a high-quality experience for users of all ages and abilities, a bike lane is recommended to be added. If available right-of-way exists it is preferred to upgrade the facility to a protected or buffered bike lane.
Convert 63rd Ave to Including On-Street Bike Lanes with Parking on One Side	Reallocate space to include on-street bike lanes by eliminating on-street parking on the east side of 63rd Ave from Olive Ave to Mountain View Rd. Recommended cross section includes a 4 FT southbound bike lane; two 11 FT travel lanes; 4 FT northbound travel lane; 3 FT buffer; and 7 FT on-street parking on west side.
Create Low-Stress Bicycle Connection on Mountain View Rd	Create regional access to Glendale Community College and Sahuaro Ranch Park by converting Mountain View Rd to: • 67th Ave to 63rd Ave – Shared lanes • Sahuaro Ranch Park Shared Use Path to 59th Ave – Signed bike route • 59th Ave to 43rd Ave – Shared lanes
Provide Regional Low Stress Connections	Create regional access to Glendale Community College and Sahuaro Ranch Park via a network of low-stress bicycle facilities, including: • Bike Lanes on 61st Ave (Alice Ave to Olive Ave) and Alice Ave (63rd Ave to 61st Ave) • Shared Lanes on 65th Ave, 61st Ave (south of Alice Ave), and on 57th Ave)

Figure 5.10. Recommended Glendale Community College Area Improvements



Safety Improvements

Safety concerns often deter people from choosing to walk or bike. With increased emphasis on the benefits of active transportation, proving safe environments is increasingly important. Overcoming these barriers is possible – with education and strategic improvements to the built environment, more people will view walking and biking as safe and convenient options. The following presents recommended targeted safety infrastructure improvements to address corridors and intersections that have historically had the highest rates of pedestrian- and bicycle-related crashes in the City.

Bethany Home Rd: Grand Ave to 43rd Ave	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should evaluate: • Application and feasibility of installing a HAWK crossing with offset pedestrian median crossing near 45th Ave to reduce jaywalking • Sight distance and lighting issues that may attribute to left and right hook crashes • Consolidating driveways from 45th Ave to 43rd Ave • Bicycle facility accommodations • Safety measures within the Carol Peck Elementary school zone • Signage applications to warn motorists of high pedestrian activity crossing areas • Targeted enforcement measures and education campaigns
Bethany Home Rd: 67th Ave to 59th Ave	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should include a 1000 FT buffer study area to include the high number of pedestrian and bicycle crashes that have occurred on intersecting streets. The RSA should evaluate: • Evaluation of pre- and post- HAWK crossing safety conditions • Traffic and pedestrian crossing signal timing to accommodate pedestrian crossings • Need for right-turn-on-red restrictions at signalized intersections • Sight distance issues that may attribute to left and right hook crashes at intersections • Lighting conditions to determine need for additional lighting at intersections • Consider installing high visibility crosswalks • Consolidating driveways on 67th Ave north of Bethany Home Rd and on Bethany Home Rd from 61st Ave to 59th Ave • Bicycle facility accommodations • Signage applications to warn motorists of high pedestrian activity crossing areas • Adjustments to speed limits on Bethany Home Rd • Targeted enforcement measures and education campaigns
Glendale Ave: 51st Ave to 43rd Ave	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should evaluate: • Application and feasibility of installing a HAWK crossing with pedestrian refuge island near 45th Ave to reduce jaywalking • Sight distance and lighting issues that may attribute to left and right hook crashes. • Need for right-turn-on-red restrictions at signalized intersections • Sight distance and lighting issues that may attribute to left and right hook crashes • Consider installing high visibility crosswalks • Consolidating driveways from 47th Ave to 43rd Ave • Bicycle facility accommodations • Signage applications to warn motorists of high pedestrian activity crossing areas • Adjustments to speed limits on Glendale Ave • Targeted enforcement measures and education campaigns

Glendale Ave/67th Ave Intersection	Implement recommendations provided by the Glendale Ave/67th Ave Intersection Roadway Safety Assessment. Additional improvements to consider include: • Need for right-turn-on-red restrictions at signalized intersections • Application and feasibility of installing a HAWK crossing with pedestrian refuge island north of the QuikTrip entrance • Signage applications to warn motorists of high pedestrian activity crossing areas • Targeted enforcement measures and education campaigns
Northern Ave/59th Ave Intersection	Implement recommendations provided by the Northern Ave/59th Ave Intersection Roadway Safety Assessment. Additional improvements to consider include: • Realignment of intersection or addition of curb extensions to address sight distance issues that may attribute to left and right hook crashes • Need for right-turn-on-red restrictions at signalized intersections • Consider installing high visibility crosswalks • Consolidating driveways on Northern Ave • Signage applications to warn motorists of high pedestrian activity crossing areas • Targeted enforcement measures and education campaigns
Bell Rd/59th Ave Intersection	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should evaluate: • Application and feasibility of installing pedestrian refuge island on all legs • Sight distance and lighting issues that may attribute to left and right hook crashes • Traffic and pedestrian crossing signal timing to accommodate crossing areas • Consider installing high visibility crosswalks • Targeted enforcement measures and education campaigns
Camelback Rd/ 67th Ave Intersection	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should evaluate: • Realignment of intersection or addition of curb extensions to address sight distance issues or safety concerns that may attribute to left and right hook crashes • Consider installing high visibility crosswalks • Bicycle facility accommodations • Lighting conditions to determine need for additional pedestrian lighting • Signage applications to warn motorists of high pedestrian activity crossing areas • Targeted enforcement measures and education campaigns
Maryland Ave/67th Ave Intersection	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should evaluate: • Sight distance and lighting issues that may attribute to left and right hook crashes • Traffic and pedestrian crossing signal timing to accommodate pedestrian crossings • Consider installing high visibility crosswalks • Bicycle facility accommodations Targeted enforcement measures and education campaigns
Peoria Ave/67th Ave Intersection	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should evaluate: • Sight distance and lighting issues that may attribute to left and right hook crashes • Traffic and pedestrian crossing signal timing to accommodate pedestrian crossings • Installation of pedestrian refuge island • Consider installing high visibility crosswalks • Bicycle facility accommodations Targeted enforcement measures and education campaigns
Thunderbird Rd: 59th Ave to 51st Ave	Conduct a Roadway Safety Assessment (RSA) to develop safety related improvement strategies to address pedestrian and bicycle safety issues. The RSA should evaluate: • Application and feasibility of installing a HAWK crossing with pedestrian refuge island west of 52nd Ave to reduce jaywalking • Traffic and pedestrian crossing signal timing to accommodate pedestrian crossings • Sight distance and lighting issues at intersections that may attribute to left and right hook crashes. • Sight distance and lighting issues that may attribute to left and right hook crashes • Consider installing high visibility crosswalks • Bicycle facility accommodations • Signage applications to warn motorists of high pedestrian activity crossing areas Targeted enforcement measures and education campaigns

Canal and Trail Improvements and Maintenance

In addition to providing new and enhanced facilities, it is imperative that the City maintain its existing canal and trail facilities that provide vital recreational and regional active transportation links. Through numerous stakeholder and community outreach events, a variety of safety and maintenance concerns were identified. Recommended canal and trail maintenance and safety improvements include:

- Ongoing replacement of broken lights, trimming of overgrown shrubs and vegetation, and regular pavement maintenance to keep the trails in good repair for biking, walking and other trail activities. Specific locations in need of maintenance/repair are:
    - Skunk Creek Trail/59th Ave – replace broken lights
    - Sunset Vita Park – overgrown trees limit sight distance
    - Arizona Canal Trail/75th Ave – overgrown shrubs limit sight distance
    - Skunk Creek Trail/Union Hills Dr – overgrown shrubs limit sight distance
- Address safety concerns that may create a barrier for persons to use a trail, including:
    - Skunk Creek Trail near Foothills Branch Library – steep curve with no lighting near library causes safety concerns
    - Skunk Creek Trail/69th Ave – sharp, steep curve causes safety concerns
    - Arizona Canal Trail/59th Ave – no pedestrian lighting
    - Arizona Canal Trail/51st Ave – underpass perceived as unsafe

Equestrian Accommodations

Glendale has had a long history and culture as an equestrian-friendly community with numerous horse properties and equestrian trails. Keeping Glendale horse friendly is a priority of many residents. To maintain and improve safety and connectivity for horseback riders in Glendale, the City should continue to invest in providing equine friendly roads, trails, and signals. Recommended equestrian accommodations include:

- At equestrian trail crossings, standard equestrian crossing signs at all at-grade road crossings should be installed to alert motorists.
- At arterial crossings along the Bridle Path and at key canal crossings, equestrian waiting areas and two push-button signal actuators that allow both pedestrians and equestrian riders to be detected should be considered. Most seated riders can operate a push button that is between 5 FT and 6 FT. Equestrian waiting areas allow horses to stand back from traffic until it is safe to cross a road. Consider expanding the width of the trail surface before it meets the road, forming a rectangular or fan-shaped waiting area.

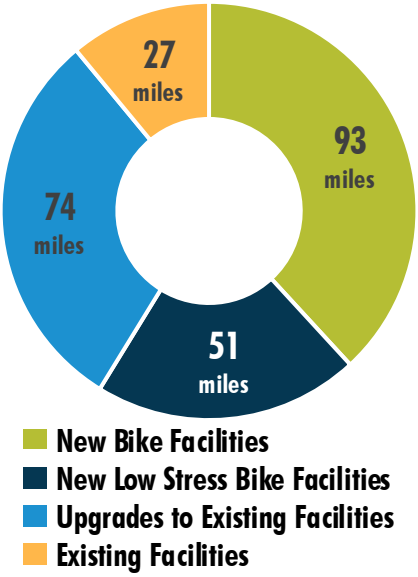
Provide equestrian staging areas at key trailheads to allow equestrians a place to park and access regional routes. Potential new equestrian staging areas to evaluate include: 67th Ave at the Arizona Canal Trail; Greenway Rd and Acoma Dr; 51st Ave south of the Arizona Canal; and off Skunk Creek Trail.

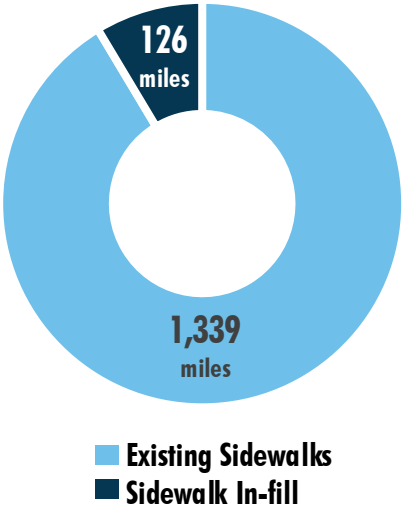
The Future Active Transportation Network

The Glendale Active Transportation Plan is an ambitious proposal for improving walking and biking in Glendale. It recommends an additional 144 centerline miles of new bicycle facilities, 74 centerline miles of improvements to existing bicycle facilities;126 miles of sidewalks, and additional improvements to make streets and their intersections safer for everyone. Should all needs be addressed, Glendale would increase their bicycle network by 134 percent and pedestrian network by nearly 10 percent.

While the ATP focuses on the identification of network gaps, Glendale has additional active transportation needs, including:

- **Arterials:** Pedestrian and bicycle facilities along arterials routes may be needed to provide the most direct and efficient route to services and destinations, as well as providing regional connections between communities.
- **New Developments:** When new developments arise, logical and accessible pedestrian and bicycle facility connections to existing or planned active transportation facilities are needed.
- **New Roadways:** Pedestrian and bicycle facilities may be needed on all new road construction and reconstruction projects to further expand the region's active transportation network.
- **Maintenance:** Ongoing facility maintenance is needed to preserve Glendale's active transportation investments.





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Figure 5.11. Future Active Transportation Network - Western Planning Area

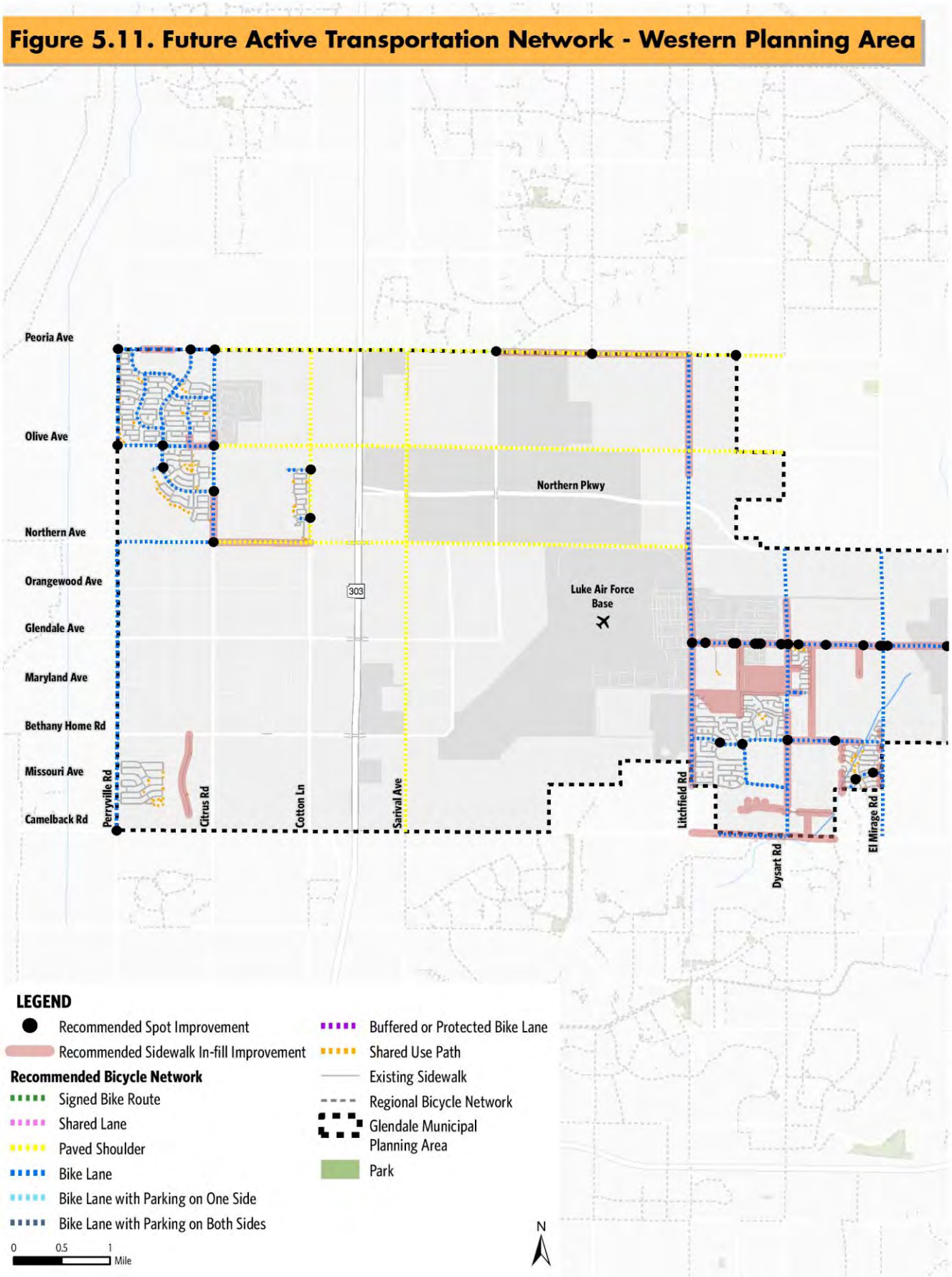


Figure 5.11. Future Active Transportation Network - Central Core

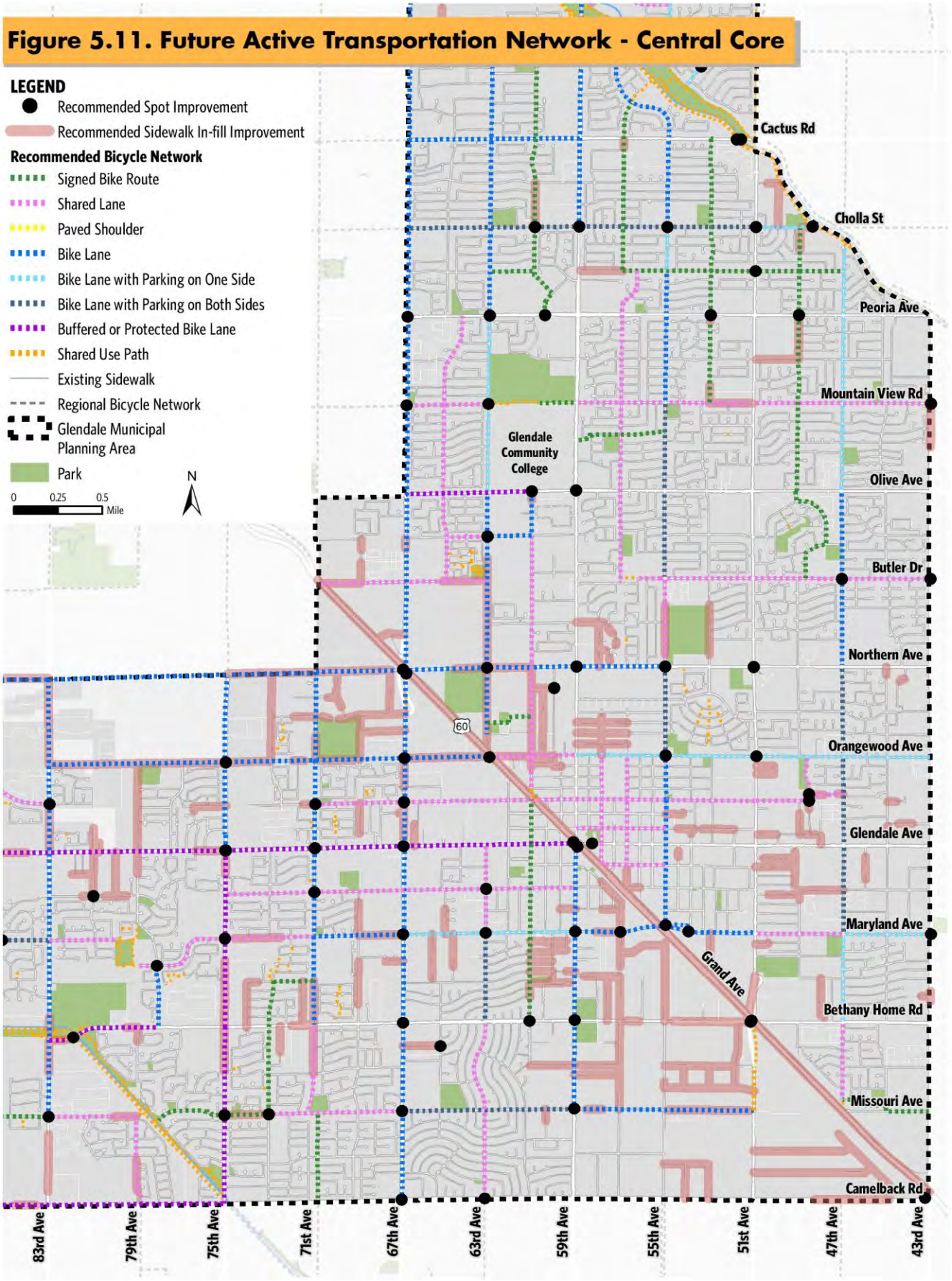


Figure 5.11. Future Active Transportation Network - Loop 101 Area

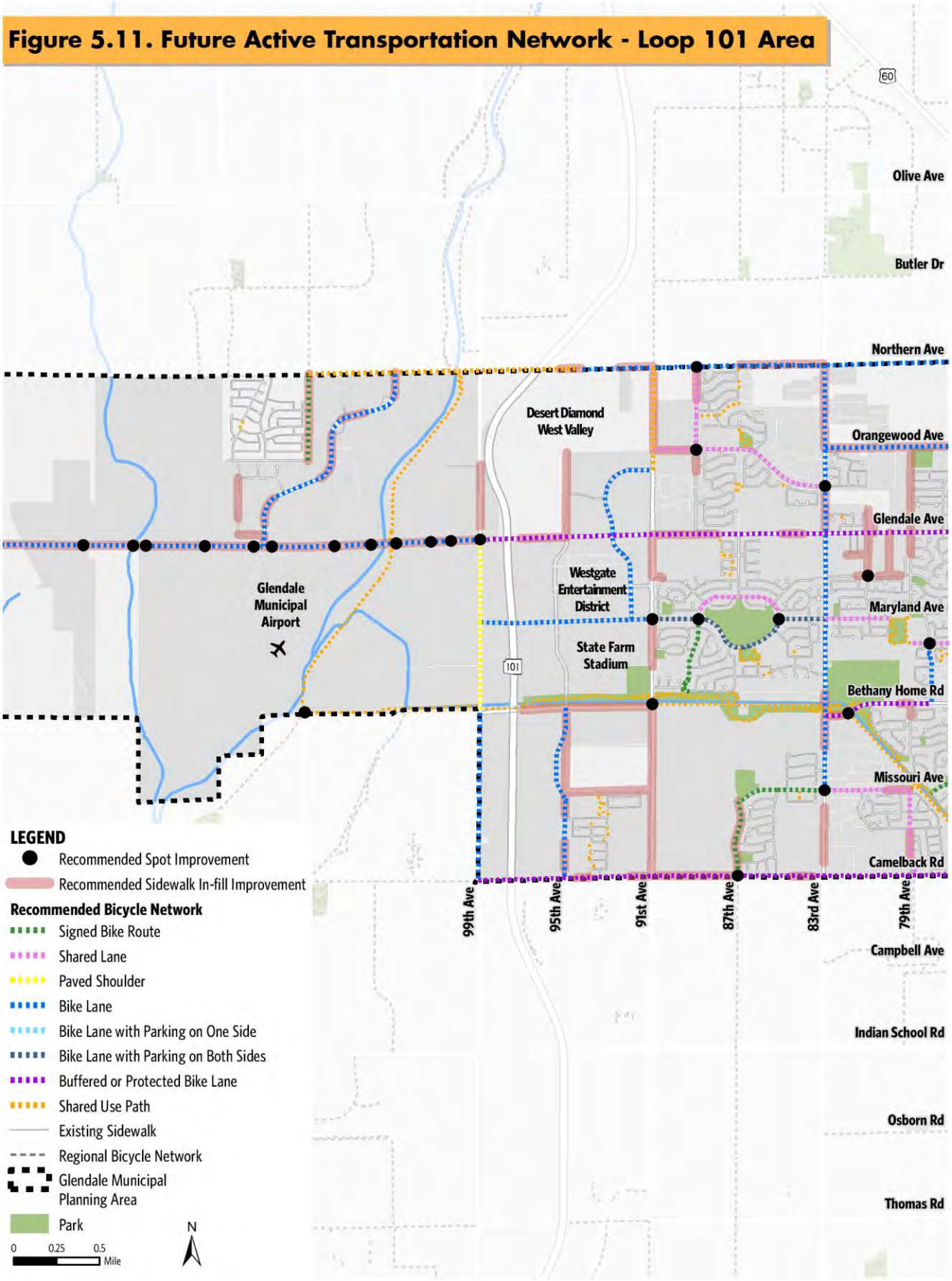
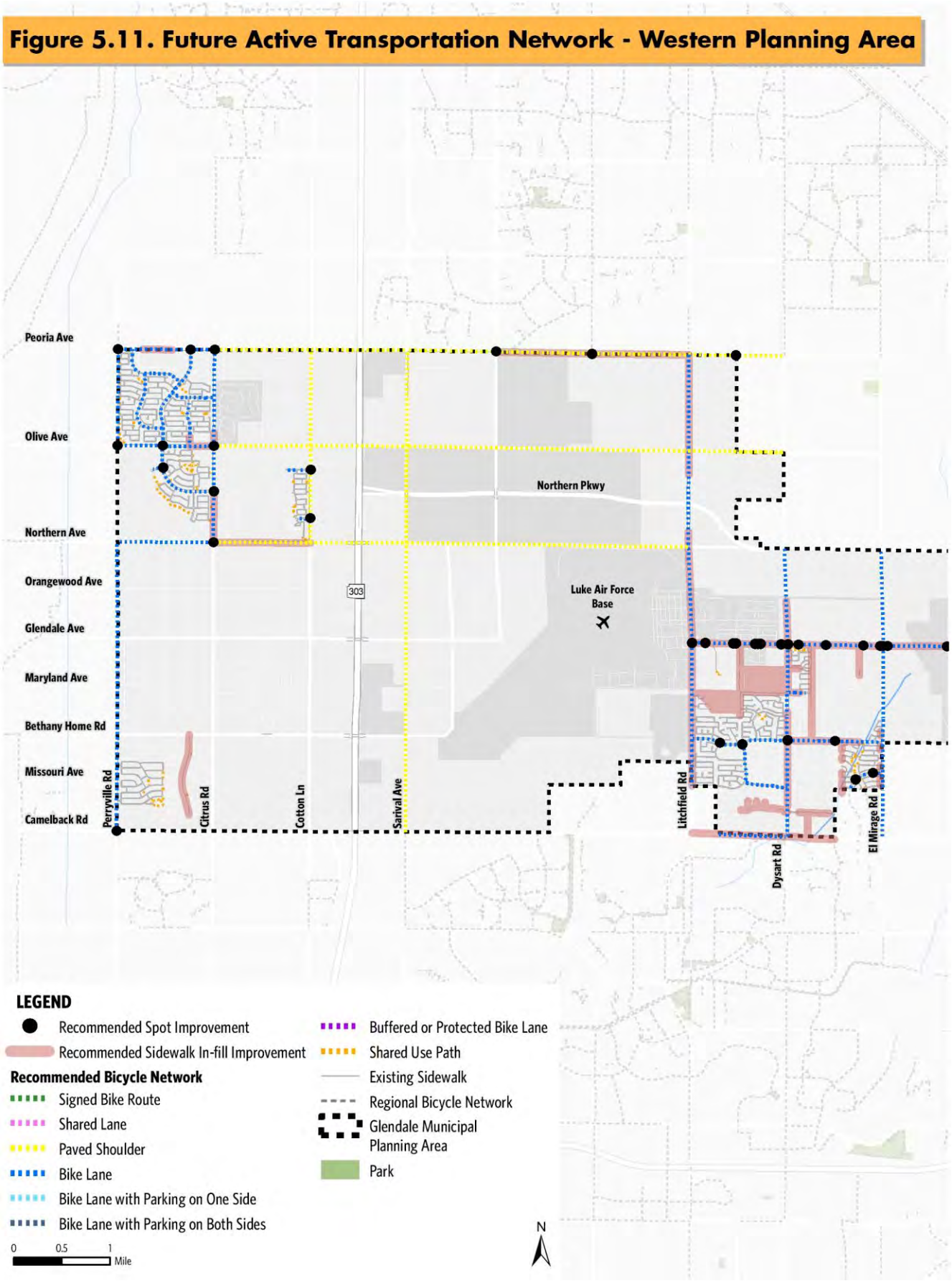


Figure 5.11. Future Active Transportation Network - Western Planning Area



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6. NEXT STEPS

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Active Transportation Recommendations

Improving conditions for walking and biking is an important priority for Glendale. Implementation of the ATP will require leadership and dedication for the investment and development of active transportation facilities. To make the ATP's vision a reality, Glendale should focus on the following overarching active transportation recommendations.



CREATE A CONNECTED, COMPLETE NETWORK TO PROVIDE MORE OPTIONS TO REACH DESTINATIONS

The transportation network should provide safe, connected infrastructure including roads, sidewalks, bicycle facilities, parking, trails and paths, and bus stops throughout the City. Barriers that inhibit pedestrian and bicycle travel should be addressed to provide more options to reach a destination.



INTERNALLY CHAMPION ACTIVE TRANSPORTATION GOALS

Glendale should dedicate a full-time staff member to champion the goals of the ATP and lead active transportation planning efforts for the City. The staff member would be reasonable for implementing improvements identified in the ATP, seeking funding sources, attending committee meetings, and coordinating with City department to incorporate ATP recommendations.



IMPROVE ACTIVE TRANSPORTATION SAFETY THROUGH EDUCATION, AWARENESS, AND ENFORCEMENT

Educate all road users about their rights and responsibilities regarding traffic laws. Ensure that all road users understand the proper uses for bicycle and pedestrian infrastructure. Work with the police, schools, and community partners to increase awareness of traffic safety through educational outreach and media campaigns.



WORK COLLABORATIVELY TO IMPLEMENT PRIORITIES

Glendale should seek opportunities to work with other jurisdictions and public, private, and non-profit sectors to fund, plan, build, and implement this plan. Glendale should also support regional active transportation efforts through data sharing, attending committee meetings, and providing technical guidance as needed.



ENCOURAGE EVERYONE TO WALK AND BIKE

Make active transportation more attractive and accessible to a variety of users - commuters, residents, recreationalists, equestrians, visitors, and people of all ages and abilities. Marketing bicycle and pedestrian attractions, infrastructure, and resources generates awareness and interest among potential users. Increased usage of active transportation modes supports economic development, healthy lifestyle choices, and community-building.

How to Implement This Plan

The completion of the ATP is the first of many necessary steps for Glendale to achieve its active transportation goals. Implementation of this plan is a long-term effort that requires dedication and commitment by Glendale staff. As shown in Figure 6.1, to implement the ATP, Glendale should follow a five-step process that includes:

- **Incorporate the ATP**: The ATP represents the ideas, issues, and needs of hundreds of public participants, as well as a shared vision for active transportation in Glendale.
- **Determine Feasibility**: Priority needs should go through a comprehensive assessment to determine what type of facility is best suited for the area, if the project is economically viable, and to conduct a fatal flaw review of environmental, right-of-way, utility, and other issues that may delay implementation.
- **Recommend Programming**: High priority, feasible improvements should be programmed, and Glendale staff should begin seeking partnerships and funding opportunities for long-term improvements. When applicable, pedestrian and bicycle improvements should be incorporated into regularly scheduled maintenance activities. Project bundling allows Glendale to address multiple gaps/issues at the same time by efficiently leveraging public funds, as well as improving funding chances.
- **Develop Annual Project List**: Projects should be reassessed and evaluated every year to create an annual project list that identifies priority active transportation investments.
- **Track and Evaluate**: Annually evaluate and report on the progress of the ATP. As the region grows and technologies and designs change, recommendations in the ATP should be reevaluated at least every five years.

Figure 6.1. Next Steps



How to Determine Feasibility

A range of factors can influence a pedestrian and bicyclists' comfort and safety. Selecting the appropriate pedestrian and bicycle facility for a roadway must balance traffic conditions, land use context, maintenance cost, and implementation cost.

Design Resources

A number of federal and state resources are available for planning and design. In addition to Glendale's roadway design standards, design resources include:

- AASHTO Guide for the Development of Bicycle Facilities
- AASHTO Guide for the Planning, Design, and Operation of Pedestrian Facilities
- FHWA Achieving Multimodal Networks
- FHWA Separated Bike Lane Planning and Design Guide
- FHWA Incorporating On-Road Bicycle Networks into Resurfacing Projects
- NACTO Designing for All Ages and Abilities
- NACTO Urban Bikeway Design Guide
- NACTO Urban Street Design Guide
- MAG Active Transportation Plan
- MAG Bicycle and Pedestrian Pathway/Railroad Crossing Recommendations
- MAG Complete Streets Guide

Roadway Modification Strategies

Many of Glendale's roadways provide adequate right-of-way to accommodate sidewalks, bike lanes or paved shoulders; however, additional right-of-way or modifications may be needed to accommodate wider or more protected facilities. As mentioned in Chapter 3, examples of roadway modification strategies to incorporate active transportation facilities include:

- Restriping and lane narrowing within the existing pavement width. Implementation of these improvements are best achieved during roadway resurfacing or restriping projects.
- Reallocating space for cross sections' elements within the existing right-of-way to realize higher quality active transportation facilities while considering the needs of motorized users.
- Locating sidewalks behind a drainage swale to avoid installing curb and gutter when there is insufficient room between the roadway and the swale.
- Widening roadways in areas where other modification strategies are not feasible. Additional right-of-way may be needed to accommodate pedestrian and bicycle facilities. Pavement widths at intersections and midblock pedestrian crossings, however, should be minimized to reduce pedestrian crossing distances.

Maintenance Considerations

Maintaining the active transportation network once it has been implemented preserves Glendale's investment and helps support the transportation needs of residents. Bicyclists and pedestrians are vulnerable to pavement/sidewalk irregularities such as cracks, potholes, broken glass, sand, etc. Unmaintained landscaping causes safety issues by obstructing bicycle lanes and sidewalks and blocking visibility. Major storms and motor vehicle crashes can leave debris, presenting hazards to pedestrians and bicyclists, which must be picked up as soon as possible.

Maintenance needs are typically identified through one of three sources: the public reporting a problem, routine inspections, or special inspections after a storm, crash, or construction project. Maintenance activities can generally be categorized into one of two types:

- Routine maintenance: performed annually or more frequently
- Major or capital maintenance: involves more intensive activities such as pavement seal coating, pavement overlays, or pavement reconstruction, or structural rehabilitation or replacement



Chasing Pavement

Integrating recommended improvements with Glendale's Pavement Management Program is a cost-effective strategy for installing on-street bicycle facilities during routine roadway maintenance and resurfacing projects. During roadway restriping and resurfacing, the existing pavement could be striped or additional pavement could be added to accommodate bike lanes and paved shoulders.

Glendale should monitor scheduled maintenance programs to ensure bicycle and pedestrian facility maintenance. Buffered sidewalks and shared use paths often require more frequent and different maintenance practices (depending on the degree and type of physical separation). During the facility design selection phase of project development, maintenance needs and costs should be considered. Routine maintenance best practices include:

- Maintaining structure quality through spot repairs, regular overlays, and long-term rehabilitation
- Sweeping and removal of garbage and debris
- Trimming overgrown vegetation
- Restriping pavement markings as needed
- Repairing damage caused by monsoons, crashes, and other unforeseen events
- Repairing and replacing signage

Living Laboratory

In order to achieve the goals and vision established in the ATP, Glendale must be willing to be creative and innovative in the design, funding, and implementation of sidewalks and bikeways. Tactical urbanism, also known as living laboratory, has been embraced as a low-cost alternative to implementing both temporary and permanent pedestrian and bicycle projects. This method of testing out longer-term infrastructure improvements sprung out of a series of citizen led efforts to "take action when confronted with the slow pace of change." It can take shape in many forms, ranging from smaller "guerilla interventions" to demonstration projects led by both community groups and cities. Examples of methods include pavement markings, pop-up bikeways, adding furniture and/or seating to create public space, and using planters as barriers for a protected bike lane.



Morgan Hill, CA conducted a two-day demonstration project to test the implementation of a buffered bike lane on Monterey Road through their downtown core.

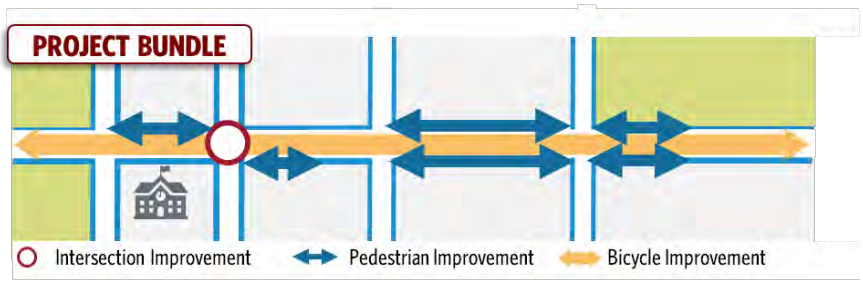


Bentonville, AR tested three projects throughout the city to determine the feasibility and evaluate the usage of two-way protected bike lanes at key locations. Cyclists and motorists were surveyed to understand issues, concerns, and to evaluate support for the project.

Glendale should embrace tactical urbanism as a strategy of implementing pedestrian and bicycle improvements. Tactical urbanism projects have low costs and have the potential to garner excitement around active transportation. Tactical urbanism projects also allow residents to envision an environment with improved active transportation facilities and to test out improvements. Several grant programs exist to fund living laboratories, including:

- Walton Family Foundation
- John S. and James L. Knight Foundation
- National Association of Realtors

Develop Project Bundles



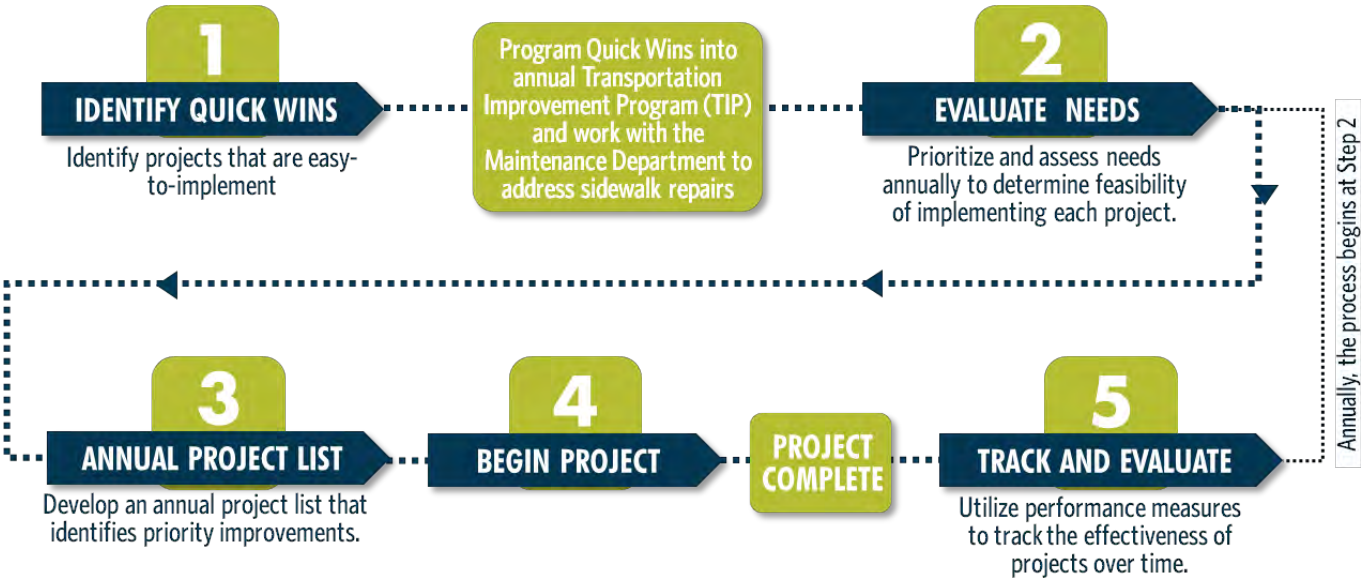
Project "bundling" is a concept that refers to grouping adjacent improvements into a single active transportation improvement project – for instance, combining intersection, sidewalk gap, and bicycle gap improvements along a section of roadway to provide a complete active transportation network along the corridor. Project bundling allows Glendale to address multiple gaps/issues at the same time by efficiently leveraging public funds, as well as improving funding chances. Appendix A provides detailed information on potential project bundles.

Developing the Annual Project List

Ultimately, the ATP presents a high-level, planning-scale evaluation of improvement needs in Glendale. All identified improvement needs presented in the ATP requires a comprehensive assessment process that includes:

- Constructability Audit – potential projects will need to undergo a thorough feasibility review of right-of-way, environmental, design, and cost constraints
- Coordination Review – identify potential stakeholder and/or private partnership opportunities

Once prioritized and evaluated, the list of projects will need to go through an annual review process to create an annual active transportation project list based on available funding. To communicate with the public and maintain a transparent project selection process, a status tracker to clearly convey the status of the projects is recommended. The flowchart below provides guidance on the annual project list development process.



How to Track and Evaluate Progress

Performance measures help to track the ATP's progress and effectiveness over time. Table 6.1 outlines a wide range of suggested performance measures to assess the success of the ATP, track changes in the built environment, identify trends in travel behaviors, and provide performance information to decision-makers. Tracking and reporting the progress of performance measures provides more transparency while building momentum and public support.

Table 6.1. Performance Measures

Performance Measure	Desired Trend
Total linear miles of walking and bicycling facilities	Increasing mileage of walking and bicycling facilities as a whole and broken down by sub areas
Percent of critical facilities and community destinations (i.e., parks, libraries, hospitals, etc.) that have active transportation facilities within 1/2 mile	Increasing percentage of active transportation facilities adjacent to critical facilities and community destinations
Percent of bus stops that provide active transportation connections	Increasing number of active transportation facilities that provide access to bus stops
Number pedestrian- and bicycle-related crashes that are classified as fatal or severe injury	Reduction in pedestrian- and bicycle-related crashes and serious and fatal injury crashes
Number of intersections and crossings that are treated with safety and accessibility improvements	Increasing number of intersections, crossings, driveways, etc. that provide pedestrian, bicycle, and ADA facilities
Percentage of households within 1/2 mile of a low stress, all ages and abilities bicycling facility and walking facility, or both	Increasing percentage of households within1/2 mile to a high comfort biking or walking facility
Miles of active transportation facilities within 1/2 mile of employment centers	Increasing number of active transportation facilities that provide access to employment centers
Percent of sidewalks and bikeways completed in areas of highest bicyclist and pedestrian demand	Increasing number of active transportation facilities in areas of high latent demand
Percent of sidewalks and bikeways completed in areas with high socioeconomic and health inequity	Increasing percentage of completed projects in areas with high socioeconomic and health inequity
Total dollars spent on active transportation capital and maintenance projects	Increasing annual programmed and funded active transportation projects
"Bicycle Friendly Community" and "Walk Friendly Community" designation from League of American Bicyclists	Higher designation status for "Bicycle Friendly Community" and "Walk Friendly Community"
Total dollars spent on active transportation capital and maintenance projects	Increasing annual programmed and funded active transportation projects

Bicycle and Pedestrian Counts

Currently, Glendale does not collect data to track how often pedestrian and bike facilities are used. Collecting data will enable the City to evaluate trends, such as increase/decrease in usage, peak travel periods, and high activity locations. Better data on pedestrian and cyclist travel can help to determine where investments are most needed. It also helps to quantify the benefits of walking and biking. Ultimately, better data will make active transportation projects more competitive for funding opportunities. Glendale should coordinate with MAG to include priority locations within Glendale to be included in future regional pedestrian and bicycle counts program. The City should establish a bike and pedestrian counts program, in combination with existing counts collection, to track and monitor the number of pedestrian and bicyclists that use facilities in Glendale. A number of counters are available that track bicyclist and pedestrian activity. Types of technology available to conduct counts are: wireless sensors, thermal imaging, app based counting software, etc.

League of American Bicyclists

Since 1995, the League of American Bicyclists have awarded "Bicycle Friendly Community" (BFC) and "Walk Friendly Community" designations to communities across the Nation. As of 2018, more than 850 communities have applied, and 431 have been awarded a Bronze, Silver, Gold or Platinum designation. Glendale submitted an application for the BFC program in 2012 and received an Honorable Mention for having some of the key building blocks of creating a BFC in place. The completion of the ATP moves the City one step closer to being awarded a BFC. To be awarded with a Bronze-level BFC award, Glendale should take the following steps and measures:

- Adopt a Complete Streets policy and guidance to ensure that roadways accommodate all users, regardless of age, ability, or mode of transportation.
- Continue to expand the bicycle network to achieve 25 percent of all total roadway mileage having a bicycle facility, 20 percent of high-speed roadways with a bicycle facility, and increased access to public transportation.
- Address major pedestrian and bicycle safety concerns and work with law enforcement to promote safety and bicycling skills
- Hire a full-time Active Transportation Program Manager dedicate to work on active transportation issues and concerns.
- Incorporate bicycle facility design guidance and standards into the City's roadway design standards.
- Develop local ordinance to ensure that bicyclists are protected (i.e., parking in bike lanes, helmet usage, etc.).
- Continue to participate in Bike Month and Bike to Work events.
- Provide pedestrian and bicycle education opportunities to schools, children, adults, and the general public.
- The City has participated in Safe Routes to School programs in the past and should continue and enhance a consistent bicycle education program at more than 50% of local elementary, middle, and high schools.

Funding Ideas

Various funding sources are available to fund maintenance of existing sidewalks and bicycle facilities, construct new sidewalks and shared use paths, and to develop new on-street bicycle facilities. Most funding sources are competitive and require the preparation of applications. Some applications may be more competitive if Glendale collaborates with local and regional agencies to jointly prepare and team on the construction of improvements. Table 6.2 provides a list of funding sources that may be applicable for needs identified in the ATP.

Federal, State, and Local Funding

Federal funds from the Surface Transportation Block Grant Program (STBGP) are allocated to the state and distributed proportional to population, allowing funding to get to as many different types of communities as possible. Program funding for transportation alternatives (TA) is included within the STBGP. Federal transportation spending can vary and tends be dependent on economic factors. Glendale should be creative in obtaining different sources of funding in order to implement projects. Municipal funds may also be used to construct bicycle and pedestrian facilities.

Roadway Safety Assessments (RSAs)

Road Safety Assessments (RSAs) are an active approach to improving road safety. Annually, MAG sets aside $300,000 to complete RSAs for member agencies. Glendale should pursue obtaining MAG RSA funds to conducted studies at key corridors and intersections identified in Chapter 5. The RSAs can help produce designs that reduce the number and severity of crashes, identify key human factor considerations, and integrate multimodal roadway safety concerns.

Public Private Partnership (P3)

Public Private Partnerships (P3s) are contractual agreements that can leverage funds from both sectors for infrastructure projects and facilities. Where municipal budgets fall short, private revenue can fill the gaps. Agreements may include funding, design, construction, operation, and/or management with terms agreed upon by the two entities.

Developers

It is anticipated that as new residential and commercial developments arise, developers will build additional facilities that will expand and enhance the active transportation network including:

- Constructing pedestrian and bicycle facilities to their property and, as applicable, provide connections to nearest facilities.
- Ensuring pedestrian connectivity through the end of cul-de-sacs to shorten trip distances for walking and bicycling.
- Installing shaded and secure bicycle parking facilities at residential, office, school, commercial, and recreational developments.

Table 6.2: Funding Options

Funding Opportunity	Overview
Glendale General Fund	The General Fund includes all sources of revenue the City receives that are not designated for a specific purpose. The General Fund budget includes sub-funds that had previously been segregated. These funds were established to track revenues and expenditures related to specific events or activities.
Federal Transit Administration (FTA) 5310 and 5311 Grant Programs	FTA Section 5310 and 5311 can be utilized for improving pedestrian and bicycle access to public transportation facilities, such as building an accessible path to a bus stop.
FTA Livability Grant Programs	Provides financial assistance to States, municipalities, transit agencies, and other public bodies to improve public transportation. Can be used for bicycle and pedestrian support facilities, such as bicycle parking, bike racks on buses, pedestrian amenities, and educational materials.
Surface Transportation Block Grant (STBG) Program	Provides states with flexible funds for a variety of highway, road, bridge, and transit projects.
Transportation Alternatives Program (TAP) - Recreational Trails Program	These grants are for short-term campaigns that will increase or preserve investments in active transportation in communities where program choices are being made on how to spend federal, state, and local funding.
Transportation Infrastructure Finance and Innovation Act (TIFIA) Loans	The TIFIA program provides credit assistance for qualified projects of regional and national significance.
Achieving Transportation Accessibility Now (ATAN)	MAG, in partnership with Valley Metro, developed a short-term strategy to improve accessibility to bus stops. In January 2017 MAG's Regional Council approved the use of $2.5 million of transit funding to improve accessibility at transit stops in the MAG region.
Congestion Mitigation/Air Quality (CMAQ) Program	Program provides federal funds to projects and programs that help nonattainment and maintenance areas comply with air quality standards.
Community Development Block Grants	Funds local development activities in low to moderate-income communities, such as affordable housing, anti-poverty programs, and infrastructure development. Can be used to build sidewalks and recreational facilities.
Highway Safety Improvement Program (HSIP) Infrastructure	Helps communities achieve significant reductions in traffic fatalities and serious injuries on all public roads. Program funds safety projects that are consistent with the State's Strategic Highway Safety Plan. This program includes the Railroad-Highway Crossings and High-Risk Rural Roads programs.
MAG Design Assistance	The MAG Design Assistance Program was initiated in 1996 to encourage the development of pedestrian facilities according to the MAG Pedestrian Policies and Design Guidelines. The intent of the program has been to stimulate integration of facilities into the planning and design of all types of infrastructure and development.
Safe Routes to School (SRTS) Program	SRTS helps make walking and biking to school a safe and more appealing method of transportation for students. MAG allocates $400,000 per year to fund qualifying SRTS projects that would involve only non-infrastructure projects. Noninfrastructure projects include educational, encouragement, and enforcement activities.
Recreational Trails Program (RTP)	Administered at the state level by Arizona State Parks, the RTP provides funds to develop and maintain recreational trails and trail-related facilities for both nonmotorized and motorized recreational trail uses.

Policy and Program Considerations

In addition to improving Glendale's active transportation network through engineering solutions, a comprehensive approach that integrates policy, programmatic, education, enforcement, and encouragement elements is the most effective approach to creating and sustaining a pedestrian- and bicycle-friendly network. The following are policy and program concepts to consider.



ENGINEERING

Engineering considerations include general policies, strategies, and design concepts to help improve the physical environment for walking and biking in Glendale.

- Update the City's Capital Improvement Programs to incorporate the projects and programs recommended in the ATP.
- Upgrade roadway design standards to include bicycle facilities.
- Include paint striping maintenance for bicycle facilities as an annual, on-going line items in the City's transportation budget.
- Review and coordinated re-paving schedule to identify opportunities to include active transportation improvements.
- Program and seek funding to implement high priority improvement projects, particularly those that provide direct connections to critical facilities such as schools, employment centers, and public transportation. As funding is secured, engineering assessments will be needed to identify what type of facility is most suitable.
- During the design process, evaluate flexible street designs that allow Glendale to install a range of pedestrian and bicycle facility types to complement the corridor and surrounding land uses.
- Partner with Valley Metro to provide amenities at bus stops and to improve pedestrian and bike connections to transit stops, regional transit centers, and park-and-ride lots
- Retrofit existing facilities to incorporate ADA compliant facilities, as needed.
- Explore options and policies to include electric bikes and scooters within the active transportation system.
- In high pedestrian activity areas, evaluate pedestrian crossing times to ensure that pedestrians have ample time to cross.
- Recommend a minimum green time at intersections that do not possess pedestrian or bicycle detection and have a high ped/bike usage. Shorter green times should be utilized only when ped/bike detection is available but not activated by a pedestrian/bicyclist.
- Implement a bike rack program which allows for bike parking to be provided within the street right-of-way.
- Work with the Parks and Recreation Department to install signage on routes that connect to parks and trailheads.
- Collaborate with MAG and local jurisdictions to incorporate wayfinding for regional bicycle routes.

EDUCATION

Education programs equip people with the knowledge, skills and confidence to walk and bike.

- Develop maps and guides that show existing networks and highlight walking and bicycling routes for accessing key destinations throughout the city.
- Consider implementing a Glendale Active Transportation Program with a focus on education, training, and promotion of active transportation. Educational outreach might focus:
    - Schools, perhaps as part of Safe Routes to Schools program
    - Bicycle skills training and safety awareness training for all roadway users
    - Bicycle repair workshops
- Create an Active Transportation webpage to host maps, education efforts, and information on planned and completed active transportation improvements.
- Partner with other jurisdiction and community groups to develop regional campaigns that address unsafe and illegal behaviors of motorists, pedestrians, and bicyclists while encouraging mutual respect among all road users and encouraging active transportation.
- Feature rotating traffic safety and educational messages on the City's website and social media channels.
- Champion and support regional initiatives aimed at driver awareness of pedestrians and bicycles, particularly at intersections with high pedestrian and bicycle related crashes.
- Support or partner with advocacy groups such as Coalition of Arizona Bicyclists in offering bicycle-skills trainings for children and adults.

ENCOURAGEMENT

Encouragement programs foster a culture that supports and encourages active transportation

- Support Bicycle Safety Month in May and Back to School Safety Month in August by organizing local events throughout Glendale.
- Sponsor "Open Street" events once a year. Open Street events involve the closure of section of a road for a half or full day and allows people to enjoy walking and biking in a car-free environment.
- Support the "Bicycle Friendly Community" and "Bicycle Friendly Business" designation program through the League of American Bicyclists.
- Host launch parties when new bicycle facilities have been constructed.
- Provide bikes and gear such as helmets, bike locks, or bike lights to encourage safe riding. Giveaways are most commonly used to incentivize or reward participation in another program activity.



ENFORCEMENT

Enforcement builds safe and responsible behaviors on the road and builds respect among all road users

- Partner with Police Department to strategize and implement targeted enforcement, possibly in areas with high number of crashes or near schools.
- During future planning projects, encourage project participation of local law enforcement in stakeholder meetings, where they can provide insight into trends and issues. During the meetings, active transportation professionals and advisory groups can also provide insight on unsafe behaviors to help police offers evaluate the best methods of enforcement.
- Implement a comprehensive safety campaign that includes education, encouragement, and enforcement components.
- Support MAG's inter-agency working group that brings together staff from MAG member agencies and other partners to strategize about where targeted enforcement efforts are most needed.
- Publicize any enforcement actions through partners and media in order to achieve maximum driver behavior change.
- Continue to train officers in bicycle safety enforcement issues and principles.

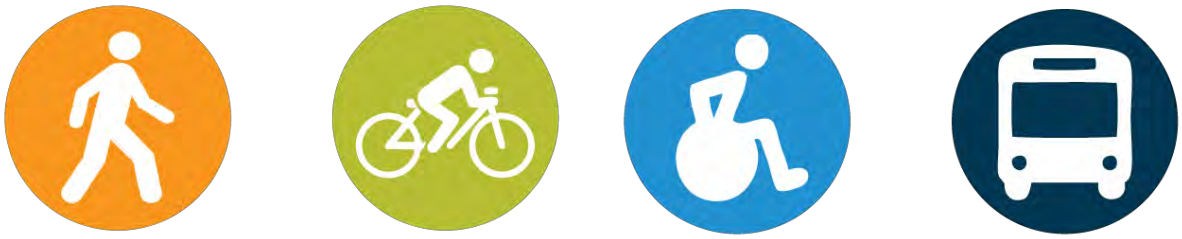


EVALUATION

Evaluation efforts seek to monitor progress and evaluate investments to continually improve active transportation

- Establish an annual bicycle and pedestrian count review program to determine baseline mode share conditions and subsequent changes.
- Develop a set of performance measures to annually evaluate and report on the progress of the ATP. Performance metrics can be compiled into a Bicycle and Pedestrian Report Card to easily illustrate accomplishments and general trends.

WALK. BIKE. CONNECT



glendale **active** transportation plan

the way we move

Appendix A. Recommended Improvement Projects

Pedestrian Network Improvement Recommendations

Intersection Improvements

On Road	Location	Project Description	Comments	Potential Constraints	Prioritization Score	Planning Cost
Peoria Ave	53rd Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops.	No	51.5	$5,000
Solano Dr	65th Ave	Install sidewalk and ADA ramp at NW corner		No	51	$12,400
Peoria Ave	49th Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops. If recommended bicycle facilities installed, provides bicycle connection across arterial.	No	50	$5,000
Peoria Ave	60th Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops.	No	49	$5,000
Thunderbird Rd	56th Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods, to hospital, and to transit stops.	No	47.5	$5,000
Myrtle Ave	48th Dr	Install sidewalk and ADA ramp at NW corner		No	46	$16,000
Orangewood Ave	67th Ave	Install sidewalk and ADA ramp at SE corner, NE corner	Project is currently in the design phase.	No	45.5	$24,700
75th Ave	Maryland Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to the school. If recommended bicycle facilities installed, provides bicycle connection across arterial. Project is currently in the design phase.	No	45.5	$5,000
Nicolet Ave	48th Dr	Install sidewalk and ADA ramp at SE corner, NE corner	Project should be combined with recommended sidewalk infill	No	45	$17,700
Missouri Ave	73rd Ave	Install sidewalk and ADA ramp at SW corner		No	42.5	$14,100
59th Ave	Arrowhead Lakes Dr	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods. If recommended bicycle facilities installed, provides bicycle connection across arterial.	No	42	$5,000
79th Ave	Saint John Rd	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops. Increases access to Arrowhead Mall.	No	40.5	$5,000
Missouri Ave	75th Ave	Conduct engineering analysis to determine feasibility/need for crosswalk	Evaluate the need for an additional crosswalk on south leg of intersection.	No	40.5	$3,000
79th Ave	Arrowhead Towne Center Dr	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to transit stops. Increases access to Arrowhead Mall.	No	40	$5,000
Glendale Ave	El Mirage Rd	Install sidewalk and ADA ramp at NW corner, SW corner, SE corner, NE corner	Potential for improvement to be developer driven.	No	39	$51,200

On Road	Location	Project Description	Comments	Potential Constraints	Prioritization Score	Planning Cost
Sack Dr	67th Ave	Install sidewalk and ADA ramp at NE corner	Potential for improvement to be developer driven.	No	39	$11,500
Maryland Ave	57th Ave	Install sidewalk and ADA ramp at SW corner	Potential for improvement to be developer driven.	No	38.5	$8,900
Augusta Ave	59th Ln	Install sidewalk and ADA ramp at SW corner		No	36.5	$8,900
Ocotillo Rd	81st Ave	Install sidewalk and ADA ramp at NW corner		No	36	$8.900
Eugie Ave	57th Dr	Conduct engineering analysis to determine feasibility/need for crosswalk	If pedestrian counts warrant, install crosswalk on Eugie St to improve access to the hospital.	No	35.5	$3,000
51th Ave	Oraibi Dr	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods and to the Skunk Creek Linear Park. Coordinate with Phoenix to address improvement needs.	No	35	$3,000
Camelback Rd	87th Ave	Install sidewalk and ADA ramp at NW corner	Coordinate with Phoenix to address improvement needs.	No	34.5	$10,600
Utopia Rd	57th Ave	Install sidewalk and ADA ramp at SE corner, NE corner	Potential for improvement to be developer driven.	No	34	$17,700
55th Ave	Voltaire Dr	Conduct engineering analysis to determine feasibility/need for crossing	If pedestrian counts warrant, install crossing to improve access to the hospital.	Yes	32.5	$5,000
Bethany Home Rd	127th Ave	Install sidewalk and ADA ramp at SW corner, NE Corner		No	32	$17,700
75th Ave	Grovers Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods, the transit center, and to Arrowhead Mall.	No	31.5	$5,000
Northern Ave	Citrus Rd	Install sidewalk and ADA ramp at NW corner, SW corner, SE corner, NE corner	Coordinate with MCDOT to address improvement needs	Yes	30	$28,400
Cholla St	61st Ave	Conduct engineering analysis to determine feasibility/need for crosswalk	If pedestrian counts warrant, install crosswalk on Cholla St	No	29.5	$3,000
Olive Ave	Citrus Rd	Install sidewalk at NW corner, SE corner, NE corner	Coordinate with MCDOT to address improvement needs.	Yes	29.5	$21,300
Desert Cove Ave	51st Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods. If recommended bicycle facilities installed, provides bicycle connection.	No	29.5	$5,000
Peoria Ave	Citrus Rd	Install sidewalk at SW corner		Yes	29	$21,100
Union Hills Dr	73rd Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods. If recommended bicycle facilities installed, provides bicycle connection.	No	29	$5,000

On Road	Location	Project Description	Comments	Potential Constraints	Prioritization Score	Planning Cost
Bethany Home Rd	137th Ave	Conduct engineering analysis to determine feasibility/need for crosswalk	Evaluate need for crosswalks on Bethany Home Rd. Coordinate with MCDOT to address improvement needs.	No	27.5	$3,000
Peoria Ave	Reems Rd	Install sidewalk at SE corner		Yes	27	$8,900
Bethany Home Rd	135th Ave	Conduct engineering analysis to determine feasibility/need for crosswalk	Evaluate need for crosswalks on Bethany Home Rd.	No	26.5	$3,000
Bell Rd	West of 70th Ave	Conduct engineering analysis to determine feasibility/need for crossing west of the bridge	Provides a connection between neighborhoods. If recommended bicycle facilities installed, provides bicycle connection to Skunk Creek Linear Park.	No	26.5	$5,000
Rattler Way	Wigwam Creek Blvd	Conduct engineering analysis to determine feasibility/need for crosswalk	Evaluate need for crosswalks on Wigwam Creek Blvd. Coordinate with MCDOT to address improvement needs.	No	25.5	$,3000
Wigwam Creek Blvd	124th Ave	Conduct engineering analysis to determine feasibility/need for crosswalk	Evaluate need for crosswalks on Wigwam Creek Blvd. Coordinate with MCDOT to address improvement needs.	No	25.5	$3,000
Greenway Rd	71st Ave	Conduct engineering analysis to determine feasibility/need for crosswalk	Evaluate need for crosswalks on Greenway Rd to improve access to Thunderbird Paseo Park. All curb ramps should be upgraded to ADA standards.	No	24.5	$3,000
Northern Ave	89th Ave	Conduct engineering analysis to determine feasibility/need for crossing	Provides a connection between neighborhoods. If recommended bicycle facilities installed, provides bicycle connection. Coordinate with the City of Peoria to address improvement needs.	No	23	$5,000
					Total	**$395,000**

Cost estimates include construction of sidewalk only and do not include costs for design, curb ramps, right-of-way acquisition, utility relocation, and other ancillary costs.

Sidewalk Infill Improvements

Recommendations are identified by northbound and eastbound direction. North-south roadways show recommendations by direction coming from south towards north. East-west roadways show recommendations by direction coming from west towards east.

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
Bethany Home Rd	52nd Ave	51st Ave - W of	Left	0.04	No	73.5	$8,000
Bethany Home Rd	57th Ave - E of	56th Ave - W of	Left	0.07	No	73	$13,000
67th Ave	Glendale Ave - N of	Myrtle Ave	Both	0.09	No	71.5	Currently Underway
Bethany Home Rd	55th Ave - E of	52nd Dr - W of	Both	0.03	No	71.5	$13,000
75th Ave	Bethany Home Rd	Griffin Ave	Right	0.38	No	71.5	$70,000
Union Hills Dr	53rd Ave - W of	52nd Ave	Left	0.17	No	71.5	$31,000
Greenway Rd	59th Ave	57th Ave	Right	0.30	No	71.5	$56,000
Union Hills Dr	55th Ave	55th Ave - E of	Left	0.04	No	71	$8,000
67th Ave	Glendale Ave - N of	Palmaire Ave - S of	Right	0.11	No	70.5	$21,000
67th Ave	Northview Ave	Orangewood Ave	Right	0.16	No	70.5	$31,000
75th Ave	Grovers Ave - N of	Union Hills Dr	Both	0.45	Yes	70	$166,000
Litchfield Rd	Glendale Ave	Lightning St - S of	Left	0.37	No	70	$69,000
Bethany Home Rd	52nd Dr - W of	52nd Ave	Right	0.33	No	69.5	$62,000
59th Ave	Behrend Dr - N of	Beardsley Rd	Right	0.06	No	69.5	$12,000
75th Ave	Village Dr - S of	Grovers Ave - N of	Right	0.28	Yes	68.5	$53,000
Bethany Home Rd	71st Dr	71st Ave	Both	0.06	No	68.5	$24,000
67th Ave	Sack Dr	Utopia Rd - S of	Right	0.13	No	68.5	$25,000
Bethany Home Rd	79th Dr	79th Ave	Left	0.07	Yes	67.5	$13,000
Northern Ave	63rd Ave - E of	61st Ave	Right	0.13	Yes	66.5	$24,000
43rd Ave	Berridge Ln - S of	Berridge Ln	Both	0.03	No	66.5	$11,000
75th Ave	Fleetwood Ln - N of	Glendale Ave - S of	Right	0.09	No	66.5	$17,000
Bethany Home Rd	72nd Ave	71st Dr	Right	0.06	No	66.5	$12,000
67th Ave	Alice Ave - S of	Alice Ave	Left	0.06	No	66	$12,000
Camelback Rd	91st Ave - W of	91st Ave	Left	0.13	No	66	$24,000
59th Ave	Union Hills Dr	Wescott Dr - N of	Right	0.24	No	66	$46,000
Camelback Rd	47th Ave - W of	45th Ave	Left	0.42	No	65.5	$78,000
55th Ave	Bethany Home Rd	Claremont St - N of	Both	0.44	No	65.5	$165,000
Dysart Rd	Ocotillo Rd	Glendale Ave	Left	0.25	No	65.5	$47,000
Thunderbird Rd	65th Ave	65th Ave - E of	Right	0.05	No	65	$10,000
Beardsley Rd	63rd Ave	61st Ave	Left	0.25	No	64.5	$47,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
Northern Ave	68th Ave - W of	68th Ave	Right	0.17	No	64	$31,000
43rd Ave	Hatcher Rd - N of	Mountain View Rd - S of	Left	0.24	No	63.5	$44,000
43rd Ave	Cavalier Dr	Keim Dr - S of	Both	0.03	No	63.5	$12,000
75th Ave	Montebello Ave	Bethany Home Rd	Left	0.25	Yes	63.5	$47,000
Thunderbird Rd	63rd Ave - W of	Arizona Canal Trail	Right	0.34	No	63	$63,000
Missouri Ave	59th Ave - E of	57th Ave - W of	Left	0.09	No	63	$18,000
Northern Ave	63rd Ave	63rd Ave - E of	Both	0.13	No	63	$47,000
83rd Ave	Grand Canal Trail	Cavalier Dr	Left	0.07	No	63	$13,000
Orangewood Ave	Grand Ave - W of	Grand Ave	Left	0.24	No	62.5	$45,000
Maryland Ave	69th Ave	67th Ave	Left	0.25	No	62.5	$47,000
Glendale Ave	Litchfield Rd - E of	Litchfield Rd - E of	Left	0.06	No	62.5	$11,000
Glendale Ave	Dysart Rd	El Mirage Rd	Both	0.99	No	62.5	$365,000
48th Ave	Bethany Home Rd - N of	Rose Ln	Right	0.23	No	62	$44,000
Dysart Rd	Maryland Ave	Ocotillo Rd	Left	0.25	No	62	$46,000
Litchfield Rd	Glendale Ave - N of	Northern Ave	Both	0.63	No	62	$232,000
Camelback Rd	Dysart Rd	129th Ave	Both	0.21	No	62	$77,000
Thunderbird Rd	67th Ave	65th Ave - E of	Left	0.09	No	62	$17,000
48th Ave	Glendale Ave - N of	Glenn Dr - S of	Right	0.05	No	61.5	$9,000
Glendale Ave	Litchfield Rd - E of	Litchfield Rd - E of	Right	0.06	No	61.5	$12,000
Litchfield Rd	Super Sabre St	Glendale Ave	Both	0.49	No	61.5	$181,000
Beardsley Rd	67th Ave - E of	63rd Ave	Left	0.25	No	61.5	$47,000
Union Hills Dr	75th Ave - E of	70th Ave	Right	0.51	Yes	61	$96,000
83rd Ave	Montebello Ave - N of	Bethany Home Rd	Both	0.13	No	61	$48,000
Glendale Ave	95th Ave - W of	95th Ave	Left	0.27	No	61	$50,000
Orangewood Ave	67th Ave	Grand Ave - W of	Both	0.25	No	60.5	$94,000
Bethany Home Rd	83rd Ave	83rd Ave - E of	Left	0.14	No	60.5	$26,000
83rd Ave	Montebello Ave	Montebello Ave - N of	Left	0.05	No	60	$10,000
Glendale Ave	135th Ave - E of	Dysart Rd	Both	0.31	No	60	$116,000
Glendale Ave	137th Ave - E of	135th Ave - E of	Left	0.44	No	59.5	$81,000
Dysart Rd	Glendale Ave	S of Canal	Both	0.45	No	59.5	$167,000
54th Ave	Glenn Dr - S of	Glenn Dr	Left	0.03	No	59	$5,000
Grand Ave	Bethany Home Rd	Glendale Ave	Left	1.42	Yes	59	$262,000
Grand Ave	61st Ave - S of	Northern Ave	Left	1.75	Yes	59	$323,000
Northern Ave	70th Ave	68th Ave - W of	Both	0.14	No	59	$53,000
61st Ave	Golden Ln - N of	Olive Ave	Left	0.05	No	59	$9,000
91st Ave	Ocotillo Rd	Glendale Ave	Right	0.25	No	59	$46,000
91st Ave	Camelback Rd	Camelback Rd - N of	Both	0.10	No	59	$36,000
51st Ave	Grand Ave	51st Ave	Both	0.23	Yes	58.5	$86,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
Glendale Ave	137th Ave - W of	137th Ave - E of	Both	0.08	No	58.5	$29,000
Bethany Home Rd	Dysart Rd	129th Ave	Both	0.25	No	58.5	$92,000
Missouri Ave	51st Ave - W of	51st Ave - W of	Both	0.27	No	58	$101,000
55th Ave	Luke Ave	Luke Ave - N of	Right	0.04	No	58	$7,000
63rd Ave	Northern Ave	Northern Ave - N of	Both	0.12	No	58	$45,000
45th Ave	Northern Ave - S of	Northern Ave	Left	0.04	No	58	$9,000
Camelback Rd	91st Ave	87th Ave	Left	0.50	No	58	$94,000
83rd Ave	Orange Dr - S of	Missouri Ave - S of	Left	0.32	No	58	$60,000
48th Dr	Myrtle Ave - N of	Nicolet Ave	Both	0.03	No	57.5	$11,000
Mountain View Rd	53rd Ave	51st Ave	Left	0.24	No	57.5	$44,000
Gardenia Ave	51st Ave	50th Dr - W of	Both	0.08	No	57.5	$30,000
Ocotillo Rd	57th Ave	56th Ave	Right	0.07	No	57.5	$14,000
83rd Ave	Glendale Ave - N of	Palmaire Ave - N of	Left	0.12	No	57.5	$23,000
Dysart Rd	Colter St - N of	Rose Ln - N of	Right	0.98	No	57.5	$182,000
Beardsley Rd	61st Ave	61st Ave - E of	Left	0.09	No	57.5	$18,000
63rd Ave	Hayward Ave - S of	Northern Ave	Right	0.25	No	57	$47,000
63rd Ave	Northern Ave - N of	Butler Dr - S of	Left	0.26	No	57	$49,000
Royal Palm Rd	Grand Ave	Grand Ave	Both	0.10	No	57	$37,000
Glenn Dr	54th Dr - E of	54th Ave	Right	0.10	No	57	$19,000
Maryland Ave	57th Ave - W of	57th Ave	Left	0.14	No	57	$26,000
Market St	57th Dr	Ocotillo Rd	Both	0.06	No	57	$22,000
57th Ave	Maryland Ave	McLellan Rd - S of	Right	0.07	No	57	$13,000
69th Ave	Maryland Ave	Ocotillo Rd - S of	Right	0.19	No	57	$35,000
Glendale Ave	91st Ave	87th Ave - W of	Right	0.38	No	57	$71,000
Camelback Rd	84th Ln - E of	83rd Ave - W of	Left	0.12	No	57	$23,000
Orangewood Ave	71st Ave	69th Ave	Left	0.25	No	56.5	$47,000
Estero Ln	Grand Ave	Morten Ave	Left	0.09	No	56.5	$18,000
Palmaire Ave	55th Ave	54th Dr - E of	Both	0.05	No	56.5	$19,000
State Ave	51st Ave	50th Dr	Both	0.10	No	56.5	$37,000
Lamar Rd	51st Ave	49th Ave	Both	0.22	No	56.5	$83,000
Glendale Ave	85th Ln	85th Ln - E of	Left	0.12	No	56.5	$23,000
Litchfield Rd	Claremont St - S of	Super Sabre St	Left	0.20	No	56.5	$37,000
Litchfield Rd	Missouri Ave	Peck Dr - N of	Left	0.74	No	56.5	$137,000
Greenway Rd	57th Ave	56th Ave - E of	Right	0.15	No	56.5	$28,000
50th Ave	Cholla St	Sunnyside Dr	Both	0.24	No	56	$89,000
63rd Ave	Butler Dr	Alice Ave - S of	Right	0.12	No	56	$23,000
Royal Palm Rd	Grand Ave	67th Ave	Left	0.07	No	56	$13,000
53rd Ave	Northern Ave - N of	Laurie Ln - S of	Left	0.28	No	56	$52,000
63rd Ave	Northern Ave - N of	Butler Dr	Left	0.11	No	56	$21,000
Northern Ave	71st Ave	70th Ave	Left	0.12	No	56	$22,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
91st Ave	Camelback Rd - N of	Bethany Home Rd	Right	0.89	No	56	$165,000
91st Ave	Rovey Ave - N of	Maryland Ave	Left	0.28	No	56	$51,000
Bethany Home Rd	127th Ave	125th Ave	Left	0.25	No	56	$47,000
Bethany Home Rd	129th Ave	127th Ave	Right	0.25	No	56	$46,000
66th Ave	Greenway Rd	Paradise Ln	Both	0.51	No	56	$188,000
61st Ave	Cholla St	Sunnyside Dr	Right	0.25	Yes	55.5	$47,000
Patrick Ln	67th Ave - W of	67th Ave	Left	0.25	No	55.5	$48,000
69th Ave	Orangewood Ave	Belmont Ave	Both	0.16	No	55.5	$59,000
San Miguel Ave	59th Ave	57th Ave	Both	0.25	No	55.5	$94,000
Morten Ave	59th Ave	57th Ave	Both	0.29	No	55.5	$107,000
Vista Ave	59th Ave	57th Ave	Both	0.29	No	55.5	$107,000
48th Dr	Myrtle Ave - S of	Myrtle Ave - S of	Right	0.02	No	55.5	$4,000
83rd Ave	Palmaire Ave - N of	Nicolet Ave - S of	Right	0.11	No	55.5	$20,000
83rd Ave	Belmont Ave - N of	Northern Ave	Both	0.27	No	55.5	$100,000
83rd Ave	Vista Ave- N of	Vista Ave - N of	Right	0.07	No	55.5	$13,000
135th Ave	Glendale Ave - S of	Glendale Ave	Both	0.05	No	55.5	$21,000
Camelback Rd	129th Ave	127th Ave	Left	0.28	No	55.5	$52,000
Grand Ave	Glendale Ave	Myrtle Ave	Both	0.32	Yes	55	$118,000
68th Ave	Frier Dr - N of	Northern Ave	Both	0.11	No	55	$40,000
Palmaire Ave	54th Dr - E of	54th Dr - E of	Right	0.05	No	55	$10,000
Glenn Dr	53rd Ave - W of	53rd Ave - W of	Right	0.05	No	55	$10,000
Glenn Dr	53rd Ave - E of	52nd Ave	Right	0.11	No	55	$22,000
55th Ave	Northern Ave - N of	Butler Dr - S of	Right	0.28	No	55	$52,000
57th Ave	Missouri Ave - N of	Bethany Home Rd	Right	0.42	No	55	$77,000
57th Ave	Ocotillo Rd	Lamar Rd - S of	Both	0.10	Yes	55	$37,000
46th Ave	Ocotillo Rd - N of	Glendale Ave - S of	Both	0.12	No	55	$46,000
Ocotillo Rd	Krall St	47th Ave	Left	0.27	No	55	$50,000
57th Ave	Bethany Home Rd	Rose Ln - S of	Both	0.13	No	55	$47,000
45th Ave	Palmaire Ave - S of	Palmaire Ave - S of	Left	0.03	No	55	$6,000
Market St	58th Ave	Ocotillo Rd	Left	0.04	No	55	$9,000
Market St	57th Dr	57th Ave	Right	0.06	No	55	$11,000
69th Ave	Keim Dr	Citrus Way - S of	Right	0.24	No	55	$45,000
65th Ave	Maryland Ave	Sierra Vista Dr - S of	Both	0.05	No	55	$18,000
75th Ave	Aurora Dr - S of	Aurora Dr - S of	Left	0.04	No	54.5	$8,000
63rd Ave	Morten Ave	Hayward Ave - S of	Both	0.13	No	54.5	$47,000
Glenn Dr	51st Ave	51st Ave - E of	Right	0.05	No	54.5	$10,000
Berridge Ln	43rd Ave - W of	43rd Ave	Right	0.05	No	54.5	$10,000
57th Ave	Missouri Ave	Missouri Ave - N of	Right	0.09	No	54.5	$16,000
53rd Ave	Peoria Ave - S of	Peoria Ave	Right	0.04	No	54.5	$7,000
Royal Palm Rd	59th Ave	58th Ave - E of	Right	0.15	No	54.5	$29,000
83rd Ave	Orangewood Ave	Vista Ave - N of	Both	0.12	No	54.5	$46,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
57th Ave	Utopia Rd - S of	Beardsley Rd	Right	0.52	No	54.5	$97,000
Thunderbird Paseo Park Dr	Eugie Ave - S of	Eugie Ave - S of	Right	0.10	No	54	$18,000
69th Ave	Butler Dr	Butler Dr - N of	Left	0.11	No	54	$22,000
Butler Dr	65th Ave - W of	63rd Ave	Right	0.25	No	54	$47,000
Frier Dr	68th Ave	67th Ave	Both	0.08	No	54	$32,000
Rose Ln	59th Ave	58th Ave	Left	0.08	No	54	$14,000
57th Dr	Rose Ln - S of	Rose Ln	Both	0.06	No	54	$21,000
McLellan Rd	58th Ave	57th Ave	Right	0.10	No	54	$19,000
Solano Dr	67th Ave	65th Ave	Left	0.21	No	54	$40,000
Bethany Home Rd	95th Ave - W of	91st Ave	Right	0.76	No	54	$140,000
61st Ave	Augusta Ave	Northern Ave	Left	0.12	Yes	53.5	$23,000
Glenn Dr	Palmaire Ave - W of	Palmaire Ave - W of	Both	0.05	No	53.5	$21,000
Nicolet Ave	48th Dr - E of	48th Dr - E of	Right	0.07	No	53.5	$13,000
54th Dr	Lamar Rd - N of	Glendale Ave - S of	Left	0.03	No	53.5	$7,000
50th Dr	State Ave	Gardenia Ave - S of	Both	0.03	No	53.5	$10,000
Belmont Ave	59th Ave	57th Ave	Both	0.29	No	53.5	$107,000
50th Dr	State Ave - N of	Gardenia Ave	Right	0.02	No	53.5	$5,000
Gardenia Ave	50th Dr - W of	50th Dr	Left	0.02	No	53.5	$4,000
Nicolet Ave	48th Dr	48th Dr - E of	Both	0.02	No	53.5	$6,000
48th Dr	Nicolet Ave	Northview Ave - S of	Right	0.02	No	53.5	$4,000
48th Ln	Glenn Dr	Myrtle Ave - S of	Left	0.12	No	53.5	$22,000
44th Dr	44th Dr - W of	44th Dr	Left	0.03	No	53.5	$6,000
71st Ave	Bethany Home Rd	Claremont St	Left	0.25	No	53.5	$47,000
Northern Ave	87th Ave	83rd Ave	Right	0.49	No	53.5	$91,000
Litchfield Rd	Peck Dr - N of	Claremont St - S of	Both	0.06	No	53.5	$22,000
57th Ave	Shaw Butte Dr - N of	Cactus Rd	Right	0.04	No	53	$8,000
Orangewood Ave	Grand Ave	59th Ln	Both	0.33	No	53	$121,000
Northern Ave	63rd Ave - W of	63rd Ave - W of	Left	0.17	Yes	53	$31,000
61st Ave	Morten Ave	Augusta Ave	Both	0.25	No	53	$94,000
61st Ave	Gardenia Ave - S of	Orangewood Ave	Right	0.06	No	53	$12,000
61st Ave	Orangewood Ave	Morten Ave	Both	0.13	No	53	$47,000
57th Ave	Rose Ln - N of	Maryland Ave	Left	0.19	No	53	$36,000
Krall St	49th Ave - W of	49th Ave	Left	0.05	No	53	$10,000
Flynn Ln	51st Ave - E of	49th Ave	Both	0.18	No	53	$66,000
Rose Ln	58th Ave	57th Dr - E of	Both	0.10	No	53	$36,000
57th Dr	Bethany Home Rd	Rose Ln - S of	Right	0.19	No	53	$36,000
Orange Dr	55th Ave	54th Ave	Left	0.12	No	53	$23,000
59th Ln	Gardenia Ave	Orangewood Ave	Right	0.04	No	53	$9,000
58th Ave	Orangewood Ave	Vista Ave - S of	Left	0.03	No	53	$7,000
Glen Harbor Blvd	Glendale Ave	Glenn Dr	Left	0.07	No	53	$13,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
Frier Dr	69th Ave - E of	68th Ave - W of	Both	0.08	No	52.5	$30,000
Frier Dr	68th Ave - W of	68th Ave	Right	0.04	No	52.5	$8,000
68th Ave	Frier Dr	Frier Dr - N of	Right	0.10	No	52.5	$20,000
Missouri Ave	55th Ave - E of	55th Ave - E of	Left	0.19	Yes	52.5	$36,000
Claremont St	57th Ave	55th Ave	Both	0.27	No	52.5	$99,000
Palmaire Ave	67th Ave - W of	67th Ave	Both	0.03	No	52.5	$11,000
58th Ave	Vista Ave	Frier Dr - S of	Both	0.15	No	52.5	$57,000
61st Ave	Berridge Ln	Keim Dr	Right	0.08	No	52.5	$15,000
Keim Dr	61st Ave	60th Ave	Right	0.13	No	52.5	$24,000
Glendale Ave	76th Dr	75th Ave	Right	0.17	Yes	52.5	$32,000
Utopia Rd	57th Ave	54th Ln - W of	Both	0.16	Yes	52.5	$58,000
Angela Dr	75th Ave	71st Dr - E of	Both	0.49	No	52	$181,000
Northern Ave	75th Ave	71st Ave	Both	0.50	No	52	$185,000
Rose Ln	57th Dr - E of	57th Ave	Right	0.03	No	52	$6,000
57th Ave	Bethany Home Rd - N of	Rose Ln	Left	0.12	No	52	$23,000
54th Ave	Camelback Rd	Orange Dr - S of	Both	0.11	No	52	$41,000
57th Ave	McLellan Rd - S of	Market St	Right	0.07	No	52	$14,000
McLellan Rd	59th Dr	59th Ave	Both	0.09	No	52	$32,000
Keim Dr	60th Ave	59th Ave - W of	Left	0.07	No	52	$13,000
Montebello Ave	61st Ave	59th Ave	Right	0.25	No	52	$47,000
60th Ave	Keim Dr	Rose Ln - S of	Right	0.04	No	52	$7,000
Rose Ln	59th Ave - W of	59th Ave	Left	0.05	Yes	52	$10,000
65th Ave	Sierra Vista Dr - S of	McLellan Rd - N of	Left	0.09	No	52	$18,000
99th Ave	Glendale Ave	Desert River Blvd - S of	Left	0.17	No	52	$31,000
Grovers Ave	70th Dr	67th Ave	Left	0.46	No	51.5	$86,000
66th Dr	Glendale Ave	Glendale Ave - N of	Both	0.10	No	51.5	$39,000
Orangewood Ave	72nd Ave - W of	71st Ave - W of	Left	0.07	No	51.5	$13,000
71st Ave	Orangewood Ave	Frier Dr - S of	Right	0.14	No	51.5	$26,000
69th Ave	Belmont Ave	Frier Dr	Left	0.09	No	51.5	$18,000
Glenn Dr	48th Dr - E of	48th Ave - W of	Right	0.01	No	51.5	$3,000
53rd Ave	Grand Ave	Grand Ave - N of	Both	0.12	No	51.5	$45,000
Stella Ln	51st Ave	50th Dr	Right	0.07	Yes	51.5	$14,000
71st Ave	Montebello Ave	Bethany Home Rd	Left	0.27	No	51.5	$51,000
Rose Ln	61st Ave	60th Ave	Both	0.13	No	51.5	$47,000
Ocotillo Rd	80th Ave - W of	80th Ave	Left	0.06	No	51.5	$12,000
79th Ave	Glendale Ave	Nicolet Ave - N of	Left	0.38	No	51.5	$70,000
80th Ave	Ocotillo Rd	Glendale Ave	Both	0.25	No	51.5	$92,000
135th Ave	Ocotillo Rd	Ocotillo Rd - N of	Both	0.04	No	51.5	$14,000
79th Ave	Solano Dr - N of	Bethany Home Rd	Left	0.07	No	51.5	$14,000
Cholla St	49th Ave	49th Ave - E of	Right	0.08	No	51	$15,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
49th Ave	Brown St	Peoria Ave	Left	0.24	No	51	$44,000
Krall St	51st Ave - E of	49th Ave - W of	Left	0.10	No	51	$18,000
Montebello Ave	59th Ave - E of	57th Ave	Right	0.22	No	51	$42,000
Belmont Ave	69th Ave	67th Ave	Both	0.25	No	51	$93,000
Augusta Ave	61st Ave	59th Ln	Right	0.13	No	51	$24,000
60th Ave	Orangewood Ave	Augusta Ave	Both	0.38	No	51	$141,000
60th Ave	Missouri Ave - S of	Missouri Ave	Right	0.06	No	51	$12,000
Citrus Way	67th Dr - W of	67th Ave	Right	0.10	No	51	$19,000
65th Dr	Maryland Ave - S of	Maryland Ave	Left	0.13	No	51	$24,000
Camelback Rd	95th Ave - E of	93rd Ave - W of	Left	0.07	No	51	$14,000
91st Ave	Orangewood Ave	Orangewood Ave - N of	Left	0.28	No	51	$53,000
99th Ave	Desert River Blvd - S of	Desert River Blvd	Right	0.19	No	51	$35,000
61st Ave	Greenbriar Dr - S of	Greenbriar Dr	Both	0.12	No	51	$43,000
53rd Ave	Phelps Rd	Bell Rd	Both	0.14	No	51	$53,000
83rd Ave	Campo Bello - N of	Union Hills Dr - S of	Left	0.34	No	50.5	$63,000
Orangewood Ave	75th Ave	75th Ave - E of	Both	0.13	No	50.5	$48,000
Frier Dr	69th Ave	69th Ave - E of	Left	0.05	No	50.5	$9,000
68th Ave	Frier Dr - N of	Frier Dr - N of	Left	0.06	No	50.5	$11,000
53rd Ave	Mountain View Rd	Cinnabar Ave - N of	Right	0.10	No	50.5	$19,000
Grand Ave	51st Ave	43rd Ave	Left	1.35	Yes	50.5	$250,000
52nd Ave	Luke Ave	Bethany Home Rd	Left	0.38	No	50.5	$70,000
Gardenia Ave	59th Ln	59th Dr	Right	0.04	No	50.5	$8,000
59th Ln	Gardenia Ave	Gardenia Ave	Both	0.02	No	50.5	$7,000
58th Ave	Northern Ave	Royal Palms Rd	Both	0.25	No	50.5	$92,000
79th Ave	Fleetwood Ln - S of	Glendale Ave	Left	0.13	No	50.5	$24,000
135th Ave	Ocotillo Rd - N of	Glendale Ave - S of	Right	0.17	No	50.5	$32,000
Ocotillo Rd	Dysart Rd	129th Ln - W of	Right	0.11	No	50.5	$21,000
52Nd Ave	Union Hills Dr	Union Hill Dr - N of	Left	0.06	No	50.5	$12,000
Sack Dr	67th Ave	67th Ave - E of	Left	0.18	No	50.5	$34,000
Palmaire Ave	45th Ave - E of	43rd Ave - W of	Right	0.08	No	50	$15,000
49th Ave	Ocotillo Rd	Glendale Ave - S of	Both	0.19	Yes	50	$70,000
53rd Ave	Camelback Rd	Orange Dr	Both	0.20	No	50	$73,000
Keim Dr	59th Ave - W of	59th Ave - W of	Right	0.02	No	50	$4,000
Keim Dr	60th Ave	59th Ave - W of	Both	0.01	No	50	$4,000
Camelback Rd	Litchfield Rd	Dysart Rd	Both	0.99	No	50	$368,000
61st Ave	Bell Rd - N of	Greenbriar Dr - S of	Both	0.13	No	50	$48,000
69th Ave	Alice Ave - S of	Alice Ave	Right	0.06	No	49.5	$12,000
53rd Ave	Grand Ave - N of	Maryland Ave	Left	0.12	Yes	49.5	$24,000
73rd Ave	Orangewood Ave	Frier Dr - N of	Left	0.30	No	49.5	$55,000
Missouri Ave	75th Ave	73rd Ave	Right	0.25	Yes	49.5	$47,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
71st Dr	Bethany Home Rd	Claremont St	Right	0.25	No	49.5	$47,000
Marlette Ave	61st Ave	60th Ave	Both	0.13	No	49.5	$47,000
Marlette Ave	60th Ave	59th Dr	Both	0.08	No	49.5	$30,000
Orangewood Ave	78th Ave - W of	75th Ave	Left	0.49	No	49.5	$91,000
81st Ave	Ocotillo Rd	Glendale Ave	Left	0.25	No	49.5	$46,000
Ocotillo Rd	81st Ave - W of	81st Ave	Left	0.06	No	49.5	$12,000
52nd Ave	Union Hills Dr - N of	Wescott Dr - S of	Both	0.13	No	49.5	$47,000
Sunnyside Dr	51st Ave - E of	50th Ave	Right	0.06	No	49	$12,000
Montebello Ave	55th Ave	52nd Ave	Both	0.40	No	49	$147,000
70th Ave	Northern Ave - S of	Northern Ave	Both	0.06	No	49	$22,000
87th Ave	Camelback Rd	Missouri Ave	Left	0.44	No	49	$82,000
Glen Harbor Blvd	Glenn Dr - N of	Northview Ave - W of	Left	0.17	No	49	$31,000
Litchfield Rd	Northern Ave	Northern Ave - N of	Both	0.16	No	49	$60,000
Desert Cove Ave	59th Ave	57th Ave	Right	0.24	Yes	48.5	$45,000
66th Dr	Glendale Ave - N of	Glendale Ave - N of	Left	0.02	No	48.5	$4,000
Brown St	51st Ave	49th Ave	Left	0.24	Yes	48.5	$45,000
Loma Ln	58th Ave	57th Dr - W of	Both	0.04	No	48.5	$15,000
57th Dr	Northern Ave - N of	Loma Ln - S of	Left	0.02	No	48.5	$4,000
57th Dr	Loma Ln - N of	Loma Ln - N of	Left	0.02	No	48.5	$4,000
57th Dr	Royal Palms Rd - S of	Royal Palms Rd - S of	Both	0.04	No	48.5	$16,000
Nicolet Ave	47th Ave - W of	47th Ave	Right	0.03	No	48.5	$6,000
Citrus Rd	Olive Ave	Olive Ave - N of	Both	0.14	No	48.5	$51,000
Claremont St	61st Ave	60th Ave	Both	0.13	No	48.5	$47,000
Orangewood Ave	82nd Dr - E of	78th Ave - W of	Both	0.41	No	48.5	$151,000
81st Ave	Glendale Ave	Palmaire Ave - N of	Right	0.13	No	48.5	$25,000
123rd Dr	Rancho Dr - N of	Palo Verde Dr	Right	0.04	No	48.5	$8,000
49th Ave	Tuckey Ln - N of	Krall St	Left	0.03	No	48	$5,000
54th Ave	Orange Dr - S of	Orange Dr	Right	0.09	No	48	$17,000
Luke Ave	55th Ave	51st Ave	Both	0.51	No	48	$188,000
57th Ave	McLellan Rd - S of	McLellan Rd - S of	Left	0.03	No	48	$6,000
65th Dr	Maryland Ave - S of	Maryland Ave - S of	Right	0.05	No	48	$9,000
65th Dr	Rose Ln - N of	Rose Ln - N of	Both	0.07	No	48	$28,000
64th Dr	Rose Ln - N of	Maryland Ave	Both	0.18	No	48	$69,000
125th Ave	Glendale Ave - S of	Glendale Ave	Both	0.25	No	48	$92,000
125th Ave	Glendale Ave - S of	Glendale Ave - S of	Left	0.06	No	48	$11,000
66th Ln	Caribe Ln	Port Au Prince Ln	Left	0.09	No	48	$17,000
Phelps Rd	53rd Ave	51st Ave	Left	0.25	No	48	$47,000
Orangewood Ave	73rd Ave - W of	73rd Ave	Left	0.13	No	47.5	$24,000
Frier Dr	72nd Dr - E of	71st Ave	Right	0.13	No	47.5	$24,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
Olive Ave	181st Ave	Citrus Rd	Left	0.25	No	46.5	$47,000
Stella Ave	61st Ave	60th Ave	Both	0.13	Yes	46.5	$48,000
123rd Dr	San Miguel Ave - N of	El Nido Ln	Right	0.03	No	46.5	$6,000
Colter St	Dysart Rd - E of	Pasadena Ave - E of	Right	0.31	No	46.5	$58,000
Orange Dr	54th Ave	53rd Ave	Both	0.13	No	46	$49,000
Citrus Rd	Northern Ave	Butler Ave	Both	0.53	No	46	$196,000
Ocotillo Rd	91st Ave	91st Ave - E of	Left	0.06	No	46	$11,000
Glen Harbor Blvd	Vista Ave	Vista Ave - N of	Both	0.08	No	46	$32,000
Maryland Ave	135th Ave	Dysart Rd	Left	0.50	No	46	$92,000
129th Ave	Maryland Ave	Glendale Ave	Right	0.50	No	46	$93,000
Torno Ct	Colter St	Colter St - N of	Right	0.07	No	46	$13,000
Pajaro Ct	Orange Ct	Colter St - N of	Right	0.10	No	46	$19,000
Orange Ct	Pajaro Ct - W of	Pajaro Ct	Right	0.08	No	46	$15,000
Maryland Ave	Dysart Rd	Sierra Vista Dr	Left	0.11	No	46	$20,000
Glenn Dr	48th Dr - W of	48th Dr - W of	Right	0.02	Yes	45.5	$5,000
69th Ave	Glenn Dr	Myrtle Ave	Left	0.17	Yes	45.5	$32,000
Frier Dr	71st Ave - E of	70th Ave - W of	Left	0.04	No	45.5	$7,000
Glen Harbor Blvd	Northern Ave - S of	Northern Ave	Both	0.06	No	45.5	$21,000
135th Ave	Maryland Ave	Ocotillo Rd	Right	0.23	No	45.5	$43,000
Tuckey Ln	134th Ave	Dysart Rd	Both	0.36	No	45.5	$132,000
McLellan Rd	134th Ave	131st Dr	Both	0.32	No	45.5	$118,000
Sierra Vista Dr	134th Ave	131st Dr	Both	0.32	No	45.5	$117,000
134th Ave	Maryland Ave	Tuckey Ln	Both	0.16	No	45.5	$59,000
131st Dr	Maryland Ave	Tuckey Ln	Both	0.16	No	45.5	$59,000
134th Dr	Maryland Ave	Ocotillo Rd	Both	0.23	No	45.5	$86,000
Lamar Rd	81st Ln - W of	81st Dr - E of	Right	0.08	No	45.5	$16,000
Ocotillo Rd	135th Ave	Dysart Rd	Both	0.50	No	45.5	$184,000
135th Ave	Claremont St - S of	Maryland Ave	Left	0.25	No	45.5	$46,000
Maryland Ave	Stella Ave	135th Ave	Both	0.43	No	45.5	$159,000
54th Ave	Wescott Dr - S of	Taro Ln	Both	0.20	Yes	45.5	$76,000
57th Ave	Union Hills Dr - N of	Utopia Rd - S of	Left	0.15	No	45.5	$28,000
54th Ave	Luke Ave	Montebello Ave	Both	0.12	No	45	$46,000
Glendale Ave	El Mirage Rd	99th Ave	Both	2.97	Yes	45	$1,099,000
Glen Harbor Blvd	Northview Ave - N of	Vista Ave	Left	0.20	No	45	$38,000
Glen Harbor Blvd	106th Ave - W of	106th Ave	Both	0.04	No	45	$14,000
Glen Harbor Blvd	Northern Ave - S of	Northern Ave - S of	Right	0.09	No	45	$17,000
129th Ave	Bethany Home Rd	Marlette Ave	Both	0.35	No	45	$131,000
Colter St	Pajaro Ct	Dysart Rd - W of	Right	0.31	No	45	$57,000
Claremont St	138th Dr - E of	135th Ave	Both	0.28	No	45	$105,000
Marlette Ave	138th Dr - E of	135th Ave	Both	0.33	No	45	$121,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
Stella Ln	138th St - E of	135th Ave	Both	0.37	No	45	$136,000
Myrtle Ave	72nd Ave - E of	71st Ave	Left	0.07	Yes	44.5	$13,000
Tom Murray Ave	Camelback Rd	Colter St - N of	Both	0.49	Yes	44.5	$181,000
Maryland Ave	74th Ave - W of	73rd Ave - E of	Right	0.26	Yes	44.5	$49,000
Ocotillo Rd	80th Ave	79th Dr - W of	Left	0.06	No	44.5	$12,000
Missouri Ave	80th Ave - E of	79th Ave - W of	Left	0.09	No	44.5	$18,000
126th Dr	Kristi Ln - N of	Montebello Ave - S of	Left	0.04	No	44.5	$7,000
126th Dr	Solano Dr - N of	Palo Verde Dr - S of	Left	0.04	No	44.5	$7,000
126th Dr	Rattler Wy	San Miguel Ave	Left	0.04	No	44.5	$8,000
123rd Dr	Missouri Ave - N of	San Juan Ave	Right	0.09	No	44.5	$17,000
Glen Harbor Blvd	106th Ave	106th Ave - E of	Right	0.11	No	44	$21,000
Colter St	Dysart Rd - W of	Dysart Rd	Left	0.12	No	44	$23,000
Peoria Ave	185th Dr - E of	183rd Ave	Right	0.31	Yes	43.5	$57,000
Orangewood Ave	83rd Ave	82nd Dr - E of	Left	0.10	No	43.5	$19,000
Colter St	51st Ave	51st Ave - E of	Both	0.56	Yes	43	$207,000
95th Ave	Glendale Ave	Cabela Dr - N of	Left	0.49	Yes	43	$91,000
129th Ave	Marlette Ave	Maryland Ave	Right	0.14	No	43	$27,000
Ora Vista Ct	Colter St	Colter St - N of	Right	0.06	Yes	43	$12,000
91st Ave	Bethany Home Rd	Bethany Home Rd - N of	Right	0.03	Yes	43	$5,000
Pinnacle Peak Rd	71st Ave	69th Dr	Right	0.21	No	42.5	$40,000
81st Ave	Olive Ave	Olive Ave - N of	Right	0.10	No	42.5	$18,000
89th Ave	State Ave - S of	Orangewood Ave	Left	0.13	No	42.5	$24,000
Northern Ave	Loop 101 - E of	93rd Ave - E of	Right	0.13	No	42.5	$24,000
Pasadena Ave	51st Ave	Tom Murray Ave	Both	0.68	Yes	42	$251,000
Myrtle Ave	45th Ave	45th Ave - E of	Right	0.05	Yes	42	$10,000
89th Ave	Frier Dr - N of	Northern Ave	Left	0.20	No	42	$37,000
Pinnacle Peak Rd	69th Dr	67th Ave - W of	Right	0.21	No	41.5	$40,000
Pinnacle Peak Rd	72nd Ave	71st Ave	Right	0.17	No	41.5	$31,000
Myrtle Ave	76th Dr	75th Ave	Right	0.18	Yes	41.5	$34,000
71st Ave	Carol Ann Way - S of	Sherri Jean Ln	Both	0.19	No	41.5	$71,000
73rd Ave	Frier Dr - N of	Northern Ave	Both	0.20	No	41	$76,000
Orangewood Ave	91st Ave	89th Ave	Right	0.25	No	41	$47,000
Beck Ln	72nd Ave - W of	67th Ave	Both	0.58	Yes	41	$214,000
Butler Dr	Grand Ave	69th Ave	Both	0.23	Yes	41	$84,000
Claremont St	74th Ave	73rd Dr - W of	Left	0.06	No	40.5	$12,000
79th Ave	Reade Ave - S of	Colter St - S of	Left	0.18	Yes	40.5	$33,000
Glenn Dr	110th Ave	Glen Harbor Blvd	Right	0.15	No	40	$29,000
Glenn Dr	110th Ave - E of	Glen Harbor Blvd - W of	Left	0.04	No	40	$7,000
95th Ave	Camelback Rd - N of	Georgia Ave - S of	Right	0.16	No	40	$30,000

On Road	From	To	Side of Road	Miles	Potential Constraints	Prioritization Score	Planning Cost
52nd Dr	Piute Ave - N of	Oraibi Dr	Left	0.10	No	39.5	$18,000
110th Ave	Northview Ave - S of	Northview Ave	Left	0.18	No	39	$34,000
110th Ave	Glenn Dr - N of	Northview Ave - S of	Right	0.13	No	39	$25,000
55th Ave	Wahalla Ln - S of	Wahalla Ln - N of	Left	0.06	No	39	$12,000
55th Ave	Wahalla Ln - N of	Beardsley Rd	Both	0.18	No	39	$65,000
Peoria Ave	Reems Rd	Bullard Ave	Both	1.00	No	38.5	$369,000
Peoria Ave	Bullard Ave	Litchfield Rd	Right	1.00	No	38	$185,000
Northern Ave	Citrus Rd	Cotton Ln	Both	1.00	No	38	$372,000
107th Ave	Orangewood Ave	Northern Ave	Both	0.50	No	36.5	$187,000
95th Ave	Georgia Ave - N of	Bethany Home Rd	Both	0.62	No	36	$231,000
Missouri Ave	79th Ave - W of	79th Ave	Both	0.03	Yes	35.5	$12,000
79th Ave	Nicolet Ave - N of	Orangewood Ave	Both	0.12	Yes	35.5	$45,000
Missouri Ave	95th Ave	91st Ave	Both	0.50	No	34	$187,000
181st Ave	Medlock Ave	Bethany Home Rd	Left	0.85	No	33	$157,000
Northern Ave	93rd Ave - E of	91st Ave	Right	0.20	Yes	32.5	$38,000
Litchfield Rd	Olive Ave - S of	Peoria Ave	Both	1.24	Yes	32.5	$460,000
Peoria Ave	136th Ave - E of	Milgard Way- W of	Left	0.02	Yes	30.5	$3,000
				124.6 miles			**23.1 Million**

Cost estimates include construction of sidewalk only and do not include costs for design, curb ramps, right-of-way acquisition, utility relocation, and other ancillary costs.

Railroad Crossings

On Road	Cross Street	Improvement Description	Approximate Length (FT)	Prioritization Score	Planning Cost Estimate
Glendale Ave	59th Ave	Install ADA compliant pedestrian facilities at railroad crossing.	35	48	$13,400
59th Ave	Glendale Ave	Install ADA compliant pedestrian facilities at railroad crossing.	30	48	$12,500
Bethany Home Rd	Grand Ave	Install ADA compliant pedestrian facilities at railroad crossing.	90	39	$23,000
Northern Ave	Grand Ave	Install ADA compliant pedestrian facilities at railroad crossing.	120	34.5	$28,200
67th Ave	Grand Ave	Install ADA compliant pedestrian facilities at railroad crossing.	60	32	$17,700
Bethany Home Rd	Grand Ave	Install ADA compliant pedestrian facilities at railroad crossing.	80	31	$21,200
Camelback Rd	43rd Ave	Coordinate with the City of Phoenix to install ADA compliant pedestrian facilities at railroad crossing.	20	29	$10,700
Orangewood Ave	Grand Ave	Install ADA compliant pedestrian facilities at railroad crossing.	120	22	$28,200
Peoria Ave	136th Ave - E of	Install ADA compliant pedestrian facilities at railroad crossing.	90	12	$23,000
				Total	**$177,900**

Cost estimates include construction of sidewalk and placement of curb ramps only and do not include costs for design, right-of-way acquisition, utility relocation, and other ancillary costs.

Transit Connections

On Road	Cross Street	Improvement Description	Prioritization Score
Glendale Ave	Litchfield Rd - E of	Install sidewalk to bus stop	32.5
Glendale Ave	Dysart Rd - E of	Install sidewalk to bus stop	32
Glendale Ave	Dysart Rd - W of	Install sidewalk to bus stop	29
Glendale Ave	135th Ave - W of	Install sidewalk to bus stop	26.5
Glendale Ave	Glen Harbor Blvd - W of	Install sidewalk to bus stop	25
Glendale Ave	135th Ave - E of	Install sidewalk to bus stop	24.5
Glendale Ave	135th Ave - E of	Install sidewalk to bus stop	24.5
Glendale Ave	125th Ave - E of	Install sidewalk to bus stop	23
Glendale Ave	Glen Harbor Blvd - W of	Install sidewalk to bus stop	23
Glendale Ave	Glen Harbor Blvd - W of	Install sidewalk to bus stop	23
Glendale Ave	El Mirage Rd - W of	Install sidewalk to bus stop	22
Glendale Ave	129th Ave - E of	Install sidewalk to bus stop	21
Glendale Ave	El Mirage Rd - E of	Install sidewalk to bus stop	21
Glendale Ave	Glen Harbor Blvd - E of	Install sidewalk to bus stop	21
Glendale Ave	99th Ave - W of	Install sidewalk to bus stop	21
Glendale Ave	El Mirage Rd - E of	Install sidewalk to bus stop	20
Glendale Ave	Glen Harbor Blvd - W of	Install sidewalk to bus stop	20
Glendale Ave	New River Rd - W of	Install sidewalk to bus stop	20
Glendale Ave	New River Rd - E of	Install sidewalk to bus stop	19
Glendale Ave	New River Trail - E of	Install sidewalk to bus stop	19
Glendale Ave	99th Ave - W of	Install sidewalk to bus stop	19

**Planning level cost estimates provided in the sidewalk infill improvements section*

Bicycle Network Improvement Recommendations

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
47th Ave	Northern Ave	Butler Dr	0.48	Shared Lanes	If ample surface width available, install bike lane	53	Yes	62.5	$2,100
47th Ave	Butler Dr	Olive Ave	0.47	Shared lanes	Potential for bike lanes if on-street parking is converted to a bike lane	40	Yes	61.5	$2,000
61st Ave	Bethany Home Rd	Glendale Ave	0.99	Signed Bike Route		42	Yes	57	$800
55th Ave	Orangewood Ave	Northern Ave	0.49	Bike Lanes with Parking on Both Sides		50	Yes	56	$5,900
Missouri Ave	67th Ave	59th Ave	0.95	Bike Lanes with Parking on Both Sides		45	Yes	55.5	$11,400
47th Ave	Orangewood Ave	Northern Ave	0.49	Bike Lanes with Parking on Both Sides		40	Yes	55	$5,900
Maryland Ave	51st Ave	47th Ave	0.48	Shared Lanes		42	Yes	53.5	$2,100
Maryland Ave	69th Ave	67th Ave	0.24	Bike Lanes		40	Yes	53.5	$2,900
63rd Ave	Olive Ave	Peoria Ave	0.99	Bike Lanes with Parking on One Side		40	Yes	53.5	$11,900
55th Ave	Thunderbird Rd	Acoma Rd	0.46	Bike Lanes		40	Yes	53	$5,600
Orangewood Ave	51st Ave	43rd Ave	0.97	Bike Lanes with Parking on One Side		40	Yes	52.5	$11,700
Maryland Ave	67th Ave	59th Ave	0.96	Bike Lanes with Parking on One Side		40	Yes	52	$11,600
Mountain View Rd	59th Ave	55th Ave	0.49	Shared Lanes		40	Yes	52	$2,100
Maryland Ave	47th Ave	43rd Ave	0.48	Bike Lanes with Parking on One Side		40	Yes	51.5	$5,800
63rd Ave	Bethany Home Rd	Maryland Ave	0.48	Bike Lanes with Parking on Both Sides		44	Yes	51	$5,800
63rd Ave	Maryland Ave	Glendale Ave	0.48	Shared Lanes		40	Yes	51	$2,100
47th Ave	Olive Ave	Peoria Ave	0.97	Bike Lanes with Parking on One Side		40	Yes	51	$11,700
Union Hills Dr	59th Ave - E of	51st Ave	0.91	Bike Lanes	Reduce travel lanes to create bike lanes	90	No	51	$187,100

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
47th Ave	Maryland Ave	Glendale Ave	0.48	Bike Lanes with Parking on Both Sides		40	Yes	50.5	$5,800
55th Ave	Eugie Ave	Thunderbird Rd	0.19	Bike Lanes		40	Yes	50.5	$2,300
55th Ave	Northern Ave	Butler Dr	0.47	Shared Lanes		40	Yes	50	$2,000
Acoma Rd	59th Ave	51st Ave	0.98	Shared Lanes		40	Yes	50	$4,200
63rd Ave	Union Hills Dr	Utopia Rd	0.49	Bike Lanes		44	Yes	50	$5,900
Maryland Ave	79th Ave	75th Ave	0.45	Shared Lanes		36	Yes	50	$2,000
Deer Valley Rd	75th Ave	67th Ave	0.97	Upgrade sidewalk to a shared use path to serve as a low stress facility	Bike lanes current exist	65	No	50	$358,600
Missouri Ave	73rd Ave	67th Ave	0.74	Shared Lanes		40	Yes	49.5	$3,200
63rd Ave	Camelback Rd	Bethany Home Rd	1.00	Shared Lanes		44	Yes	49.5	$4,300
63rd Ave	Paradise Ln	Bell Rd	0.41	Bike Lanes	Remove on-street parking, no houses face street.	40	Yes	49	$5,000
51st Ave	Bell Rd	Union Hills Rd	1.05	Bike Lanes	Widen shared use path for low stress option	70	Yes	49	$800
47th Ave	Marshall Ave	Bethany Home Rd	0.43	Shared Lanes		40	Yes	48	$1,900
55th Ave	Mountain View Rd	Peoria Ave	0.48	Bike Lanes	Remove on-street parking, no houses face street	40	Yes	48	$5,800
51st Ave	Union Hills Rd	Beardsley Rd	0.91	Bike Lanes	Widen shared use path for low stress option	73	Yes	48	$700
47th Ave	Bethany Home Rd	Rose Ln	0.23	Bike Lanes with Parking on One Side		50	Yes	47	$2,800
71st Ave	Ocotillo Rd	Glendale Ave	0.23	Bike Lanes or Shared Lanes	If on-street parking permitted, make shared lanes	60	Yes	47	$2,800
Mountain View Rd	67th Ave	63rd Ave	0.44	Shared Lanes		40	Yes	47	$1,900
Cholla St	67th Ave	59th Ave	0.94	Bike Lanes with Parking on Both Sides		40	Yes	47	$11,300
Mountain View Rd	51st Ave	43rd Ave	0.98	Shared Lanes		40	Yes	46.5	$4,200
48th Ln	Butler Dr	Olive Ave	0.68	Signed Bike Route		40	Yes	46.5	$500
Maryland Ave	59th Ave	57th Ave	0.24	Bike Lanes		40	Yes	46	$2,900
Orangewood Ave	55th Ave	51st Ave	0.50	Bike Lanes with Parking on One Side		40	Yes	46	$6,000

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
55th Ave	Butler Dr	Olive Ave	0.47	Bike Lanes with Parking on Both Sides		53	Yes	46	$5,700
63rd Ave	Peoria Ave	Cactus Rd	0.97	Bike Lanes		40	Yes	46	$11,700
Butler Dr	51st Ave	43rd Ave	0.98	Shared Lanes		40	Yes	45.5	$4,200
71st Ave	Glendale Ave	Myrtle Ave	0.23	Bike Lanes or Shared Lanes	If on-street parking permitted, make shared lanes	60	Yes	45	$2,800
63rd Ave	Bell Rd	Greenbriar Dr	0.32	Bike Lanes	Remove on-street parking, no houses face street	40	Yes	45	$3,900
Utopia Rd	59th Ave	57th Ave	0.31	Shared Lanes		64	Yes	45	$1,400
61st Ave	Northview Ave	Alice Ave	1.43	Shared Lanes		40	Yes	44.5	$6,100
Butler Dr	55th Ave	51st Ave	0.48	Shared Lanes		40	Yes	44	$2,100
Greenway Rd	71st Ave	67th Ave	0.56	Bike Lanes		40	Yes	44	$6,800
55th Ave	Sweetwater Ave	Eugie Ave	0.43	Bike Lanes with Parking on One Side		40	Yes	44	$5,200
63rd Ave	Alice Ave - S of	Olive Ave	0.36	Bike Lanes		40	Yes	43.5	$4,400
55th Ave	Olive Ave	Mountain View Rd	0.48	Bike Lanes with Parking on Both Sides		40	Yes	43.5	$5,800
Cholla St	59th Ave	51st Ave	0.98	Bike Lanes with Parking on Both Sides		40	Yes	43.5	$11,800
47th Ave	Glendale Ave	Orangewood Ave	0.48	Bike Lanes with Parking on Both Sides		53	Yes	43	$5,800
55th Ave	Peoria Ave	Desert Cove Ave	0.24	Bike Lanes	Remove on-street parking, no houses face street	40	Yes	43	$2,900
Utopia Rd	57th Ave	55th Ave	0.31	Shared Lanes		40	Yes	43	$1,400
Glendale Ave	Litchfield Rd	Dysart Rd	0.99	Buffered Bike Lanes or Paved Shoulders	Potentially may need to widen road	70	No	43	$203,500
Maryland Ave	55th Ave	54th Ave - W of	0.07	Bike Lanes		45	Yes	43	$900
49th Ave	Olive Ave	Mountain View Rd	0.50	Signed Bike Route		40	Yes	43	$400
Country Gables Dr	63rd Ave	59th Ave	0.41	Bike Lanes	Space available to widen bike lane	56	Yes	42.5	$5,000
47th Ave	Rose Ln	Maryland Ave	0.23	Bike Lanes with Parking on One Side		40	Yes	41.5	$2,800
Mountain View Rd	55th Ave	51st Ave	0.47	Shared Lanes		32	Yes	41.5	$2,000

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
63rd Ave	Country Gables Dr	Greenway Rd	0.44	Bike Lanes	Change signs to remove parking	56	Yes	41	$5,300
55th Ave	Bell Rd	Union Hills Dr	1.06	Bike lanes	Remove on street parking	36	Yes	41	$12,800
57th Ave	Union Hills Rd - N of	Beardsley Rd	0.91	Shared Lanes		48	Yes	41	$3,900
Sweetwater Ave	Columbine Dr - W of	51st Dr - W of	0.41	Bike Lanes with Parking on One Side		40	Yes	41	$5,000
Patrick Ln	71st Ave	67th Ave	0.49	Bike Lanes		45	Yes	41	$5,900
Myrtle Ave	67th Ave	62nd Ave	0.57	Shared Lanes		40	Yes	40.5	$2,500
71st Ave	W Arrowhead Loop Rd	Deer Valley Rd	0.62	Shared Lanes		40	Yes	40.5	$2,700
Sweetwater Ave	67th Ave	59th Ave	0.94	Bike Lanes with Parking on One Side	Large lot homes, parking on both sides may not be needed	40	Yes	40	$11,300
Maryland Ave	85th Ave	83rd Ave	0.25	Bike Lanes with Parking on Both Sides		45	Yes	40	$3,000
79th Ave	Bethany Home Rd	Discovery Ave	0.34	Bike Lanes		45	Yes	39.5	$4,100
Discovery Ave	79th Dr - W of	79th Ave	0.09	Shared Lanes		42	Yes	39.5	$400
57th Ave	Talavi Blvd	Bell Rd	0.15	Bike Lanes	Remove on street parking	60	Yes	39	$1,800
63rd Ave	Greenway Rd	Paradise Ln	0.48	Bike Lanes with Parking on One Side		40	Yes	39	$5,800
Grover Ave	63rd Ave	59th Ave	0.39	Bike Lanes with Parking on One Side		40	Yes	39	$4,700
47th Ave	Peoria Ave	45th Dr - N of	0.36	Bike Lanes with Parking on One Side		40	Yes	39	$4,400
Arrowhead Loop Rd	71st Ave	67th Ave	0.49	Shared Lanes		40	Yes	39	$2,100
71st Ave	Deer Valley Rd	Hillcrest Blvd	0.24	Shared Lanes		40	Yes	39	$1,100
49th Ave	Mountain View Rd	Peoria Ave	0.50	Signed Bike Route		40	Yes	39	$400
63rd Ave	Northern Ave	Alice Ave - S of	0.61	Bike Lanes		30	Yes	38.5	$7,400
Rose Garden Ln	75th Ave	71st Ave	0.39	Shared Lanes		45	Yes	38.5	$1,700
Paradise Ln	59th Ave - E of	51st Ave - W of	0.85	Bike Lanes	If on-street parking permitted, make shared lanes	30	Yes	38	$10,200
Cholla St	51st Ave	49th Ave - E of	0.31	Bike Lanes with Parking on One Side		40	Yes	38	$3,800

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Arrowhead Loop Rd	67th Ave	63rd Ave	0.40	Shared Lanes		54	Yes	38	$1,700
Bethany Home Rd	125th Ave	El Mirage Rd	0.22	Bike Lanes		70	Yes	38	$200
49th Ave	Peoria Ave	Cholla St	0.49	Signed Bike Route		40	Yes	38	$400
Behrend Dr	63rd Ave	59th Ave	0.49	Bike Lanes	Possible Buffered Bike Lane	54	Yes	37.5	$5,900
71st Ave	Myrtle Ave	Orangewood Ave	0.25	Bike Lanes/Shared Lanes	On-street parking on west side	65	Yes	37.5	$3,000
Frier Dr	62nd Ave	61st Ave	0.12	Signed Bike Route		28	Yes	37	$100
Greenway Rd	71st Ave	Grandview Rd	0.55	Bike Lanes		44	Yes	37	$6,600
73rd Ave	Union Hills Dr	Julie Dr	0.08	Signed Bike Route		32	Yes	37	$100
57th Ave	Hartford Ave - S of	Union Hills Dr	0.94	Shared Lanes		40	Yes	37	$4,000
Oraibi Dr	55th Ave	51st Ave	0.41	Shared Lanes		40	Yes	37	$1,800
Deer Valley Rd	Arrowhead Loop Rd	59th Ave	0.30	Shared Lanes		55	Yes	37	$1,300
Hillcrest Blvd	71st Ave	67th Ave	0.51	Bike Lanes		40	Yes	37	$6,200
63rd Ave	Utopia Rd	Beardsley Rd	0.49	Bike Lanes		56	Yes	36	$5,900
Missouri Ave	85th Ave	83rd Ave	0.26	Signed Bike Route		46	Yes	36	$200
55th Ave	Cholla Rd	Desert Hills Dr - S of	0.49	Bike Lanes	Remove on-street parking, no houses face street	40	Yes	35.5	$5,900
Frier Dr	63rd Ave	62nd Ave	0.12	Signed Bike Route		25	Yes	35	$100
62nd Ave	Frier Dr	Frier Dr	0.03	Signed Bike Route		20	Yes	35	$100
Grandview Dr	73rd Dr - W of	73rd Ave	0.10	Bike Lanes		44	Yes	35	$1,200
Deer Valley Rd	67th Ave	N Arrowhead Loop Rd	0.67	Shared Lanes		55	Yes	35	$2,900
Arrowhead Loop Rd	63rd Ave	Deer Valley Rd	0.85	Shared Lanes		54	Yes	35	$3,600
Grandview Dr	73rd Ave	Bell Rd - S of	0.34	Bike Lanes		44	Yes	34	$4,100
73rd Dr	Morrow Dr	71st Ln	0.32	Signed Bike Route		32	Yes	34	$300
Julie Dr	Morrow Dr	73rd Ave	0.08	Signed Bike Route		32	Yes	34	$100
Utopia Rd	67th Ave	63rd Ave	0.36	Shared Lanes		56	Yes	34	$1,600
Utopia Rd	63rd Ave	59th Ave	0.54	Shared Lanes		56	Yes	34	$2,300

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
51st Ave	Beardsley Rd	Potter Dr	0.37	Shared Lanes		30	Yes	34	$1,600
Hillcrest Blvd	Via Montoya Dr - N of	71st Ave	0.69	Bike Lanes		40	Yes	34	$8,300
Emil Rovey Pkwy	89th Ave	83rd Ave	0.82	Shared Lanes		48	Yes	34	$3,500
55th Ave	Acoma Rd	Country Gables Dr	0.24	Bike Lanes with Parking on One Side		40	Yes	34	$2,900
55th Ave	Greenway Rd	Paradise Ln	0.48	Bike Lanes with Parking on One Side		40	Yes	33	$5,800
Maryland Ave	91st Ave	89th Ave	0.25	Bike Lanes with Parking on Both Sides	Currently striped, install signage	48	Yes	33	$200
63rd Ave	Greenbriar Dr	Grovers Ave	0.26	Bike Lanes with Parking on One Side		40	Yes	33	$3,200
55th Ave	Desert Cove Ave	Cholla St	0.24	Bike Lanes with Parking on One Side		40	Yes	32.5	$2,900
63rd Ave	Cactus Rd	Sweetwater Ave	0.47	Bike Lanes		40	Yes	32	$5,700
Glendale Ave	New River Rd	99th Ave	0.79	Buffered Bike Lanes or Paved Shoulders	Potentially may need to widen road	80	No	32	$162,400
Glendale Ave	129th Ave	El Mirage Rd	0.73	Buffered Bike Lanes or Paved Shoulders	Potentially may need to widen road	70	No	32	$150,100
62nd Ave	Behrend Dr	Beardsley Rd	0.08	Bike Lanes		55	Yes	31.5	$1,000
Grovers Ave	58th Ln	55th Ave	0.46	Bike Lanes with Parking on One Side		40	Yes	31	$5,600
Utopia Rd	75th Ave	67th Ave	1.06	Shared Lanes		56	Yes	31	$4,500
71st Ave	Hillcrest Blvd	Patrick Ln	0.52	Bike Lanes		46	Yes	31	$6,300
85th Ave	Maryland Ave	85th Ave	0.09	Shared Lanes		40	Yes	31	$400
55th Ave	Cholla St	Altadena Ave	0.19	Bike Lanes	Remove on-street parking, no houses face street	40	Yes	30.5	$2,300
71st Ave	Beardsley Rd	W Arrowhead Loop Rd	0.34	Bike Lanes		32	Yes	30.5	$4,100
Talavi Blvd	57th Ave	56th Ave	0.13	Bike Lanes	Remove on street parking	60	Yes	30	$100
Utopia Rd	79th Ave	75th Ave	0.52	Bike Lanes	Currently striped, add signage	56	Yes	30	$400
Maryland Ave	89th Ave	85th Ave	0.60	Bike Lanes with Parking on Both Sides		45	Yes	30	$7,200

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Melinda Ln	59th Ave	52nd Ave	0.89	Shared Lanes		40	Yes	29	$3,800
McLellan Rd	89th Ave	85th Ave	0.41	Shared Lanes		40	Yes	29	$1,800
Glendale Ave	115th Ave	Glen Harbor Blvd	0.73	Buffered Bike Lanes or Paved Shoulders	Potentially may need to widen road	70	No	29	$150,100
Glendale Ave	Glen Harbor Blvd	New River Rd	0.44	Buffered Bike Lanes or Paved Shoulders	Potentially may need to widen road	75	No	29	$90,500
71st Ln	Topeka Dr	Utopia Rd	0.05	Signed Bike Route		32	Yes	28	$100
Grovers Ave	55th Ave - E of	51st Ave - W of	0.40	Bike Lanes with Parking on One Side		38	Yes	28	$4,800
Glendale Ave	El Mirage Rd	115th Ave	0.97	Buffered Bike Lanes or Paved Shoulders	Potentially may need to widen road	70	No	28	$199,400
Mohawk Ln	59th Ave	56th Ave	0.43	Signed Bike Route		45	Yes	28	$400
55th Ave	Country Gables Dr	Greenway Rd	0.27	Bike Lanes with Parking on One Side		40	Yes	28	$3,300
Desert Cove Ave	49th Ave	47th Ave	0.24	Signed Bike Route		40	Yes	27.5	$200
89th Ave	Maryland Ave	89th Ave	0.10	Shared Lanes		40	Yes	27	$500
Missouri Ave	87th Ave	85th Ave	0.25	Signed Bike Route		46	Yes	27	$200
71st Ave	Patrick Ln	Pinnacle Peak Rd	0.25	Bike Lanes		31	Yes	26	$3,000
52nd Ave	Abraham Ln	Potter Dr	0.44	Shared Lanes		40	Yes	25	$1,900
Peoria Ave	Sarival Ave	Reems Rd	0.88	Paved Shoulders	Coordinate with Peoria to address improvement needs.	80	No	24	$26,400
Perryville Rd	Olive Ave	Peoria Ave	0.97	Bike Lanes	Coordinate with MCDOT to address improvement needs.	35	Yes	23	$11,700
			73.9 miles						**$2.05 Million**

Planning cost estimates include:

- *Signed Bike Route - costs for sign placement every ½ mile only*
- *Shared Lanes - costs for pavement markings placement every 250 FT only*
- *Bike Lane-*
    - *Bike lanes fit within current configuration – includes cost to add bike lanes in both directions*
    - *Corridor less than 0.50 miles – includes obliterating existing striping and restriping corridor*
    - *Corridor greater than 0.50 miles – includes slurry seal or micro seal and restriping corridor*
    - *Modifications to raised medias included in planning cost estimates*
    - *Costs do not include right-of-way acquisition, utility relocation, additional pavement, etc.*

New Bicycle Facilities

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
67th Ave	Camelback Rd	Glendale Ave	2.01	Bike Lanes	Narrow travel or center turn lane	65	No	76.5	$413,100
Glendale Ave	67th Ave	59th Ave	0.97	Protected Bike Lane	Narrow travel or center turn lane	65	No	75.5	$199,400
Bell Rd	83rd Ave	75th Ave	0.76	Protected Bike Lane	Major investment and roadway construction required	90	No	75	$2,280,000
59th Ave	Bethany Home Rd	Maryland Ave	0.5	Bike Lanes	Narrow travel or center turn lane	65	No	74	$102,800
59th Ave	Maryland Ave	Glendale Ave	0.5	Bike Lanes	Narrow travel or center turn lane	65	No	70.5	$102,800
Bell Rd	75th Ave	67th Ave	1	Protected Bike Lane	Major investment and roadway construction required	90	No	70	$3,811,500
67th Ave	Northern Ave	Cholla St	2.51	Bike Lanes	Narrow travel or center turn lane	65	No	69	$515,900
Northern Ave	59th Ave	55th Ave	0.5	Bike Lanes	Narrow travel or center turn lane	65	No	68.5	$121,500
Glendale Ave	75th Ave	67th Ave	1	Protected Bike Lane	Reduce travel lanes and median	85	No	67.5	$3,000,000
55th Ave	Maryland Ave	Myrtle Ave	0.71	Bike Lanes		42	Yes	67.5	$8,600
59th Ave	Missouri Ave	Bethany Home Rd	0.5	Bike Lanes	Narrow travel or center turn lane	65	No	66	$102,800
59th Ave	Sweetwater Ave	Thunderbird Rd	0.5	Bike Lanes	Major investment to narrow median	70	No	65.5	$2,345,300
59th Ave	Thunderbird Rd	Acoma Rd	0.47	Bike Lanes	Narrow travel or center turn lane	65	No	65	$114,300
Union Hills Dr	67th Ave	Skunk Creek Trail	0.63	Bike Lanes	Major investment to narrow median	80	No	65	$1,890,000
Missouri Ave	59th Ave	51st Ave	1.02	Bike Lanes	Narrow travel or center turn lane	40	Yes	64.5	$44,900
67th Ave	Glendale Ave	Northern Ave	0.98	Bike Lanes	Narrow travel or center turn lane	65	No	64.5	$238,200
59th Ave	Greenway Rd	Kings Ave	0.72	Bike Lanes	Narrow travel or center turn lane	65	No	64	$175,000
Ocotillo Rd	75th Ave	59th Ave	1.97	Shared Lanes		40	Yes	64	$8,400
Thunderbird Rd	67th Ave	59th Ave	0.97	Bike Lanes	Narrow travel or center turn lane	65	No	63.5	$235,800

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Northern Ave	63rd Ave	59th Ave	0.51	Bike Lanes	Narrow travel or center turn lane	65	No	63	$143,100
Myrtle Ave	Grand Ave	58th Ave	0.38	Shared Lanes		42	Yes	62	$1,700
Union Hills Dr	Skunk Creek Trail	59th Ave	0.34	Bike Lanes	Major investment to narrow median	60	No	62	$1,020,000
61st Ave	Alice Ave	Olive Ave	0.24	Bike Lanes		70	Yes	61.5	$2,900
Olive Ave	67th Ave	61st Ave	0.71	Protected Bike Lane	Narrow travel or center turn lane	65	No	61.5	$172,600
59th Ave	Cholla St	Sweetwater Ave	0.99	Bike Lanes	Narrow travel or center turn lane	65	No	61	$240,600
67th Ave	Greenway Rd	Bell Rd	0.91	Bike Lanes	Narrow travel or center turn lane	65	No	61	$187,100
Cactus Rd	67th Ave	59th Ave	0.96	Bike Lanes	Reduce median west of 63rd Ave	65	No	61	$197,300
59th Ave	Acoma Rd	Greenway Rd	0.52	Bike Lanes	Narrow travel or center turn lane	70	No	60.5	$126,400
75th Ave	Bell Rd	Grovers Ave	0.5	Bike Lanes	Requires median reduction	80	No	60	$1,500,000
Union Hills Dr	79th Ave	75th Ave	0.5	Bike Lanes	Major investment to narrow median	70	No	60	$1,500,000
Camelback Rd	99th Ave	75th Ave	3	Buffered or Protected Bike Lane		70	No	59.5	$616,500
59th Ave	Union Hills Dr	Beardsley Rd	0.99	Bike Lanes	Widen shared use path for low stress option	100	Yes	59	$10,900
Glenn Dr	59th Ave	55th Ave	0.51	Shared Lanes		50	Yes	59	$2,200
Mountain View Rd	59th Ave - W of	59th Ave	0.21	Signed Bike Route		30	Yes	58	$200
58th Ave	Lamar Dr	Glendale Ave	0.11	Shared Lanes		55	Yes	58	$500
67th Ave	Thunderbird Rd	Greenway Rd	1	Bike Lanes	Narrow travel or center turn lane	65	No	58	$205,500
Myrtle Ave	55th Ave	51st Ave	0.51	Shared Lanes		40	Yes	57.5	$2,200
65th Ave	Butler Dr	Peoria Ave	1.52	Shared Lanes			Yes	57.5	$6,500
Union Hills Dr	75th Ave	67th Ave	1	Bike Lanes	Major investment to narrow median	70	No	57	$3,000,000
67th Ave	Bell Rd	Grovers Ave - S of	0.39	Bike Lanes	Narrow travel or center turn lane	80	No	57	$32,000
75th Ave	Union Hills Dr	Beardsley Rd	1.09	Bike Lanes	Requires median reduction	65	No	57	$224,000
67th Ave	Cholla St	Thunderbird Rd	1.49	Bike Lanes	Narrow travel or center turn lane	65	No	57	$306,200
Myrtle Ave	62nd Ave	Grand Ave	0.15	Shared Lanes		40	Yes	56.5	$700

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
58th Ave	Glendale Ave	Orangewood Ave	0.5	Shared Lanes		30	Yes	56.5	$2,200
55th Ave	Myrtle Ave	Orangewood Ave	0.25	Bike Lanes		30	Yes	56.5	$3,000
Maryland Ave	75th Ave	71st Ave	0.5	Shared Lanes		20	Yes	56.5	$2,200
57th Ave	Evergreen Rd	Desert Cove Ave	1.79	Shared Lanes			Yes	56.5	$7,600
Maryland Ave	83rd Ave	80th Ave - E of	0.37	Shared Lanes		45	Yes	56	$1,600
57th Ave	Lamar Rd	Palmaire Ave	0.29	Shared Lanes		50	Yes	56	$1,300
Vogel Ave	59th Ave	55th Ave	0.51	Signed Bike Route			Yes	56	$400
61st Ave	Glendale Ave	Myrtle Ave	0.25	Signed Bike Route		35	Yes	55.5	$200
53rd Ave	Mountain View Rd	Cactus Rd	1.49	Signed Bike Route			Yes	55.5	$1,100
67th Ave	Grovers Ave - S of	Union Hills Dr	0.61	Bike Lanes	Reduce travel lanes and median	65	No	55	$1,830,000
67th Ave	Union Hills Dr	Utopia Rd	0.5	Bike Lanes	Reduce travel lanes and median	85	No	55	$1,500,000
Union Hills Dr	83rd Ave	79th Ave	0.5	Bike Lanes	Reduce travel lanes and median	120	No	55	$1,500,000
83rd Ave	Bell Rd	Union Hills Dr	1.06	Bike Lanes	Bridge at New River	65	No	55	$217,900
Greenway Rd	67th Ave	63rd Ave	0.47	Bike Lanes		65	No	55	$33,900
Orangewood Ave	59th Ave	55th Ave	0.5	Bike Lanes with Parking on One Side		40	Yes	54.5	$6,000
67th Ave	Utopia Rd	Arrowhead Loop Rd	0.82	Bike Lanes	Narrow travel or center turn lane	85	No	54	$230,100
Northern Ave	71st Ave	63rd Ave	0.97	Bike Lanes	Reduce travel lanes and median	65	No	54	$2,910,000
Litchfield Rd	Glendale Ave	Northern Ave	0.99	Bike Lanes		26	Yes	54	$11,900
Glendale Ave	99th Ave	75th Ave	3	Protected Bike Lane	Narrow travel or center turn lane	38	No	54	$841,500
75th Ave	Deer Valley Rd	74th Ave - W of	0.49	Bike Lanes		40	Yes	53.5	$21,600
51st Ave	Missouri Ave	Montebello Ave	0.25	Shared Use Path	Requires access management	65	No	53.5	$92,400

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Eugie Ave	Arizona Canal Trail/Thunderbird Paseo Park Dr	55th Ave	0.55	Shared Lanes			Yes	53.5	$2,400
Orangewood Ave	Grand Ave	59th Ave	0.48	Shared Lanes		25	Yes	52.5	$2,100
Missouri Ave	75th Ave	73rd Ave	0.25	Signed Bike Route		20	Yes	52.5	$200
Bethany Home Rd	Grand Canal Trail	79th Ave	0.49	Bike Lanes or Buffered Bike Lanes	Narrow travel or center turn lane	65	No	52.5	$35,300
79th Ave	Camelback Rd	Missouri Ave	0.5	Shared Lanes		40	Yes	52.5	$2,200
Maryland Ave	99th Ave	91st Ave	1	Bike Lanes		65	Yes	52	$12,000
59th Ave	Beardsley Rd	Melinda Ln	0.98	Bike Lanes	Widen shared use path for low stress option	100	Yes	52	$11,800
Myrtle Ave	58th Ave	55th Ave	0.36	Shared Lanes		30	Yes	51.5	$1,600
Myrtle Ave	51st Ave	48th Dr	0.3	Shared Lanes		40	Yes	51.5	$1,300
Dysart Rd	Camelback Rd	Bethany Home Rd	1	Bike Lanes	Install bike lanes on west side	78	Yes	51.5	$243,000
73rd Ave	Missouri Ave	Claremont St - N of	0.76	Signed Bike Route		30	Yes	51.5	$600
75th Ave	Glendale Ave	Glenn Dr - S of	0.11	Bike Lanes	Narrow travel or center turn lane	90	No	51.5	$8,000
67th Ave	Deer Valley Rd	Hillcrest Blvd - N of	0.36	Bike Lanes	Narrow travel or center turn lane	75	No	51	$26,000
71st Ave	Camelback Rd	Missouri Ave	0.5	Signed Bike Route		40	Yes	51	$400
57th Ave	Bell Rd	Hartford Ave	0.17	Shared Lanes		40	Yes	51	$800
Lamar Rd	58th Ave	55th Ave	0.36	Shared Lanes		42	Yes	51	$1,600
61st Ave	Myrtle Ave	Northview Ave	0.08	Shared Use Path		40	No	50.5	$29,600
75th Ave	Beardsley Rd	Rose Garden Ln	0.41	Bike Lanes	Requires median reduction	65	No	50.5	$1,230,000
75th Ave	Grovers Ave	Union Hills Dr	0.5	Bike Lanes	Requires median reduction	60	No	50	$1,500,000

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Eugie Ave	Arizona Canal Trail/Thunderbird Paseo Park Dr	55th Ave	0.55	Shared Lanes			Yes	53.5	$2,400
Orangewood Ave	Grand Ave	59th Ave	0.48	Shared Lanes		25	Yes	52.5	$2,100
Missouri Ave	75th Ave	73rd Ave	0.25	Signed Bike Route		20	Yes	52.5	$200
Bethany Home Rd	Grand Canal Trail	79th Ave	0.49	Bike Lanes or Buffered Bike Lanes	Narrow travel or center turn lane	65	No	52.5	$35,300
79th Ave	Camelback Rd	Missouri Ave	0.5	Shared Lanes		40	Yes	52.5	$2,200
Maryland Ave	99th Ave	91st Ave	1	Bike Lanes		65	Yes	52	$12,000
59th Ave	Beardsley Rd	Melinda Ln	0.98	Bike Lanes	Widen shared use path for low stress option	100	Yes	52	$11,800
Myrtle Ave	58th Ave	55th Ave	0.36	Shared Lanes		30	Yes	51.5	$1,600
Myrtle Ave	51st Ave	48th Dr	0.3	Shared Lanes		40	Yes	51.5	$1,300
Dysart Rd	Camelback Rd	Bethany Home Rd	1	Bike Lanes	Install bike lanes on west side	78	Yes	51.5	$243,000
73rd Ave	Missouri Ave	Claremont St - N of	0.76	Signed Bike Route		30	Yes	51.5	$600
75th Ave	Glendale Ave	Glenn Dr - S of	0.11	Bike Lanes	Narrow travel or center turn lane	90	No	51.5	$8,000
67th Ave	Deer Valley Rd	Hillcrest Blvd - N of	0.36	Bike Lanes	Narrow travel or center turn lane	75	No	51	$26,000
71st Ave	Camelback Rd	Missouri Ave	0.5	Signed Bike Route		40	Yes	51	$400
57th Ave	Bell Rd	Hartford Ave	0.17	Shared Lanes		40	Yes	51	$800
Lamar Rd	58th Ave	55th Ave	0.36	Shared Lanes		42	Yes	51	$1,600
61st Ave	Myrtle Ave	Northview Ave	0.08	Shared Use Path		40	No	50.5	$29,600
75th Ave	Beardsley Rd	Rose Garden Ln	0.41	Bike Lanes	Requires median reduction	65	No	50.5	$1,230,000

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Litchfield Rd	Missouri Ave	Bethany Home Rd	0.49	Bike Lanes or Shared Use Path		26	Yes	48.5	$5,900
71st Ave	Missouri Ave	Bethany Home Rd	0.5	Shared Lanes		60	Yes	48.5	$2,200
79th Ave	Union Hills Dr	Utopia Rd	0.62	Bike Lanes	Striped as paved shoulder	65	Yes	48	$2,700
Nicolet Ave	48th Dr	47th Ave	0.19	Shared Lanes		28	Yes	48	$900
83rd Ave	Missouri Ave	Bethany Home Rd - N of	0.5	Bike Lanes or Shared Use Path	Narrow travel or center turn lane	65	No	48	$102,800
55th Ave	56th Ave	Bell Rd	0.18	Bike Lanes	Remove on street parking	60	Yes	48	$2,200
Orangewood Ave	71st Ave	Grand Ave	1	Bike Lanes	Potentially may need to widen road	40	No	48	$99,000
75th Ave	Rose Garden Ln	Deer Valley Rd	0.5	Bike Lanes	Requires median reduction	80	No	47.5	$102,800
Bethany Home Rd	127th Ave	125th Ave	0.25	Bike Lanes		45	No	47.5	$3,000
Paradise Ln	63rd Ave	59th Ave	0.5	Bike Lanes	If on-street parking permitted, make shared lanes	40	Yes	47	$6,000
57th Ave	Palmaire Ave	Orangewood Ave	0.33	Shared Lanes		32	Yes	47	$1,400
Saint John Rd	79th Ave	75th Ave	0.57	Shared Lanes			Yes	47	$2,500
Missouri Ave	83rd Ave	79th Ave	0.48	Shared Lanes		25	Yes	46.5	$2,100
67th Ave	Hillcrest Blvd - N of	Pinnacle Peak Rd - N of	0.64			70	No	46	$236,600
83rd Ave	Emil Rovey Pkwy	Northern Ave	0.71	Bike Lanes		40	Yes	46	$8,600
Alice Ave	65th Ave	63rd Ave	0.23	Shared Lanes			Yes	46	$1,000
135th Ave	Missouri Ave	Bethany Home Rd	0.4	Bike Lanes		36	Yes	45.5	$4,800
Paradise Ln	67th Ave	63rd Ave	0.47	Bike Lanes	If on-street parking permitted, make shared lanes	40	Yes	45	$5,700
Butler Dr	56th Dr	55th Ave	0.15	Shared Lanes		40	Yes	44.5	$700
Missouri Ave	135th Ave	Dysart Rd	0.45	Bike Lanes		36	Yes	44.5	$5,400
Maryland Ave	71st Ave	69th Ave	0.25	Bike Lanes	Potentially may need to widen road	45	No	44.5	$375,000

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Acoma Rd	Hearn Rd - W of	Country Gables Dr	0.49	Signed Bike Route		40	Yes	44.5	$400
99th Ave	Maryland Ave	Glendale Ave	0.48	Paved Shoulders		45	No	44	$14,400
Grovers Ave	75th Ave	71st Dr - E of	0.5	Signed Bike Route		30	Yes	43	$400
73rd Ave	Grovers Ave	Union Hills Dr	0.5	Signed Bike Route		28	Yes	43	$400
El Mirage Rd	Oregon Ave - N of	Keim Dr - N of	0.75	Bike Lanes or Shared Use Path	Potentially may need to widen road	45	No	43	$74,300
Bethany Home Rd	83rd Ave	Grand Canal Trail	0.14	Bike Lanes or Buffered Bike Lanes	Narrow travel or center turn lane	65	No	43	$10,100
Kings Ave	67th Ave	63rd Ave	0.47	Signed Bike Route		40	Yes	43	$400
91st Ave	Zanjero Blvd	Orangewood Ave	0.12	Shared Use Path		60	No	43	$44,400
Alice Ave	63rd Ave	61st Ave	0.25	Bike Lanes			Yes	43	$3,000
Desert Cove Ave	57th Ave	49th Ave	1	Signed Bike Route		40	Yes	42.5	$800
57th Ave	Desert Cove Ave	Marshall Ranch Dr	0.96	Signed Bike Route			Yes	42.5	$700
99th Ave	Bethany Home Rd	Maryland Ave	0.53	Paved Shoulders		48	No	42	$15,900
Paradise Ln	69th Ave	67th Ave	0.25	Signed Bike Route		22	Yes	42	$200
71st Ave	Orangewood Ave	Northern Ave	0.5	Bike Lanes	Under construction	60	Yes	41.5	$6,000
Claremont St	73rd Ave	71st Ave	0.23	Signed Bike Route		30	Yes	41.5	$200
Grovers Ave	70th Dr - W of	67th Ave	0.49	Signed Bike Route		50	Yes	41	$400
Orangewood Ave	83rd Ave	75th Ave	1	Bike Lanes	Potentially may need to widen road	35	No	41	$99,000
El Mirage Rd	Keim Dr - N of	Glendale Ave	0.75	Bike Lanes or Shared Use Path	Potentially may need to widen road	25	No	41	$74,300
Orangewood Ave	75th Ave	71st Ave	0.51	Bike Lanes	Potentially may need to widen road	40	No	40.5	$50,500
Missouri Ave	77th Ln	75th Ave	0.39	Signed Bike Route		40	Yes	40	$300
95th Ave	Camelback Rd	Bethany Home Rd	1.01	Bike Lanes	New Road/Under Construction	60	Yes	40	$12,200
Arrowhead Lakes Dr	59th Ln	Melinda Ln	0.82	Signed Bike Route		40	Yes	40	$600

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
87th Ave	Camelback Rd	Missouri Ave	0.44	Signed Bike Route		25	Yes	39.5	$400
63rd Ave	Morten Ave	Northern Ave - S of	0.34	Bike Lanes		30	Yes	39.5	$4,100
89th Ave	Rovey Ave	Maryland Ave	0.41	Signed Bike Route		40	Yes	39	$300
71st Ave	Greenway Rd	Paradise Ln	0.34	Signed Bike Route		45	Yes	39	$300
89th Ave	Frier Dr - S of	Northern Ave	0.29	Shared Lanes		25	Yes	38	$1,300
Zanjero Blvd	Glendale Ave	91st Ave	0.6	Bike Lanes	Narrow travel or center turn lane	60	No	38	$123,300
93rd Ave	Maryland Ave	Glendale Ave	0.52	Bike Lanes or Shared Use Path	Reduce travel lanes and median	60	No	38	$1,560,000
New Road	73rd Ave	73rd Ave - E of	0.23	Shared Use Path		250	No	37	$85,100
Olive Ave	Perryville Rd	Cortessa Pkwy	0.47	Bike Lanes		50	Yes	37	$5,700
Paradise Ln	71st Ave	69th Ave	0.25	Signed Bike Route		45	Yes	37	$200
Dysart Rd	Maryland Ave	Northern Pkwy	1.49	Bike Lanes or Shared Use Path			Yes	37	$17,900
74th Ave	Irma Ln	Rose Garden Ln	0.16	Signed Bike Route		30	Yes	36.5	$200
Burton Ave	Cortessa Pkwy - W of	Cortessa Pkwy	0.12	Bike Lanes		38	Yes	36.5	$1,500
Mountain View Rd	Cortessa Pkwy	Cortessa Pkwy - E of	0.08	Bike Lanes		38	Yes	36.5	$1,000
Desert Cove Ave	63rd Ave	61st Ave	0.25	Signed Bike Route			Yes	36.5	$200
89th Ave	Emil Rovey Pkwy	Frier Dr - S of	0.19	Shared Lanes		45	Yes	36	$900
59th Ln	Arrowhead Loop Rd	Arrowhead Lakes Dr	0.27	Signed Bike Route		32	Yes	36	$200
99th Ave	Camelback Rd	Bethany Home Rd	0.97	Bike Lanes	Narrow travel or center turn lane		No	36	$199,400
Olive Ave	Citrus Rd	Cotton Ln	1	Paved Shoulders		26	No	35	$30,000
Rovey Ave	89th Ave	88th Ln - W of	0.06	Signed Bike Route		35	Yes	35	$100
Northern Ave	99th Ave - W of	91st Ave - W of	0.71	Bike Lanes or Shared Use Path		70	No	35	$146,000
Orangewood Ave	91st Ave	89th Ave	0.25	Shared Lanes		25	Yes	35	$1,100
Redfield Rd	67th Ave	Crocus Dr	0.46	Signed Bike Route		40	Yes	35	$400

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Peoria Ave	Citrus Rd	Cotton Ln	1	Paved Shoulders		30	No	35	$30,000
Tonopah Dr	55th Ave	51st Ave	0.51	Signed Bike Route		32	Yes	35	$400
56th Ave	55th Ave	Mohawk Ln	0.11	Signed Bike Route		40	Yes	35	$100
91st Ave	Orangewood Ave	Northern Ave	0.48	Shared Use Path			No	35	$177,500
Sherri Jean Ln	Greenway Rd	71st Ave	0.2	Signed Bike Route			Yes	35	$200
Olive Ave	Citrus Rd - W of	Citrus Rd	0.06	Bike Lanes		100	Yes	34	$800
Cotton Ln	Olive Ave	Peoria Ave	1	Paved Shoulders		60	No	34	$30,000
55th Ave	Country Gables Dr	Caribbean Ln	0.21	Bike Lanes with Parking on One Side		40	Yes	34	$2,600
Sarival Ave	Glendale Ave	Northern Ave	1	Paved Shoulders		26	No	34	$30,000
55th Ave	56th Ave	Tonopah Dr	0.05	Signed Bike Route		40	Yes	34	$100
185th Dr	Kolina Ln - S of	Peoria Ave	0.07	Bike Lanes		38	Yes	33	$900
Sarival Ave	Northern Ave	Olive Ave	1	Paved Shoulders		26	No	33	$30,000
Sarival Ave	Northern Ave	Olive Ave	1	Paved Shoulders		26	No	33	$30,000
Denham Dr	184th Ave	Cortessa Pkwy	0.1	Bike Lanes		38	Yes	32.5	$1,200
181st Ave	Olive Ave	Olive Ave - N of	0.08	Bike Lanes		38	Yes	32.5	$1,000
73rd Ave	Angela Dr - S of	Grovers Ave	0.26	Signed Bike Route		25	Yes	32	$200
Sarival Ave	Olive Ave	Peoria Ave	1	Paved Shoulders		26	No	32	$30,000
Mountain View Rd	180th Ave - E of	Citrus Rd	0.08	Bike Lanes		36	Yes	32	$1,000
Peoria Ave	Cortessa Pwy	Citrus Rd	0.25	Bike Lanes or Shared Use Path		48	Yes	32	$3,000
Northern Ave	107th Ave	99th Ave - W of	0.87	Bike Lanes or Shared Use Path		60	No	32	$10,500
Sarival Ave	Bethany Home Rd	Glendale Ave	1	Paved Shoulders		26	No	32	$30,000
Sarival Ave	Camelback Rd	Bethany Home Rd	1	Paved Shoulders		26	No	31	$30,000
Peoria Ave	Perryville Rd	Cortessa Pwy	0.75	Bike Lanes or Shared Use Path	Potentially may need to widen road	38	No	31	$74,300
Peoria Ave	Litchfield Rd	Dysart Rd	1	Paved Shoulders		45	No	31	$30,000
Litchfield Rd	Northern Ave	Olive Ave	0.99	Bike Lanes or Shared Use Path		26	Yes	31	$11,900
Olive Ave	Cotton Ln	Sarival Ave	1	Paved Shoulders		26	No	31	$30,000

On Road	From	To	Miles	Recommended Bike Facility	Comments	Pavement Width (FT)	Fits in Pavement	Prioritization Score	Planning Cost
Peoria Ave	Cotton Ln	Sarival Ave	1	Paved Shoulders		30	No	31	$30,000
Cotton Ln	Northern Ave	Olive Ave	1	Paved Shoulders		60	No	31	$30,000
Glen Harbor Blvd	Glendale Ave	Northern Ave	1.47	Bike Lanes or Shared Use Path			Yes	31	$17,700
Peoria Ave	Reems Rd	Bullard Ave	1	Paved Shoulders		26	No	30	$30,000
Peoria Ave	Bullard Ave	Litchfield Rd	1	Paved Shoulders		36	No	30	$30,000
Citrus Rd	Olive Ave	Mountain View Rd	0.5	Bike Lanes	Potentially may need to widen road	36	No	30	$49,500
185th Ave	Cheryl Dr - S of	Burton Ave	0.06	Bike Lanes		38	Yes	29	$800
Burton Ave	185th Ave - W of	184th Ave - E of	0.12	Bike Lanes		38	Yes	29	$1,500
Citrus Rd	Mountain View Rd	Peoria Ave	0.5	Bike Lanes	Potentially may need to widen road	36	No	29	$49,500
Citrus Rd	Northern Ave	Olive Ave	1	Bike Lanes	Potentially may need to widen road	36	No	29	$99,000
Perryville Rd	Camelback Rd	Northern Ave	3	Bike Lanes		65	Yes	29	$36,000
El Mirage Rd	Glendale Ave	Northern Ave	1	Bike Lanes or Shared Use Path	Potentially may need to widen road	25	No	29	$99,000
107th Ave	Orangewood Dr	Northern Ave	0.5	Signed Bike Route		38	Yes	29	$400
El Mirage Rd	Camelback Rd	Oregon Dr - N of	0.5	Bike Lanes or Shared Use Path	Potentially may need to widen road	48	No	28	$49,500
Northern Ave	Citrus Rd	Litchfield Rd	4.93	Paved Shoulders		26	No	27	$147,900
Northern Ave	Perryville Rd	Citrus Rd	1	Bike Lanes	Potentially may need to widen road		No	26	$99,000
Litchfield Rd	Olive Ave	Peoria Ave	1	Bike Lanes or Shared Use Path	Potentially may need to widen road	26	No	23	$99,000
Olive Ave	Sarival Ave	Dysart Rd	4.39	Paved Shoulders		24	No	23	$131,700
			142 miles						**$47.0 Million**

Planning cost estimates include:

- *Signed Bike Route - costs for sign placement every ½ mile only*
- *Shared Lanes - costs for pavement markings placement every 250 FT only*
- *Bike Lane-*
  - *Bike lanes fit within current configuration – includes cost to add bike lanes in both directions*
  - *Corridor less than 0.50 miles – includes obliterating existing striping and restriping corridor*
  - *Corridor greater than 0.50 miles – includes slurry seal or micro seal and restriping corridor*
  - *Modifications to raised medias included in planning cost estimates*
  - *Costs do not include right-of-way acquisition, utility relocation, additional pavement, etc.*

Bicycle Treatments at Intersections

On Road	Cross Street	Direction	Prioritization Score
Maryland Ave	67th Ave	W, E	67.5
Northern Ave	55th Ave	N	66
Bethany Home Rd	61st Ave	N	64
Missouri Ave	59th Ave	W	63.5
Union Hills Dr	59th Ave	E	62.5
Missouri Ave	67th Ave	W, E	62
Glendale Ave	75th Ave	N	61
Orangewood Ave	51st Ave	W, E	57.5
Maryland Ave	43rd Ave	W	57
Greenway Rd	67th Ave	W	55.5
Maryland Ave	59th Ave	E	55
Peoria Ave	63rd Ave	S, N	54.5
63rd Ave	63rd Ave	S, N	54
Northern Ave	63rd Ave	S, N	53.5
Maryland Ave	63rd Ave	W, E, S, N	53.5
Cholla St	59th Ave	E	52.5
Ocotillo Rd	63rd Ave	N	52.5
Cholla St	51st Ave	W, E	51
Missouri Ave	73rd Ave	E	49.5
Alice Ave	63rd Ave	S, N	48
Camelback Rd	63rd Ave	N	46.5
Union Hills Dr	55th Ave	S	46.5
Union Hills Dr	57th Ave	N	46.5
Union Hills Dr	57th Ave	S	46.5
Glendale Ave	71st Ave	S, N	46
Glendale Ave	Dysart Rd	W	46
Glendale Ave	Litchfield Rd	E	46
Rose Garden Ln	75th Ave	E	45.5
Arrowhead Loop Rd	67th Ave	W	45.5
Glendale Ave	Lalomai	W, E	45.5
Grovers Ave	59th Ave	E	45.5
Maryland Ave	55th Ave	E	45
Country Gables Dr	59th Ave	W	45
Glendale Ave	135th Ave - W of	E	44.5
Behrend Dr	59th Ave	W	44.5
Cholla St	55th Ave	W, E	44
Glendale Ave	El Mirage Rd	W, E	44
71st Ave	Ocotillo Rd	N	44
Deer Valley Rd	75th Ave	E	43.5
Mountain View Rd	67th Ave	E	43.5
Mountain View Rd	63rd Ave	W	43.5
Orangewood Ave	55th Ave	N, E	43.5
Mountain View Rd	43rd Ave	W	43.5
Orangewood Ave	75th Ave	S, N	43.5
Union Hills Dr	51st Ave	W	43
Utopia Rd	63rd Ave	E	43
Myrtle Ave	71st Ave	S	42.5
Myrtle Ave	67th Ave	E	41.5
Sweetwater Ave	51st Ave	W	41
Utopia Rd	59th Ave	W	41
Utopia Rd	57th Ave	W, E, S, N	41
Bell Rd	57th Ave	S	40.5

On Road	Cross Street	Direction	Prioritization Score
Butler Dr	43rd Ave	W	40
Butler Dr	47th Ave	E	40
Acoma Rd	55th Ave	E	40
Hillcrest Blvd	71st Ave	S	39.5
Acoma Rd	59th Ave	E	39.5
Behrend Dr	63rd Ave	E	39.5
Bell Rd	51st Ave	N	39.5
Deer Valley Rd	67th Ave	W	38.5
Maryland Ave	54th Ave	W, E, N	38.5
Maryland Ave	57th Ave	W, E	38.5
Paradise Ln	59th Ave	E	38.5
Bell Rd	55th Ave	N	38.5
Missouri Ave	83rd Ave	W, E	38
Beardsley Rd	71st Ave	N	37.5
Melinda Ln	59th Ave	W, E	37.5
Glendale Ave	99th Ave	W	37
Oraibi Dr	51st Ave	W	37
Paradise Ln	55th Ave	W	35.5
63rd Ave	Country Gables Dr	W, N	35
Sweetwater Ave	55th Ave	W	34.5
Grovers Ave	63rd Ave	E	34.5
Grovers Ave	55th Ave	W, E, S, N	34.5
Grovers Ave	57th Ave	W, E, S, N	34.5
Maryland Ave	85th Ave	E, S	34
Discovery Dr	79th Ave	W, E, S	34
Emil Rovey Pkwy	83rd Ave	W	34
Behrend Dr	62nd Ave	N	34
Pinnacle Peak Rd	71st Ave	S	33.5

On Road	Cross Street	Direction	Prioritization Score
Grandview Rd	Grandview Rd	S	33.5
Utopia Rd	63rd Ave	SE	33.5
Sweetwater Ave	67th Ave	E	33
Bethany Home Rd	Dysart Rd	W	32.5
Grovers Ave	51st Ave	W	32.5
Rose Garden Ln	71st Ave	W	32
Maryland Ave	91st Ave	E	32
Maryland Ave	89th Ave	W, E	31.5
Grandview Rd	73rd Ave	E	31.5
Beardsley Rd	62nd Ave	S	31
Beardsley Rd	63rd Ave	S	30.5
Emil Rovey Pkwy	89th Ave	E	30
Paradise Ln	51st Ave	W	30
Deer Valley Rd	Deer Valley Rd	E, S	30
Beardsley Rd	63rd Ave	S	29.5
Peoria Ave	Reems Rd	W	29
Peoria Ave	Bullard Ave	E	29
Beardsley Rd	57th Ave	S	29
Olive Ave	Cortessa Pkwy	N	28
Camelback Rd	Perryville Rd	N	27
Granite Vista Loop S	Cotton Ln	W	27
Peoria Ave	Perryville Rd	S	26
Seldon Ln	Citrus Rd	W	26
Granite Vista Loop N	Cotton Ln	W	26
White Tank Mountain Blvd	183rd Ave	W	25
Peoria Ave	Cortessa Pkwy	S	25
Olive Ave	Perryville Rd	N	23

Typical cost estimates for striping thermoplastic pavement markings range from $0.50 to $2.00 per linear foot. At signalized intersections with high volumes of bicycles and/or motor vehicles, especially those with frequent bicyclist left-turns and/or motorist right-turns, bike boxes should be considered for installation. Bike boxes typically cost $5,000 per box with on-going pavement maintenance requirements. Bike boxes should be painted green to help bicyclists and motorists visually identify potential conflict areas and encourage safe driving and bicycling behavior.

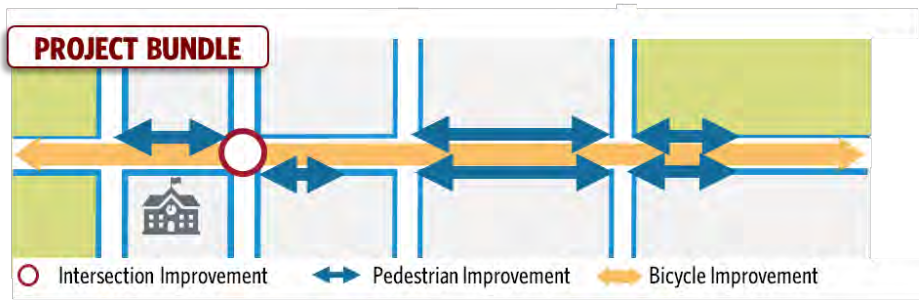
River, Canal, Freeway Crossings

On Road	Crossing	Improvement Description	Comments	Prioritization Score
Bethany Home Rd	Grand Canal Path	Conduct additional engineering analysis to determine feasibility/need for crossing on west side of canal	Provides at-grade arterial crossing connecting paved canal paths	41.5
Union Hills Dr	Skunk Creek Trail	Conduct additional engineering analysis to determine feasibility/need for crossing on west side of canal	Provides arterial crossing connecting the western paved canal path. Median pedestrian refuge island would be needed.	40.5
Thunderbird Rd	Arizona Canal Trail	Conduct additional engineering analysis to determine feasibility/need for crossing on west side of canal	Provides at-grade arterial crossing connecting paved canal paths	39
59th Ave	Arizona Canal Trail	Conduct additional engineering analysis to determine feasibility/need for crossing on south side of canal	Provides at-grade arterial crossing connecting paved canal paths	37.5
91st Ave	Grand Canal Path	Conduct additional engineering analysis to determine feasibility/need for crossing	Provides at-grade arterial crossing connecting paved canal paths. Traffic signal is current present at the intersection; however, no designated crossing exists.	37
Thunderbird Rd	Arizona Canal Trail	Conduct additional engineering analysis to determine feasibility/need for crossing on east side of canal	Provides at-grade arterial crossing connecting paved and unpaved canal paths	36
67th Ave	Arizona Canal Trail	Conduct additional engineering analysis to determine feasibility/need for crossing	Provides at-grade arterial crossing connecting paved and unpaved canal paths. Creates a long regional connection from Acoma Drive	34.5
Cactus Rd	Arizona Canal Trail	Conduct additional engineering analysis to determine feasibility/need for crossing	Provides at-grade arterial crossing connecting paved canal paths May need to widen sidewalks on Cactus Rd.	34

Regional Connections

On Road	Crossing	Improvement Description	Comments	Prioritization Score
67th Ave	Skunk Creek Trail	Install connection from sidewalks on 67th Avenue to Skunk Creek Linear Park	No connection from on-street facilities and sidewalks present to connect to trails.	34.5
Cholla St	Arizona Canal Trail	Install connection to Arizona Canal Trail	No connection from on-street facilities and sidewalks present to connect to trail.	23.5
Taro Ln	Skunk Creek Trail	Install connection to Skunk Creek Linear Park	No connection from on-street facilities and sidewalks present to connect to trail.	23
64th Ave	Grovers Ave	Install connection to Skunk Creek Linear Park	No connection from on-street facilities and sidewalks present to connect to trail.	22.5
Grandview Rd	Phelps Rd - W of	Add connection to Skunk Creek Trail	Man-made dirt path between Grandview Rd and trail currently exists. Any future extensions of Grandview Rd should include sidewalks and bicycle facilities.	22.5
Sherri Jean Ln	72nd Ave	Install connection between Sherri Jean Ln and 72nd Ave	Delineators limit access communities and to Greenway Grenada Park and the Arizona Canal Trail.	22.5
Rose Garden Ln	75th Ave - W of	Install connection to New River Trail	Man-made dirt path between Rose Garden Ln and Aspera Blvd to the New River Trail currently exists.	18.5
Pinnacle Peak Rd	74th Ave - W of	Install connection to paved trail	Man-made dirt path between Pinnacle Peak Rd and paved trail that ends at Cielo Grande currently exists.	16.5
Grovers Ave	70th Dr - W of	Install connection to connect Grovers Ave	Ditch creates barrier between Grovers Ave and trail connections to the Skunk Creek Linear Park	11.5
Ball Park Blvd	Bethany Home Rd	Install connection from Grand Canal Linear Park to Grand Canal and Ball Park Blvd	Paving and extending facilities connects regional facilities to planned developments, major regional activity centers, and the Grand Canal Linear Park	11

Project Bundles



Project "bundling" is a concept that refers to grouping adjacent improvements into a single active transportation improvement project – for instance, combining intersection, sidewalk gap, and bicycle gap improvements along a section of roadway to provide a complete active transportation network along the corridor. Project bundling allows Glendale to address multiple gaps/issues at the same time by efficiently leveraging public funds, as well as improving funding chances.

It is important to remember that all projects within the proposed network have merit and should be implemented as timely and cost-effective opportunities arise. The results of the prioritization process are not intended to preclude projects from receiving funding or from being incorporated into a new development or other roadway project.

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
67th Ave: Glendale Ave to Orangewood Ave	• 67th Ave: Palmaire Ave to Glendale Ave: Fill-in sidewalk gap on east side • 67th Ave: Glendale Ave to Myrtle Ave: Fill-in sidewalk gap on both sides • 67th Ave: Northview Ave to Orangewood Ave: Fill-in sidewalk gap on east side • Palmaire Ave: west of 67th Ave: Fill-in sidewalk gap on both sides	N/A	• 67th Ave/Orangewood Ave: Install sidewalk at SE corner, NE corner	70.5
Bethany Home Rd: 72nd Ave to 71st Ave	• Bethany Home Rd: 72nd Ave to 71st Dr: Fill-in sidewalk gap on south side • Bethany Home Rd: 71st Dr to 71st Ave: Fill-in sidewalk gap on both sides	N/A	N/A	66.5
69th Ave: Butler Dr to El	• 69th Ave: Butler Dr to Butler Dr - N of: Fill-in sidewalk gap on west side • 67th Ave: S of Alice Ave to Alice Ave: Fill-in sidewalk gap on west side • 67th Ave: S of Alice Ave to Alice Ave: Fill-in sidewalk gap on east side	N/A	N/A	66
Bell Rd: 83rd Ave to 67th Ave	N/A	• Bell Rd: 75th Ave to 67th Ave: Protected Bike Lane • Bell Rd: 83rd Ave to 75th Ave: Protected Bike Lane	• Bell Rd/70th Ave: Conduct engineering analysis to determine feasibility/need for crossing	65.5
Glendale Ave: 75th Ave and 67th Ave	• N 75th Ave: Fleetwood Ln to Glendale Ave – S of: Fill-in sidewalk gap on east side	• 75th Ave: Glendale Ave to Glenn Dr – S of: Add bike lanes • Glendale Ave: 75th Ave to 67th Ave: Buffered or protected bike lane • Glendale Ave: 67th Ave to 59th Ave: Protected Bike Lane	• Glendale Ave/75th Ave: Extend bike facility to stop bar	65.5
Missouri Ave: 59th Ave to 51st Ave	• Missouri Ave: 59th Ave to 57th Ave: Fill-in sidewalk gap on north side • Missouri Ave: 55th Ave to 55th Ave: Fill-in sidewalk gap on north side • Missouri Ave: 51st Ave to 51st Ave: Fill-in sidewalk gap on both sides	• 51st Ave: Missouri Ave to Montebello Ave: Add bike lanes • Missouri Ave: 59th Ave to 51st Ave: Add bike lanes	N/A	64.5
67th Ave: Cholla St to Greenway Rd	N/A	• 67th Ave: Thunderbird Rd to Greenway Rd: Add bike lanes • 67th Ave: Cholla St to Thunderbird Rd: Add bike lanes	• 67th Ave/Arizona Canal Trail: Conduct additional engineering analysis to determine feasibility/need for crossing	64.5
Northern Ave: 63rd Ave to 55th Ave	• Northern Ave: 63rd Ave to 61st Ave: Fill-in sidewalk gap on south side • Northern Ave: 63rd Ave to 63rd Ave – E of: Fill in sidewalk gap on both sides	• Northern Ave: 63rd Ave to 59th Ave: Add bike lanes • Northern Ave: 59th Ave to 55th Ave: Add bike lanes	N/A	63
Maryland Ave: 59th Ave to 43rd Ave	• Maryland Ave: 57th Ave – E of to 57th Ave: Fill-in gap on north side	• Maryland Ave: 55th Ave – E of to Maryland Ave: Add bike lines • Maryland Ave: 51st Ave to 47th Ave: Shared lanes • Maryland Ave: 47th Ave to 43rd Ave: Bike lanes with parking on one side • Maryland Ave: 59th Ave to 57th Ave: Add bike lanes • Maryland Ave: 55th Ave to 54th Ave – W of: Add bike lanes	• Maryland Ave: 43rd Ave: Extend bike facility to stop bar • Maryland Ave: 59th Ave: Extend bike facility to stop bar • Maryland Ave: 55th Ave: Extend bike facility to stop bar • Maryland Ave: 54th Ave: Extend bike facility to stop bar • Maryland Ave: 57th Ave: Extend bike facility to stop bar • Maryland Ave: 57th Ave: Install sidewalk at SW corner	62.5
55th Ave: Myrtle Ave to Orangewood Ave	N/A	• 55th Ave: Myrtle Ave to Orangewood Ave: Add bike lanes • 55th Ave: Maryland Ave to Myrtle Ave: Add bike lanes • 55th Ave: Orangewood Ave to Northern Ave: Bike lanes with parking on both sides	• Orangewood Ave/55th Ave: Extend bike facility to stop bar	60
Olive Ave: 67th Ave to 61st Ave	N/A	• Olive Ave: 67th Ave to 61st Ave: Protected bike lane	• Olive Ave/ 61st Ave: Conduct RSA	59.5
Eagle Pass area: Camelback Rd to Rose Ln	• 60th Ave: Missouri Ave – S of to Missouri Ave: Fill-in sidewalk gap on east side • Missouri Ave: 75th Ave to 73rd Ave	• 71st Ave: Camelback Rd to Missouri Ave: Signed bike route • 73rd Ave: Missouri Ave to Claremont St – N of: Signed bike route • Claremont St: 73rd Ave to 71st Ave: Signed bike route • Missouri Ave: 75th Ave to 73rd Ave: Signed bike route • Missouri Ave: 77th Ln to 75th Ave: Signed bike route • Missouri Ave: 67th Ave to 59th Ave: Bike lanes with parking on both sides • Missouri Ave: 73rd Ave to 67th Ave: Shared lanes	• Missouri Ave/59th Ave: Extend bike facility to stop bar • Missouri Ave/67th Ave: Extend bike facility to stop bar • Missouri Ave/ 73rd Ave: Extend bike facility to stop bar • Missouri Ave/73rd Ave: Install sidewalk at SW corner • Missouri Ave/75th Ave: Conduct engineering analysis to determine feasibility/crossing need	59
91st Ave: Camelback Rd to Camelback Rd – N of	• 91st Ave: Camelback Rd to Camelback Rd – N of: Fill-in sidewalk gap on both sides • 91st Ave: Camelback Rd to Bethany Home Rd: Fill-in sidewalk gap on east side • Bethany Home Rd: 95th Ave to 91st Ave: Fill-in sidewalk gap on south side • 91st Ave: Bethany Home Rd to Bethany Home Rd: Fill-in sidewalk gap on east side	N/A	• Grand Canal/91st Ave: Conduct additional engineering analysis to determine feasibility/need for crossing	59
Glendale Ave/59th Ave	N/A	N/A	• Glendale Ave/59th Ave: Install ADA compliant pedestrian facilities at railroad crossing • 59th Ave/Glendale Ave: Install ADA compliant pedestrian facilities at railroad crossing	59
59th Ave: Sweetwater Ave to Thunderbird Rd	N/A	• 59th Ave: Cholla St to Sweetwater Ave: Add bike lanes • 59th Ave: Sweetwater Ave to Thunderbird Rd: Add bike lanes	• 59th Ave/Arizona Canal Trail: Conduct additional engineering analysis to determine feasibility/need for crossing	58.5

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
59th Ave: Thunderbird Rd to Kings Ave	N/A	59th Ave: Thunderbird Rd to Kings Ave: Add bike lanes	59th Ave/Monte Cristo Ave: Conduct additional engineering analysis to determine feasibility/need for crossing	58.5
55th Ave: Sweetwater Ave to Paradise Ln	N/A	• 55th Ave: Voltaire Dr to Thunderbird: Add bike lanes • 55th Ave: Country Gables Dr to Caribbean Ln: Bike lanes with parking on one side • 55th Ave: Thunderbird Rd to Acoma Rd: Add bike lanes • 55th Ave: Greenway Rd to Paradise Ln: Add bike lanes with parking on one side • 55th Ave: Eugie Ave to Thunderbird Rd: Add bike lanes • 55th Ave: Sweetwater Ave to Eugie Ave: Bike Lanes with Parking on one side • 55th Ave: Country Gables Dr to Greenway Rd: Bike lanes with parking on one side • 55th Ave: Acoma Rd to Country Gables Dr: Bike lanes with parking on one side • Sweetwater Ave: Columbine Dr to 51st Dr: Bike lanes with parking on one side	• Sweetwater Ave/51st Ave: Extend bike facility to stop bar • Sweetwater Ave/55th Ave: Extend bike facility to stop bar • 55th Ave/Voltaire Dr: Conduct engineering analysis to determine feasibility/need for crossing	57.5
Union Hills Dr: 83rd Ave to 67th Ave	Union Hills Dr: 75th Ave to 70th Ave: Fill-in sidewalk gap on south side	• Union Hills Dr: 83rd Ave to 67th Ave: Add bike lanes	N/A	57
Union Hills Dr: 67th Ave to 59th Ave	N/A	• Union Hills Dr: 67th Ave to 59th Ave: Add bike lanes	Union Hills Dr/Skunk Creek Trail: Conduct engineering analysis to determine feasibility/need for crossing	57
63rd Ave: Morten Ave to Sweetwater Ave	• Estero Ln: Grand Ave to Morten Ave: Fill-in sidewalk gap on north side • 63rd Ave: Morten Ave to Hayward Ave: Fill-in sidewalk gap on both sides • 63rd Ave: Hayward Ave to Northern Ave: Fill-in sidewalk gap on east side • 63rd Ave: Butler Dr to Alice Ave: Fill-in sidewalk gaps on east side • 63rd Ave: Northern Ave to S of Butler Dr: Fill-in sidewalk gap on west side • 63rd Ave: Northern Ave to Northern Ave – N of: Fill-in sidewalk gap on both sides • 63rd Ave: Northern Ave to Butler Dr: Fill-in sidewalk gap on west side	• 63rd Ave: Morten Ave to Olive Ave: Shared lanes • 63rd Ave: Olive Ave to Peoria Ave: Bike lanes with parking on one side • 63rd Ave: Peoria Ave to Sweetwater Ave: Shared lanes	• Peoria Ave/63rd Ave: Extend bike facility to stop bar • Northern Ave/63rd Ave: Extend bike facility to stop bar • Alice Ave/63rd Ave: Extend bike facility to stop bar	57
Maryland Ave: 67th Ave to 59th Ave	• 61st Ave: Berridge Ln to Keim Dr: Fill-in sidewalk gap on east side	• 61st Ave: Bethany Home Rd to Glendale Ave: Signed bike route • Maryland Ave: 67th Ave to 59th Ave: bike lanes with parking on one side	• Bethany Home Rd/61st Ave: Extend bike facility to stop bar • Maryland Ave/63rd Ave: Extend bike facility to stop bar	57
63rd Ave: Camelback Rd to Bethany Home Rd	N/A	• 63rd Ave: Camelback Rd to Bethany Home Rd: Shared lanes • 63rd Ave: Bethany Home Rd to Maryland Ave: Bike lanes with parking on both sides • 63rd Ave: Maryland Ave to Glendale Ave: Shared lanes	• Ocotillo Rd/63rd Ave: Extend bike facility to stop bar • Camelback Rd/63rd Ave: Extend bike facility to stop bar	57
58th Ave: Glendale Ave to Orangewood Ave	• 57th Ave: Ocotillo Rd to Lamar Rd – S of: Fill-in sidewalk gap on both sides • Ocotillo Rd: 57th Ave to 56th Ave: Fill-in sidewalk gap on south side	• 57th Ave: Lama Rd to Orangewood Ave: Shared lanes • 58th Ave: Lama Rd to Orangewood Ave: Shared lanes • 59th Ave: Glendale Ave to Glenn Dr: Add bike lanes • Glenn Dr: 59th Ave to 55th Ave: Shared lanes • Lamar Rd: 58th Ave to 55th Ave: Shared lanes	N/A	56.5
Camelback Rd: 99th Ave to 75th Ave	• Camelback Rd: 84th Ln to 83rd Ave – W of: Fill-in sidewalk gap on north side • Camelback Rd: 91st Ave to 87th Ave: Fill-in sidewalk gap on north side • Camelback Rd: 95th Ave to 93rd Ave – W of: Fill-in sidewalk gap on north side • Camelback Rd: 91st Ave to 91st Ave: Fill-in sidewalk gap on north side	• Camelback Rd: 99th Ave to 75th Ave: Buffered or protected bike lane	• Camelback Rd/87th Ave: Install sidewalk at NW corner	56.5
Camelback Rd: Litchfield Rd to 127th Ave	• Camelback Rd: Litchfield Rd to 129th Ave: Fill-in sidewalk gap on both sides • Camelback Rd: 129th Ave to 127th Ave: Fill-in sidewalk gap on north side	N/A	N/A	
61st Ave: Glendale Ave to Alice Ave	• 61st Ave: Morten Ave to Augusta Ave: Fill-in sidewalk gap on both sides • 61st Ave: Augusta Ave to Northern Ave: Fill-in sidewalk on west side • 61st Ave: Gardenia Ave to Morten Ave: Fill-in sidewalk gap on east side • 61st Ave: Orangewood Ave to Morten Ave: Fill-in sidewalk gap on both sides	• 61st Ave: Glendale Ave to Myrtle Ave: Signed bike route • 61st Ave: Myrtle Ave to Northview Ave: Add bike lanes • 61st Ave: Northview Ave to Alice Ave: Shared lanes	N/A	55
83rd Ave: 55th to Union Hills Dr	• 83rd Ave: Campo Bello Dr to Union Hills Dr – S of: Fill-in sidewalk gap on west side	• 83rd Ave: Bell Rd to Union Hills Dr: Add bike lanes	N/A	55
Orangewood Ave: Grand Ave to 55th Ave	• Orangewood Ave: Grand Ave to 59th Ln: Fill-in sidewalk gap on both sides	• Orangewood Ave: Grand Ave to 55th Ave: Shared lanes	• Orangewood Ave/75th Ave: Extend bike facility to stop bar	54.5

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
Myrtle Ave: 55th Ave to 48th Dr	N/A	• Myrtle Ave: 55th Ave to 48th Dr: Shared lanes	• Nicolet Ave/48th Dr: Install sidewalk at SE corner, NE corner	54.5
Northern Ave: 71st Ave to 63rd Ave	• Northern Ave: 68th Ave to 68th Ave: Fill-in sidewalk gap on south side • Northern Ave: 63rd Ave to 63rd Ave: Fill-in sidewalk gap on north side • Northern Ave: 70th Ave to 68th Ave: Fill-in sidewalk gap on both sides • Northern Ave: 71st Ave to 70th Ave: Fill-in sidewalk gap on north side	• Northern Ave: 71st Ave to 63rd Ave: Add bike lanes	N/A	54
Solano Dr: 67th Ave to 65th Ave	• Solano Dr: 67th Ave to 65th Ave: Fill-in sidewalk gap on north side	N/A	• Solano Dr/65th Ave: Install sidewalk at NW corner	54
Maryland Ave: 79th Ave – W of to 67th Ave	• Maryland Ave: 69th Ave to 67th Ave: Fill-in sidewalk gap on north side • Maryland Ave: 74th Ave to 73rd Ave: Fill-in sidewalk gap on south side	• 79th Ave: Bethany Home Rd to Discovery Dr: Add bike lanes • Discovery Dr: 79th Dr to 79th Ave: Shared lanes • Maryland Ave: 71st Ave to 69th Ave: Add bike lanes • Maryland Ave: 75th Ave to 71st Ave: Shared lanes • Maryland Ave: 69th Ave to 67th Ave: Add bike lanes • Maryland Ave: 79th Ave to 75th Ave: Shared lanes	• Maryland Ave/67th Ave: Extend bike facility to stop bar • 75th Ave/Maryland Ave: Conduct engineering analysis to determine feasibility/need for crossing • Discovery Dr/79th Ave: Extend bike facility to stop bar	53.5
Thunderbird Rd: 67th Ave to 59th Ave	• Thunderbird Rd: 65th Ave to 65th Ave: Fill-in sidewalk gap on south side • Thunderbird Rd: 63rd Ave to Arizona Canal Trail: Fill-in sidewalk gap on south side • Thunderbird Rd: 67th Ave to 65th Ave: Fill-in sidewalk gap on north side	• Thunderbird Rd: 67th Ave to 59th Ave: Add bike lanes	• Thunderbird Rd/56th Ave: Conduct engineering analysis to determine feasibility/need for crossing • Thunderbird Rd/Thunderbird Rd: Conduct additional engineering analysis to determine feasibility/need for crossing • Thunderbird Rd/Arizona Canal Trail: Conduct additional engineering analysis to determine feasibility/need for crossing • Thunderbird Rd/Thunderbird Rd: Conduct additional engineering analysis to determine feasibility/need for crossing	53.5
Myrtle Ave: 71st Ave to Grand Ave	• Myrtle Ave: 72nd Ave to 71st Ave: Fill-in sidewalk gap on north side • 69th Ave: Glenn Dr to Myrtle Ave: Fill-in sidewalk ap on west side	• Myrtle Ave: 71st Ave to Grand Ave: Shared lanes	• Myrtle Ave/67th Ave: Extend bike facility to stop bar	53
71st Ave: Missouri Ave to Northern Ave	• 71st Ave: Orangewood Ave to Frier Dr: Fill-in sidewalk gap on east side • 71st Ave: Montebello Ave to Bethany Home Rd: Fill-in sidewalk gap on west side • 71st Ave: Bethany Home Rd to Claremont St: Fill-in sidewalk gap on west side	• 71st Ave: Missouri Ave to Northern Ave: Add bike lanes • 71st Ave: Missouri Ave to Bethany Home Rd: Shared lanes	• Glendale Ave/71st Ave: Extend facility to stop bar • 71st Ave/Ocotillo Rd: Extend facility to stop bar • Myrtle Ave/71st Ave: Extend facility to stop bar	52.5
Butler Dr: 56th Dr to 43rd Ave	N/A	• Butler Dr: 56th Dr to 43rd Ave: Shared lanes	• Butler Dr/43rd Ave: Extend facility to stop bar • Butler Dr/47th Ave: Extend facility to stop bar	52.5
67th Ave: Union Hills Dr to Deer Valley Rd	• 67th Ave: Sack Dr to Utopia Rd: Fill-in sidewalk gap on east side • Sack Dr: 67th Ave to 67th Ave: Fill-in sidewalk gap on north side	• 67th Ave: Union Hills Dr to Deer Valley Rd: Add bike lanes	• Sack Dr/67th Ave: Install sidewalk at NE corner	52
Greenway Rd: 67th Ave to 63rd Ave	N/A	• Greenway Rd: 67th Ave to 63rd Ave: Add bike lanes	• 63rd Ave/Greenway Rd: Conduct engineering analysis to determine feasibility/need for equestrian crossing	52
Dysart Rd: Camelback Rd to Northern Pkwy	• Dysart Rd: Maryland Ave to Ocotillo Rd: Fill-in gap on west side • Dysart Rd: Ocotillo Rd to Glendale Ave: Fill-in sidewalk gap on west side • Dysart Rd: Colter St to Rose Ln: Fill-in sidewalk gap on both sides	• Dysart Rd: Camelback Rd to Bethany Home Rd: Add bike lanes • Dysart Rd: Maryland Ave to Northern Pkwy: Add bike lanes	N/A	51.5
55th Ave: Northern Ave to Butler Dr	• 55th Ave: Northern Ave to Butler Dr: Fill-in sidewalk gap on east side	• 55th Ave: Northern Ave to Butler Dr: Shared lanes • 55th Ave: Mountain View Rd to Peoria Ave: Add bike lanes • 55th Ave: Olive Ave to Mountain View Rd: Bike lanes with parking on both sides • 55th Ave: Butler Dr to Olive Ave: Bike lanes with parking on both sides • 55th Ave: Peoria Ave to Desert Cove Ave: Add bike lanes • 55th Ave: Desert Cove Ave to Cholla St: Bike lanes with parking on both sides	• Northern Ave/55th Ave: Extend bike facility to stop bar	51.5
67th Ave: Bell Rd to Union Hills Dr	N/A	• 67th Ave: Bell Rd to Union Hills Dr: Add bike lanes	• 67th Ave/Skunk Creek Trail: Install connection from sidewalks on 67th Ave to Skunk Creek linear park	51

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
53rd Ave: Mountain View Rd to Cactus Rd	• Brown St: 51st Ave to 49th Ave: Fill-in sidewalk gap on north side • 49th Ave: Brown St to Peoria Ave: Fill-in sidewalk gap on west side • 53rd Ave: Peoria Ave to Peoria Ave: Fill-in sidewalk gap on east side	• 53rd Ave: Mountain View Rd to Cactus Rd: Signed bike route • 61st Ave: Peoria Ave to Cholla St: Signed bike route • Desert Cove Ave: 63rd Ave to 61st Ave: Signed bike route	• Peoria Ave/53rd Ave: Conduct engineering analysis to determine feasibility/need for crossing • Peoria Ave/49th Ave: Conduct engineering analysis to determine feasibility/need for crossing • Peoria Ave/60th Ave: Conduct engineering analysis to determine feasibility/need for crossing	50.5
75th Ave: Bell Rd to Union Hills Dr	• 75th Ave: Grovers Ave to Union Hills Dr: Fill-in sidewalk gap on both sides • 75th Ave: Village Dr to Grovers Ave: fill-in sidewalk gap on east side	• 75th Ave: Bell Rd to Union Hills Dr: Add bike lanes	N/A	50
83rd Ave: Missouri Ave to Northern Ave	• 83rd Ave: Glendale Ave to Palmaire Ave: Fill-in sidewalk gap on west side • 83rd Ave: Palmaire Ave to Nicolet Ave - S of: Fill-in sidewalk gap on east side • 83rd Ave: Belmont Ave to Northern Ave: Fill-in sidewalk gap on both sides • 83rd Ave: Orangewood Ave to Vista Ave - N of: Fill-in sidewalk gap on both sides • 83rd Ave: Vista Ave to Vista Ave: Fill-in sidewalk gap on east side • 83rd Ave: Montebello Ave to Bethany Home Rd: Fill-in sidewalk gap on both sides • 83rd Ave: Montebello Ave to Montebello Ave - N of: Fill-in sidewalk gap on west side • 83rd Ave: Grand Canal Trail to Cavalier Dr: Fill-in sidewalk gap on west side • 83rd Ave: Orange Dr to Missouri Ave: Fill-in sidewalk gap on west side	• 83rd Ave: Maryland Ave to Emil Rovey Pkwy: Add bike lanes • 83rd Ave: Missouri Ave to Maryland Ave: Add bike lanes • 83rd Ave: Emil Rovey Pkwy to Northern Ave		50
59th Ave: Union Hills Dr to Melinda Ln	• 59th Ave: Behrend Dr to Beardsley Rd: Fill-in sidewalk gap on east side • 59th Ave: Union Hills Dr to Wescott Dr: Fill-in sidewalk gap on east side	• 59th Ave: Union Hills Dr to Melinda Ln: Add bike lanes	• 59th Ave/Arrowhead Lakes Dr: Conduct engineering analysis to determine feasibility/need for crossing	50
Area surrounding 81st Ave: Ocotillo Rd to Palmaire Ave – N of	• 81st Ave: Ocotillo Rd to Glendale Ave: Fill-in sidewalk gap on west side • Ocotillo Rd: 81st Ave to 81st Ave: Fill-in sidewalk gap on north side • 80th Dr: Glendale Ave to Glendale Ave - N of: Fill-in sidewalk gap on east side • 81st Ave: Glendale Ave to Palmaire Ave - N of: Fill-in sidewalk gap on east side • Lamar Rd: 81st Ln to 81st Dr: Fill-in sidewalk gap on south side • Lamar Rd: 81st Dr to 81st Ave: Fill-in sidewalk gap on both sides	N/A	• Ocotillo Rd/81st Ave: Install sidewalk at NW corner	49.5
Litchfield Rd: Camelback Rd – N of to Peoria Ave	• Litchfield Rd: Glendale Ave to Northern Ave: Fill-in sidewalk gap on both sides • Litchfield Rd: Northern Ave to Northern Ave - N of: Fill-in sidewalk gap on both sides • Litchfield Rd: Glendale Ave to Lightning St: Fill-in sidewalk gap on west side • Litchfield Rd: Super Sabre St to Glendale Ave: Fill-in sidewalk gap on both sides • Litchfield Rd: Claremont St to Super Sabre St: Fill-in sidewalk gap on west side • Litchfield Rd: Peck Dr to Claremont St: Fill-in sidewalk gap on both sides • Litchfield Rd: Missouri Ave to Peck Dr: Fill-in sidewalk gap on west side • Litchfield Rd: Olive Ave to Peoria Ave: Fill-in sidewalk gap on both sides	• Litchfield Rd: Bethany Home Rd to Peoria Ave: Add bike lanes	N/A	49
61st Ave: Cholla St to Sweetwater Ave	• 61st Ave: Cholla St to Sunnyside Dr: Fill-in sidewalk gap on east side	• 61st Ave: Cholla St to Sweetwater Ave: Signed bike route	N/A	49
Eugie Ave: Arizona Canal Trail/Thunderbird Paseo Dr to 55th Ave	• Thunderbird Paseo Dr: Eugie Ave to Eugie Ave: Fill-in sidewalk gap on east side	• Eugie Ave: Arizona Canal Trail/Thunderbird Paseo Dr to 55th Ave: Shared lanes	• Eugie Ave/57th Dr: Conduct engineering analysis to determine feasibility/need for crosswalk	49
Mountain View Rd: 59th Ave - W of to 43rd Ave	• Mountain View Rd: 53rd Ave to 51st Ave: Fill-in sidewalk gap on north side	• Mountain View Rd: 59th Ave – W of to 59th Ave: Signed bike route • Mountain View Rd: 59th Ave to 43rd Ave: Shared lanes	• Mountain View Rd/43rd Ave: Extend bike facility to stop bar	48.5
Butler Dr: Grand Ave to 63rd Ave	• Butler Dr: 65th Ave to 63rd Ave: Fill-in sidewalk gap on south side • Butler Dr: Grand Ave to 69th Ave: Fill-in sidewalk gap on both sides	• Butler Dr: Grand Ave to 63rd Ave: Shared lanes	N/A	48.5
63rd Ave: Paradise Ln to Grovers Ave	N/A	• 63rd Ave: Paradise Lane to Grovers Ave: Add bike lanes • 63rd Ave: Greenbriar Dr to Grovers Ave: Bike Lanes with Parking on One Side	• 63rd Ave/63rd Ave: Extend bike facility to stop bar	48
75th Ave: Beardsley Rd to Hillcrest Blvd	• 75th Ave: Aurora Dr to Aurora Dr - S of: Fill-in sidewalk gap on west side	• 75th Ave: Rose Garden Ln to Deer Valley Rd: Add bike lanes • 75th Ave: Deer Valley Rd to Hillcrest Blvd: Shared lanes • 75th Ave: Beardsley Rd to Rose Garden Ln: Add bike lanes	N/A	47.5
Area surrounding 60th Ave: Gardenia Ave to Augusta Ave	• Gardenia Ave: 61st Ave to 59th Ln: Fill-in sidewalk gap on south side • 59th Ln: Gardenia Ave to Orangewood Ave: Fill-in sidewalk gap on east side • Gardenia Ave: 59th Ln to 59th Dr: Fill-in sidewalk gap on east side • 59th Ln: Gardenia Ave to Gardenia Ave: Fill-in sidewalk gap on both sides • Augusta Ave: 61st Ave to 59th Ln: Fill-in sidewalk gap on south side • 60th Ave: Orangewood Ave to Augusta Ave: Fill-in sidewalk gap on both sides	N/A	Augusta Ave/59th Ln: Install sidewalk at SW corner	47.5

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
48th Dr: Myrtle Ave to Orangewood Ave	• 48th Dr: Myrtle Ave to Nicolet Ave: Fill-in sidewalk gap on both sides • Nicolet Ave: 48th Dr to 48th Dr - E of: Fill-in sidewalk gap on south side • Nicolet Ave: 48th Dr to 48th Dr - E of: Fill-in sidewalk gap on both sides • 48th Dr: Nicolet Ave to Northview Ave - S of: Fill-in sidewalk gap on east side • 48th Dr: Myrtle Ave to Myrtle Ave - S of: Fill-in sidewalk gap on east side • Nicolet Ave: 47th Ave to 47th Ave: Fill-in sidewalk gap on south side	• 48th Dr: Myrtle Ave to Orangewood Ave: Shared lanes • Nicolet Dr: 48th Dr to 47th Ave: Shared lanes	• Myrtle Ave/48th Dr: Install sidewalk at NW corner	47
Area surrounding Maryland Ave: 91st Ave to 80th Ave – E of	N/A	• 85th Ave: Maryland Ave to 85th Ave: Shared lanes • 89th Ave: Rovey Ave to Maryland Ave: Signed bike route • 89th Ave: Maryland Ave to 89th Ave: Shared lanes • Maryland Ave: 83rd Ave to 80th Ave - E of: Shared lanes • Maryland Ave: 91st Ave to 83rd: Bike Lanes with Parking on Both Sides • McLellan Rd: 89th Ave to 85th Ave: Shared lanes • Rovey Ave: 89th Ave to 88th Ln - W of: Signed bike route	• Maryland Ave/ 85th Ave: Extend bike facility to stop bar • Maryland Ave/91st Ave: Extend bike facility to stop bar • Maryland Ave/ 89th Ave: Extend bike facility to stop bar	47
Desert Cove Ave: 57th Ave to 49th Ave	• 49th Ave: Shangri La Rd to Cholla St: Fill-in sidewalk gap on east side • Desert Cove Ave: 59th Ave to 57th Ave: Fill-in sidewalk gap on south side	• 49th Ave: Butler Dr to Cholla St: Signed bike route • Desert Cove: 57th Ave to 47th Ave: Signed bike route	• Desert Cove Ave/51st Ave: Conduct engineering analysis to determine feasibility/need for crossing	46.5
Cholla St: 67th Ave to 49th Ave – E of	• Cholla St: 49th Ave to 49th Ave – E of: Fill-in sidewalk gap on south side	• Cholla St: 51st Ave to 49th Ave: Bike Lanes with Parking on One Side • Cholla St: 67th Ave to 51st Ave: Bike Lanes with Parking on Both Sides	• Cholla St/59th Ave: Extend bike facility to stop bar • Cholla St/ 51st Ave: Extend bike facility to stop bar • Cholla St/ 55th Ave: Extend bike facility to stop bar • Cholla St/61st Ave: Conduct engineering analysis to determine feasibility/need for crosswalk • Cholla St/Arizona Canal Trail: Install connection to Arizona Canal Trail	46.5
Rose Garden Ln: 75th Ln – W of to 75th Ave	• Rose Garden Ln: 75th Ln to 75th Ave: Fill-in sidewalk gap on south side	N/A	• Rose Garden Ln/75th Ave - W of: Install connection to New River Trail	46.5
Area surrounding Missouri Ave: 87th Ave to 79th Ave	• Missouri Ave: 80th Ave to 79th Ave - W of: Fill-in sidewalk gap on north side • Missouri Ave: 79th Ave to 79th Ave: Fill-in sidewalk gap on both sides • 87th Ave: Camelback Rd to Missouri Ave: Fill-in sidewalk gap on west side • 79th Ave: Reade Ave to Colter St - S of: Fill-in sidewalk gap on west side	• 79th Ave: Camelback Rd to Missouri Ave: Shared lanes • 87th Ave: Camelback Rd to Missouri Ave: Signed bike route • Missouri Ave: 83rd Ave to 79th Ave: Shared lanes • Missouri Ave: 87th to 83rd Ave: Signed bike route	• Missouri Ave/83rd Ave: Extend bike facility to stop bar	46
Saint John Rd/Grovers Ave: 79th Ave to 71st Dr – E of	• Angela Dr: 75th Ave to 71st Dr - E of: Fill-in sidewalk gap on both sides	• Grovers Ave: 75th Ave to 71st Dr - E of: Signed bike route • Saint John Rd: 79th Ave to 75th Ave: Shared lanes	• 75th Ave/Grovers Ave: Conduct engineering analysis to determine feasibility/need for crossing • Grovers Ave/70th Dr - W of: Install connection to connect Grovers Ave	45
79th Ave: Saint John Rd to Union Hills Dr	N/A	• 79th Ave: Saint John Rd – S o f to Union Hills Dr: Add bike lanes • 79th Ave: Union Hills Dr to 79th Ave: Shared lanes	• 79th Ave/Saint John Rd: Conduct engineering analysis to determine feasibility/need for crossing • 79th Ave/Campo Bello Dr: Conduct engineering analysis to determine feasibility/need for crossing	45
99th Ave: Bethany Home Rd to Glendale Ave	• 99th Ave: Glendale Ave to Desert River Blvd - S of: Fill-in sidewalk gap on west side • 99th Ave: Desert River Blvd to Desert River Blvd: Fill-in sidewalk gap on east side	• 99th Ave: Bethany Home Rd to Glendale Ave: Add paved shoulders	N/A	44
57th Ave: Talavi Blvd to Union Hills Dr	N/A	• 57th Ave: Talavi Blvd to Union Hills Dr: Remove on street parking	• Union Hills Dr/57th Ave: Extend bike facility to stop bar • Bell Rd/57th Ave: Extend bike facility to stop bar	44
Orangewood Ave: 83rd Ave to Grand Ave	• Orangewood Ave: 75th Ave to 75th Ave - E of • Orangewood Ave: 72nd Ave to 71st Ave - W of • Orangewood Ave: 71st Ave to 69th Ave • Orangewood Ave: 67th Ave to Grand Ave - W of • Orangewood Ave: Grand Ave to Grand Ave • Orangewood Ave: 73rd Ave to 73rd Ave • Orangewood Ave: 78th Ave to 75th Ave • Orangewood Ave: 82nd Dr to 78th Ave - W of • Orangewood Ave: 83rd Ave to 82nd Dr - E of	• Orangewood Ave: 71st Ave to Grand Ave: Add bike lanes • Orangewood Ave: 83rd Ave to 75th Ave: Add bike lanes • Orangewood Ave: 75th Ave to 71st Ave: Add bike lanes	• Orangewood Ave/Grand Ave: Install ADA compliant pedestrian facilities at railroad crossing	44
Northern Ave/Grand Ave	N/A	N/A	• Northern Ave/Grand Ave: Install ADA compliant pedestrian facilities at railroad crossing • Northern Ave/67th Ave: Install ADA compliant pedestrian facilities at railroad crossing	43.5

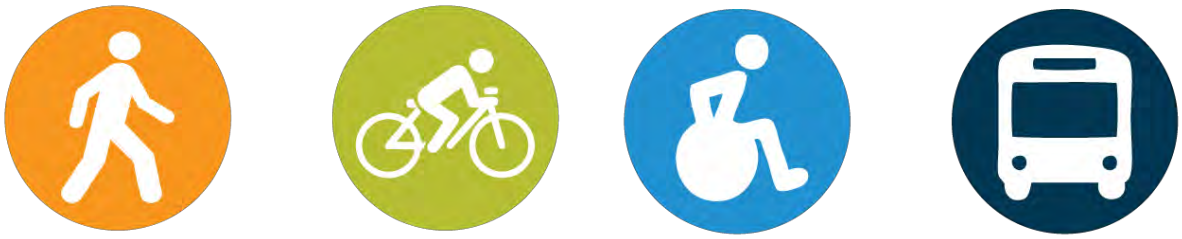
General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
Glendale Ave: Litchfield Rd to El Mirage Rd	• Glendale Ave: E of Litchfield Rd: Fill-in sidewalk gap on south side • Glendale Ave: E of Litchfield Rd: Fill-in sidewalk gap on north side • Glendale Ave: W of 137th Ave: Fill-in sidewalk gap on both sides • Glendale Ave: E of 137th Ave: Fill-in sidewalk gap on north side • Glendale Ave: 135th Ave to Dysart Rd: Fill-in sidewalk gap on both sides • Glendale Ave: Dysart Rd to El Mirage Rd: Fill-in sidewalk gap on both sides	• Glendale Ave: Dysart Rd to 129th Ave: Add bike lanes • Glendale Ave: 129th Ave to El Mirage Rd: Add bike lanes • Glendale Ave: Litchfield Rd to Dysart Rd: Add bike lanes	• Glendale Ave/Dysart Rd: Extend bike facility to stop bar • Glendale Ave/Litchfield Rd: Extend bike facility to stop bar • Glendale Ave/Lalomai: Extend bike facility to stop bar • Glendale Ave/135th Ave: Extend bike facility to stop bar • Glendale Ave/El Mirage Rd: Install sidewalk at NW corner, SW corner, SE corner, NE corner • Glendale Ave/Litchfield Rd: Install sidewalk to bus stop • Glendale Ave/Dysart Rd: Install sidewalk to bus stop • Glendale Ave/Dysart Rd: Install sidewalk to bus stop • Glendale Ave/135th Ave: Install sidewalk to bus stop • Glendale Ave/135th Ave: Install sidewalk to bus stop • Glendale Ave/135th Ave: Install sidewalk to bus stop • Glendale Ave/125th Ave: Install sidewalk to bus stop • Glendale Ave/El Mirage Rd: Install sidewalk to bus stop • Glendale Ave/129th Ave: Install sidewalk to bus stop	42
Bethany Home Rd: 83rd Ave to 79th Ave	• Bethany Home Rd: 83rd Ave to 79th Ave: Fill-in sidewalk gap on north side	• Bethany Home Rd: 83rd Ave to 79th Ave: Buffered or Protected Bike Lane	• Bethany Home Rd/Grand Canal Path: Conduct additional engineering analysis to determine feasibility/need for crossing	42
Orangewood Ave: 55th Ave to 43rd Ave	N/A	• Orangewood Ave: 55th Ave to 43rd Ave: Bike Lanes with Parking on One Side	• Orangewood Ave/51st Ave: Extend bike facility to stop bar	42
Union Hills Dr: 59th Ave – E of to 51st Ave	• Union Hills Dr: 53rd Ave to 52nd Ave: Fill-in sidewalk gap on north side	• Union Hills Dr: 59th Ave - E of to 51st Ave: Widen sidewalk to create shared use path	• Union Hills Dr/59th Ave: Extend bike facility to stop bar • Union Hills Dr/51st Ave: Extend bike facility to stop bar	42
Bethany Home Rd: Dysart Rd to El Mirage Rd	• Bethany Home Rd: 127th Ave to 125th Ave: Fill-in sidewalk gap on north side • Bethany Home Rd: 129th Ave to Maryland Ave: Fill-in sidewalk gap on south side	• Bethany Home Rd: Dysart Rd to El Mirage Rd: Add paved shoulders	• Bethany Home Rd/127th Ave: Install sidewalk at SW corner, NE Corner	41.5
Acoma Rd: 59th Ave to 51st Ave	N/A	• Acoma Rd: 59th Ave to 51st Ave: Shared lanes	• Acoma Rd/55th Ave: Extend bike facility to stop bar • Acoma Rd/59th Ave: Extend bike facility to stop bar	41.5
Grovers Ave: 70th Dr – W of to 67th Ave	• Grovers Ave: 70th Dr to 67th Ave: Fill-in sidewalk gap on north side	• Grovers Ave: 70th Dr - W of to 67th Ave: Signed bike route	N/A	41
93rd Ave: Maryland Ave to Orangewood Ave	• Orangewood Ave: 91st Ave to 89th Ave: Fill-in sidewalk gap on south side	• 91st Ave: Zanjero Blvd to Orangewood Ave: Widen sidewalk to create shared use path • 93rd Ave: Maryland Ave to Glendale Ave: Add bike lanes • Zanjero Blvd: Glendale Ave to 91st Ave: Add bike lanes • Emil Rovey Pkwy: 89th Ave to 83rd Ave: Shared lanes • Orangewood Ave: 91st Ave to 89th Ave: Shared lanes	• Emil Rovey Pkwy/83rd Ave: Extend bike facility to stop bar • Emil Rovey Pkwy/89th Ave: Extend bike facility to stop bar	41
Patrick Ln: 71st Ave to 67th Ave	• Patrick Ln: 67th Ave - W of to 67th Ave: Fill-in sidewalk gap on north side	• Patrick Ln: 71st Ave to 67th Ave: Add bike lanes	N/A	41
73rd Ave: Angela Dr – S of to Union Hills Dr	N/A	• 73rd Ave: Angela Dr – S of to Grovers Ave: Signed bike route • 73rd Ave: Grovers Ave to Union Hills Dr: Signed bike route	• Union Hills Dr/73rd Ave: Conduct engineering analysis to determine feasibility/need for crossing	40
95th Ave: Camelback Rd to Bethany Home Rd	• 95th Ave: Camelback Rd to Georgia Ave - S of: Fill-in sidewalk gap on east side • 95th Ave: Georgia Ave to Bethany Home Rd: Fill-in sidewalk gap on both sides • Missouri Ave: 95th Ave to 91st Ave: Fill-in sidewalk gap on both sides	• 95th Ave: Camelback Rd to Bethany Home Rd: Add bike lanes	N/A	40
Country Gables Dr: 63rd Ave to 59th Ave	N/A	• 63rd Ave: Country Gables Dr to Greenway Rd: Add bike lanes • Country Gables Dr: 63rd Ave to 59th Ave: Add bike lanes	• Country Gables Dr/59th Ave: Extend bike facility to stop bar • 63rd Ave/ Country Gables Dr: Extend bike facility to stop bar	40
Greenway Rd: 67th Ave to Grandview Rd	N/A	• Greenway Rd: 67th Ave to Grandview Rd: Add bike lanes	• Greenway Rd/67th Ave: Extend bike facility to stop bar • Greenway Rd/71st Ave: Conduct engineering analysis to determine feasibility/need for crosswalk	40
63rd Ave: Union Hills Dr to Utopia Rd	N/A	• 63rd Ave: Union Hills Dr to Utopia Rd: Bike Lanes	• Utopia Rd/63rd Ave: Extend bike facility to stop bar	40
Deer Valley Rd: 75th Ave to 67th Ave	N/A	• Deer Valley Rd: 75th Ave to 67th Ave: Widen sidewalk to create shared use path to create a low stress facility	• Deer Valley Rd/75th Ave: Extend bike facility to stop bar • Deer Valley Rd/67th Ave: Extend bike facility to stop bar	40

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
Utopia Rd/Oraibi Dr: 59th Ave to 51st Ave	• 55th Ave: Wahalla Ln to Wahalla Ln • 55th Ave: Wahalla Ln to Beardsley Rd • Utopia Rd: 57th Ave to 54th Ln • 52nd Dr: Piute Ave to Oraibi Dr	• Oraibi Dr: 55th Ave to 51st Ave: Shared lanes • Utopia Rd: 59th Ave to 57th Ave: Shared lanes • Utopia Rd: 57th Ave to 55th Ave: Shared lanes	• Oraibi Dr/51st Ave: Extend bike facility to stop bar • Utopia Rd/57th Ave: Install sidewalk at SE corner, NE corner • Taro Ln/Skunk Creek Trail: Install connection to Skunk Creek Linear Park	39
Grandview Rd: Phelps Rd – W of to Bell Rd	N/A	• Grandview Rd: Phelps Rd – W of to 73rd Ave – W of: Add bike lanes • Grandview Rd: 73rd Ave – E of to Bell Rd: Add bike lanes	• Grandview Rd/Grandview Rd: Extend bike facility to stop bar • Grandview Rd/73rd Ave: Extend bike facility to stop bar • Grandview Rd/Phelps Rd: Add connection to Skunk Creek Trail	38
51st Ave: Union Hills Dr to Beardsley Rd	N/A	• 51st Ave: Union Hills Rd to Beardsley Rd: Widen shared use path for low stress options	• 51st Ave/Oraibi Dr: Conduct engineering analysis to determine feasibility/need for crossing	38
51st Ave: Bell Rd to Union Hills Dr	N/A	• 51st Ave: Bell Rd to Union Hills Rd: Widen shared use path for low stress options	• Bell Rd/51st Ave: Extend bike facility to stop bar	38
Northern Ave: Citrus Rd to Cotton Ln	• Northern Ave: Citrus Rd to Cotton Ln: Fill-in sidewalk gap on both sides	N/A	• Northern Ave/Citrus Rd: Install sidewalk at NW corner, SW corner, SE corner, NE corner	38
Peoria Ave: Perryville Rd to Dysart Rd	• Peoria Ave: Reems Rd to Bullard Ave: Fill-in sidewalk gap on both sides • Peoria Ave: Bullard Ave to Litchfield Rd: Fill-in sidewalk gap on south side • Peoria Ave: 185th Dr to 183rd Ave: Fill-in sidewalk gap on south side • Peoria Ave: 136th Ave to Milgard Wy- W of: Fill-in sidewalk gap on north side	• Peoria Ave: Reems Rd to Litchfield Rd: Add paved shoulders • Peoria Ave: Perryville Rd to Citrus Rd: Add bike lanes • Peoria Ave: Citrus Rd to Reems Rd: Add paved shoulders	• Peoria Ave/Reems Rd: Extend bike facility to stop bar • Peoria Ave/Bullard Ave: Extend bike facility to stop bar • Peoria Ave/Citrus Rd: Install sidewalk at SW corner • Peoria Ave/Reems Rd: Install sidewalk at SE corner • Peoria Ave/136th Ave: Install ADA compliant pedestrian facilities at railroad crossing	37
Mountain View Rd: 67th Ave to 63rd Ave	N/A	• Mountain View Rd: 67th Ave to 63rd Ave: Shared lanes	• Mountain View Rd/ 67th Ave: Extend bike facility to stop bar • Mountain View Rd/63rd Ave: Extend bike facility to stop bar	37
55th Ave: Bell Rd to Union Hills Dr	• 55th Ave: Bell Rd to Union Hills Dr: Shared lanes	• 55th Ave: Bell Rd to Union Hills Dr: Remove on street parking	• Union Hills Dr/55th Ave: Extend bike facility to stop bar • Bell Rd/55th Ave: Extend bike facility to stop bar	37
Grovers Ave: 63rd Ave to 55th Ave	N/A	• Grovers Ave: 58th Ln to 55th Ave: Bike Lanes with Parking on One Side • Grovers Ave: 63rd Ave to 59th Ave: Bike Lanes with Parking on One Side	• Grovers Ave/59th Ave: Extend bike facility to stop bar • Grovers Ave/63rd Ave: Extend bike facility to stop bar • Grovers Ave/55th Ave: Extend bike facility to stop bar • Grovers Ave/57th Ave: Extend bike facility to stop bar	37
Area surrounding Rose Garden Ln: 75th Ave to 71st Ave	N/A	• 74th Ave: Irma Ln to Rose Garden Ln: Signed bike route • Rose Garden Ln: 75th Ave to 71st Ave: Shared lanes	• Rose Garden Ln/75th Ave: Extend bike facility to stop bar • Rose Garden Ln/71st Ave: Extend bike facility to stop bar	36.5
57th Ave: Desert Cove Ave to Marshall Ranch Dr	• 57th Ave: Shaw Butte Dr – N of to Cactus Rd: Fill-in sidewalk gap on east side	• 57th Ave: Desert Cover Ave to Marshall Ranch Dr: Signed bike route	• Cactus Rd/Arizona Canal Trail: Conduct additional engineering analysis to determine feasibility/need for crossing	36
55th Ave: Cholla St to Cactus Rd	N/A	• 55th Ave: Cholla St to Cactus Rd: Bike lanes	N/A	35.5
Northern Ave: 107th Ave to 99th Ave – E of	• Northern Ave: Loop 101 - E of to 93rd Ave - E of: Fill-in sidewalk gap on south side • Northern Ave: 93rd Ave - E of to 91st Ave: Fill-in sidewalk gap on south side	• Northern Ave: 99th Ave to 91st Ave: Add shared use path • Northern Ave: 107th Ave to 99th Ave: Add shared use path	N/A	35
Sweetwater Ave: 67th Ave to 59th Ave	• 65th Dr: Sweetwater Ave to Willow Ave - S of: Fill-in sidewalk gap on both sides	• Sweetwater Ave: 67th Ave to 59th Ave: Bike Lanes with Parking on One Side	• Sweetwater Ave/67th Ave: Extend bike facility to stop bar	35
91st Ave: Orangewood Ave to Northern Ave	• 91st Ave: Northern Ave - S of to Northern Ave: Fill-in sidewalk gap on both sides • 91st Ave: Orangewood Ave to Orangewood Ave - N of: Fill-in sidewalk gap on both sides	• 91st Ave: Orangewood Ave to Northern Ave: Widen sidewalk to create shared use path	N/A	35
Area surrounding Missouri Ave: 135th Ave to Dysart Rd	• Bethany Home Rd: Dysart Rd to 129th Ave: Fill-in sidewalk gap on both sides	• 135th Ave: Missouri Ave to Bethany Home Rd: Add bike lanes • Missouri Ave: 135th Ave to Dysart Rd: Add bike lanes	• Bethany Home Rd/Dysart Rd: Extend bike facility to stop bar • Bethany Home Rd/137th Ave: Conduct engineering analysis to determine feasibility/need for crosswalk • Bethany Home Rd/135th Ave: Conduct engineering analysis to determine feasibility/need for crosswalk	34.5
Paradise Ln: 59th Ave – E of to 51st Ave – W of	N/A	• Paradise Ln: 59th Ave to 51st Ave - W of: Add bike lanes	• Paradise Ln/59th Ave: Extend bike facility to stop bar • Paradise Ln/55th Ave: Extend bike facility to stop bar • Paradise Ln/51st Ave: Extend bike facility to stop bar	34

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
Behrend Dr: 63rd Ave to 59th Ave	N/A	• Behrend Dr: 63rd Ave to 59th Ave: Add bike lanes	• Behrend Dr/63rd Ave: Extend bike facility to stop bar • Behrend Dr/ 59th Ave: Extend bike facility to stop bar	34
Area surrounding Perryville Rd: Olive Ave to Peoria Ave	• Perryville Rd: Olive Ave to Peoria Ave: Add bike lanes • Olive Ave: Perryville Rd to Cortessa Pkwy: Add bike lanes • Olive Ave: Citrus Rd to Citrus Rd: Add bike lanes	• Olive Ave: 181st Ave to Citrus Rd: Fill-in sidewalk gap on north side	• Olive Ave/Citrus Rd: Install sidewalk at NW corner, SE corner, NE corner • Peoria Ave/Perryville Rd: Extend bike facility to stop bar • Olive Ave/Perryville Rd: Extend bike facility to stop bar	33
71st Ave: 67th Ave to Hillcrest Blvd	• 71st Ave: W Arrowhead Loop Rd to Deer Valley Rd: Shared lanes	• 71st Ave: 67TH Ave to 71st Ave: Shared lanes	• Arrowhead Loop Rd/67th Ave: Extend bike facility to stop bar • Hillcrest Blvd/71st Ave: Extend bike facility to stop bar	32.5
89th Ave: Orangewood Ave to Northern Ave	• 89th Ave: Orangewood Ave to Northern Ave: Shared lanes	• 89th Ave: Frier Dr - N of to Northern Ave: Fill-in sidewalk gap on west side • 89th Ave: State Ave to Orangewood Ave: Fill-in sidewalk gap on west side	• Northern Ave/89th Ave: Conduct engineering analysis to determine feasibility/need for crossing	32
57th Ave: Union Hills Dr - N of to Beardsley Rd	• 57th Ave: Union Hills Dr to Utopia Rd - S of: Fill-in sidewalk gap on west side • 57th Ave: Utopia Rd to Beardsley Rd: Fill-in sidewalk gap on east side	• 57th Ave: Union Hills Dr to Beardsley Rd: shared lanes	• Union Hills Dr/ 57th Ave: Extend bike facility to stop bar • Utopia Rd/57th Ave: Extend bike facility to stop bar • Beardsley Rd/57th Ave: Extend bike facility to stop bar	32
Sherri Jean Ln: 73rd Ave to 71st Ave	• 69th Ln: Caribe Ln to Port Au Prince Ln: Fill-in sidewalk gap on west side • Beck Ln: 72nd Ave to 67th Ave: Fill-in sidewalk gap on both sides • 71st Ave: Carol Ann Way to Sherri Jean Ln: Fill-in sidewalk gap on both sides • 66th Ave: Greenway Rd to Paradise Ln: Fill-in sidewalk gap on both sides • 69th Ave: Greenway Rd to Judy Lynn Ln: Fill-in sidewalk gap on both sides	• Sherri Jean Ln: 73rd Ave to 71st Ave: Signed bike route	• Sherri Jean Ln/72nd Ave: Install connection between Sherri Jean Ln and 72nd Ave	32
Waddell area between Olive Ave and Peoria Ave	• 181st Ave: Olive Ave to Olive Ave: Fill-in sidewalk gap on east side	• 181st Ave: Olive Ave to Olive Ave - N of: Add bike lanes • 185th Ave: Cheryl Dr to Burton Ave: Add bike lanes • 185th Dr: Kolina Ln to Burton Ave: Add bike lanes • Burton Ave: 185th Ave to 184th Ave - E of: Add bike lanes • Burton Ave: Cortessa Pkwy to Cortessa Pkwy: Add bike lanes • Denham Dr: 184th Ave to Cortessa Pkwy: Add bike lanes • Mountain View Rd: Cortessa Pkwy to Cortessa Pkwy - E of: Add bike lanes • Mountain View Rd: 180th Ave to Citrus Rd: Add bike lanes	• Olive Ave/Cortessa Pkwy: Extend bike facility to stop bar • Sheldon Ln/Citrus Rd: Extend bike facility to stop bar • White Tank Mountain Blvd/183rd Ave: Extend bike facility to stop bar • Peoria Ave/Cortessa Pkwy: Extend bike facility to stop bar	31
Utopia Rd: 63rd Ave to 59th Ave	N/A	• Utopia Rd: 63rd Ave to 59th Ave: Shared lanes	• Utopia Rd/63rd Ave: Extend bike facility to stop bar • Utopia Rd/59th Ave: Extend bike facility to stop bar	31
Glen Harbor Blvd: Glendale Ave to Northern Ave	• Glen Harbor Blvd: Glenn Dr to Northview Ave - W of: Fill-in sidewalk gap on west side • Glen Harbor Blvd: Northview Ave to Vista Ave: Fill-in sidewalk gap on west side • Glen Harbor Blvd: Vista Ave to Vista Ave: Fill-in sidewalk gap on both sides • Glen Harbor Blvd: 106th Ave to 106th Ave: Fill-in sidewalk gap on both sides • Glen Harbor Blvd: 106th Ave to 106th Ave: Fill-in sidewalk gap on east side • Glen Harbor Blvd: Northern Ave to Northern Ave: Fill-in sidewalk gap on east side • Glen Harbor Blvd: Northern Ave to Northern Ave: Fill-in sidewalk gap on both sides • Glen Harbor Blvd: Glendale Ave to Glenn Dr: Fill-in sidewalk gap on west side	• Glen Harbor Blvd: Glendale Ave to Northern Ave: Add bike lanes	N/A	31
62nd Ave: Behrend Dr to Beardsley Rd	N/A	• 62nd Ave: Behrend Dr to Beardsley Rd: Add bike lanes	• Behrend Dr/62nd Ave: Extend bike facility to stop bar • Beardsley/62nd Ave: Extend bike facility to stop bar	30.5
Citrus Rd: Northern Ave to Peoria Ave	• Citrus Rd: Olive Ave to Olive Ave - N of: Fill-in sidewalk gap on both sides • Citrus Rd: Northern Ave to Butler Ave: Fill-in sidewalk gap on both sides	• Citrus Rd: Northern Ave to Peoria Ave: Add bike lanes	N/A	30
Cotton Ln: Northern Ave to Peoria Ave	N/A	• Cotton Ln: Northern Ave to Peoria Ave: Add paved shoulders	• Granite Vista Loop S/Cotton Ln: Extend bike facility to stop bar • Granite Vista Loop N/Cotton Ln: Extend bike facility to stop bar	30
Utopia Rd/63rd Ave: 67th Ave to Beardsley Rd	N/A	• 63rd Ave: Utopia Rd to Beardsley Rd: Bikes Lanes • Utopia Rd: 67th Ave to 63rd Ave: Shared lanes	• Beardsley Rd/63rd Ave: Extend bike facility to stop bar	30
Utopia Rd: 63rd Ave to 59th Ave	N/A	• Utopia Rd: 63rd Ave to 59th Ave: Shared lanes	• Utopia Rd/63rd Ave: Extend bike facility to stop bar • Utopia Rd/59th Ave: Extend bike facility to stop bar	31

General Location	Sidewalk Infill Project	Bicycle Improvement	Spot Improvement	Prioritization Score
107th Ave: Orangewood Dr to Northern Ave	• 107th Ave: Orangewood Dr to Northern Ave: Fill-in sidewalk gap on both sides	• 107th Ave: Orangewood Dr to Northern Ave: Signed bike route	N/A	29
Perryville Rd: Camelback Rd to Northern Ave	N/A	• Perryville Rd: Camelback Rd to Northern Ave: Add bike lanes	• Camelback Rd/Perryville Rd: Extend bike facility to stop bar	27
71st Ave: Hillcrest Blvd to Pinnacle Peak Rd	• Pinnacle Peak Rd: 69th Dr to 67th Ave - W of: Fill-in sidewalk gap on south side • Pinnacle Peak Rd: 71st Ave to 69th Dr: Fill-in sidewalk gap on south side • Pinnacle Peak Rd: 72nd Ave to 71st Ave: Fill-in sidewalk gap on south side • Pinnacle Peak Rd: 74th Ave to 72nd Ave: Fill-in sidewalk gap on south side	• 71st Ave: Hillcrest Ave to Pinnacle Peak Rd: Add bike lanes	• Pinnacle Peak Rd/71st Ave: Extend bike facility to stop bar	27
71st Ave: Beardsley Rd to Arrowhead Loop Rd	N/A	• 71st Ave: Beardsley Rd to Arrowhead Loop Rd: Add bikes lanes	• Beardsley Rd/71st Ave: Extend bike facility to stop bar	26.5
Melinda Ln: Beardsley Rd to 59th Ave	N/A	• 51st Ave: Beardsley Rd to Potter Dr • 52nd Ave: Abraham Ln to Potter Dr • Melinda Ln: 59th Ave to 52nd Ave	• Melinda Lane/59th Ave: Extend bike facility to stop bar	24
Grovers Ave: 55th Ave – W of to 51st Ave – E of	N/A	• Grovers Ave: 55th Ave to 51st Ave: Nike Lanes with Parking on one SideGle	• Grovers Ave/51st Ave: Extend bike facility to stop bar	24
Glendale Ave: El Mirage Rd to 99th Ave	• Glendale Ave: El Mirage Rd to 99th Ave: Fill-in sidewalk gap on both sides	• Glendale Ave: El Mirage Rd to 99th Ave: Add bike lanes	• Glendale Ave/El Mirage Rd: Extend bike facility to stop bar • Glendale Ave/99th Ave: Extend bike facility to stop bar • Glendale Ave/Glen Harbor Blvd - W of: Install sidewalk to bus stop • Glendale Ave/W of New River Rd: Install sidewalk to bus stop • Glendale Ave/New River Trail: Install sidewalk to bus stop	20

WALK. BIKE. CONNECT



glendale **active** **transportation plan**

the way we move

Appendix B.
Facility Design Guidance

The Design Guidelines of the Glendale Active Transportation Plan are intended to assist in designing transportation systems that serve all users. The guidelines will help to implement the Glendale Active Transportation Plan and create a community with accessible and comfortable active transportation options for people of all ages and abilities. The design recommendations are not intended to impose standards, but to offer guidance. The guidelines listed here are a collection of national and regional design resources, best practices, methods, and guidelines which help the design of facilities and amenities. This list is not exhaustive. Glendale should consider consulting publications directly to ensure that the most current version is used. Arizona Department of Transportation and Maricopa Association of Governments should also be consulted, especially when improvements are proposed on a state-owned roadway.

Pedestrian Facilities and Treatments

To create a pedestrian network that works for all users, a series of facilities and treatments are needed to create a connected, complete, and accessible network. The following section outlines potential pedestrian facilities and treatment types to assist Glendale in the selection and design of pedestrian facilities.

Sidewalk



Sidewalks are the most fundamental element of the pedestrian network. Providing adequate and accessible facilities can lead to increased numbers of people walking, improved safety, and the creation of social space. When retrofitting gaps in the sidewalk network, locations near transit stops, schools, parks, public buildings, and other areas with high concentrations of pedestrians should be the highest priority.

Design Guidance and Considerations

- 6 FT sidewalks on collector or arterial roadways. A pedestrian through zone width of 6 FT enables two pedestrians (including wheelchair users) to travel side-by-side, or to pass each other comfortably.
- In areas of high demand, sidewalks should contain adequate width to accommodate high volumes and different walking speeds of pedestrians. For example, a minimum 8 FT wide sidewalk (10 FT preferred) is recommended adjacent to schools.
- 5 FT minimum in areas with low pedestrian volume and constrained right-of-way.
- Should be built to Americans with Disabilities Act (ADA) standards.
- High quality transitions/crossings are advised such as refuge islands, rectangular rapid flashing beacons, pedestrian hybrid beacons, high visibility crosswalks, lighting, etc.
- Minimum sidewalk clear widths should be free of sign posts, utility and signal cabinets, pull boxes and poles, mailboxes, fire hydrants and other obstructions.

Planning Cost Estimates

- $30 – $40 per Linear Foot (LF)
    - Price varies by the size of the project
    - Does not include cost for right-of-way acquisition, utility relocation, and other ancillary costs
    - Does not include 10% contingency and 20% design

Shared Use Path



Shared use paths are paved facilities physically separated from motorized vehicular traffic by an open space or barrier. Shared use paths are used by bicyclists, pedestrians, skaters, runners and others. Since shared use paths serve as pedestrian facilities, they must comply with ADA standards.

Design Guidance and Considerations

- 10 FT minimum width is recommended in most situations and will be adequate for moderate to heavy use. 12 FT width is recommended for heavy use situations with high concentrations of multiple users such as joggers, bicyclists, rollerbladers and pedestrians.
- Shared use paths along roadways are best in areas with few access points or roadways crossing the path. Where a shared use path terminates, special consideration should be given to transitions so as not to encourage unsafe wrong way riding by bicyclists.

Planning Cost Estimates

- $60 – $80 per LF
    - Price varies by the size of the project
    - Does not include 10% contingency, 20% design, and costs for right-of-way acquisition, utility relocation, and other ancillary costs

Crossings and Intersections

Crossings are potential conflict points between pedestrians and vehicles and thus require design treatments that set clear expectations for all roadway users. Every intersection in Glendale should be designed for pedestrian safety and comfort, with pedestrian enhancements appropriate for the roadway conditions. The specific type of treatment at a crossing may range from simple, marked crosswalks to full traffic signals or grade-separated crossings. Crosswalk lines should not be used indiscriminately, and appropriate selection and implementation of crossing treatments should be evaluated by an engineer or an engineering study before installation.

Marked Crosswalk



A marked crosswalk signals to motorists that they must stop for pedestrians and encourages pedestrians to cross at designated locations. Installing crosswalks alone, however, will not necessarily make crossings safer. Marked crosswalks at signalized intersections should be determined by an engineering assessment.

Design Guidance and Considerations

- Marked crosswalks at signalized and unsignalized intersections should be determined by an engineering assessment.
- Because the effectiveness of marked crossings depends entirely on their visibility, maintaining marked crossings should be a high priority. Thermoplastic markings offer more durability than conventional paint.
- Consider consolidating arterial mid-block crosswalks and make them safer through various efforts.

Planning Cost Estimates

- $770 each. Crosswalks using other materials such as brick range from $2,500 to $5,000 each.

High-Visibility Crosswalk



High-visibility crosswalks may include a variety of crosswalk striping designs, such as ladder, continental, or bar pairs. A high-visibility crosswalk is much easier for an approaching motorist to see than traditional parallel lines. Research has found that high-visibility markings were detected at about twice the distance upstream as the transverse markings during daytime conditions.

Design Guidance and Considerations

- The high-visibility markings may be supplemented with the pedestrian crossing warning signs (sign W11-2 in the MUTCD) on each approach to the crosswalk.
- Should be used at intersections with conflicts between vehicle and pedestrian movements, areas with high pedestrian traffic, and in commercial business districts.
- An engineering study should be performed before a marked crosswalk is installed at a location away from a traffic control signal or an approach controlled by a STOP or YIELD sign.

Planning Cost Estimates

- $2,500 each

Raised Crosswalk



Raised crosswalks function as an extension of the sidewalk and allow pedestrians to have a greater prominence as they cross the street. Raised crossings make the pedestrian more prominent in the driver's field of vision. Additionally, approach ramps may reduce vehicle speeds and improve motorist yielding.

Design Guidance and Considerations

- Typically installed on 2-lane or 3-lane roads with speed limits of 30 mph or less and annual average daily traffic (AADT) below about 9,000.
- The crosswalk table is typically at least 10 FT wide and designed to allow the front and rear wheels of a passenger vehicle to be on top of the table at the same time.
- Raised crossings are generally avoided on truck routes, emergency routes, and arterial streets.
- Use detectable warnings at the curb edges to alert vision-impaired pedestrians that they are entering the roadway.

Planning Cost Estimates

- $8,000 each. Costs will vary based on the width of the road, as well as drainage conditions and the type of material used.

Median Refuge Island



Median refuge islands are located at the mid-point of a marked crossing and help improve path user safety by allowing pedestrians to cross one direction of traffic at a time. Refuge islands minimize pedestrian exposure by shortening crossing distance and increasing the number of available gaps for crossing.

Design Guidance and Considerations

- May be appropriate on multi-lane roadways depending on speeds and volumes. Consider configuration with active warning beacons for improved yielding compliance.
- Appropriate at signalized or unsignalized crosswalks.
- Cut-through median refuge islands are preferred over curb ramps to better accommodate wheelchair users.
- Requires 6 FT width between travel lanes (8 -10 FT preferred to accommodate bikes with trailers and wheelchair users). Clear width of 4 FT required, but preferably same width as crosswalk.
- Refuge islands may collect road debris and may require somewhat frequent maintenance.
- If a refuge island is landscaped, the landscaping should not compromise the visibility of pedestrians crossing in the crosswalk.

Planning Cost Estimates

- $6,000 to $10,000 each. The cost for installing a raised concrete refuge island with landscaping ranges from $10,000 to $30,000.

Offset Crossings



Offset crossings share similar characteristics with standard median refuge islands with one major difference – a staggered crossing. This offset is done to help orient the pedestrian toward oncoming traffic on the approach to each leg of the crossing. Eye contact between the pedestrian and motorist is improved at an offset crossing, which improves communication between the two roadway users.

Design Guidance and Considerations

- Offset crossings can be used in place of a standard median refuge island at most locations where the latter is appropriate; however, offset median refuge islands are typically installed at mid-block locations where there is an existing center turn lane or median at least 6 FT wide.
- Offset crossings include refuge islands which may collect road debris and may require somewhat frequent maintenance. Trees and plantings must be maintained so as not to impair visibility.

Planning Cost Estimates

Costs vary widely depending on the length and type, ranging from $10,000 to $40,000.

Rectangular Rapid Flash Beacon (RRFB)



RRFBs are user-actuated amber LEDs that supplement warning signs at unsignalized intersections or midblock crosswalks. They can be activated by pedestrians manually by a push button or passively by a pedestrian detection system. The device includes two rectangular-shaped yellow indications, each with an LED-array-based light source, that flash with high frequency when activated. RRFBs have been shown to increase motor vehicle yielding compliance at crossings of multi-lane or high-volume roadways.

Design Guidance and Considerations

- Use RRFBs at high volume pedestrian crossings or at priority bicycle route crossings, including shared use paths. RRFBs should not be installed in locations with sight distance constraints that limit the driver's ability to view pedestrians as they approach the crosswalk.
- RRFBs are installed on both sides of the roadway at the edge of the crosswalk. If there is a pedestrian refuge or other median, an additional beacon should be installed in the median.
- Push buttons should be easy to identify and located on the right-hand side of the path. They should be positioned so that bicyclists do not have to dismount to activate.
- Depending on power supply, site conditions, the equipment used, and other variables, maintenance can be minimal. Solar powered RRFBs can help reduce costs.

Planning Cost Estimates

- $25,000 each

High-intensity Activated CrossWalK (HAWK)



A High-intensity Activated crossWalK (HAWK) functions similarly to a full traffic signal and offers the highest degree of motorist stopping through the use of a red signal indication. HAWKs are used to improve non-motorized crossings of major streets in locations where side street volumes do not support installation of a conventional traffic signal or where there are concerns that a conventional signal will encourage additional motor vehicle traffic on the minor street. HAWKs may also be used at midblock crossing locations.

Design Guidance and Considerations

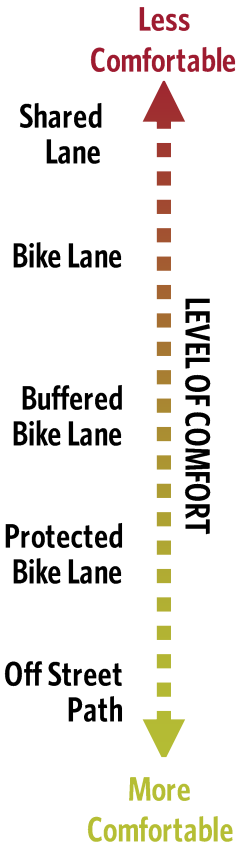
- Suitable for arterial streets where speeds are 30-45 mph and there are three or more lanes of traffic in each direction (or two lanes with a median refuge).
- Push buttons should be easy to identify and located on the right-hand side of the path. They should be positioned so that bicyclists do not have to dismount to activate.
- Each crossing, regardless of traffic speed or volume, requires additional review by a registered engineer to identify sight lines, potential impacts on traffic progression, timing with adjacent signals, capacity and safety.

Planning Cost Estimates

- $300,000 each

Bicycle Facility Types

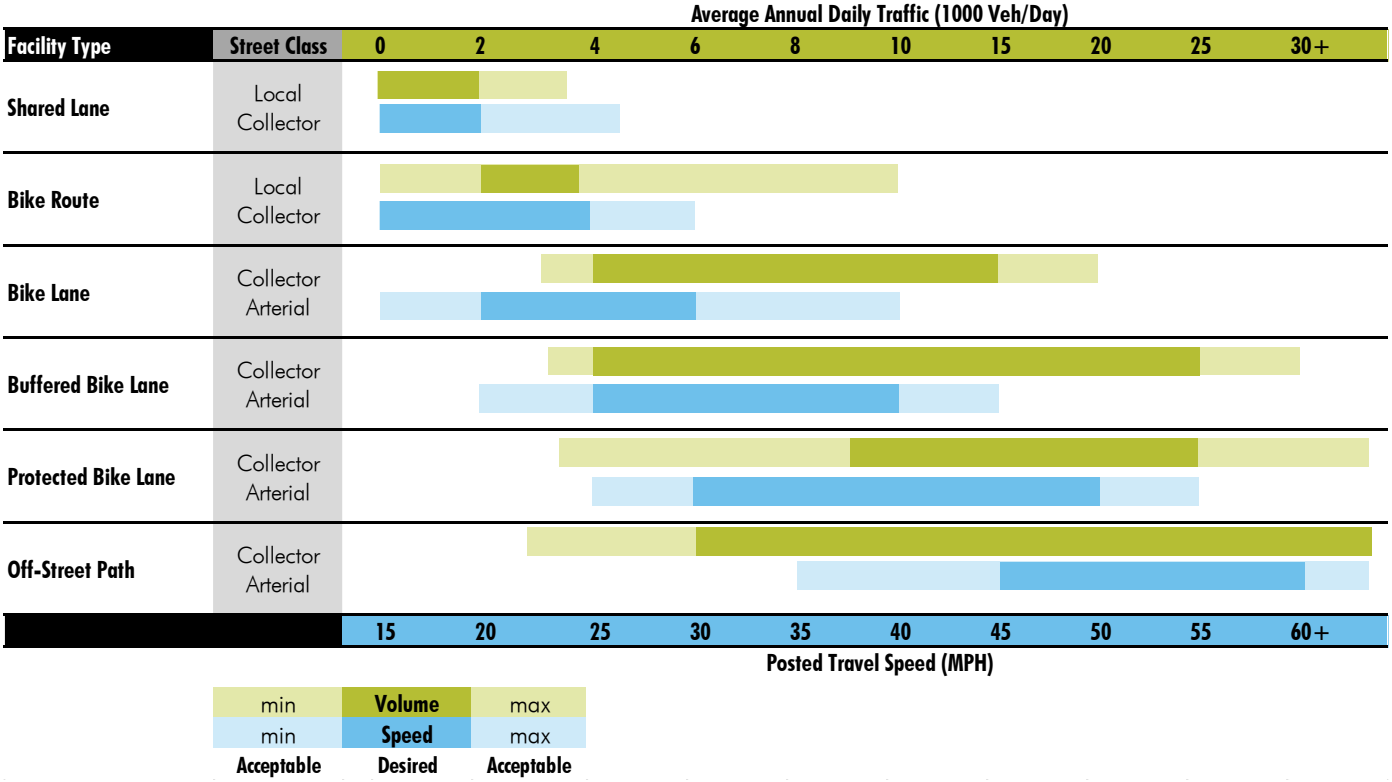
Selecting the best bikeway facility type for a given roadway can be challenging, due to the range of factors that influence bicycle users' comfort and safety. There are no 'hard and fast' rules for determining the most appropriate type of facility for a particular location – roadway speeds, volumes, right-of-way width, presence of parking, adjacent land uses, and expected user types are all critical elements of this decision. User preference varies with bicyclist's skill level, trip purpose, and individual characteristics, and no simple rule exists for determining what users prefer. However, as the level of separation increases, a facility becomes more attractive to a wider range of bicycle users—making bicycling a more viable and preferred transportation mode.



Studies find that the most significant factors influencing bicycle use are motor vehicle traffic volumes and speeds. Additionally, most bicyclists prefer facilities separated from motor vehicle traffic or located on local roads with low motor vehicle traffic speeds and volumes. Because off-street pathways are physically separated from the roadway, they are perceived as safe and attractive routes for bicyclists who prefer to avoid motor vehicle traffic.

Figure 3.1 presents contextual guidance for determining what type of bicycle facility is best provided on a particular roadway based on speeds and volumes. Furthermore, consistent use of treatments and application of bikeway facilities allow users to anticipate whether they would feel comfortable riding on a particular facility and plan their trips accordingly.

Figure 3.1. Bicycle Facility Guidance

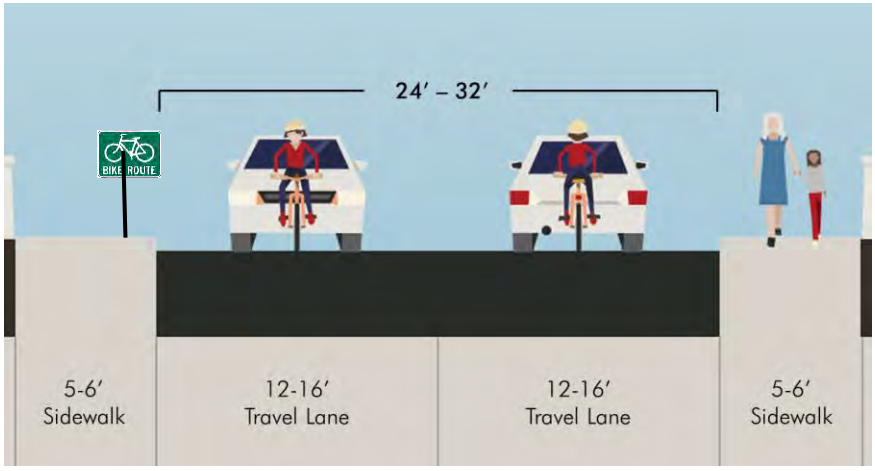


Bicycle Facility Classifications

This section summarizes the active transportation classification system and the associated facility types.

Signed Bike Route

Signed Bike Routes are low-volume streets where motorists and bicyclists share the same space. Traffic calming and other treatments along the corridor reduce vehicle speeds so that motorists and bicyclists generally travel at the same speed, creating a more comfortable environment for all users.

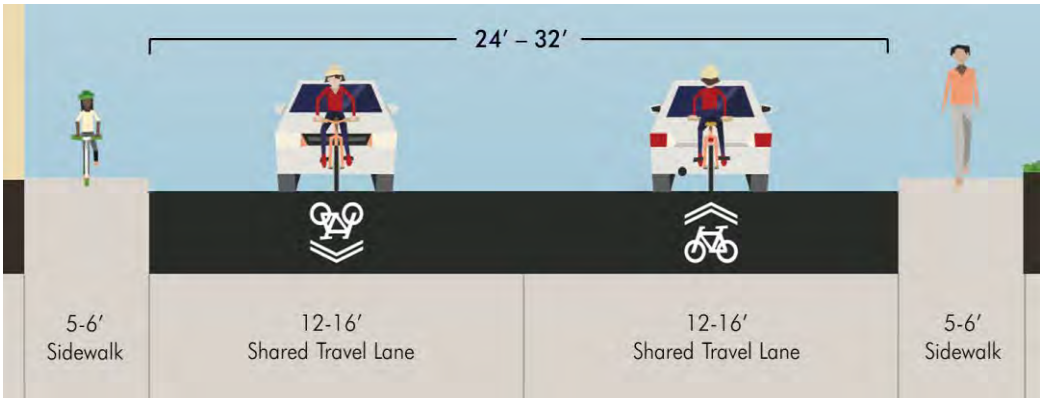


Design Guidance and Considerations

- Bike Route signage (MUTCD D11-1) should be applied at intervals frequent enough to keep bicyclists informed of changes in route direction and to remind motorists of the presence of bicyclists. Signs are commonly placed:
  - At the beginning and end of the bike route
  - At major changes in direction or at intersections with other bike routes
  - At intervals not to exceed ½ mile
- The AASHTO Guide for the Development of Bicycle Facilities recommends wide curb lanes with widths between 12 FT and 13.5 FT. Beyond this width, bike lanes may be provided.
- Bike route signage (MUTCD D11-1) costs $360 per installation.

Shared Lanes

Marked shared roadways use shared lane markings, also known as "sharrows," to designate a shared travel lane. Shared lane markings are used to encourage bicycle travel and remind motorists of the potential presence of bicyclists within a narrow travel lane.

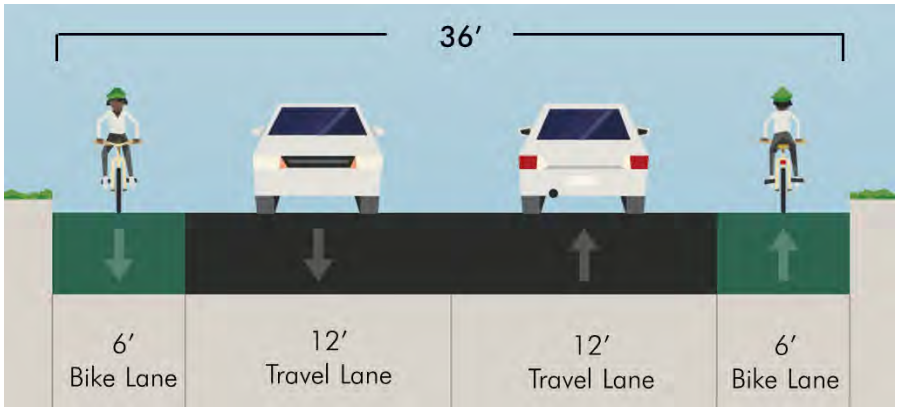


Design Guidance and Considerations

- Shared Lane Marking (SLM) stencils can serve a number of purposes, such as making motorists aware of the need to share the road with bicyclists, showing bicyclists the proper direction of travel, and, with correct placement, reminding bicyclists to keep a safe distance from parked cars to prevent "dooring" collisions.
- Place shared lane markings in the center of the effective travel lane to reduce marking wear and encourage cyclists to occupy the lane outside of the door zone of parked cars.
- Do not use on roads with speed limits higher than 35 mph.
- Markings should be placed immediately after intersections and spaced at 250 FT intervals thereafter.
- SLMs pair well with MUTCD R4-11 "Bikes May Use Full Lane" signs.
- SLMs cost $100 per location.

Bike Lane

A Bike Lane is defined as a portion of the roadway that has been designated by striping, signage, and pavement markings for the preferential or exclusive use of bicyclists.

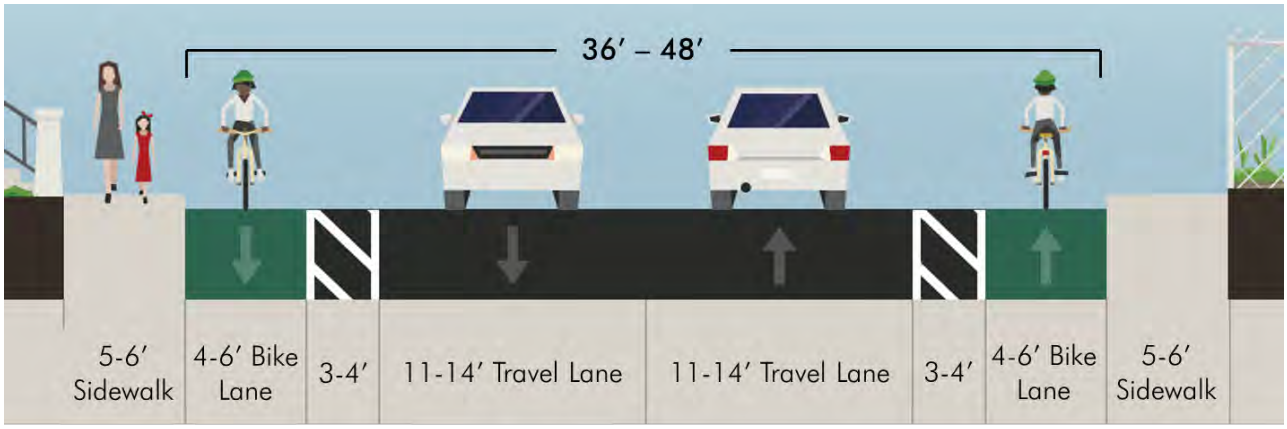


Design Guidance and Considerations

- Bike lanes are most appropriate on arterial and collector streets where higher traffic volumes and speeds warrant greater separation. Bike lanes may also increase a rider's comfort level, discourage bicyclists from riding on the sidewalk, and remind motorists that bicyclists have a right to the road.
- The desirable bike lane width adjacent to a curb face is 6 FT. The desirable ridable surface adjacent to a street edge or longitudinal joint is minimum of 4 FT. In areas where illegal parking in bike lanes is a concern, 5 FT wide bike lanes may be preferred.
- Wider bicycle lanes are desirable in certain situations such as on higher speed arterials (45 mph+) where use of a wider bicycle lane would increase separation between passing vehicles and bicyclists. Appropriate signing and stenciling is important with wide bicycle lanes to ensure motorists do not mistake the lane for a vehicle lane or parking lane.
- A bike lane next to a parking lane shall be at least 5 FT wide, unless there is a marked buffer between them. Wherever possible, minimize parking lane width in favor of increased bike lane width.
- Bike lane striping costs: $12,000 per mile
    - 10% contingency
    - 20% design
    - Does not include additional pavement, stencils, signage

Buffered Bike Lane

Buffered bike lanes are conventional bicycle lanes paired with a designated buffer space, separating the bicycle lane from the adjacent motor vehicle travel lane and/or parking lane.

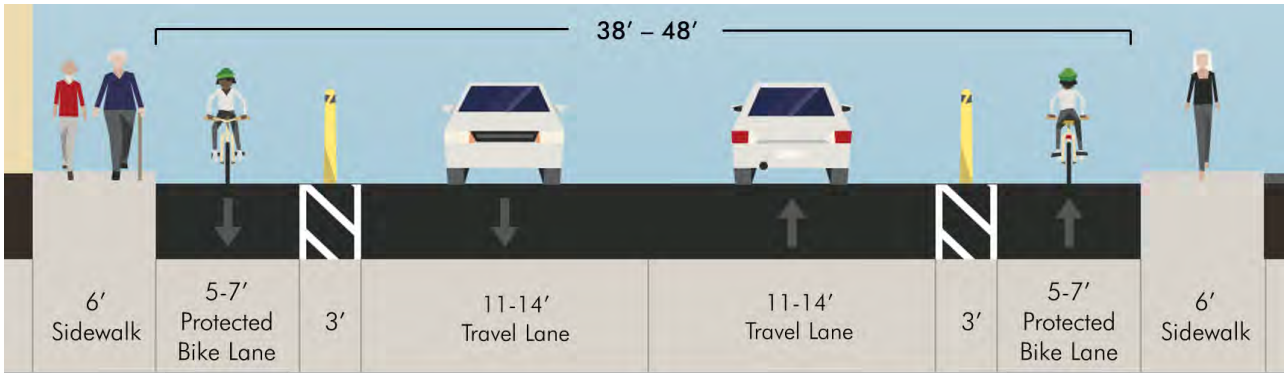


Design Guidance and Considerations

- This treatment is appropriate for bike lanes on roadways with high motor vehicle traffic volumes and speed, adjacent to parking lanes, or a high volume of truck or oversized vehicle traffic.
- Buffers should be at least 2 FT wide. If 3 FT or wider, mark with diagonal or chevron hatching. For clarity at driveways or minor street crossings, consider a dotted line for the inside buffer boundary where cars are expected to cross.
- Where bicyclist volumes are high, bicyclist speed differentials are significant, or where side-by-side riding is desired, the desired bicycle travel area width is 7 FT.
- Parking side buffers are helpful in areas with high turnover parking to reduce the risk of dooring.
- Bicycle lane wording and/or symbol and arrow markings (MUTCD 9C-3) should be used to define the bike lane and designate that portion of the street for preferential use by bicyclists.
- Color may be used at the beginning of each block to discourage motorists from entering the buffered lane.
- Buffered bike lane striping costs: $16,000 per mile
    - 10% contingency
    - 20% design
    - Does not include additional pavement, stencils, signage

Protected Bike Lane

By separating bicyclists from motor traffic, "protected" or physically separated bike lanes can offer a higher level of comfort than conventional bike lanes. The separating element can include flexible delineators, extruded curbs, on-street parking, or other barriers.

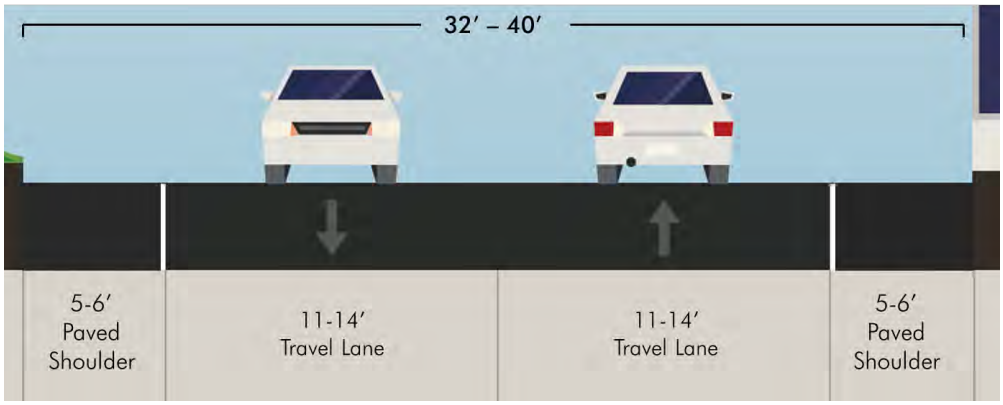


Design Guidance and Considerations

- Protected bike lanes can be accomplished by combining a painted buffer with a physical barrier such as flexible bollards, a landscaped buffer, or a parking lane.
- Dedicated and protected space for bicyclists makes a facility attractive for riders of all levels and ages.
- The minimum desired width should be 5 FT in constrained locations. In areas with high bicyclist volumes or uphill sections, the minimum desired width should be 7 FT to allow for bicyclists passing each other.
- Separation width depends on physical separation method. Minimum separation width next to parking is 3 FT to accommodate opening doors.
- Pavement markings, symbols, and/or arrow markings must be placed at the beginning of the separated bike lane and at intervals along the facility based on engineering judgment to define the bike direction.
- Protected bike lanes should be maintained in order to be free of potholes, broken glass, and other debris.
- Costs will ultimately include the striping both a bike lane, a buffer zone, and costs of the physical separator. Costs vary greatly depending on the type of separation provided. Flexible bollards typically range from $100 to $600 per pole.

Paved Shoulder

Typically found in less dense or unincorporated areas, paved shoulders are paved, striped shoulders wide enough for bicycle travel.



Design Guidance and Considerations

- Appropriate on roads with moderate to high volumes and speeds and on roadways with a large amount of truck traffic.
- Paved shoulders are generally used by commuter and long-distance recreational riders, rather than families with children or more inexperienced riders.
- Shoulder width of at least 5 FT recommended, but additional space should be provided on roads if there are higher levels of bicycle usage, if motor vehicle speeds are high or if there is a higher percentage of truck and bus traffic.
- If the shoulder has rumble strips designed to alert swerving motorists, there should still be at least four feet of "rideable" surface for bicyclists. Periodic gaps in the rumble strips should also be provided to allow bicyclists to travel across the rumble strip as needed.
- Where paved shoulders are present, accommodations should be made for bicyclists through the intersection.
- Costs: $30,000 per mile

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